

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

CK

DATE _____

2017-04-05

X891/X893 MLB Top: EVT

LAST_MODIFICATION=Mon Apr 3 13:03:06 2017

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51	80	RADIOS		06/04/2015

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-02221	1	SCH,MLB_TOP,X891	SCH	NO	COMMON
828-00863	1	PCB,MLB_TOP,X891	PCB	NO	COMMON

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB_TOP, 639-84583, ULTIMATE)	EEEE_32WJ	NO	ULTIMATE
825-7691	1	EEEE FOR (MLB_TOP, 639-83499, EXTREME)	EEEE_HP26	NO	EXTREME

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00358	1	SKYE+3GB, B0, M, DEV	U1000	CRITICAL	COMMON

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00359	339S00358	BOM_TABLE_ALTS	U1000	DDR-H, 3G, B0
339S00360	339S00358	BOM_TABLE_ALTS	U1000	DDR-S-20, 3G, B0
339S00361	339S00358	BOM_TABLE_ALTS	U1000	DDR-S-18, 3G, B0

NAND

Ultimate

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00287	1	HYNIX, 3DV3, ULTIMATE	U2600	CRITICAL	ULTIMATE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00284	335S00287	BOM_TABLE_ALTS	U2600	TOSHIBA, 1Z, ULTIMATE
335S00285	335S00287	BOM_TABLE_ALTS	U2600	TOSHIBA, 8IC3S, ULTIMATE
335S00286	335S00287	BOM_TABLE_ALTS	U2600	SANDISK, 8IC3S, ULTIMATE
335S00288	335S00287	BOM_TABLE_ALTS	U2600	SAMSUNG, 3DV4, ULTIMATE

Extreme

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00240	1	HYNIX, 3DV3, EXTREME	U2600	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00228	335S00240	BOM_TABLE_ALTS	U2600	TOSHIBA, 8IC3S, EXTREME
335S00247	335S00240	BOM_TABLE_ALTS	U2600	SANDISK, 8IC3S, EXTREME
335S00276	335S00240	BOM_TABLE_ALTS	U2600	SAMSUNG, 3DV4, EXTREME

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, Kyocera
138S00150	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, SEMCO
138S00151	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	BOM_TABLE_ALTS	ALL	0402, 16uF@1V, Kyocera
138S00163	138S00144	BOM_TABLE_ALTS	ALL	0402, 16uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	BOM_TABLE_ALTS	ALL	0201, 3uF@1V, Kyocera
138S00164	138S00139	BOM_TABLE_ALTS	ALL	0201, 3uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00145	138S00146	BOM_TABLE_ALTS	ALL	0402, 5.1uF@3V, Kyocera
138S00165	138S00146	BOM_TABLE_ALTS	ALL	0402, 5.1uF@3V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, Kyocera
138S00142	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, SEMCO
138S00166	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, Taiyo

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

Global Ferrites

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00194	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 1500HM, TDK
155S00200	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 1500HM, TY

CRITICAL PART#	COMMENT
155S00610	FERR BD, 1500HM, 01005

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S0764	118S0717	BOM_TABLE_ALTS	ALL	RES, 3.92K, 0.1%, 0201
138S0648	138S0652	BOM_TABLE_ALTS	ALL	CAP, X5R, 4.7uF, 4.3V, 6.65MM, 0402, TAIYO
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 01005
138S00049	138S0831	BOM_TABLE_ALTS	ALL	CAP, CER, X5R, 2.2UF, 20%, 6.3V, 0201

CRITICAL PART#	COMMENT
118S0717	RES, 3.92K, 0.1%, 0201
138S0652	CAP, X5R, 4.7uF, 6.3V, 0.65MM, 0402
138S0706	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 20%
132S0400	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 01005
138S0831	CAP, CER, X5R, 2.2UF, 20%, 6.3V, 0201

Global Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00710	152S00617	BOM_TABLE_ALTS	ALL	IND, MLG, 0.1uH, 20%, 6.1A, 29MM, H= .65, 1608
152S00712	152S00620	BOM_TABLE_ALTS	ALL	IND, MLG, 0.1uH, 20%, 7.2A, 17MM, H=0.8, 2012
152S00713	152S00621	BOM_TABLE_ALTS	ALL	IND, MLG, 0.47uH, 20%, 3.5A, 53MM, H= .65, 2012
152S00714	152S00622	BOM_TABLE_ALTS	ALL	IND, MLG, 1.0uH, 20%, 2.1A, 100MM, H= .65, 2012
152S00716	152S00626	BOM_TABLE_ALTS	ALL	IND, MLG, 1.5uH, 20%, 3.1A, 100MM, H= .65, 2012
152S00717	152S00631	BOM_TABLE_ALTS	ALL	IND, MLG, 1.0uH, 20%, 3.5A, 78MM, H=0.8, 2012
152S00718	152S00632	BOM_TABLE_ALTS	ALL	IND, MLG, 1.0uH, 20%, 3.2A, 68MM, H=0.8, 2016
152S00720	152S00640	BOM_TABLE_ALTS	ALL	IND, MLG, 0.47uH, 3.8A, 55MM, H=0.65MM, 2012
152S00721	152S00641	BOM_TABLE_ALTS	ALL	IND, MLG, 0.47uH, 4A, 48MM, H=0.8MM, 2012
152S00715	152S00623	BOM_TABLE_ALTS	ALL	IND, MLG, 1uH, 3.6A, 68MM, H=0.8MM, 2016
152S00653	152S00651	BOM_TABLE_ALTS	ALL	IND, 1.2uH, 3A, 2016, 0.65Z
152S00649	152S00650	BOM_TABLE_ALTS	L3340, L3341	IND, 0.47uH, 6.6A, 3225, 0.8Z

CRITICAL PART#	COMMENT
152S00617	IND, MLG, 0.1uH, 20%, 6.1A, 29MM, H= .65, 1608
152S00620	IND, MLG, 0.1uH, 20%, 7.2A, 17MM, H=0.8, 2012
152S00621	IND, MLG, 0.47uH, 20%, 3.5A, 53MM, H= .65, 2012
152S00622	IND, MLG, 1.0uH, 20%, 2.1A, 100MM, H= .65, 2012
152S00626	IND, MLG, 1.5uH, 20%, 3.1A, 100MM, H= .65, 2012
152S00631	IND, MLG, 1.0uH, 20%, 3.5A, 78MM, H=0.8, 2012
152S00632	IND, MLG, 1.0uH, 20%, 3.2A, 68MM, H=0.8, 2016
152S00640	IND, MLG, 0.47uH, 3.8A, 55MM, H=0.65MM, 2012
152S00641	IND, MLG, 0.47uH, 4A, 48MM, H=0.8MM, 2012
152S00623	IND, MLG, 1uH, 3.6A, 68MM, H=0.8MM, 2016
152S00651	IND, 1.2uH, 3A, 2016, 0.65Z
152S00650	IND, 0.47uH, 6.6A, 3225, 0.8Z

XTAL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0612	197S0446	BOM_TABLE_ALTS	Y1000	XTAL, 24M, 1612

CRITICAL PART#	COMMENT
197S0446	XTAL, 24M, 1612

Multi-Vendor Criticals

CRITICAL PART#	COMMENT
377S0106	SUPPR, TRANS, VARISTOR, 12V, 33PF, 01005
197S0446	XTAL, 24MHZ, 30PPM, 9.5PF, 50 OHM MAX, 1612
155S0576	FERR BD, 10 OHM, 50%, 750MA, 0.07 DCR, 01005
155S00168	FLTR, NOISE, 65 OHMZ, 3.40MM, 0.7-20KHZ, 0005
138S0979	CAP, CER, X5R, 10UF, 20%, 10V, 0402, H=0.65MM
138S0692	CAP, CER, X5R, 1UF, 20%, 6.3V, 0201
138S0683	CAP, CER, X5R, 1UF, 10%, 25V, 0402
138S0652	CAP, CER, X5R, 4.7UF, 20%, 6.3V, H=0.65MM, 0402
138S00070	CAP, X5R, 4.7UF, 20%, 25V, 0402
138S00014	CAP, CER, 1UF, 20%, 10V, X5R, 0201, H=0.39MM
132S0664	CAP, CER, 0.047UF, 10%, 25V, X5R, 0201
132S0663	CAP, CER, X5R, 1UF, 10%, 25V, 0402
132S06534	CAP, CER, X5R, 0.1UF, 10%, 25V, 0201
132S0436	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 01005
132S0396	CAP, CER, X5R, 1000PF, 10%, 10V, 01005
132S0316	CAP, CER, X5R, 0.1UF, 20%, 6.3V, 01005
132S0304	CAP, CER, X5R, 0.22UF, 20%, 6.3V, 0201
132S0296	CAP, CER, X5R, 1000PF, 10%, 6.3V, 01005
132S0318	CAP, CER, X5R, 820PF, 10%, 10V, 01005

CRITICAL PART#	COMMENT
132S0288	CAP, CER, X5R, 0.1UF, 10%, 10V, 0201
132S0275	CAP, CER, X5R, 470PF, 10%, 10V, 01005
132S0249	CAP, CER, X7R, 220PF, 10%, 10V, 01005
132S0245	CAP, CER, X5R, 0.01UF, 10%, 6.3V, 01005
132S00093	CAP, X5R, 0.022UF, 20%, 6.3V, 01005
132S00025	CAP, CER, X5R, 0.047UF, 20%, 6.3V, 01005
132S00008	CAP, CER, 0.1UF, 10%, 50V, X7R, 0402
131S00083	CAP, CER, NP0/C0G, 220PF, 2%, 50V, 0201
131S00004	CAP, CER, 27PF, 5%, C0G, 25V, 0201
131S00307	CAP, CER, NP0/C0G, 100PF, 5%, 10V, 01005
131S0225	CAP, CER, NP0/C0G, 15PF, 5%, 16V, 01005
131S0223	CAP, CER, NP0/C0G, 27PF, 5%, 10V, 01005
131S0220	CAP, CER, NP0/C0G, 12PF, 5%, 10V, 01005
131S0216	CAP, CER, NP0/C0G, 47PF, 5%, 16V, 01005
131S00053	CAP, CER, C0G, 220PF, 5%, 10V, 01005
118S00068	RES, MF, 1.3 MOHM, 1%, 2000PPM, 1/20W, 0201
117S0055	RES, MF, 1/20W, 2M OHM, 5, 0201, SMD
107S0257	THERMISTOR, NTC, 10K OHM, 1%, B=3435, 01005

Soft-Term Cap Sub BOMs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00155	1	SUBBOM, MLB, TOP, CAP, TYPICAL, X801	SUBBOM_CAP	CRITICAL	COMMON

Agnes Input

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	4	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2970, C2971, C2980, C2981	CRITICAL	SOFT_CAP
138S0831	4	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2970, C2971, C2980, C2981	CRITICAL	TYPICAL_CAP

Agnes Output

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	9	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3060, C3061, C3062, C3063, C3064, C3065, C3066, C3067, C3068, C3069, C3070, C3071, C3072, C3073, C3074	CRITICAL	SOFT_CAP
138S0831	9	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3060, C3061, C3062, C3063, C3064, C3065, C3066, C3067, C3068, C3069, C3070, C3071, C3072, C3073, C3074	CRITICAL	TYPICAL_CAP

Sensors

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3602, C3622	CRITICAL	SOFT_CAP
138S0831	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3602, C3622	CRITICAL	TYPICAL_CAP

RCAM B2Bs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3909, C3925, C4025	CRITICAL	SOFT_CAP
138S0831	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3909, C3925, C4025	CRITICAL	TYPICAL_CAP

Strobe B2B

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4303	CRITICAL	SOFT_CAP
138S0831	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4303	CRITICAL	TYPICAL_CAP

Audio

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4809, C4805	CRITICAL	SOFT_CAP
138S0831	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4809, C4805	CRITICAL	TYPICAL_CAP

Pearl B2B

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4613	CRITICAL	SOFT_CAP
138S0831	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4613	CRITICAL	TYPICAL_CAP

Acorn

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00160	2	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C5641, C5653	CRITICAL	SOFT_CAP
138S0979	2	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KY0	C5641, C5653	CRITICAL	TYPICAL_CAP

CODEC

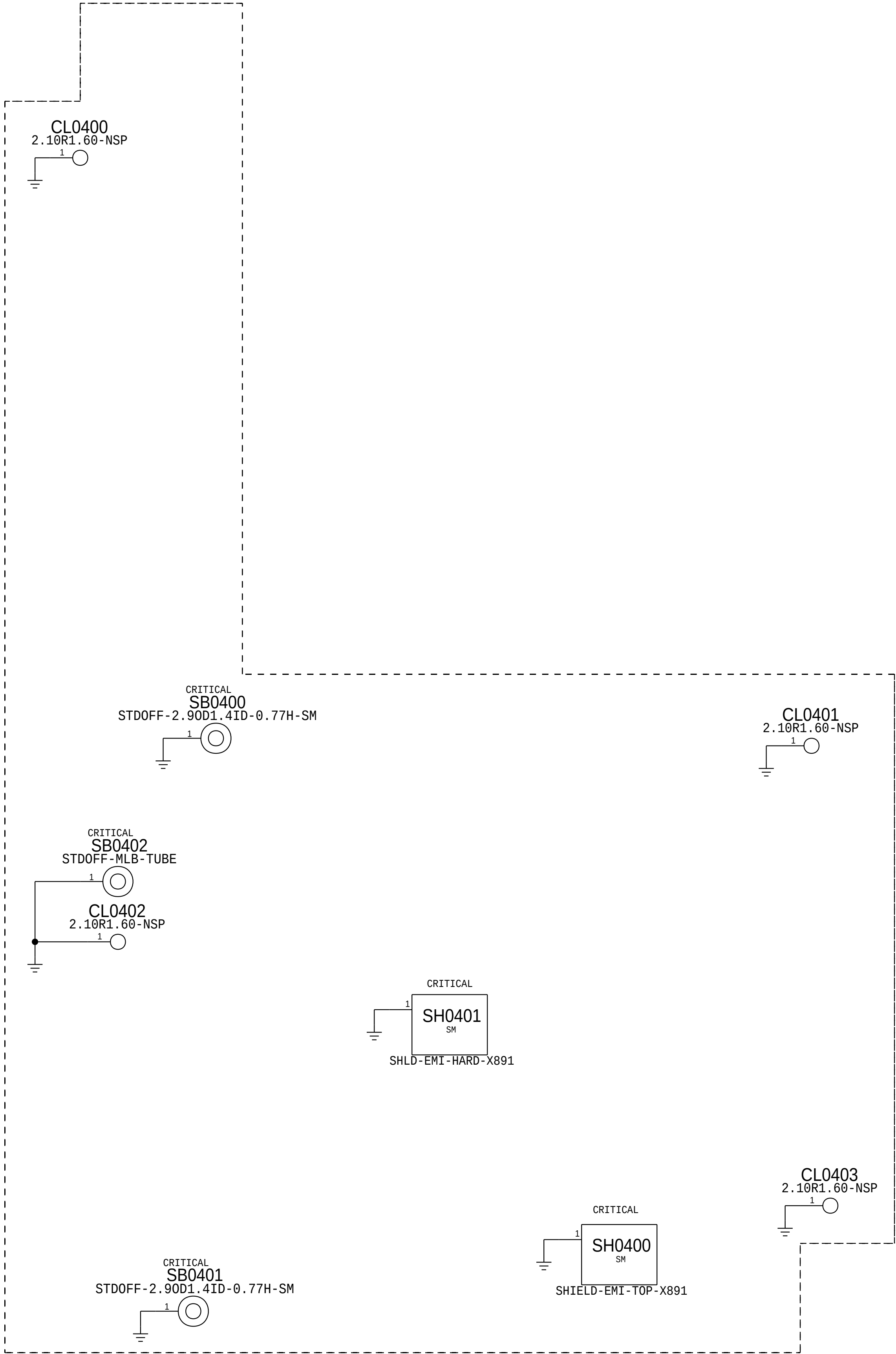
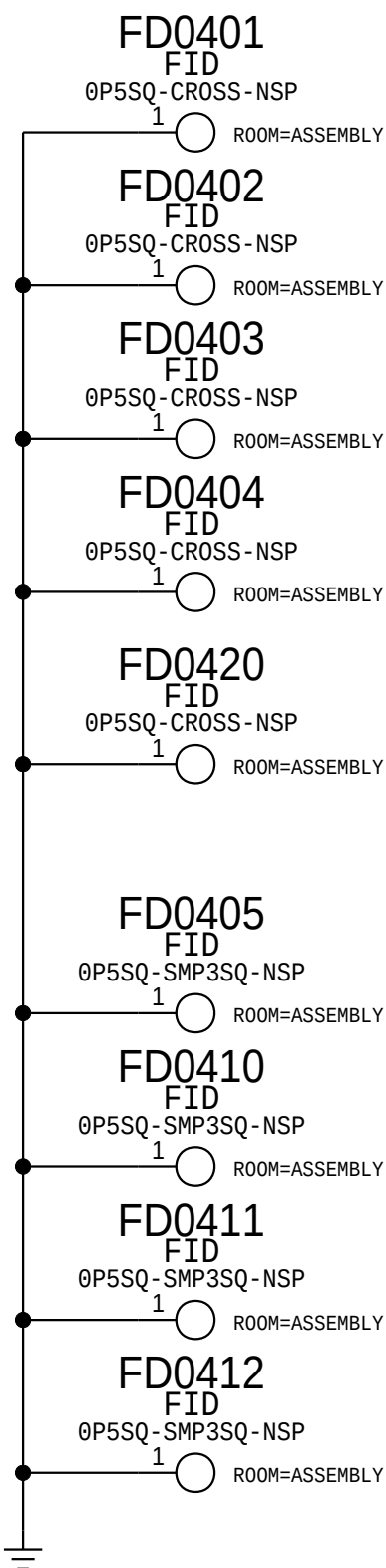
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138S00160	2	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C4811, C4808	CRITICAL	SOFT_CAP
138S0979	2	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KY0	C4811, C4808	CRITICAL	TYPICAL_CAP

Ansel

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00160	1	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C3710	CRITICAL	SOFT_CAP
138S0979	1	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KY0	C3710	CRITICAL	TYPICAL_CAP

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00156	685-00155	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM, MLB, TOP, CAP, SOFT, X801

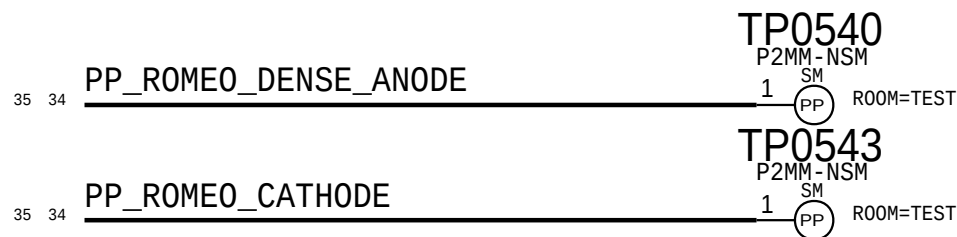
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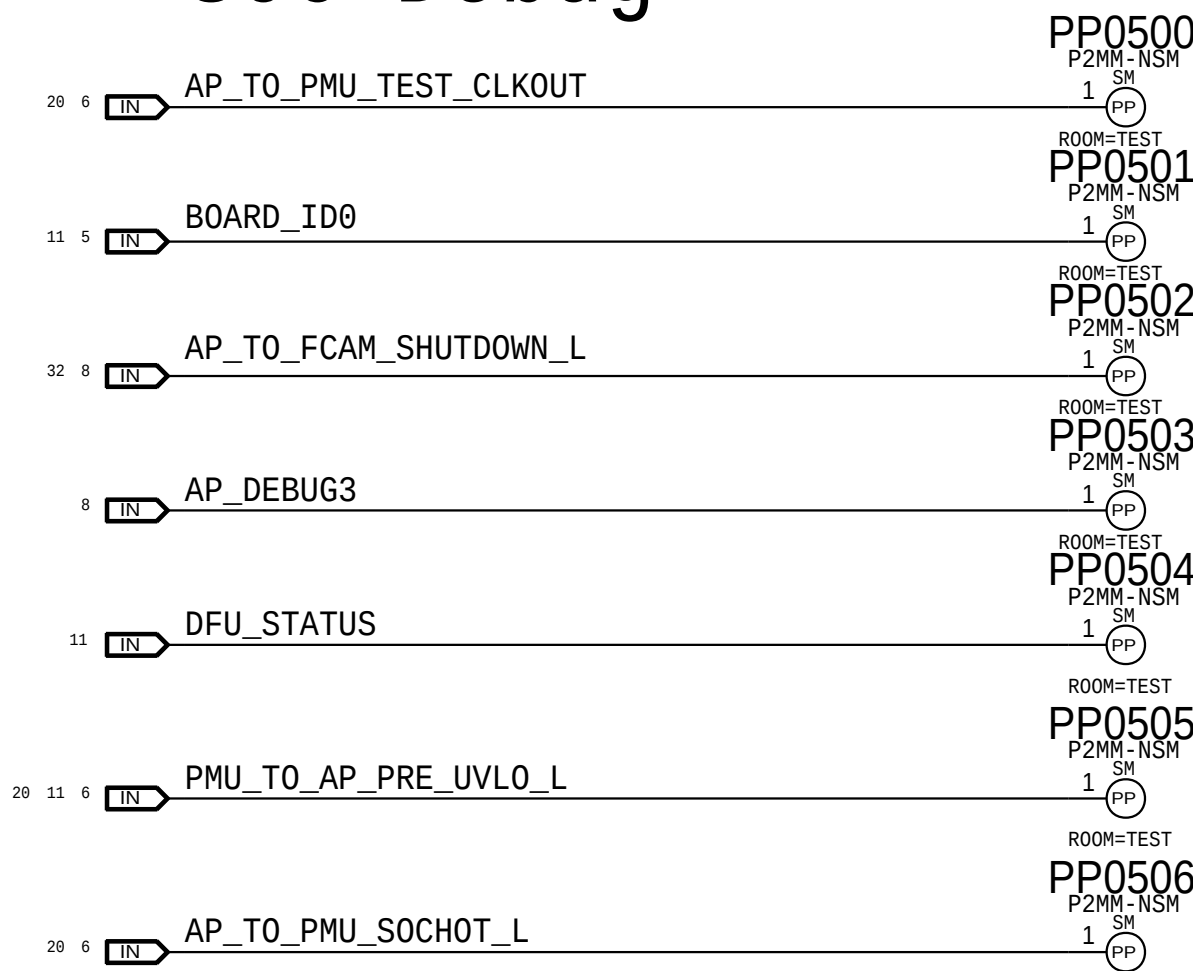
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Probe Points

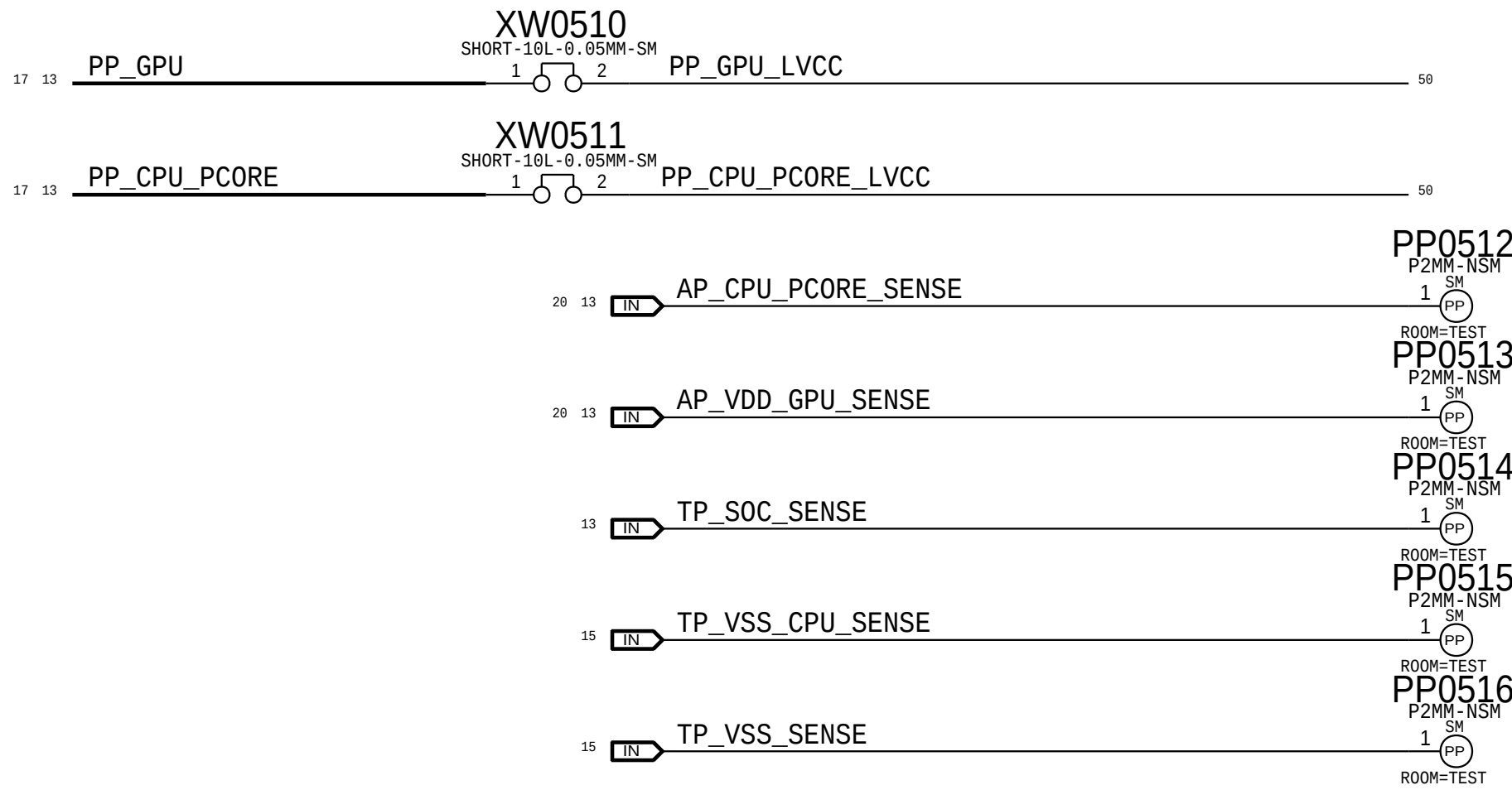
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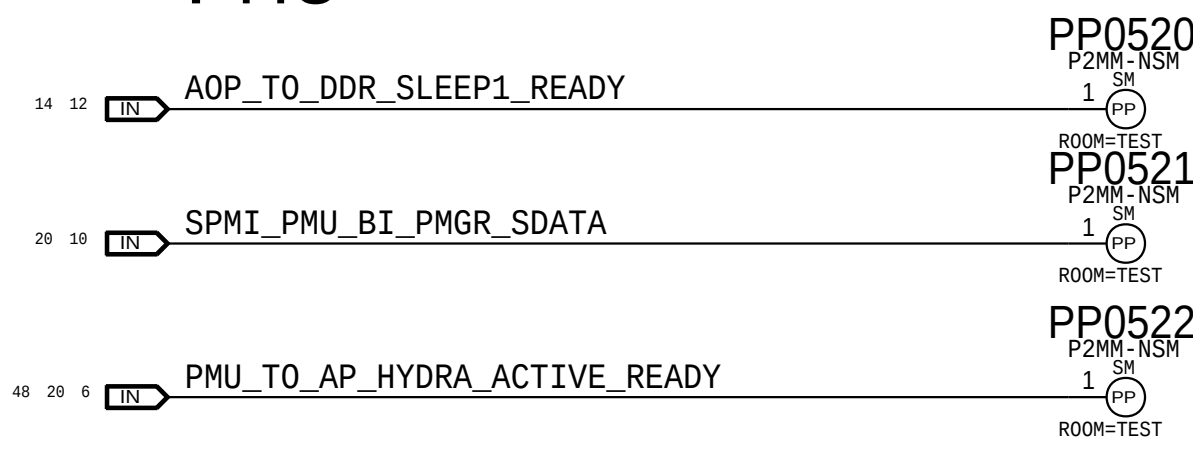
SOC Debug



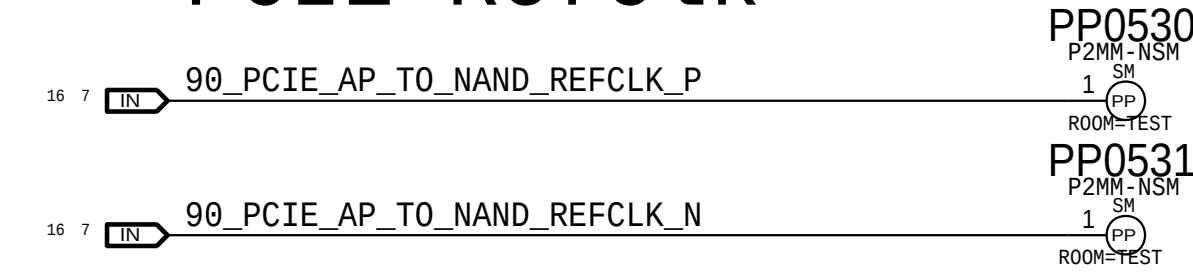
SOC CPU/GPU



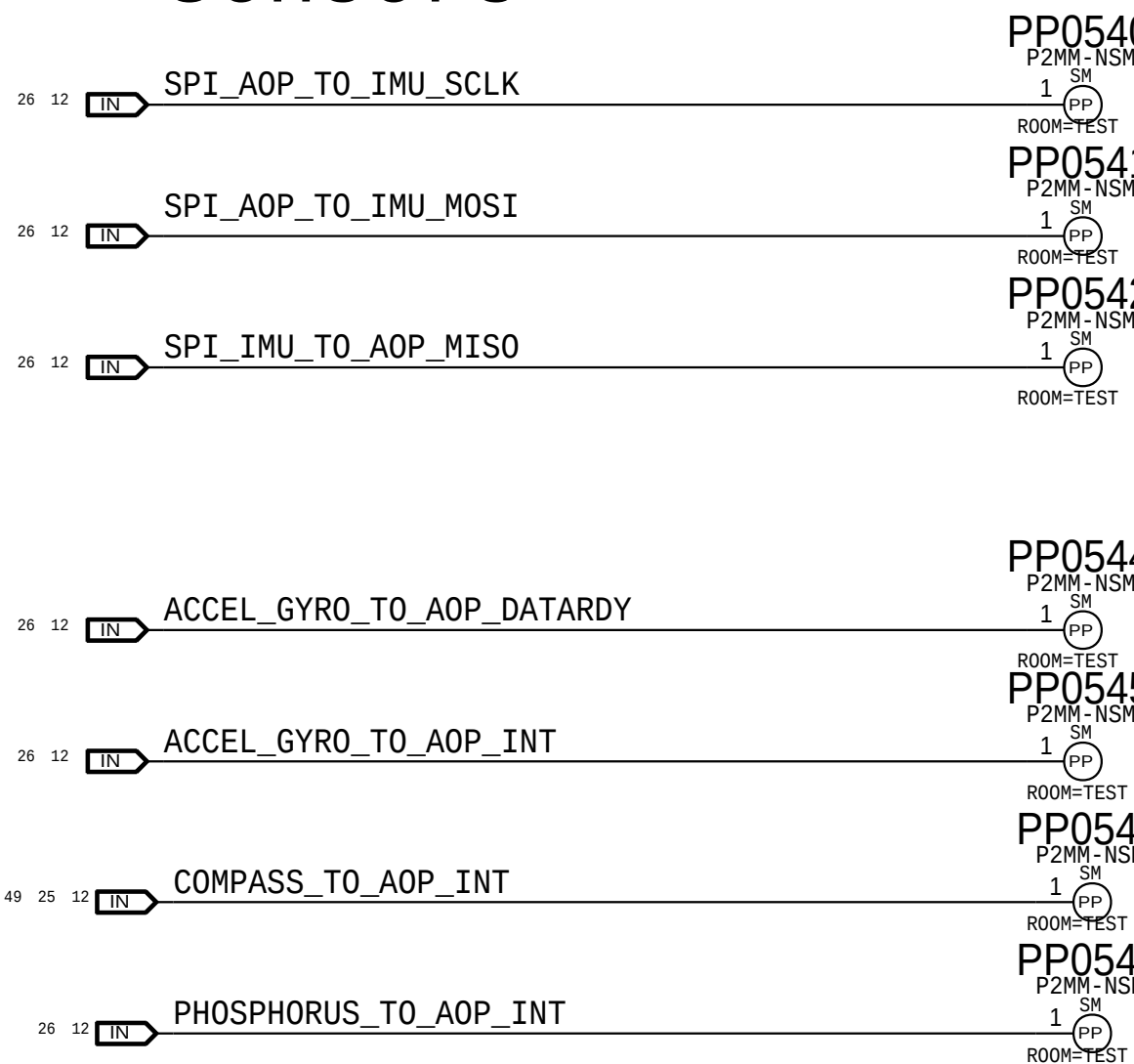
PMU



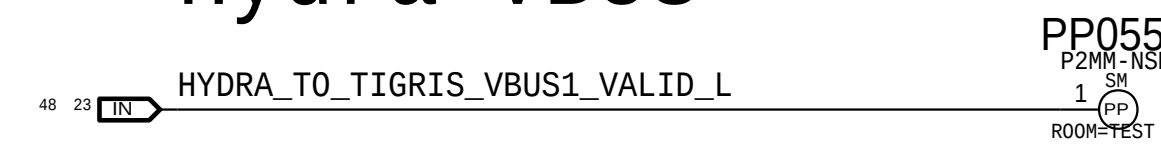
PCIE Refclk



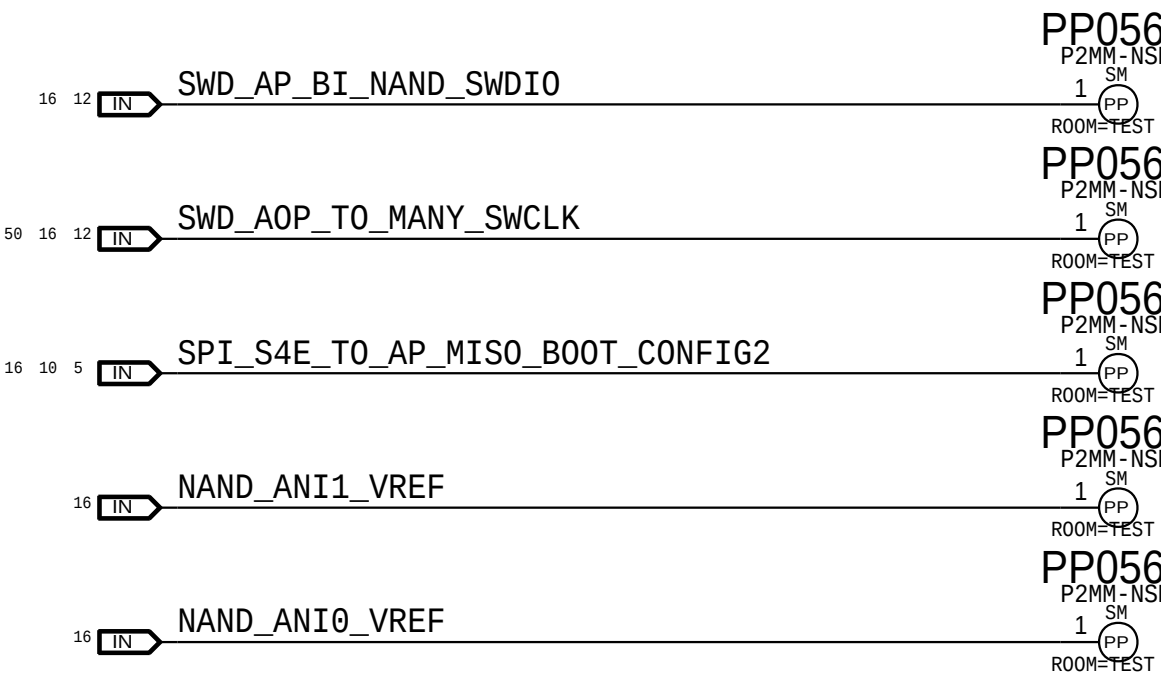
Sensors



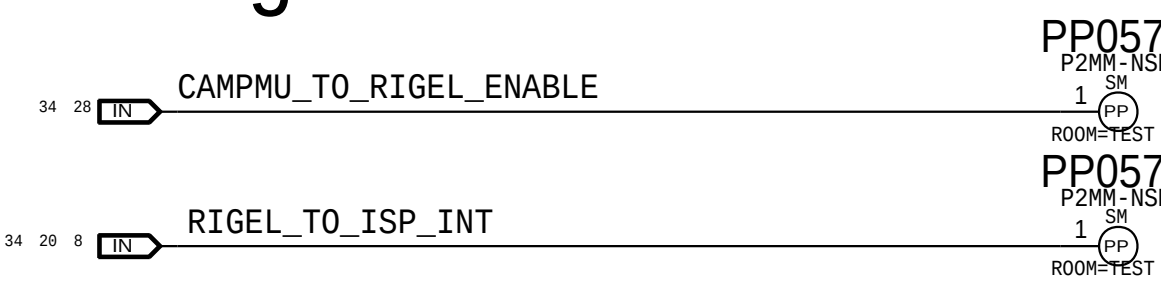
Hydra VBUS



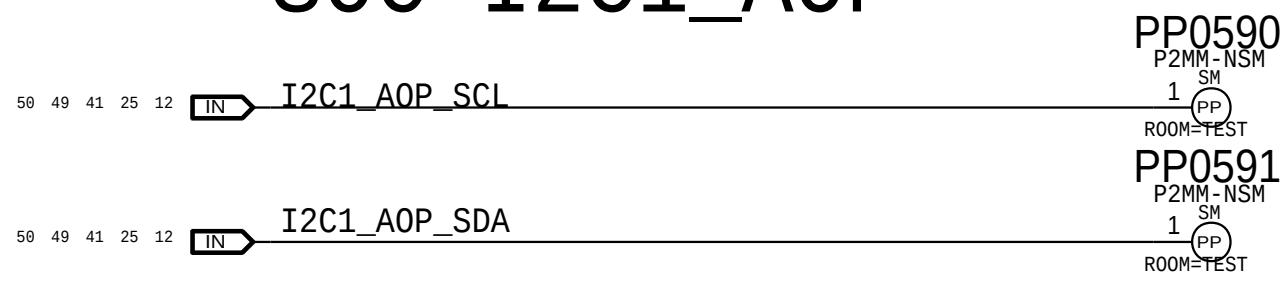
NAND



Rigel



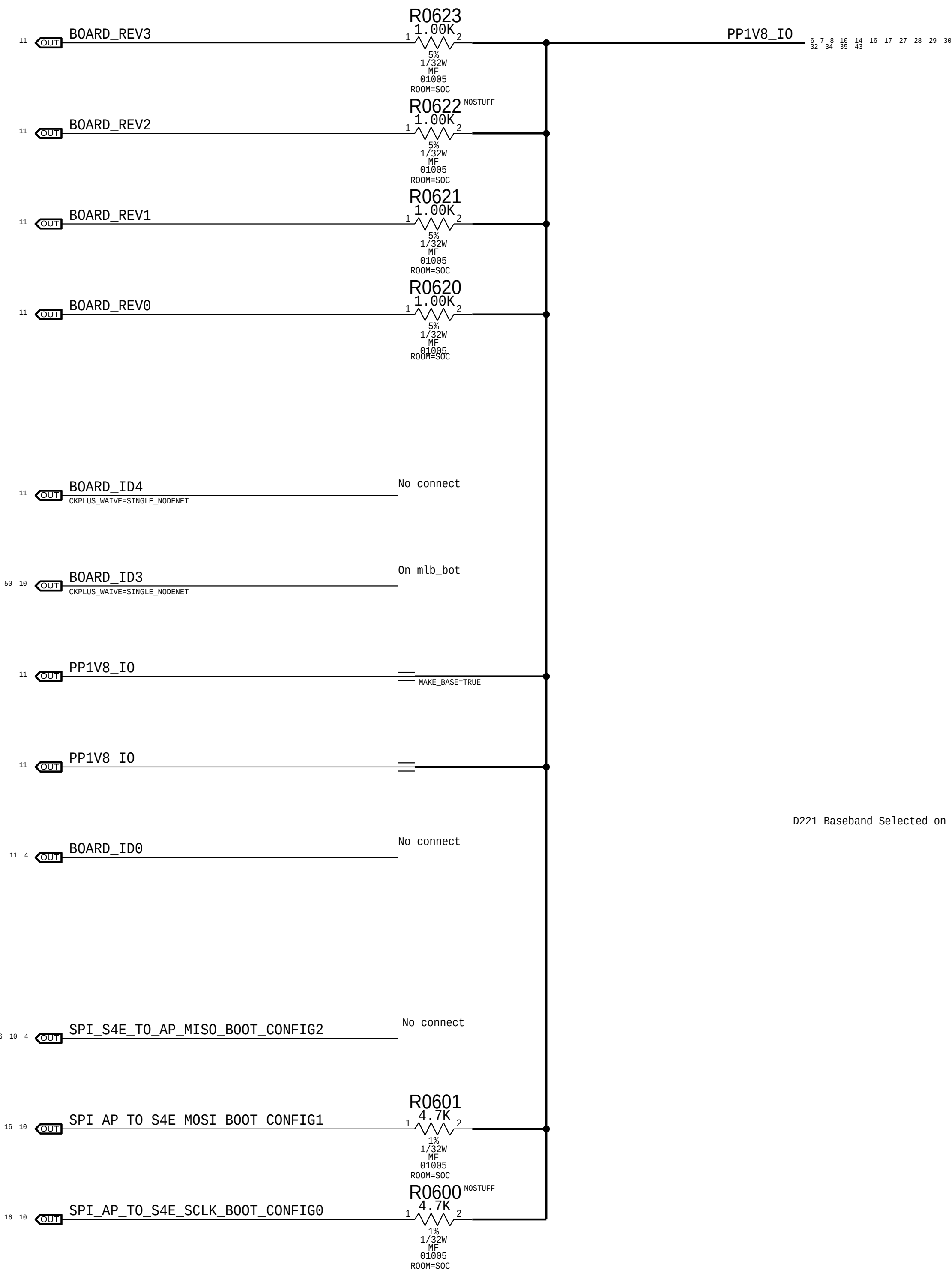
SOC I2C1_AOP



CCG2



BOOTSTRAPPING:BOARD REV
BOARD ID
BOOT CONFIG



Board Rev[3:0]				
Float=Low I PU=High				
	3	2	1	0
	Build Major		Build Minor	
Proto 1	1	1	1	1
Proto 1 (DOE6)	1	1	1	0
Proto 2	1	1	0	1
Proto 3	1	1	0	0
EVT	1	0	1	1
(Spare)	1	0	1	0
Carrier	0	1	1	1
(Spare)	0	1	1	0
DVT	0	0	1	1
(Spare)	0	0	1	0
(Spare)	0	0	0	1
PVT	0	0	0	0

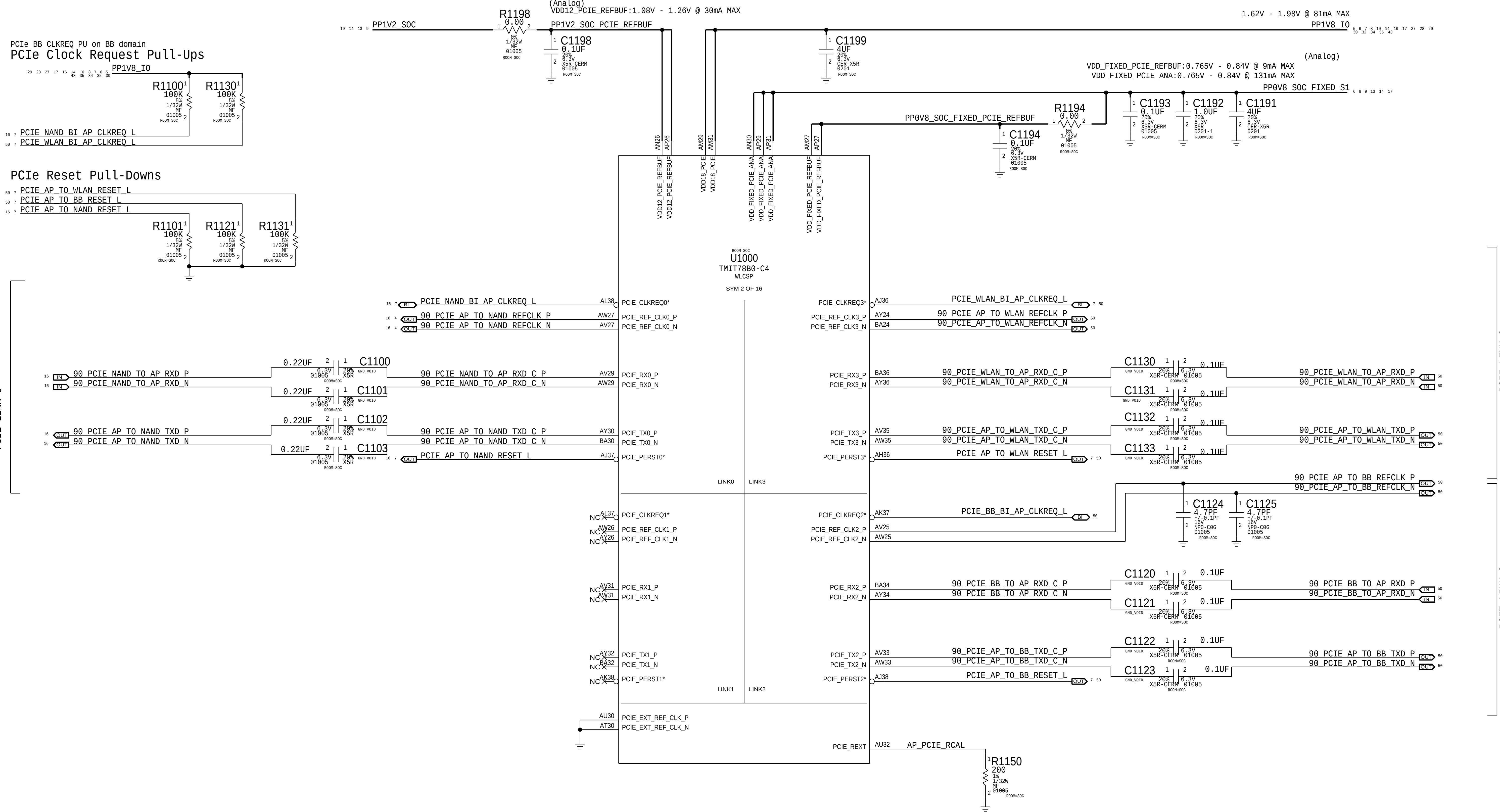
Board ID[4:0]					
Float=Low I PU=High					
	4	3	2	1	0
	X	Baseband	Form Factor		MLB/Dev
D20 MLB	0	0	0	1	0
D20 Dev	0	0	0	1	1
D21 MLB	0	0	1	0	0
D21 Dev	0	0	1	0	1
D22 MLB	0	0	1	1	0
D22 Dev	0	0	1	1	1
D201 MLB	0	1	0	1	0
D201 Dev	0	1	0	1	1
D211 MLB	0	1	1	0	0
D211 Dev	0	1	1	0	1
D221 MLB	0	1	1	1	0
D221 Dev	0	1	1	1	1

Boot Config[2:0]			
Float=Low I PU=High			
	2	1	0
SPI0	0	0	0
SPI0 Test Mode	0	0	1
SPI0 NAND	0	1	0
SPI0 NAND Test	0	1	1
Slow SPI0 Test	1	1	0

8	7
---	---



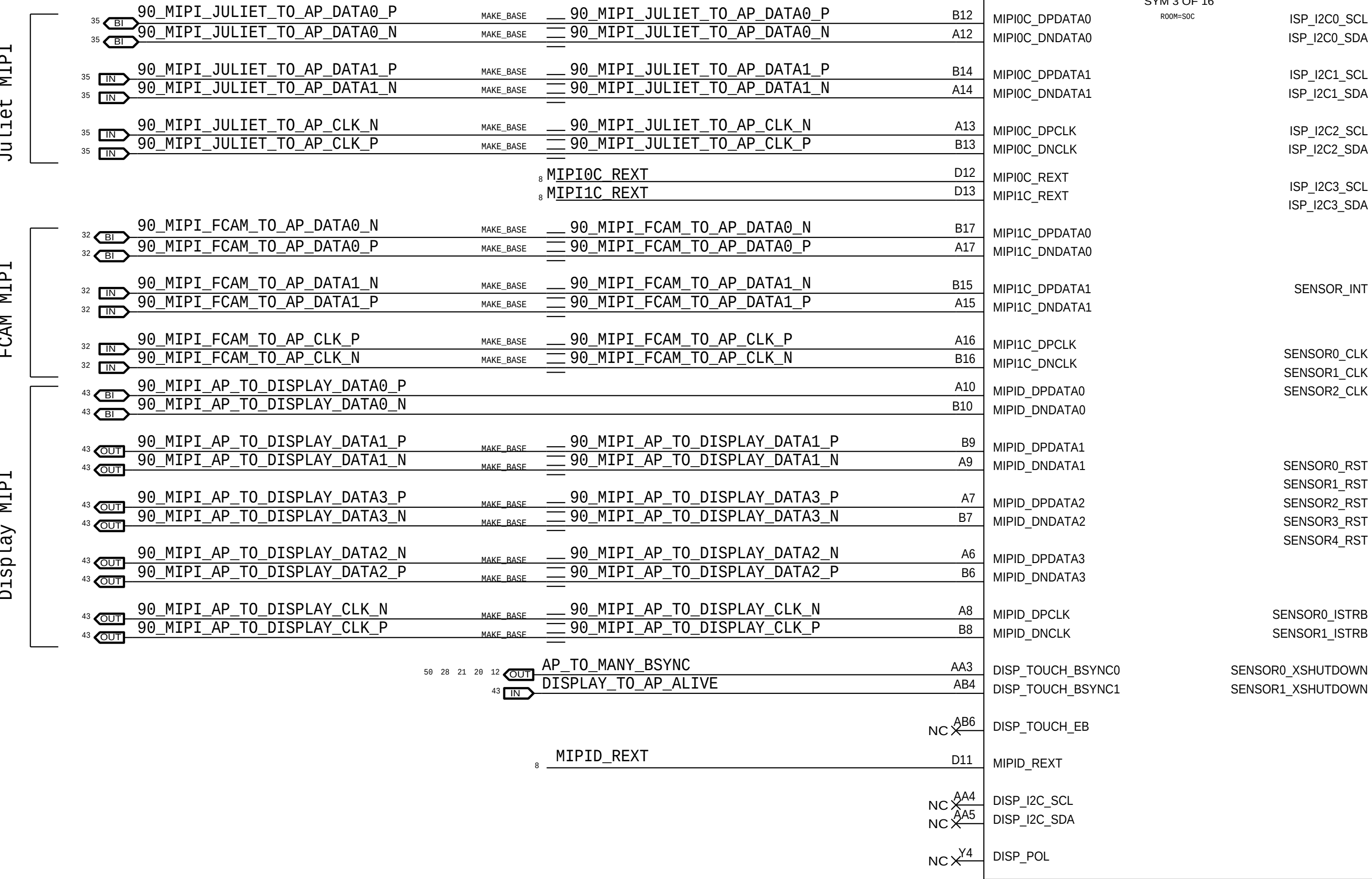
SOC - PCIE INTERFACES



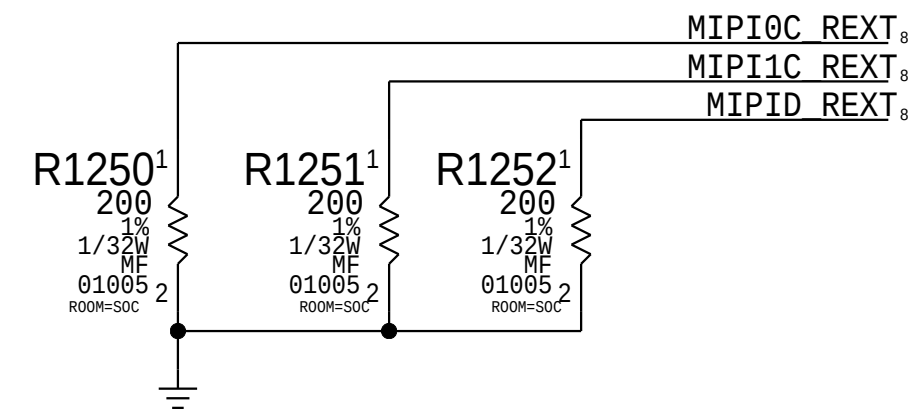
SOC - MIPI & ISP INTERFACES

MIPI Lane & Polarity Swapping

Program	Function	SOC LANE	DISTAL LANE	Lane Swap	Polarity Swap
X891	Display	Clock_P	Clock_N		✓
		Clock_N	Clock_P		
		Lane_0_P	Lane_0_N		
		Lane_0_N	Lane_0_P		
		Lane_1_P	Lane_1_N		
		Lane_1_N	Lane_1_P		
	FCAM	Lane_2_P	Lane_2_N	✓	
		Lane_2_N	Lane_2_P		
		Lane_3_P	Lane_3_N	✓	✓
		Lane_3_N	Lane_3_P		
		Lane_2_P	Lane_2_N	✓	✓
		Lane_2_N	Lane_2_P		
JULIET	Display	Clock_P	Clock_N		✓
		Clock_N	Clock_P		
		Lane_0_P	Lane_0_N		
		Lane_0_N	Lane_0_P		
		Lane_1_P	Lane_1_N		
		Lane_1_N	Lane_1_P		
	FCAM	Lane_2_P	Lane_2_N		
		Lane_2_N	Lane_2_P		
		Lane_3_P	Lane_3_N		
		Lane_3_N	Lane_3_P		
		Lane_2_P	Lane_2_N		
		Lane_2_N	Lane_2_P		

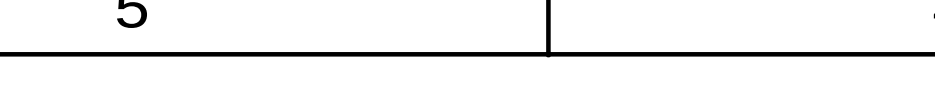
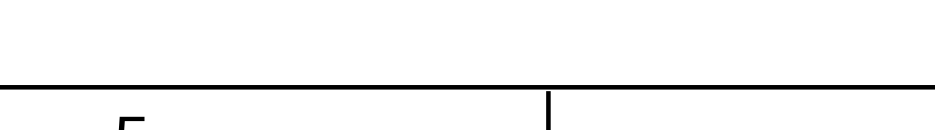
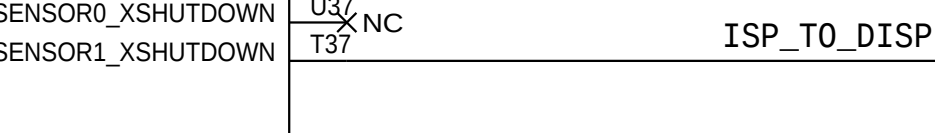
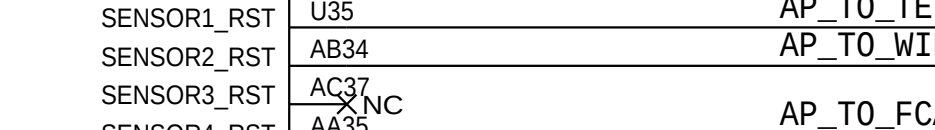
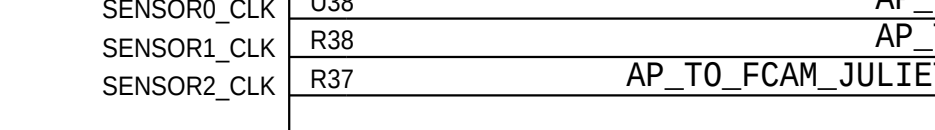
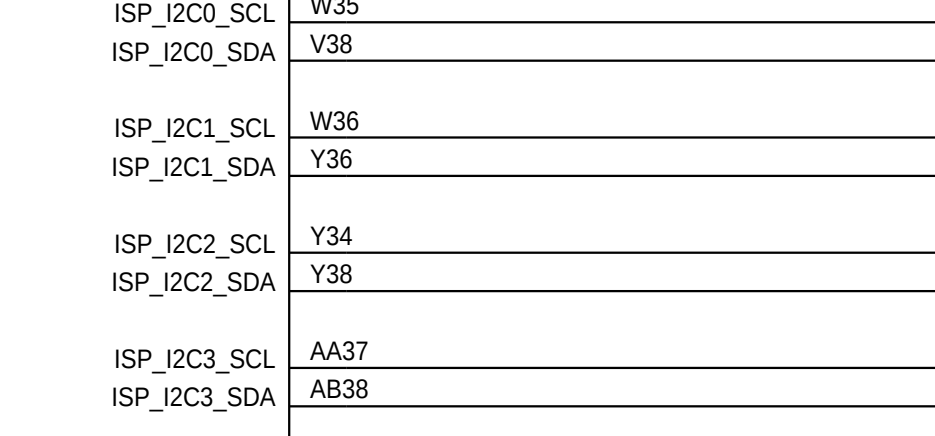
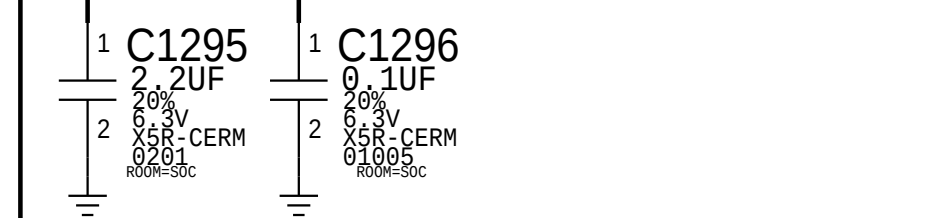


MIPI Reference

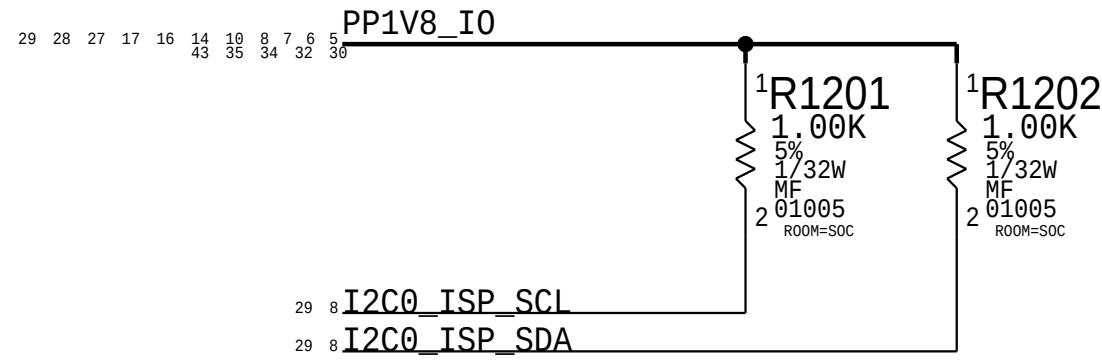


(Analog)
0.765V - 0.84V @ 40mA MAX
PP0V8_SOC_FIXED_S1

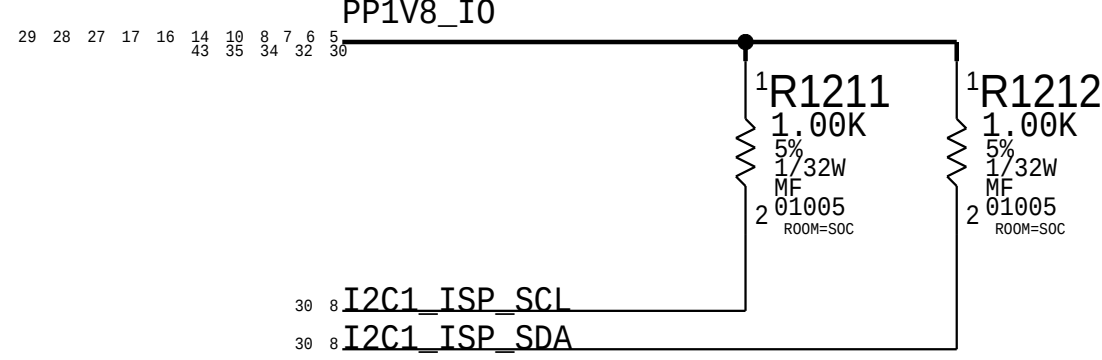
1.62V - 1.98V @ 10mA MAX
PP1V8_I0



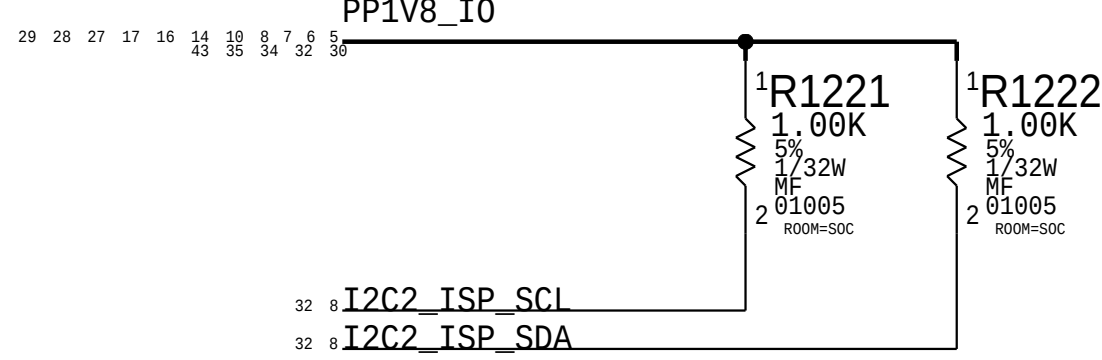
ISP I2C0



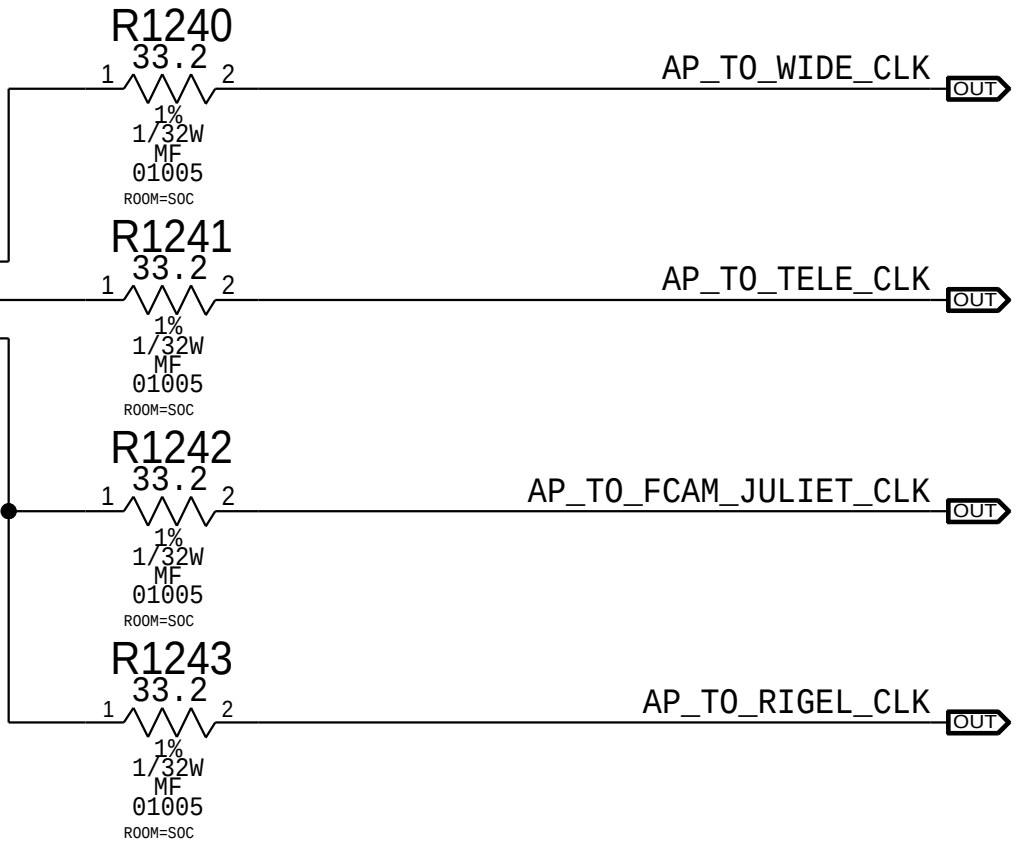
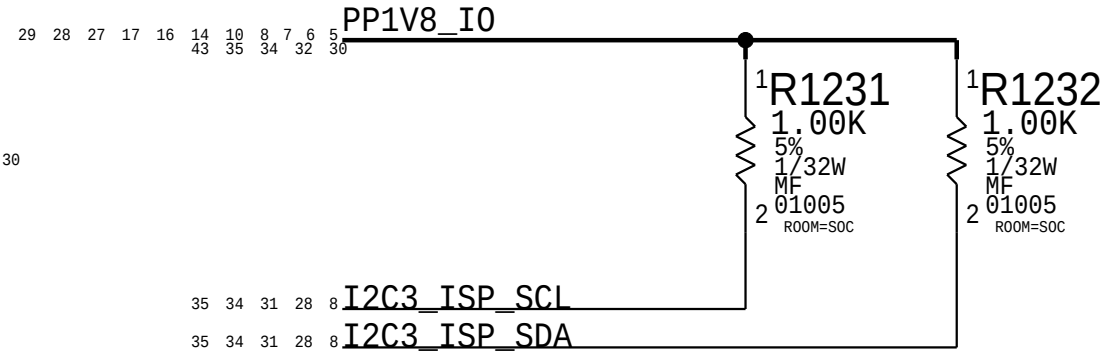
ISP I2C1



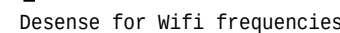
ISP I2C2



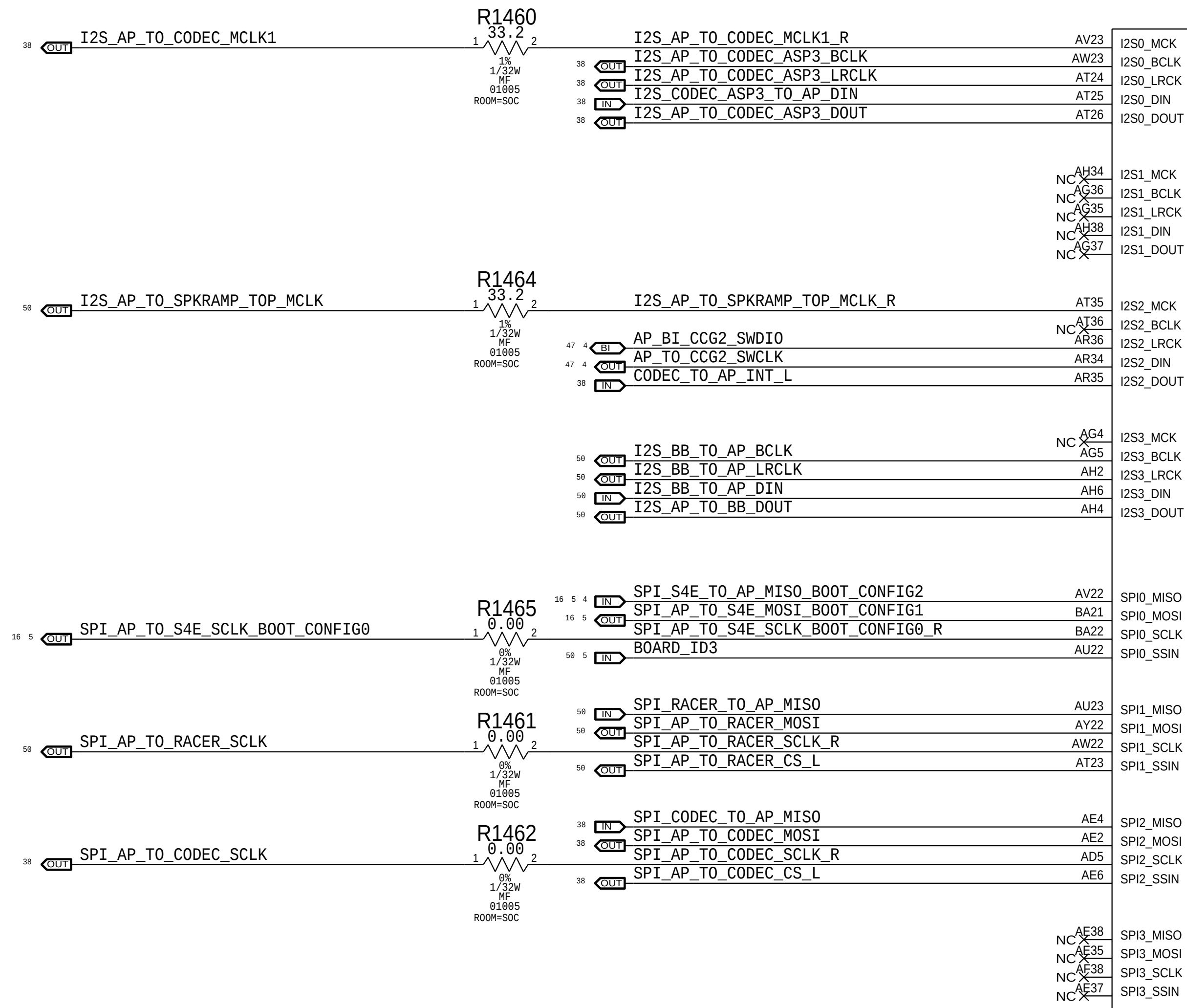
ISP I2C3



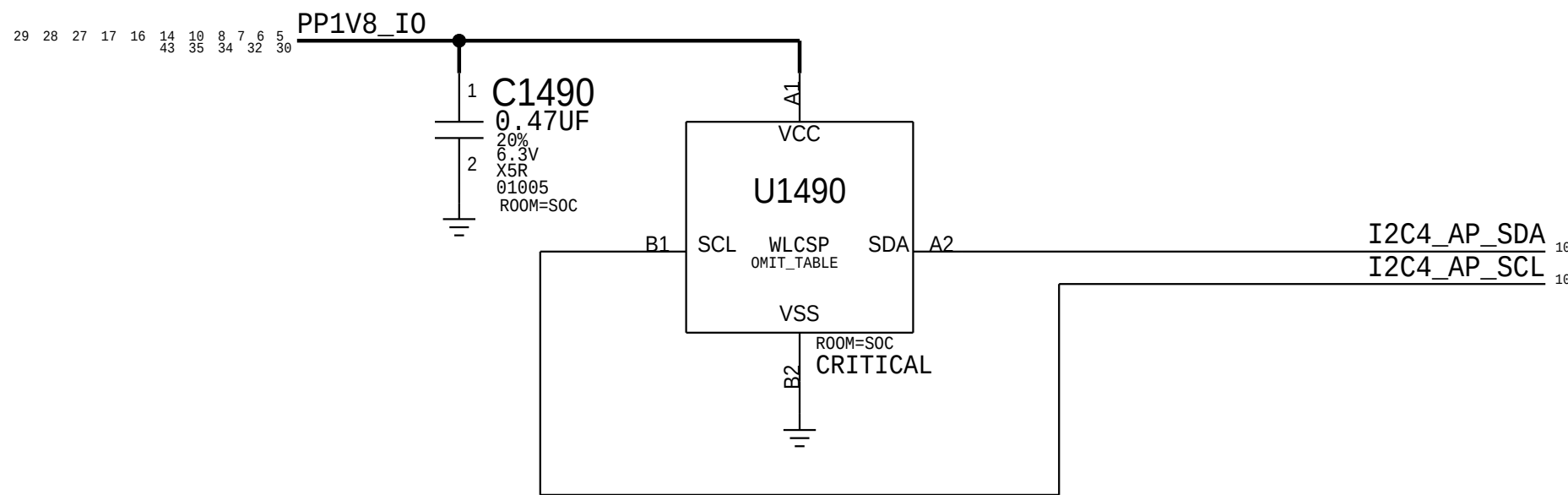
9 0F 51



SOC - SERIAL INTERFACES

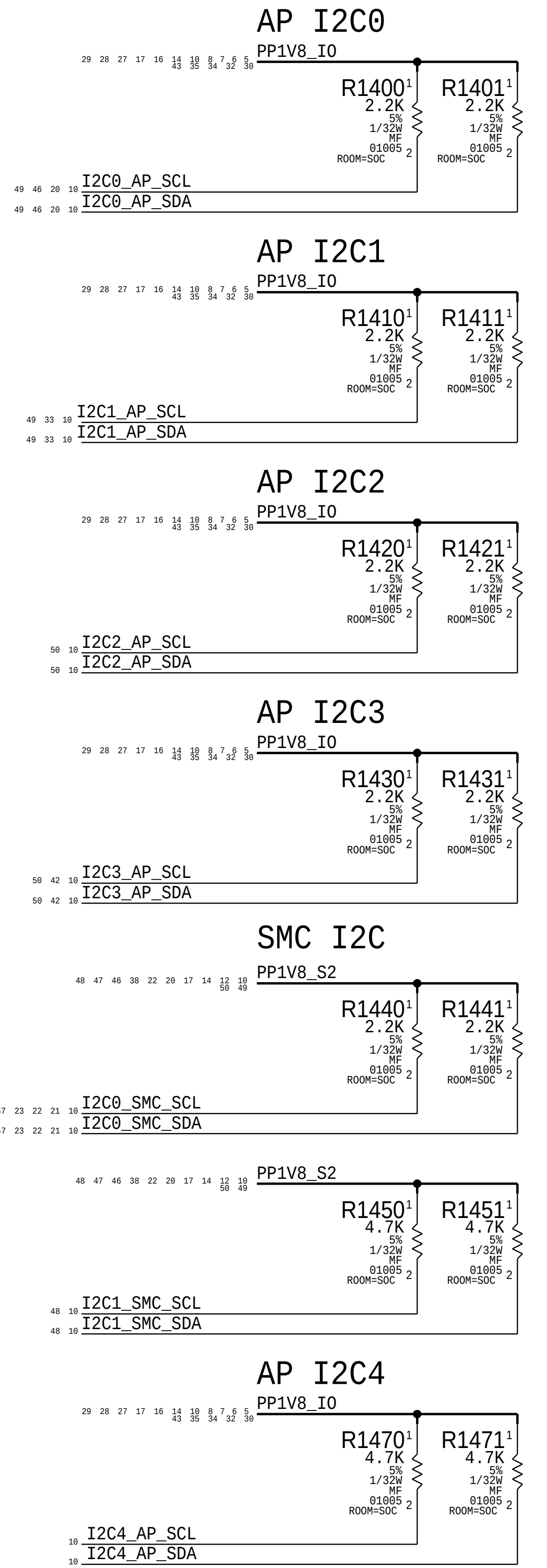


SPI: Route as Daisy-Chain. No T's Allowed
Place series terminations close to SoC Pins

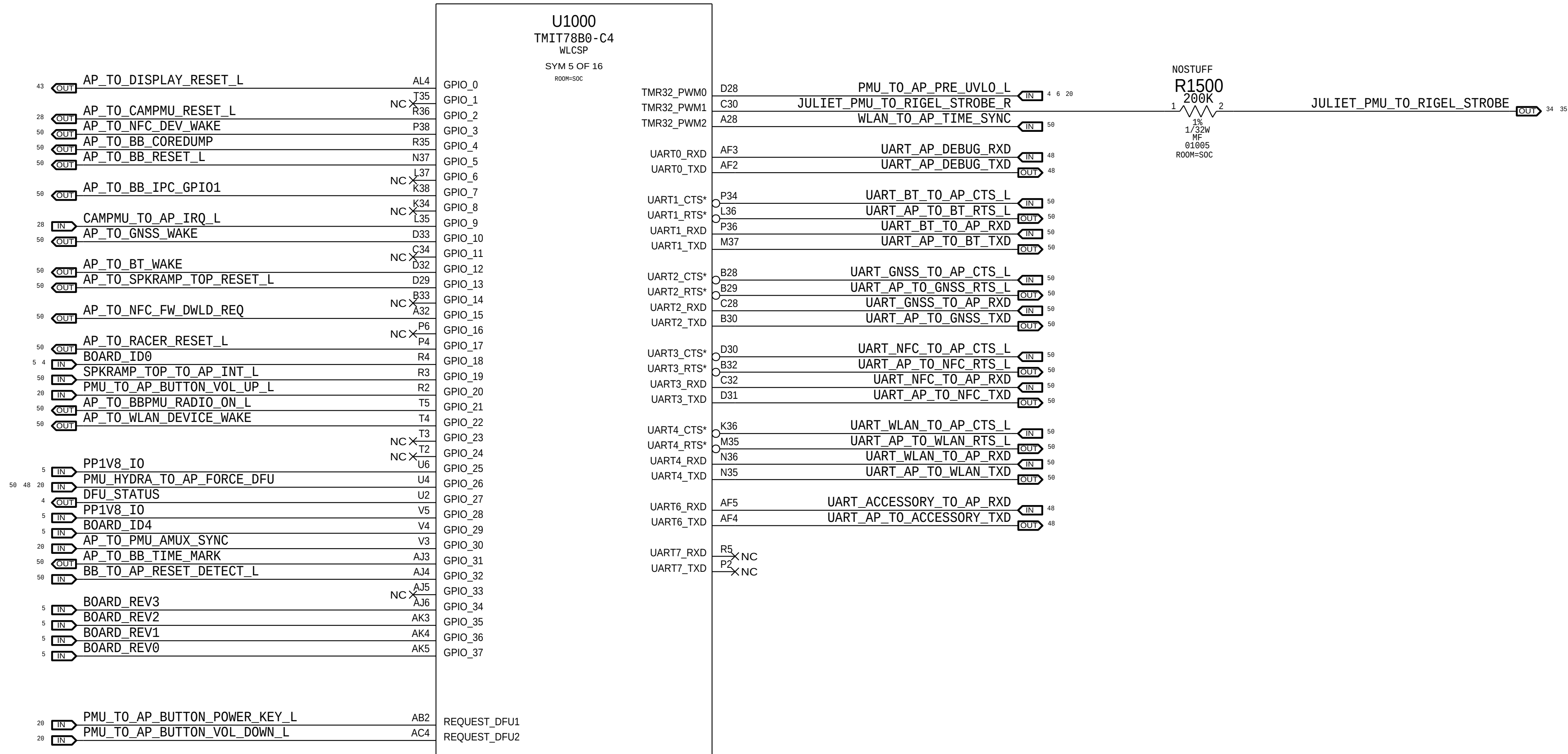


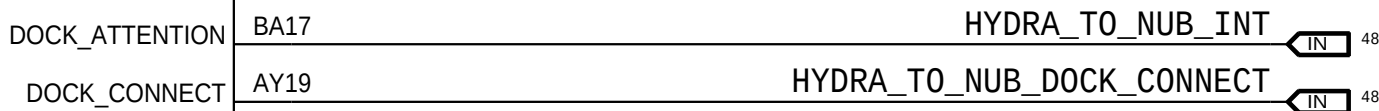
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00234	1	WLCSP	U1490	CRITICAL	COMMON

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00233	335S00234	BOM_TABLE_ALTS	U1490	U1490

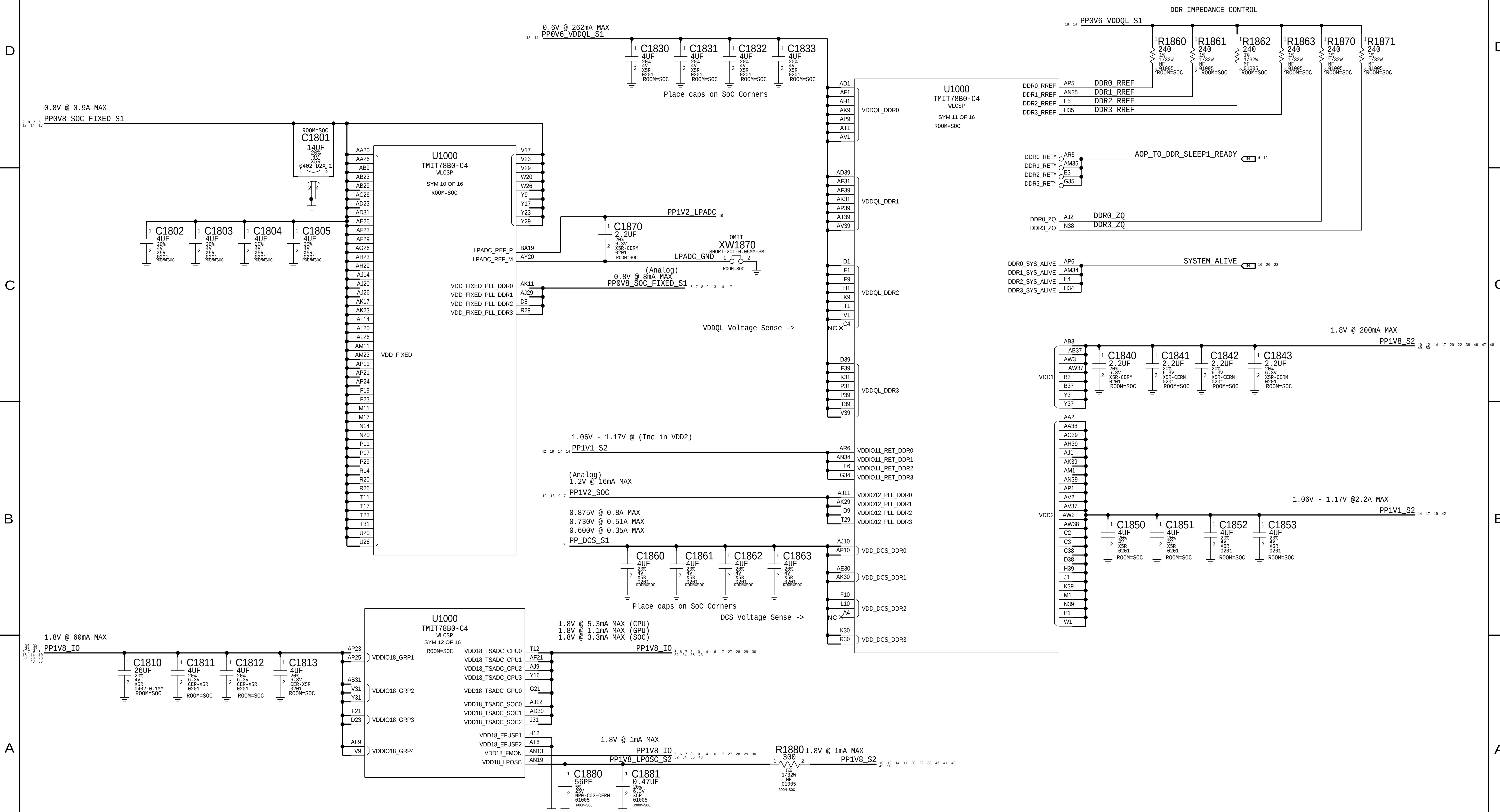


SOC - GPIO INTERFACES

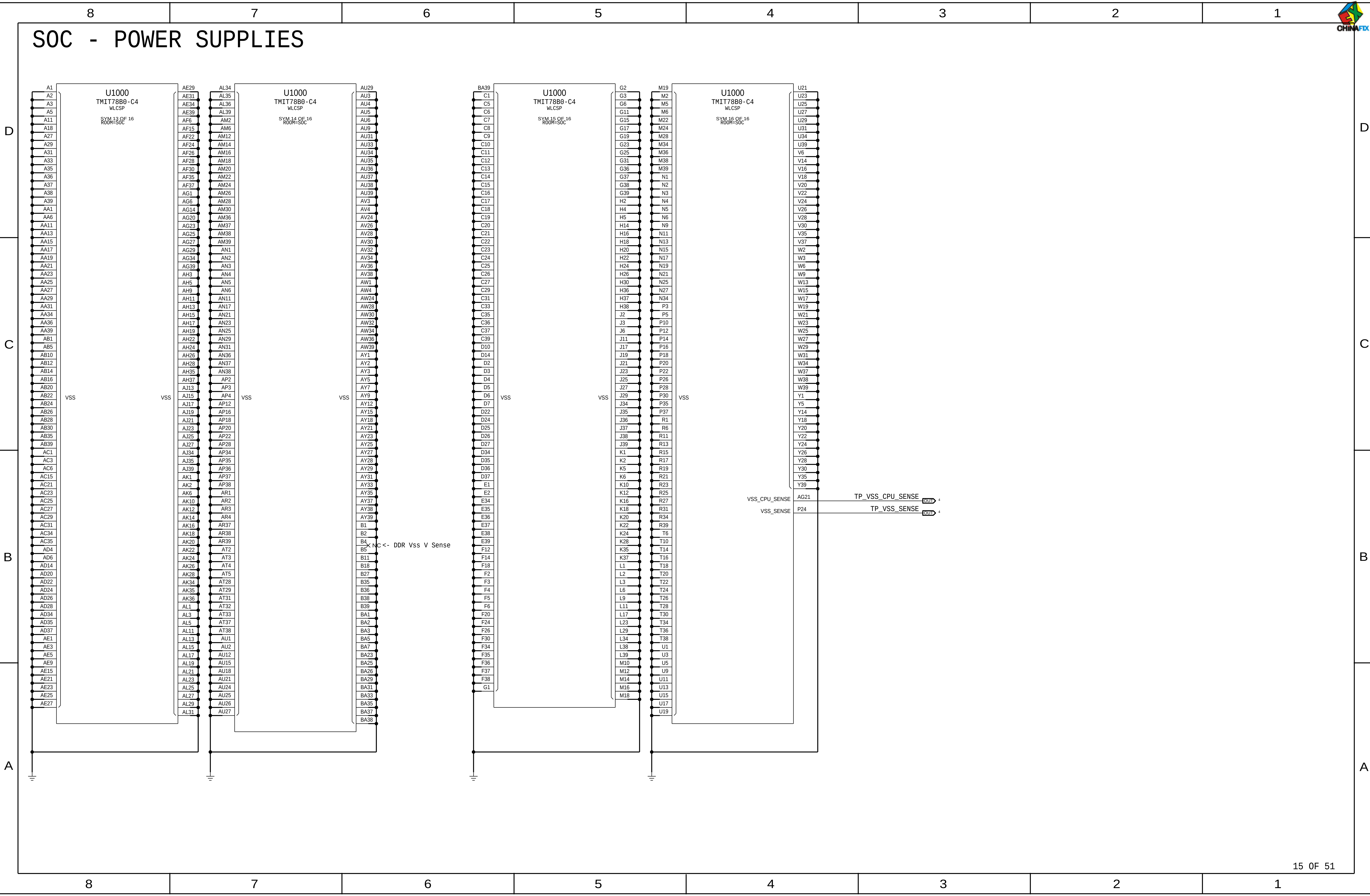




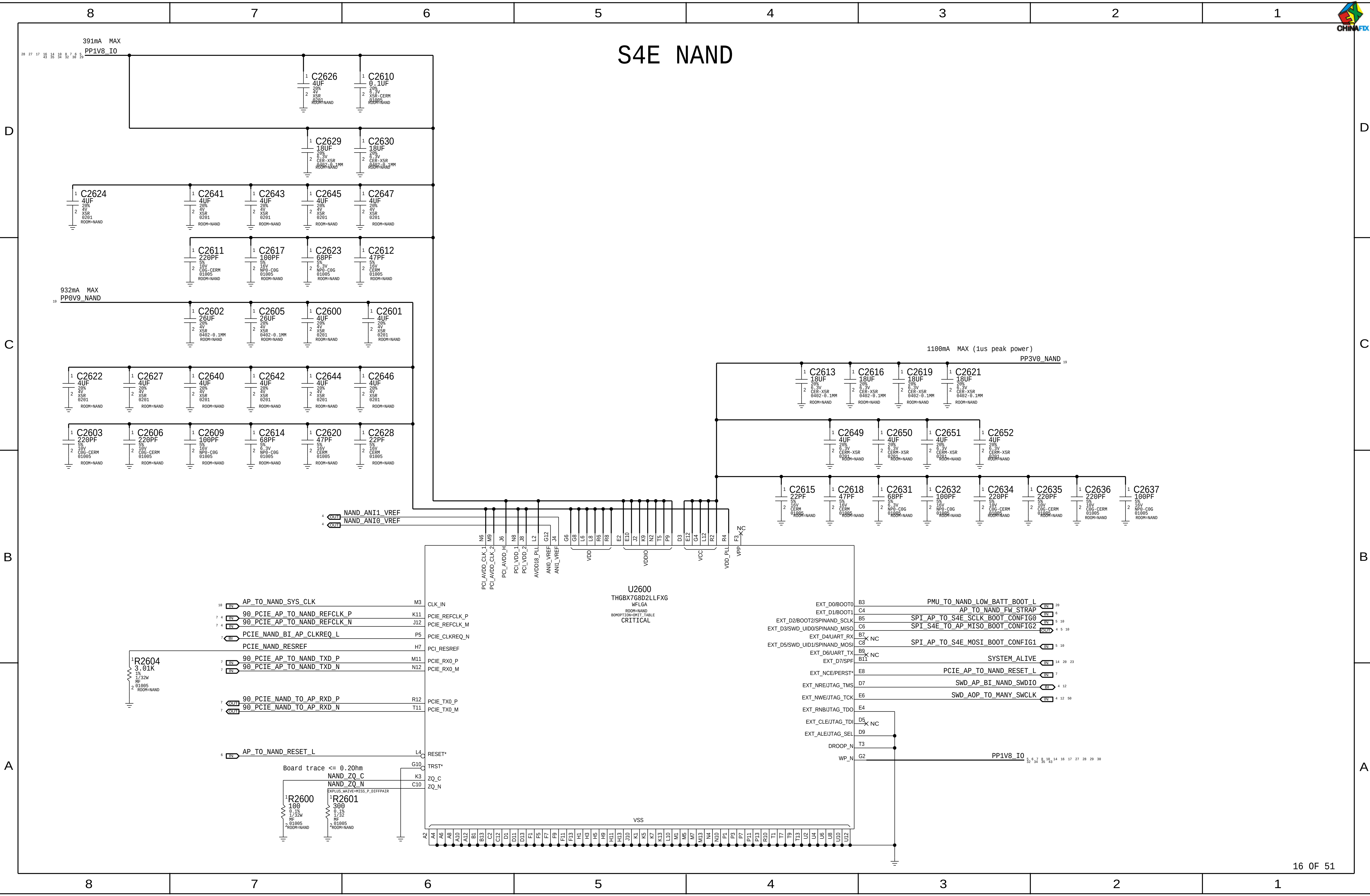
SOC - POWER SUPPLIES

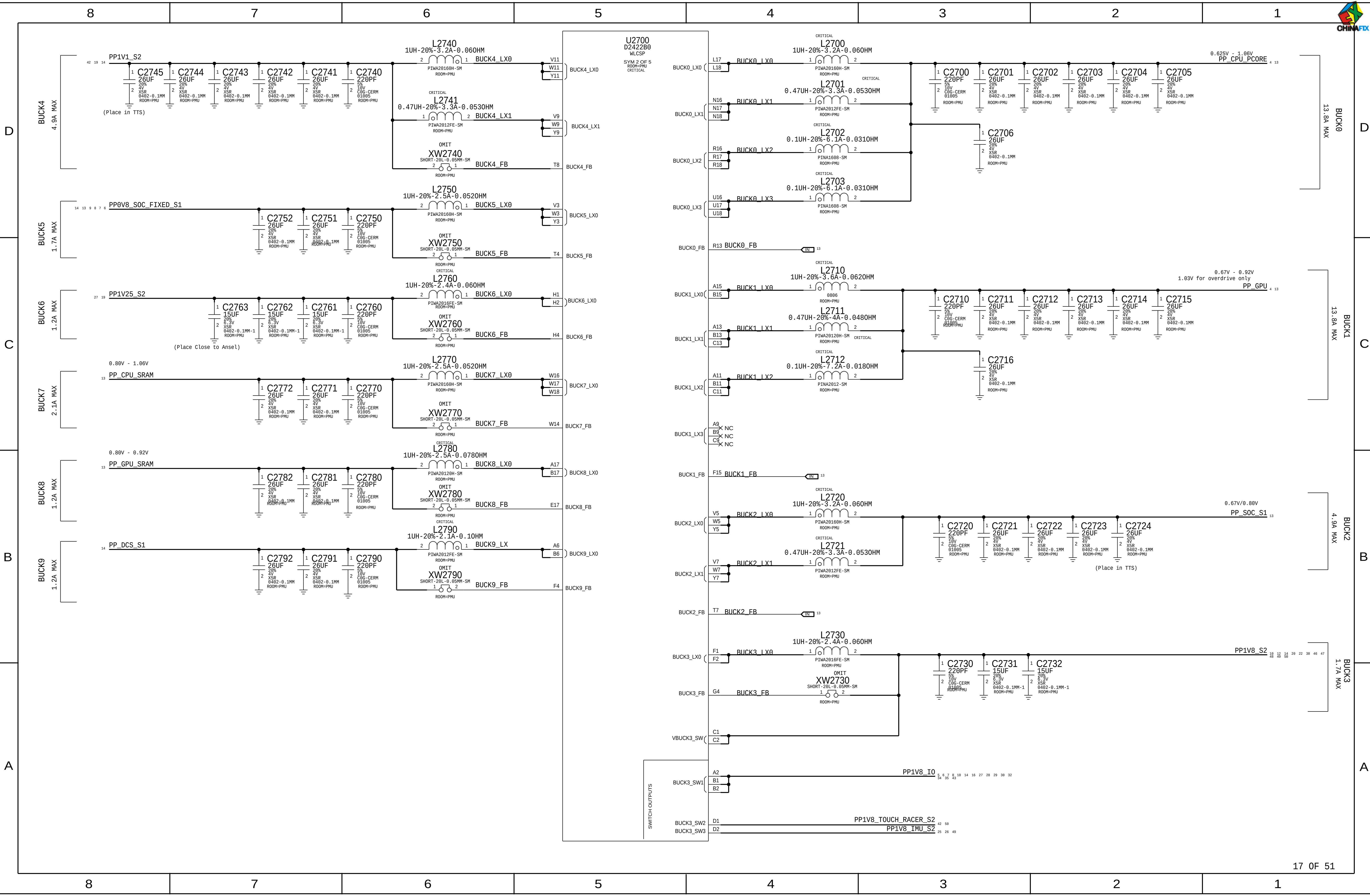


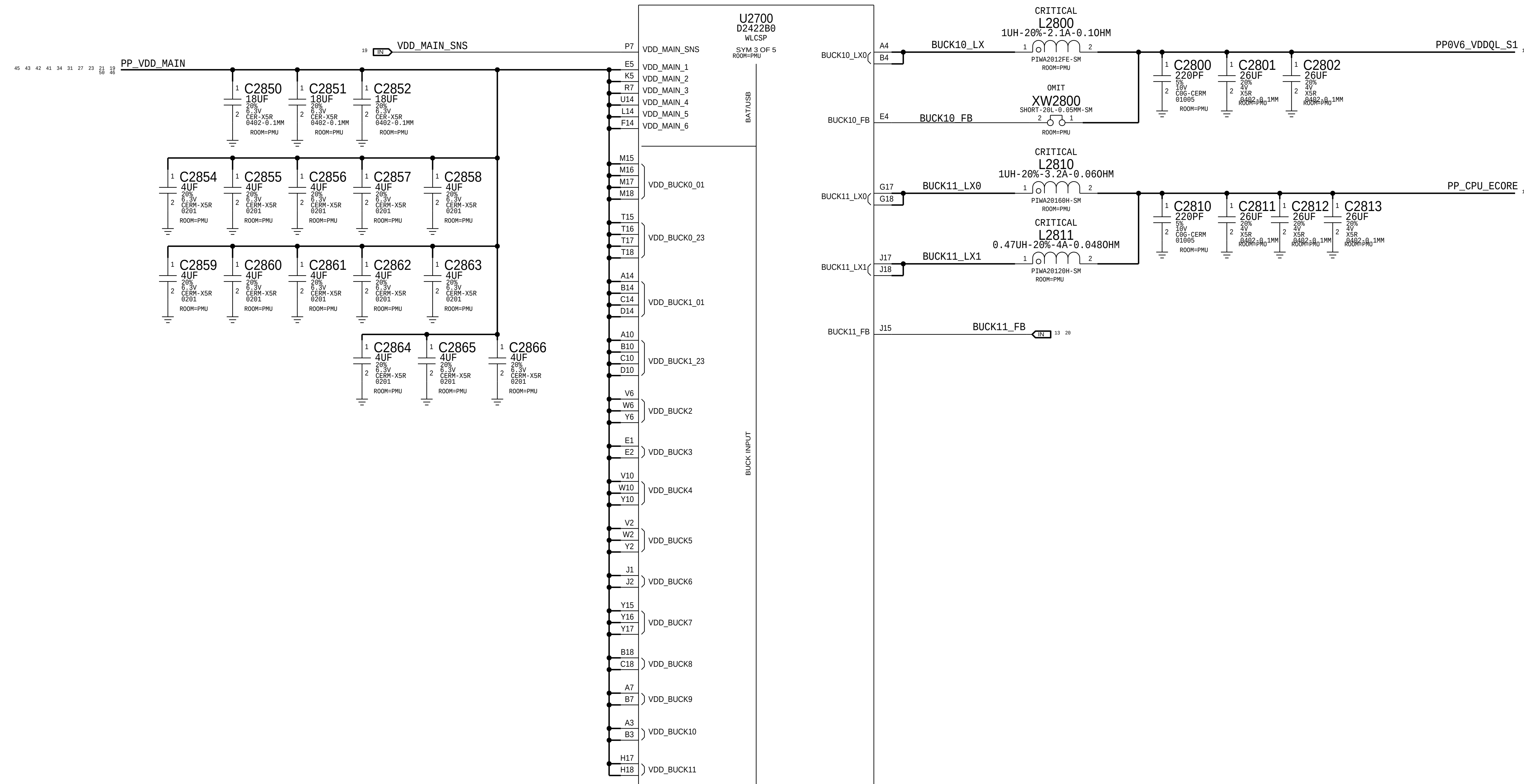
SOC - POWER SUPPLIES



S4E NAND







D

D

C

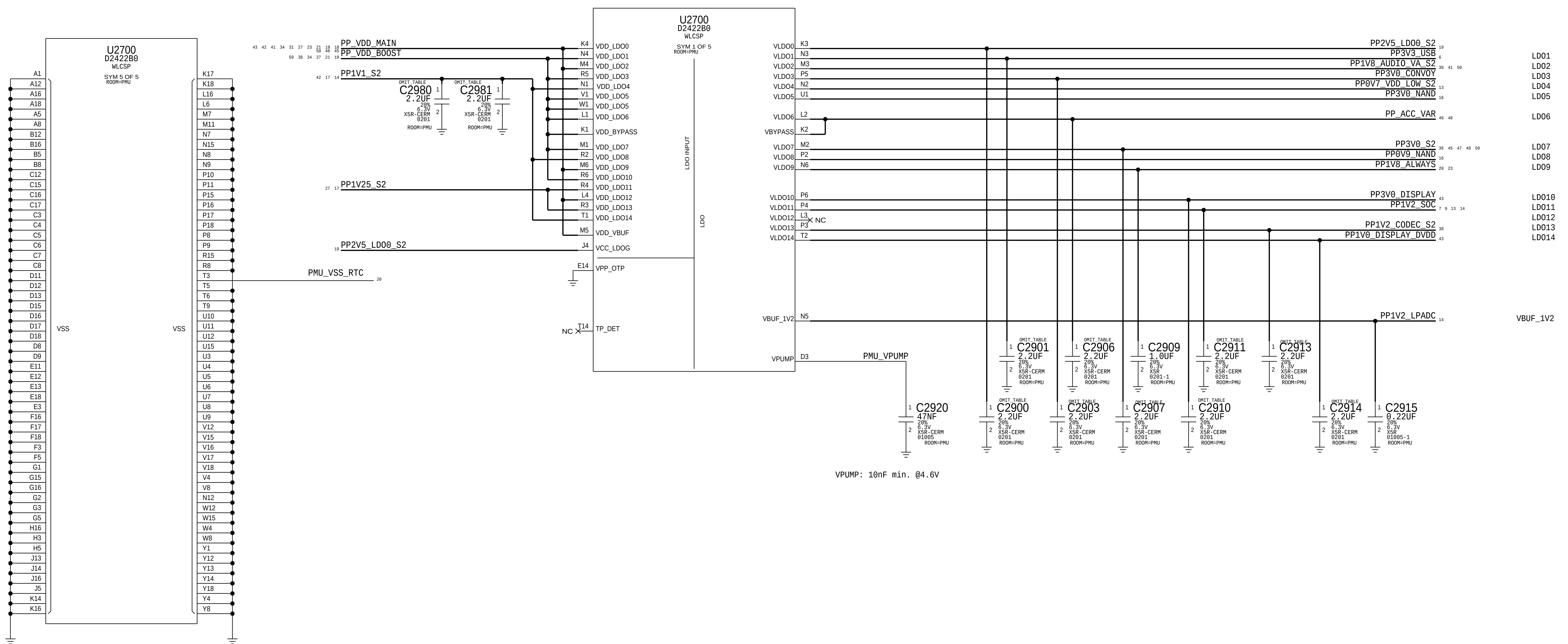
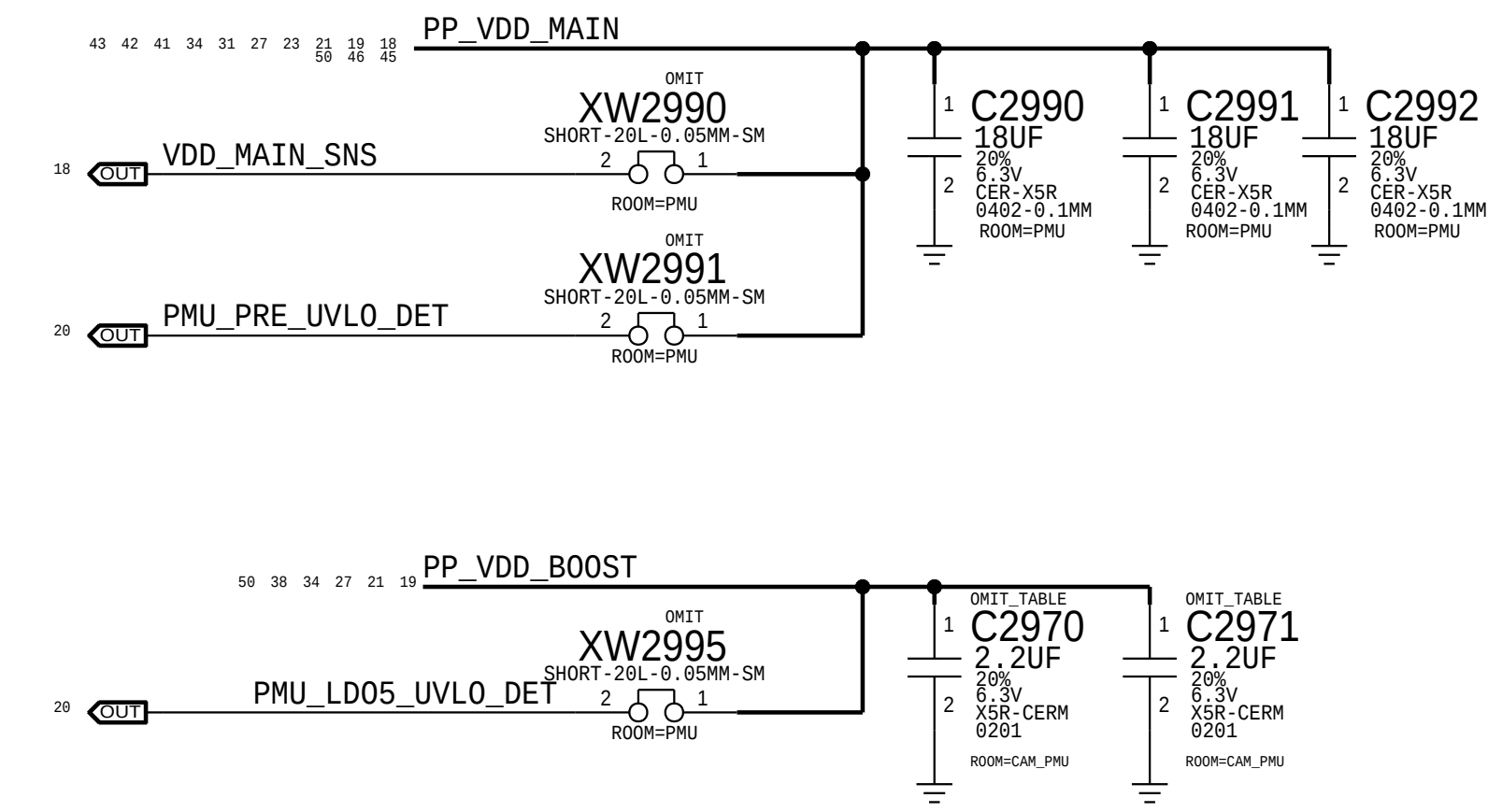
C

B

B

A

A



20 0F 51

D

D

C

C

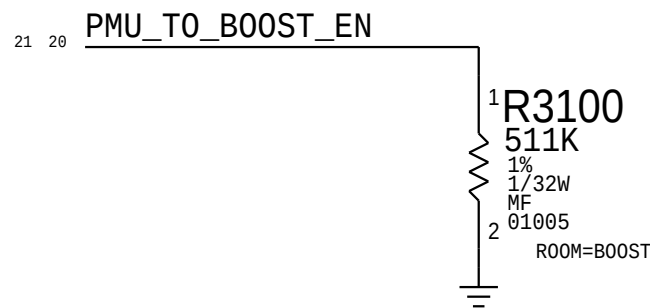
B

B

A

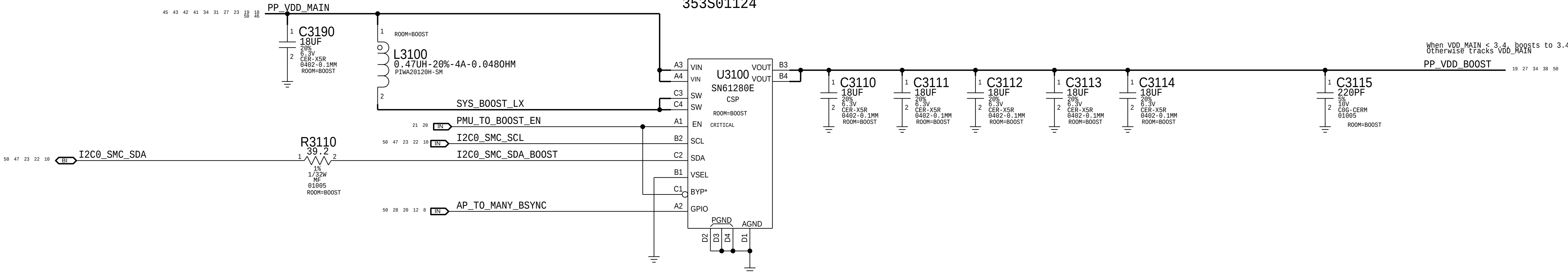
A

Boost Enable Pull



BOOST

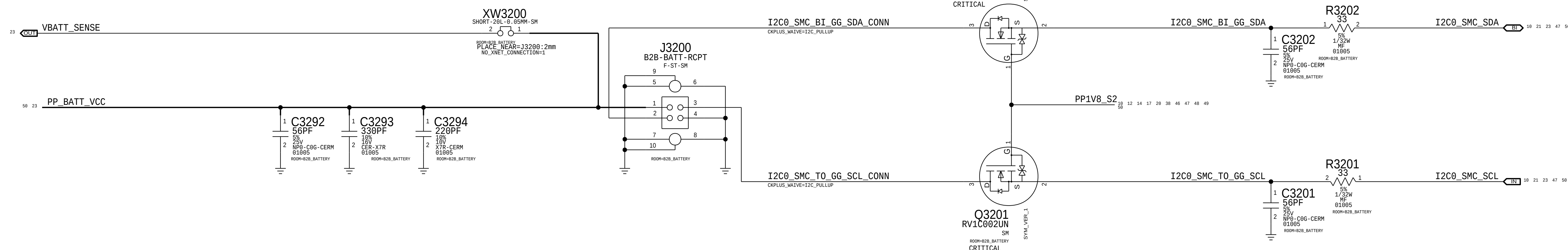
353S01124



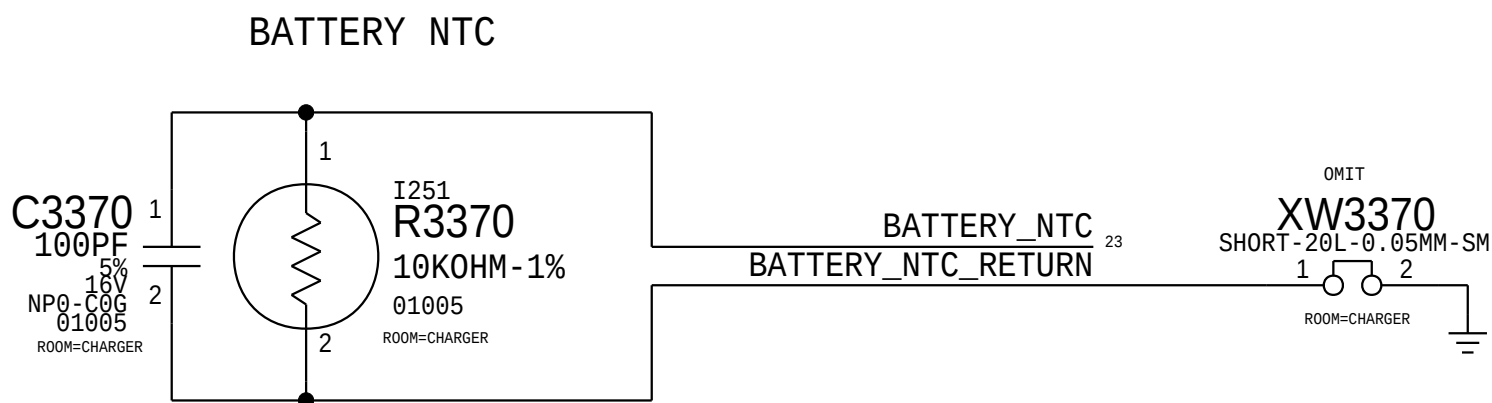
BATTERY CONNECTOR

Rcpt: 516S00232 <-- This one on MLB
Plug: 516S00233

Gas gauge I2C level translator



23 0F 51





8

7

6

5

4

3

2

1

D

D

C

C

Iktara Components on MLB Bottom

B

B

A

A

8

7

6

5

4

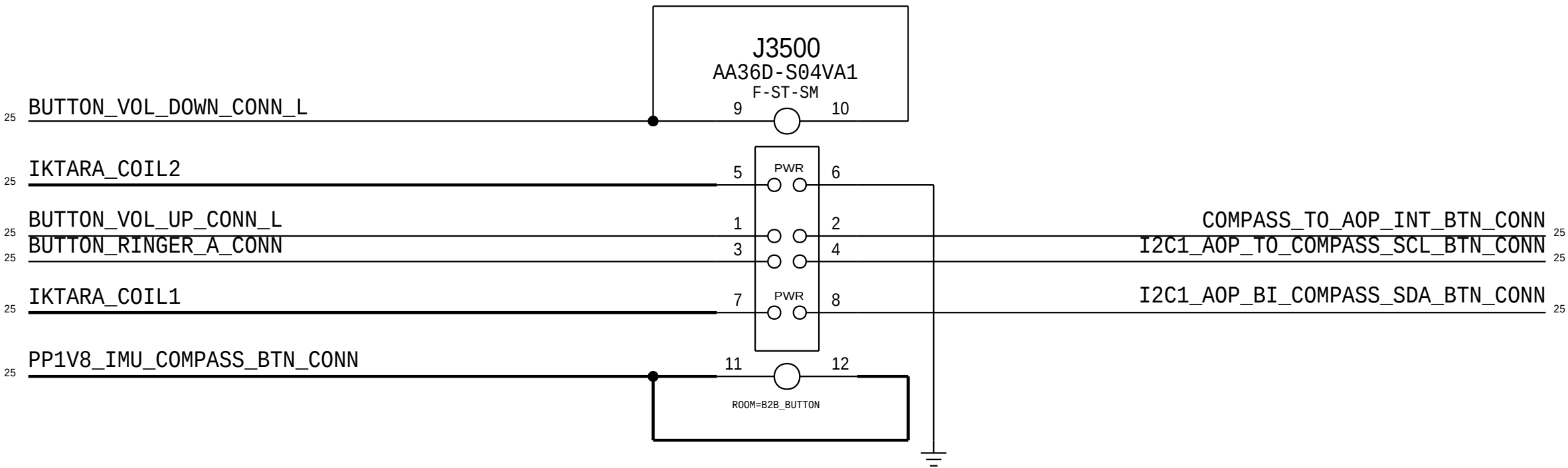
3

2

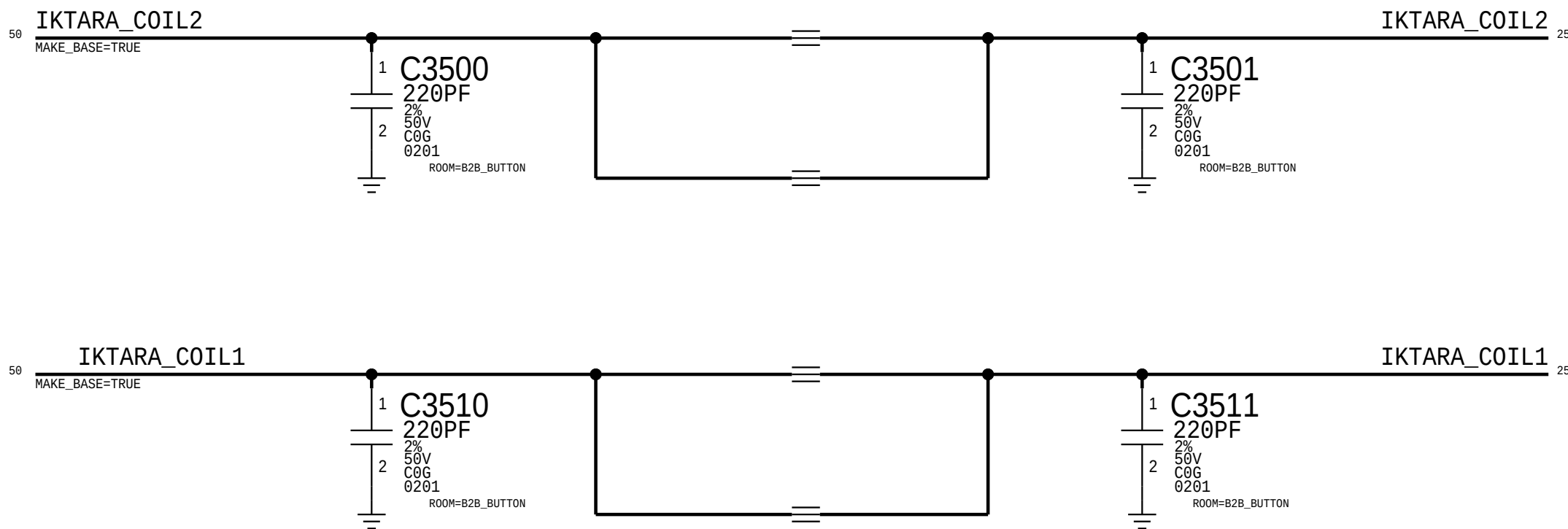
1

Cyclone + Button Connector

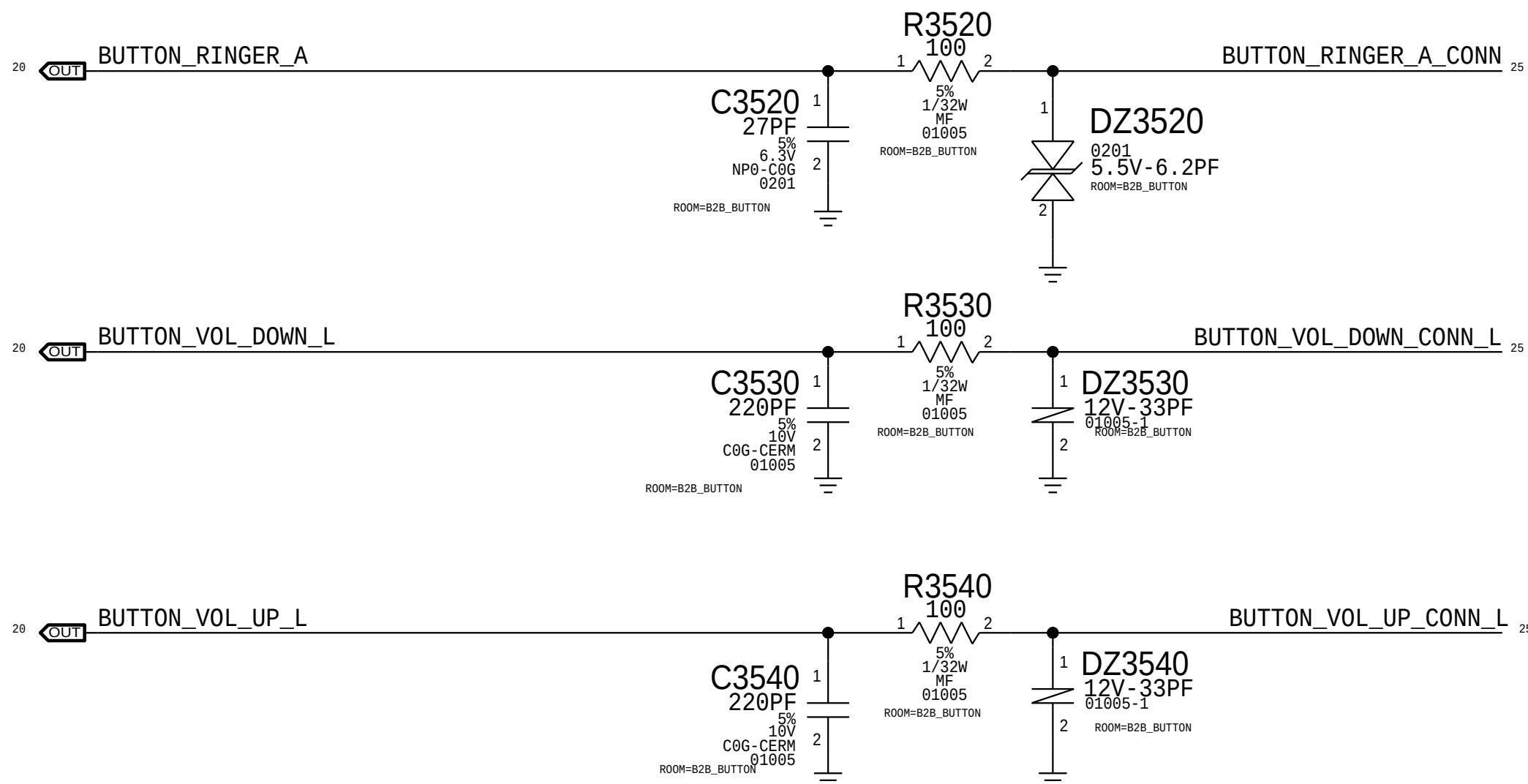
Rcpt: 516S00289 <-- This one on MLB
Plug: 516S00290



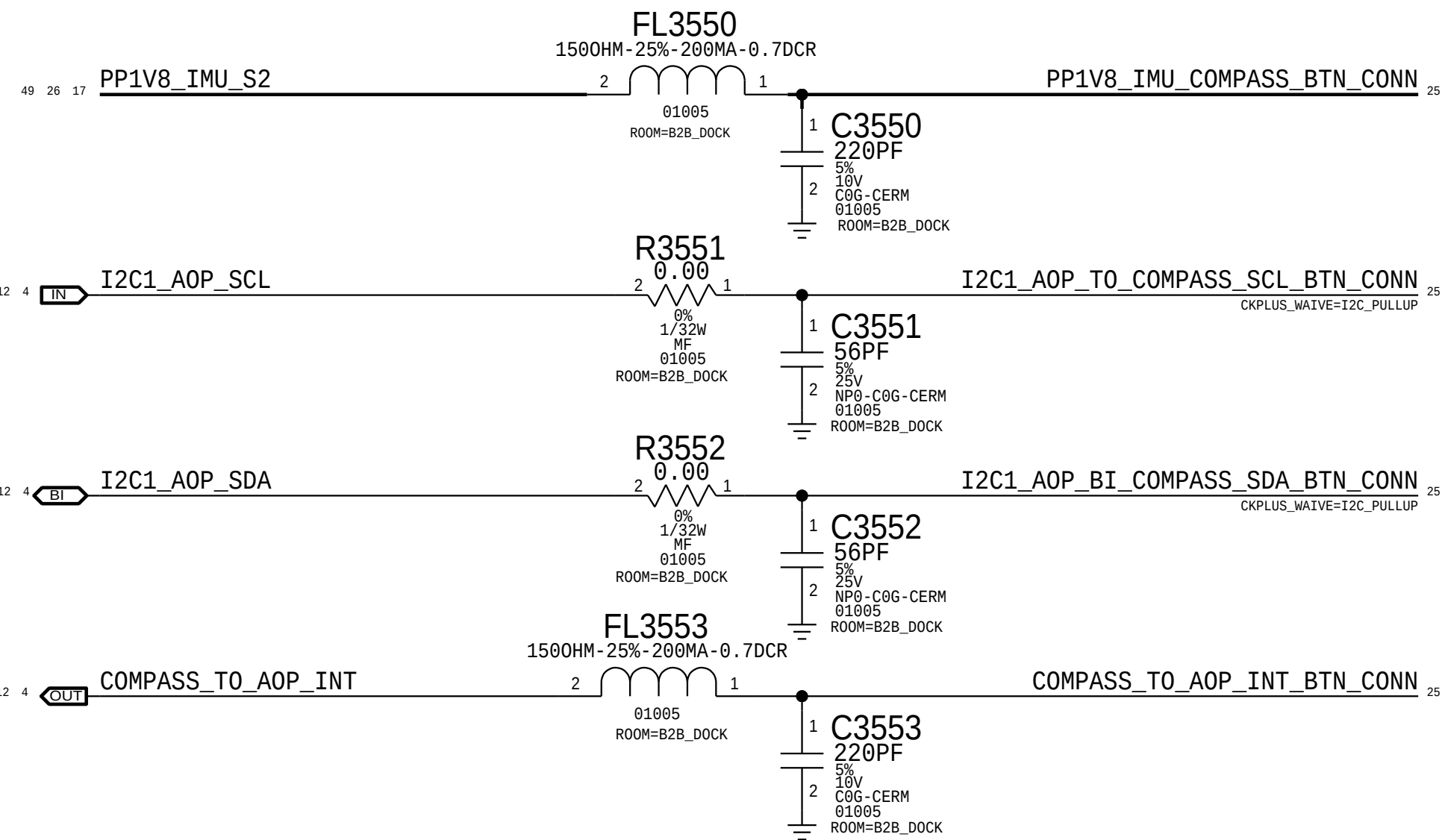
Cyclone Filtering



BUTTONS



Compass (Button Flex Location)



D

C

B

A

D

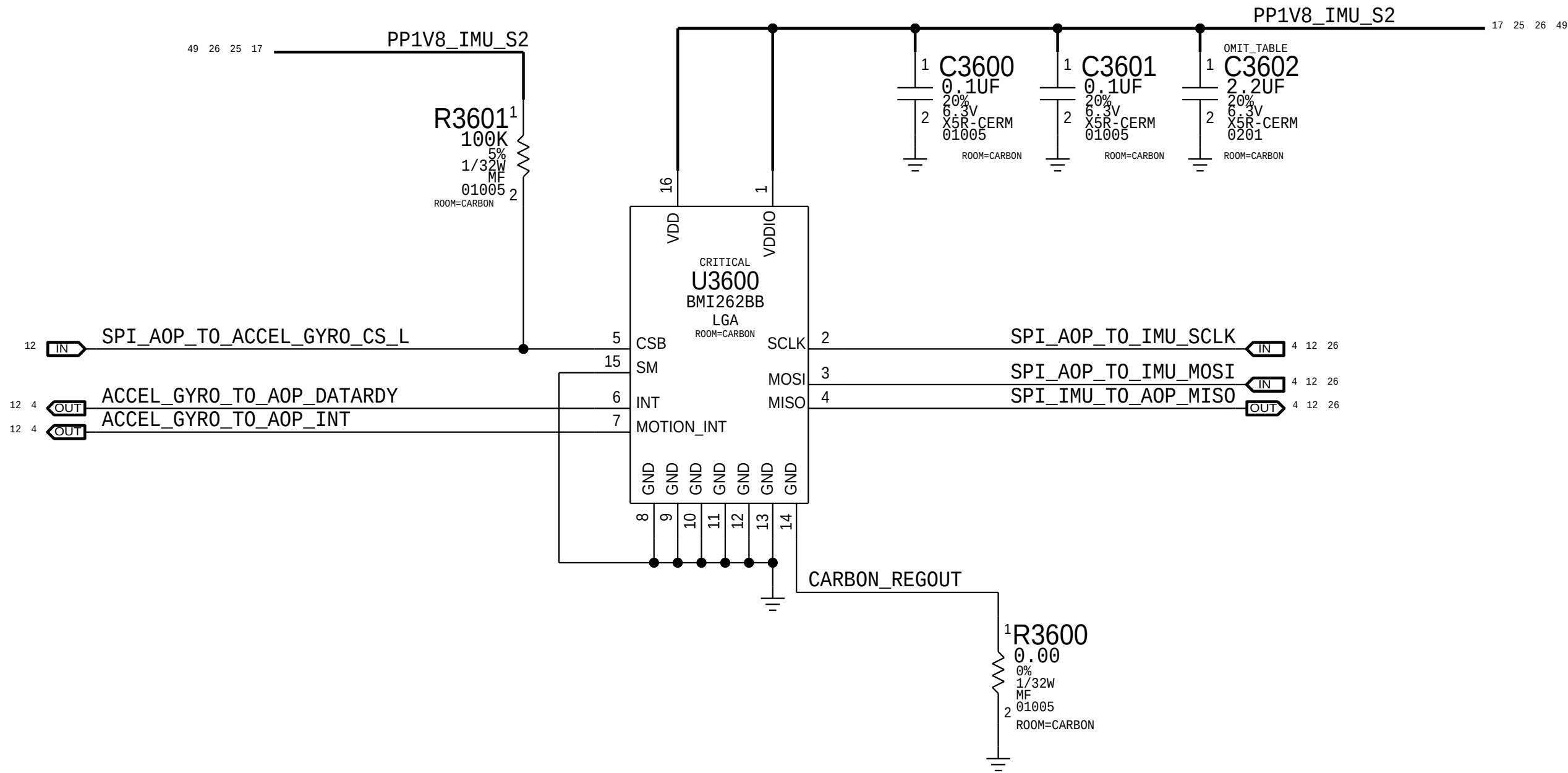
C

B

A

Graphite - Accel & Gyro

APN: 338S00304



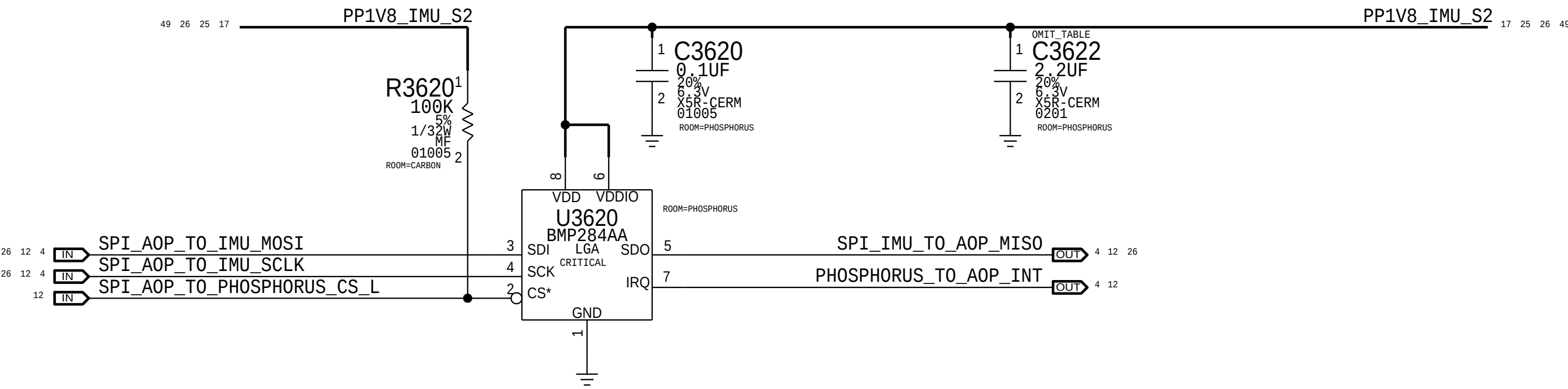
Magnesium - Compass

(On Dock or Button Flex)

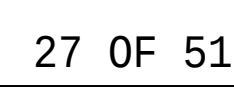
Phosphorus

BOSCH (APN:338S00188)

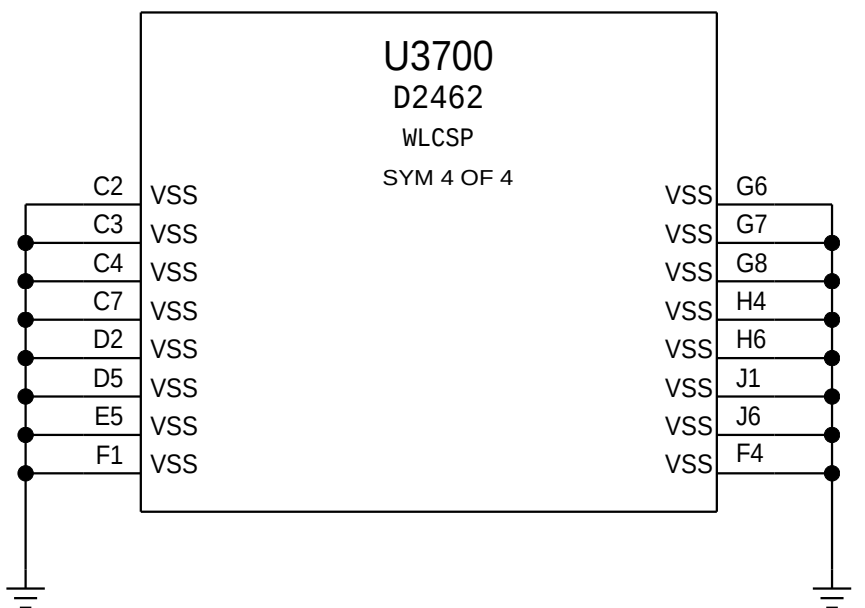
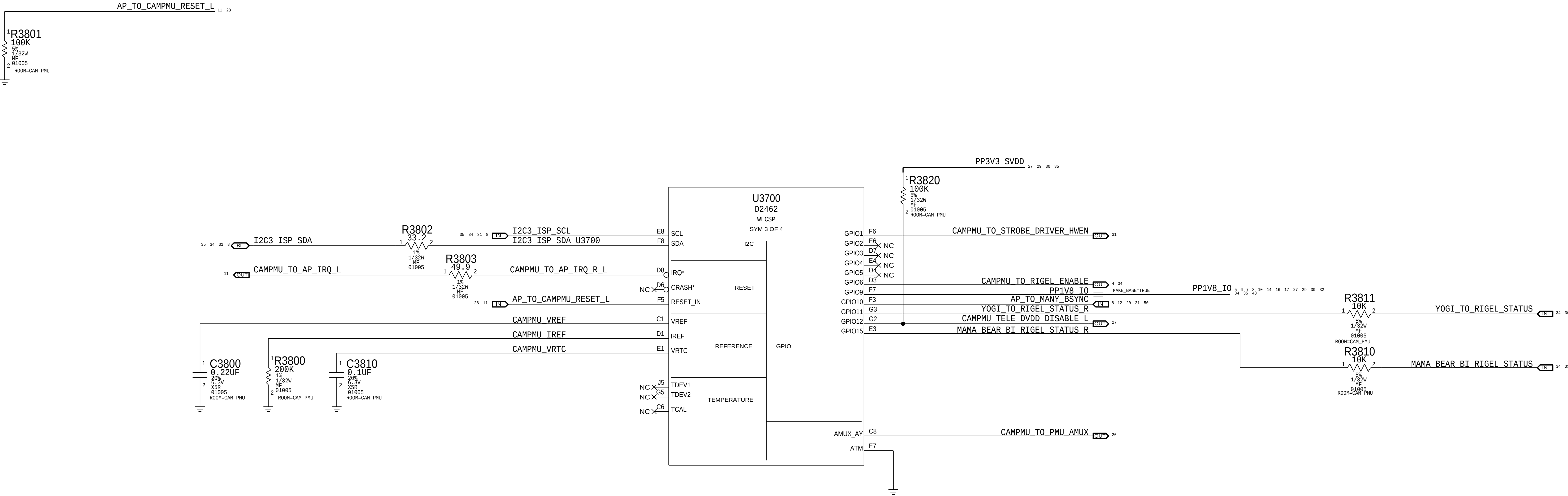
ST (APN:338S00230)



27 0F 51

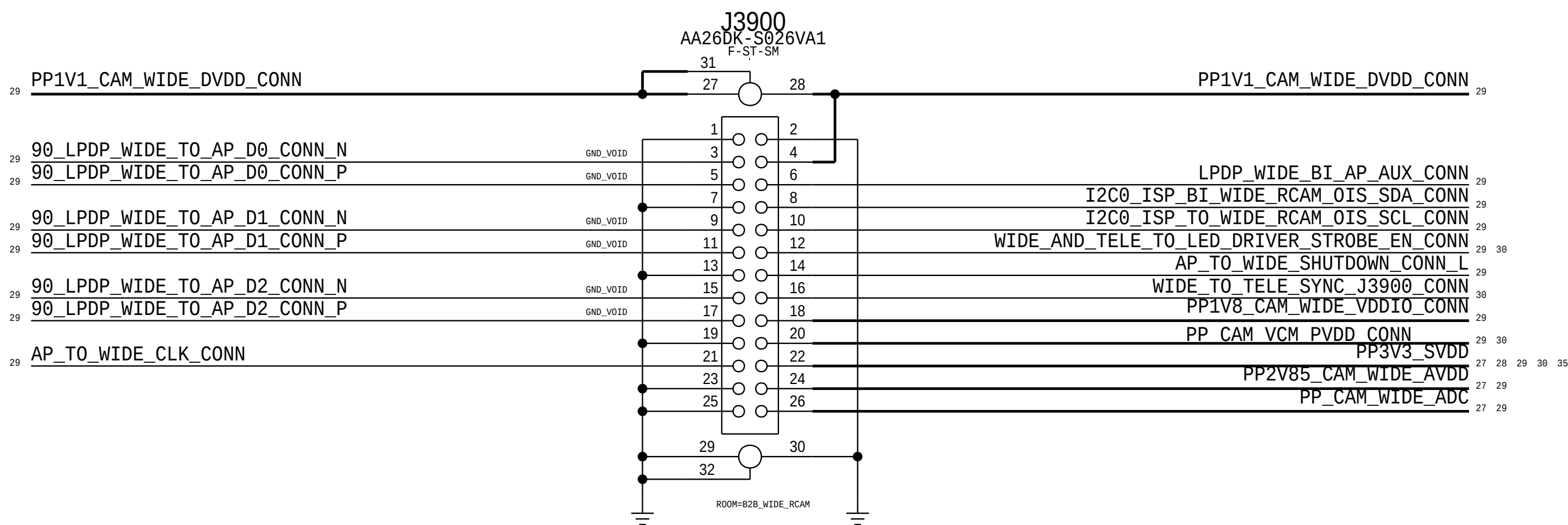


Pull Downs

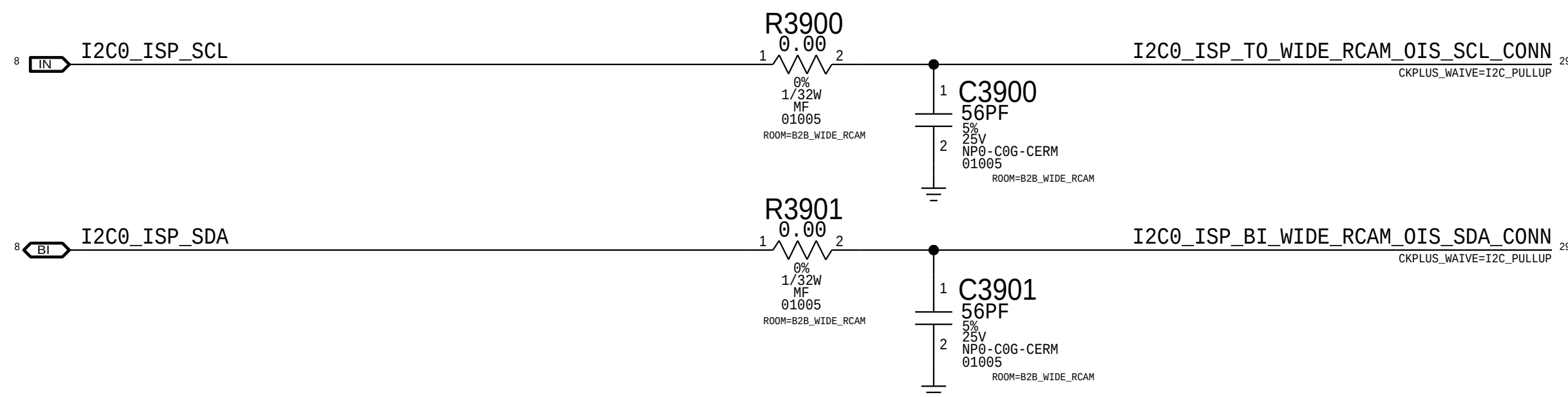


Wide Camera Connector

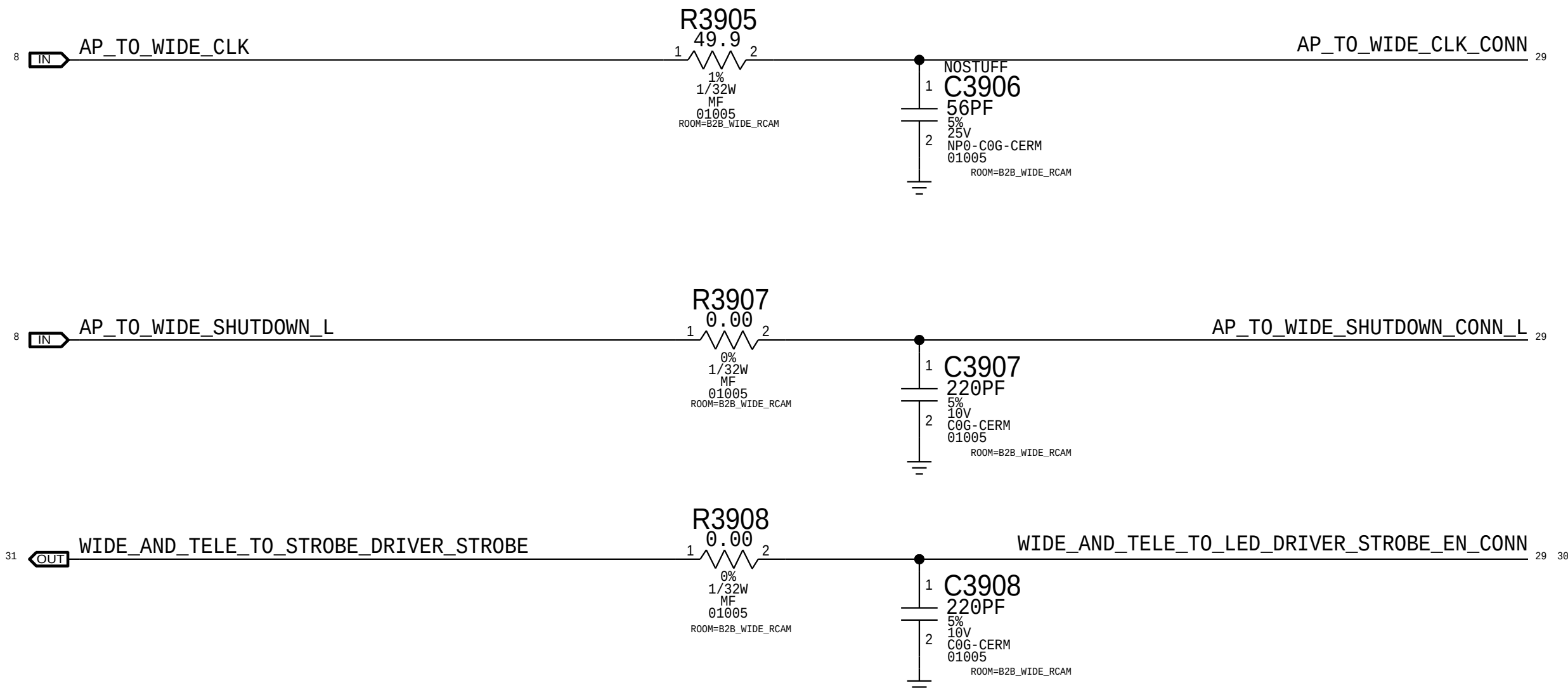
Rcpt: 516S00313 <-- This one on MLB
Plug: 516S00314



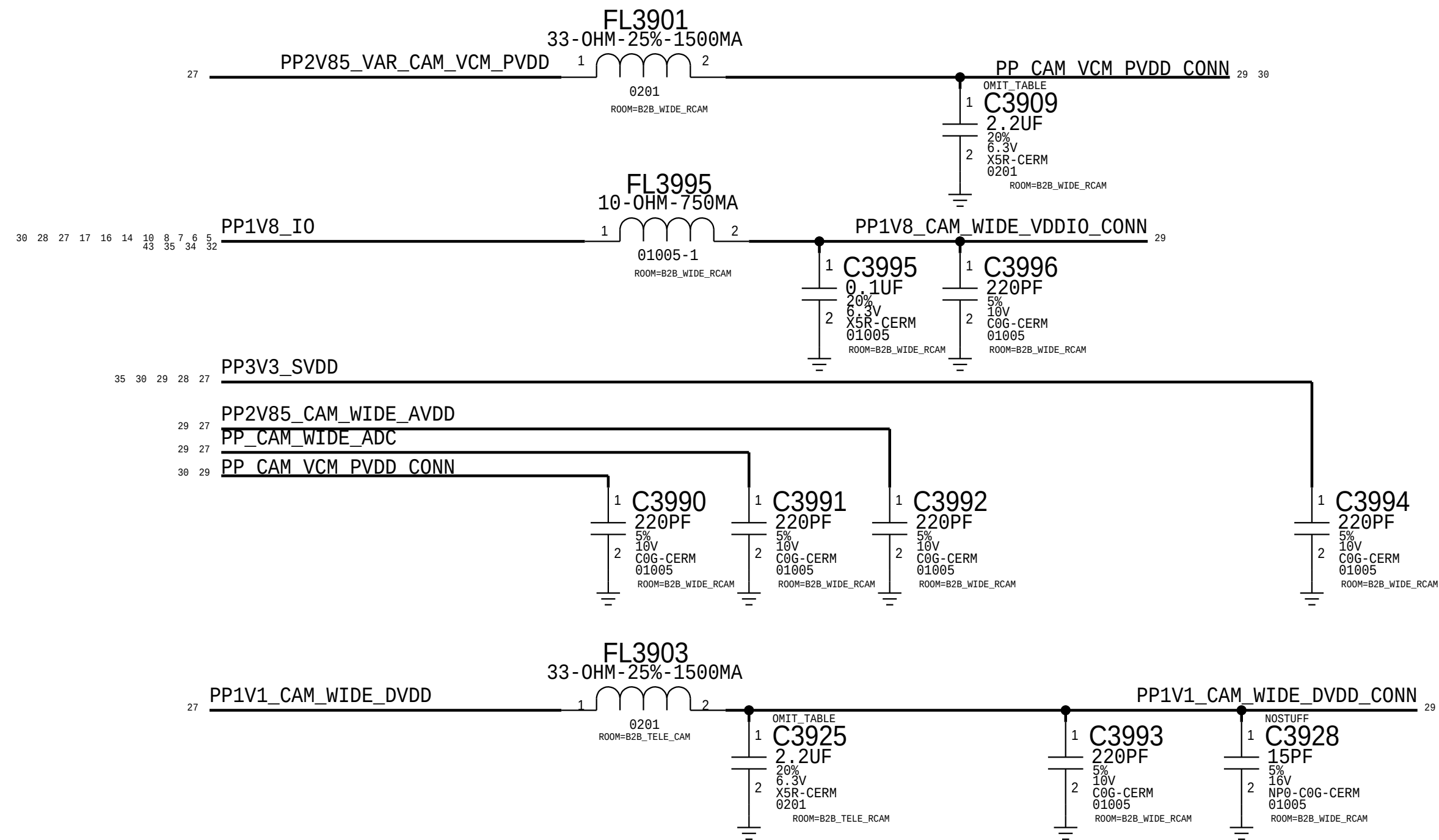
ISP I2C



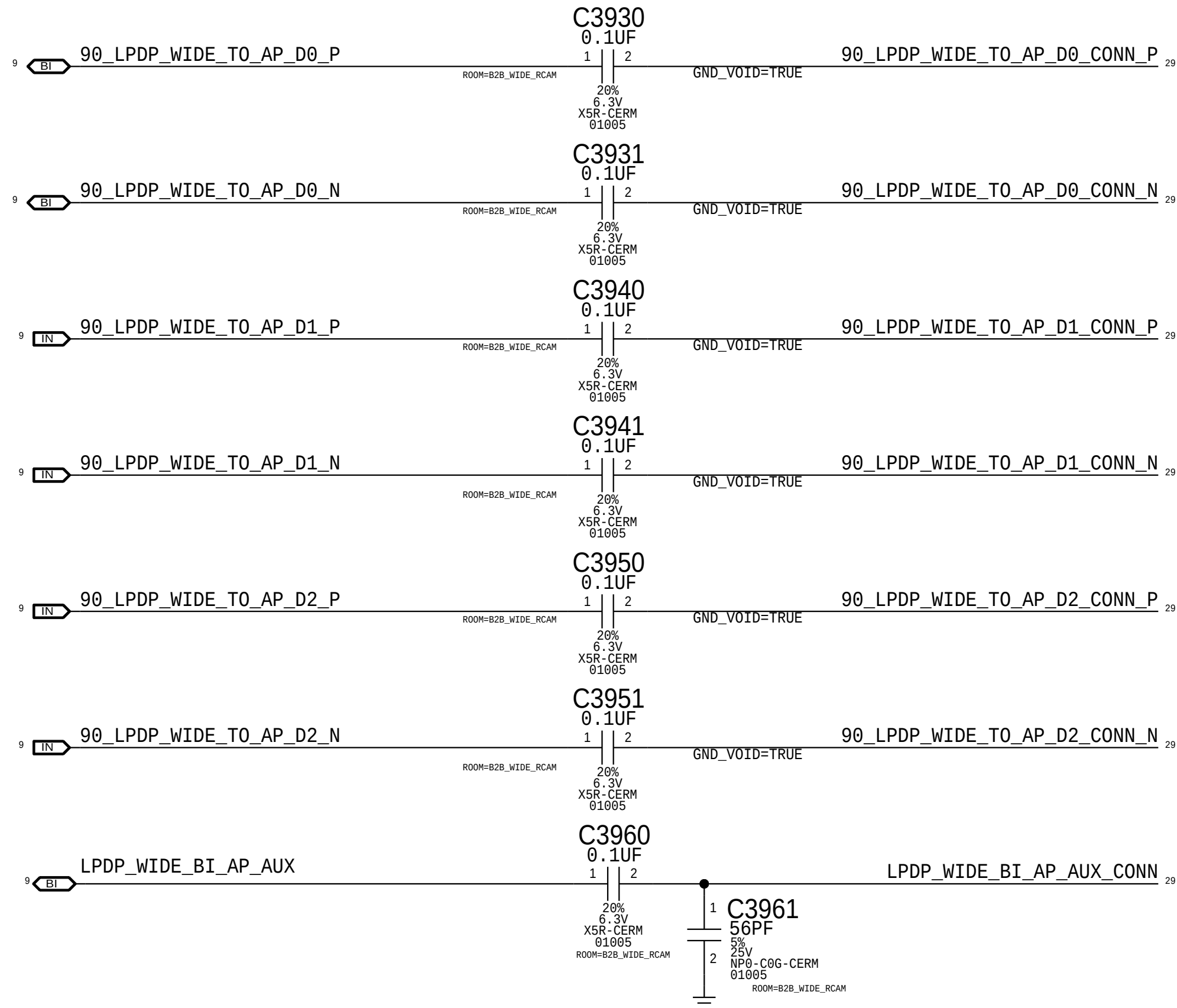
I0 Filters



Power Filtering



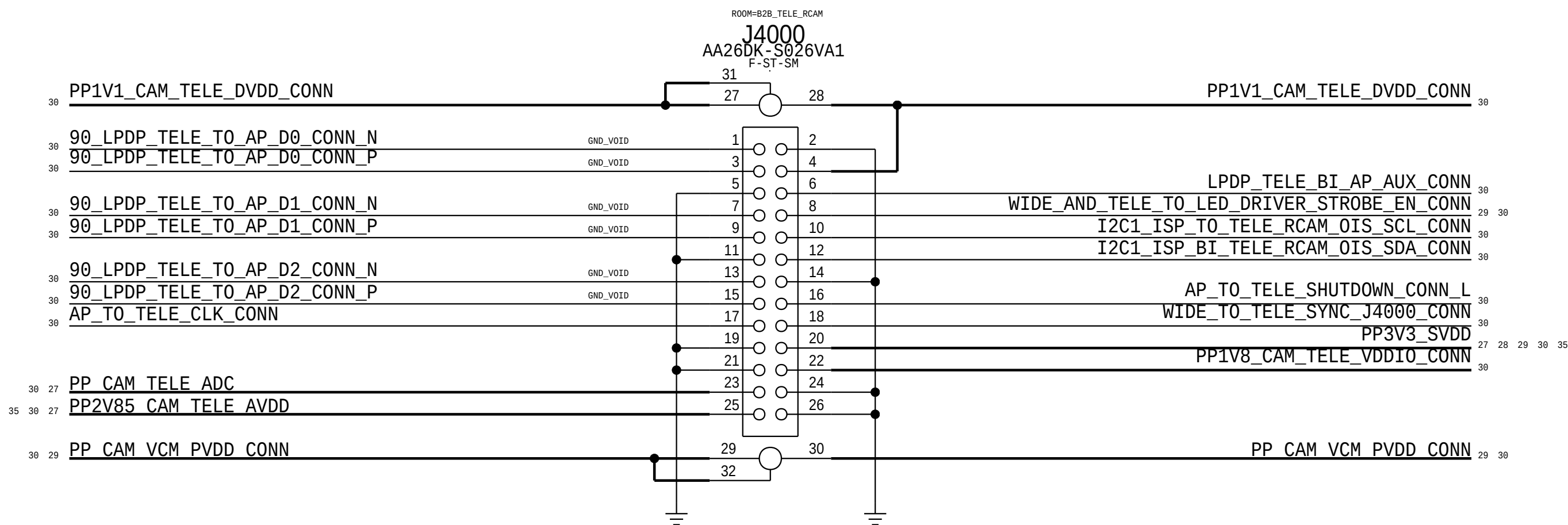
LPDP Filters



Tele Camera Connector

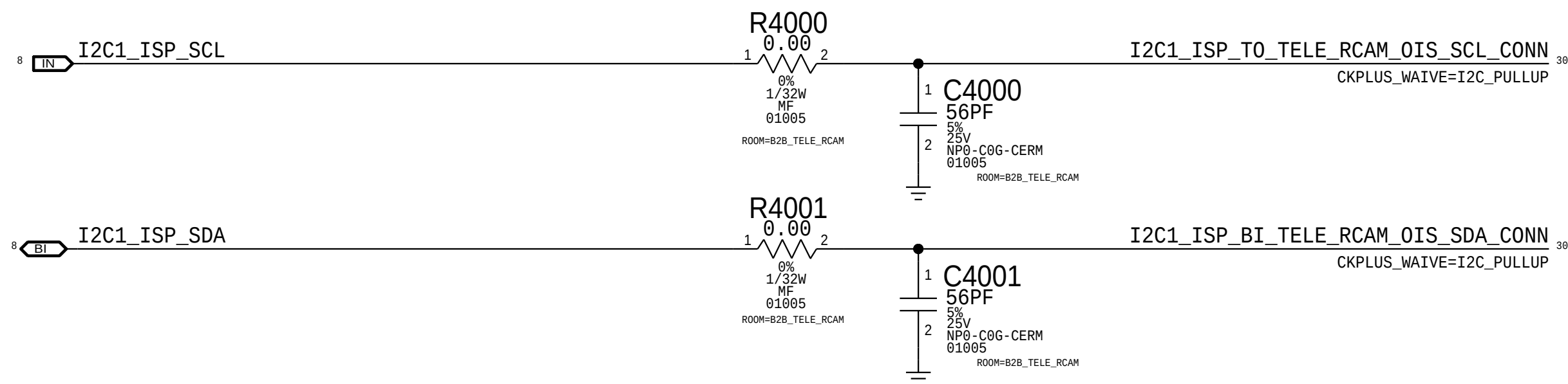
Rcpt: 516S00313 <-- This one on MLB
Plug: 516S00314

D



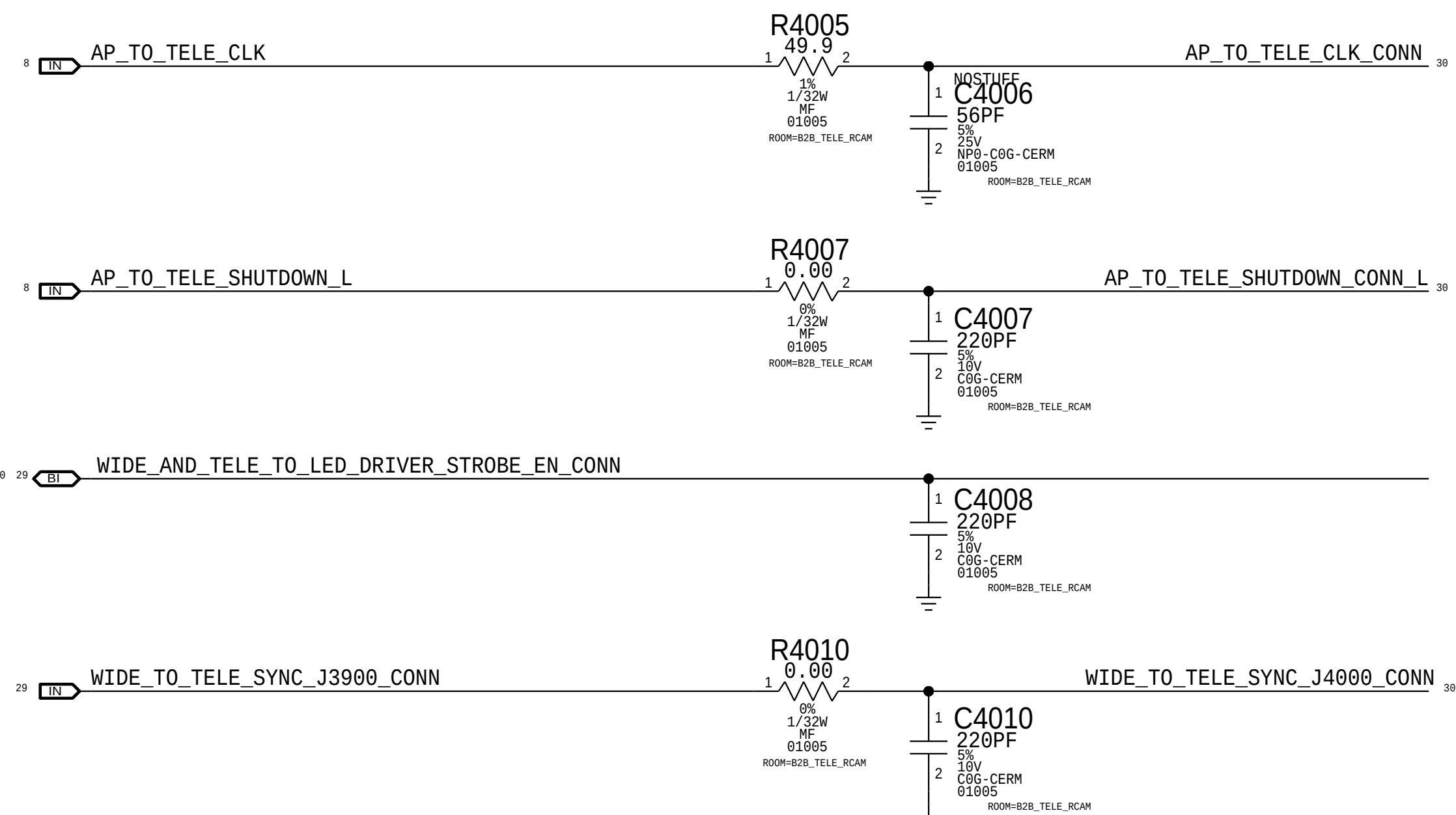
ISP I2C

C



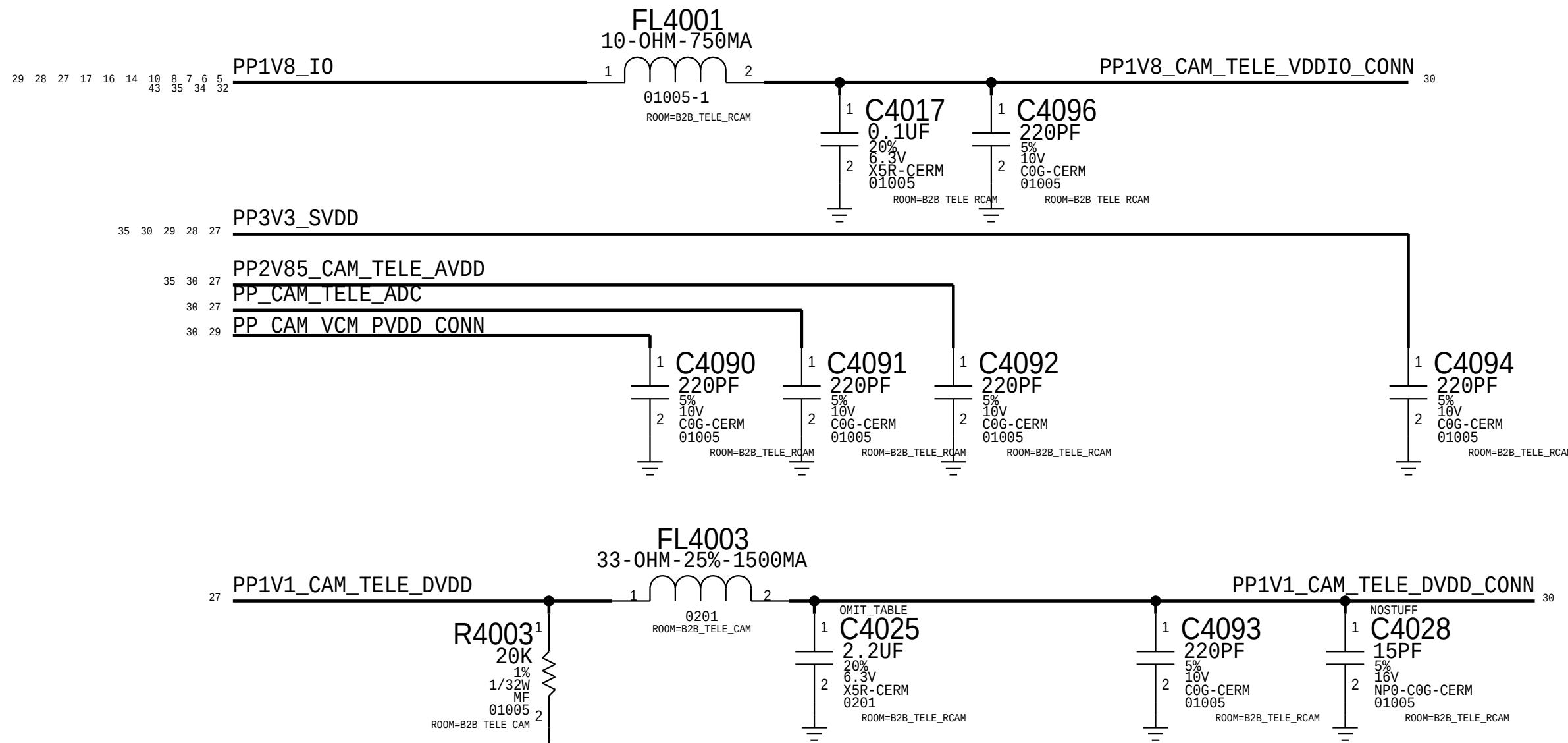
B

I/O Filters



A

Power Filtering



D

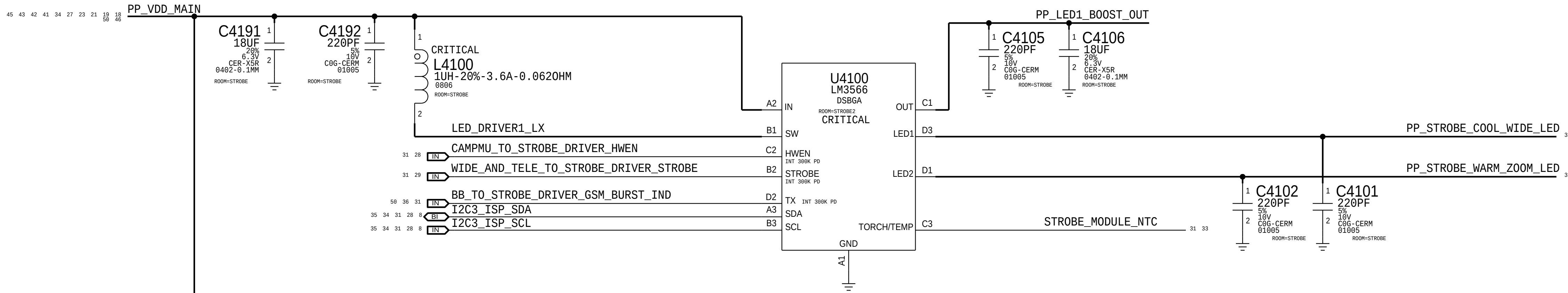
C

B

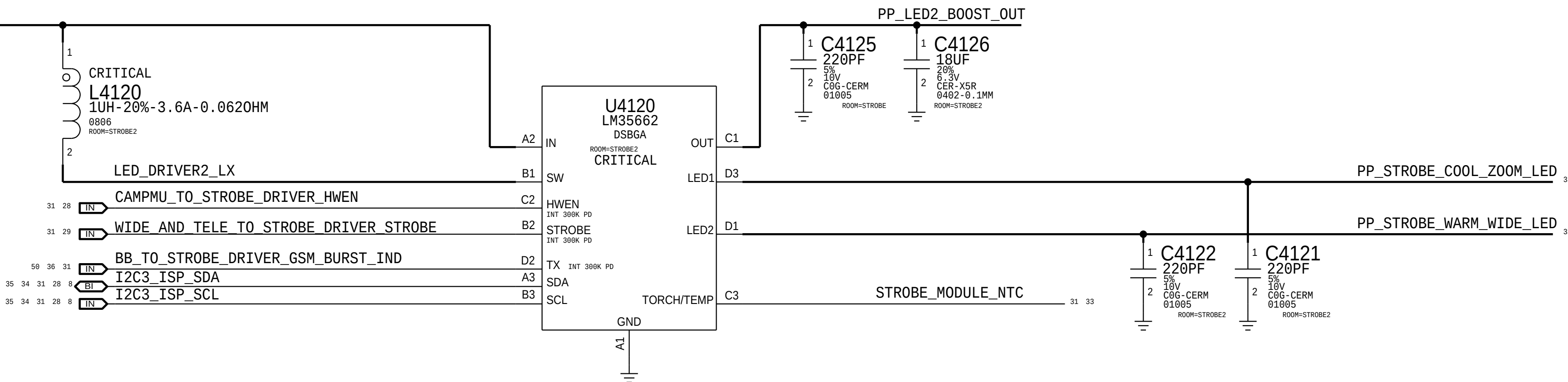
A

LED STROBE DRIVERS (NEON)

APN:353S00558
I2C Address (7-bit): 0x63



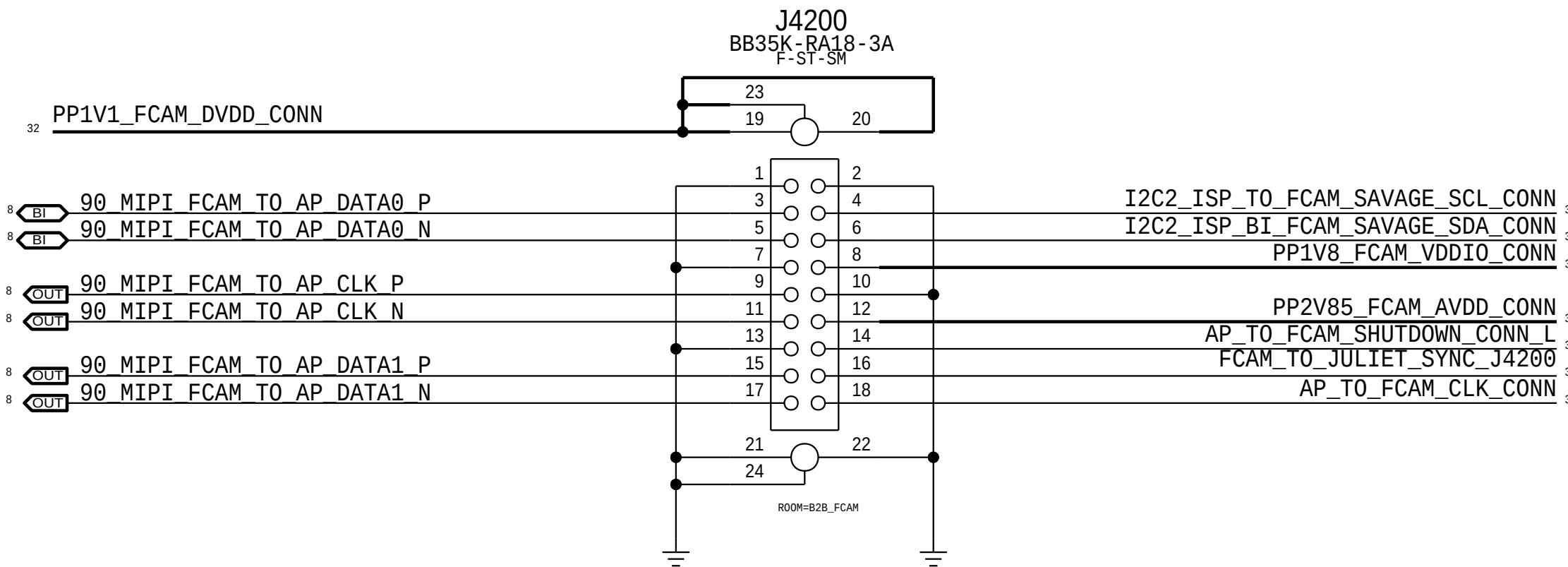
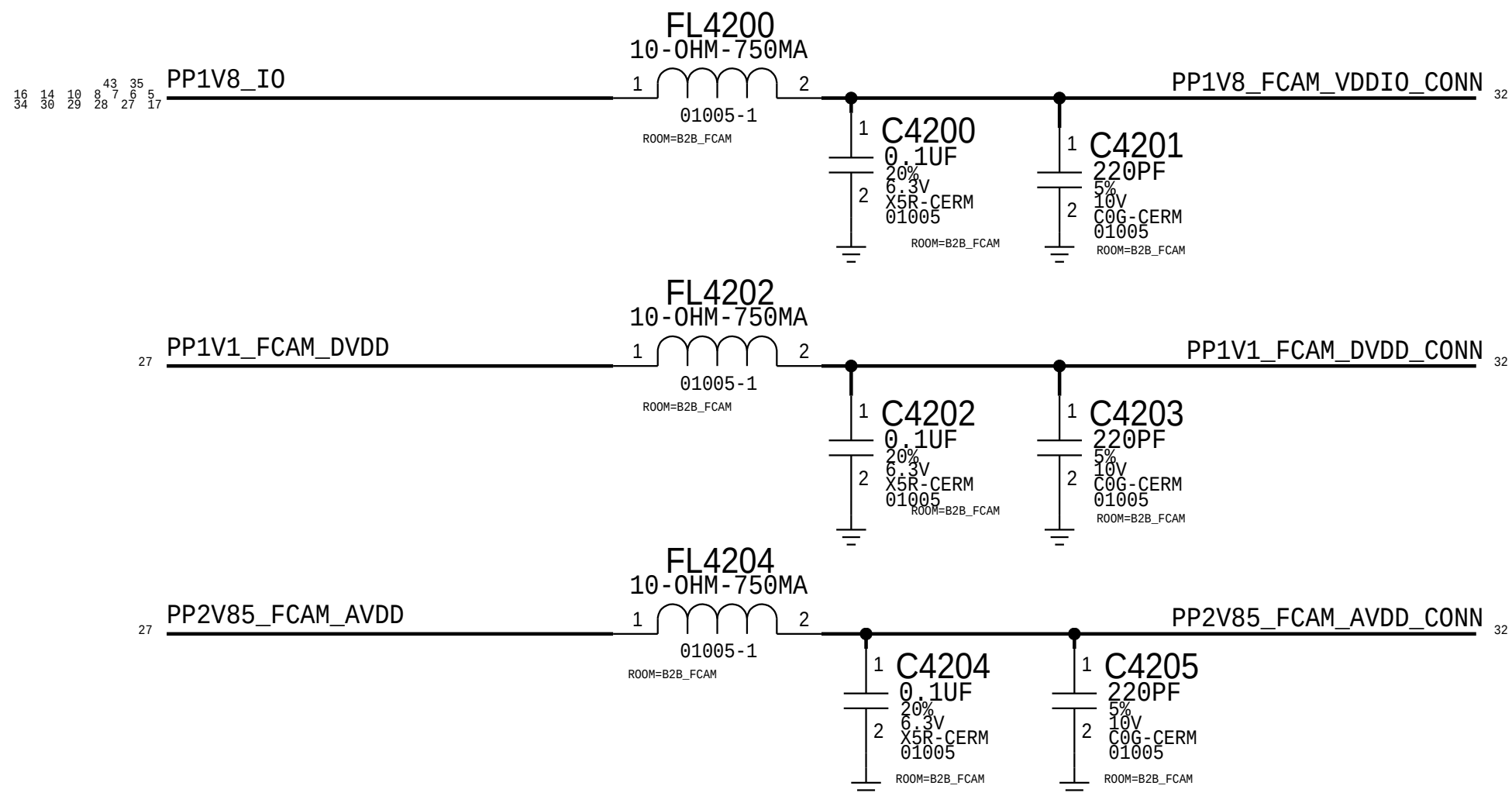
APN:353S00868
I2C Address (7-bit): 0x67



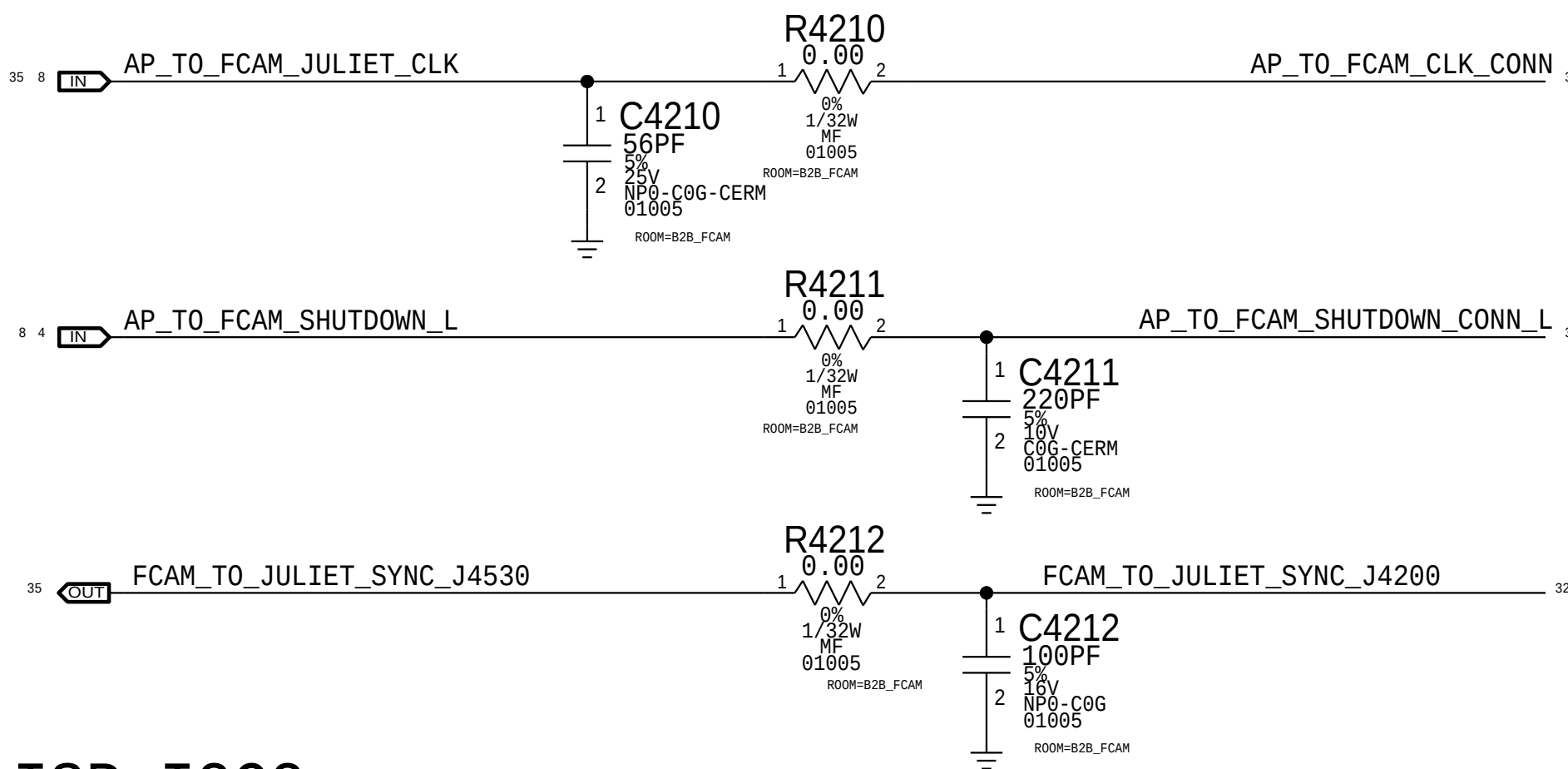
NEW HAMPSHIRE POWER

FCAM Connector

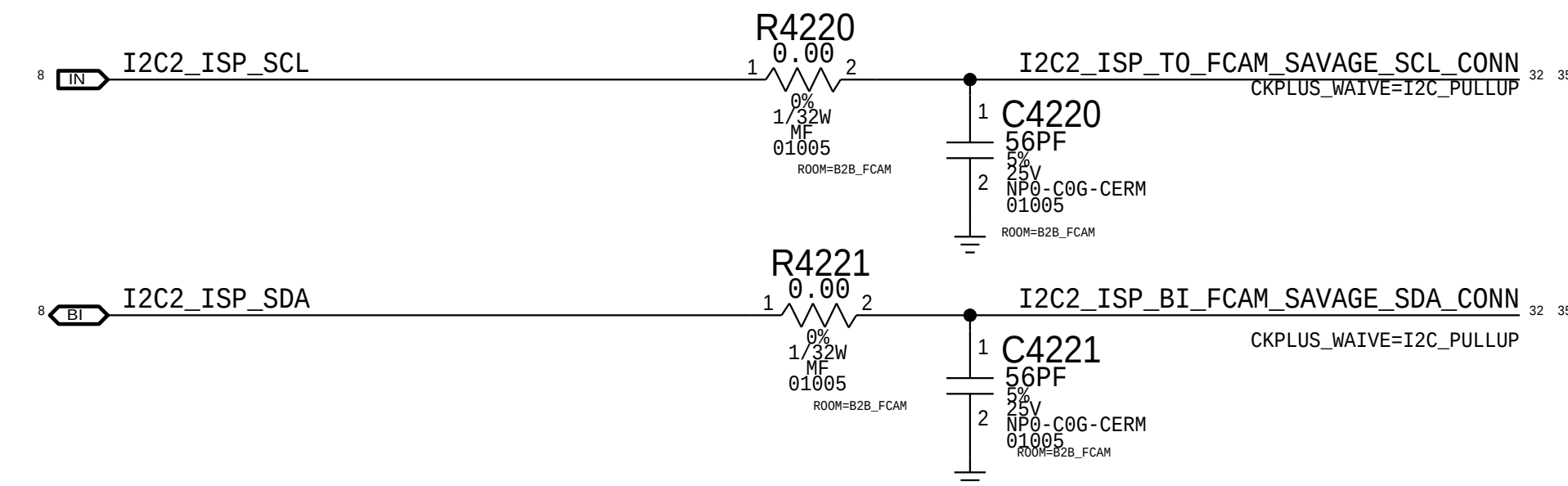
Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245



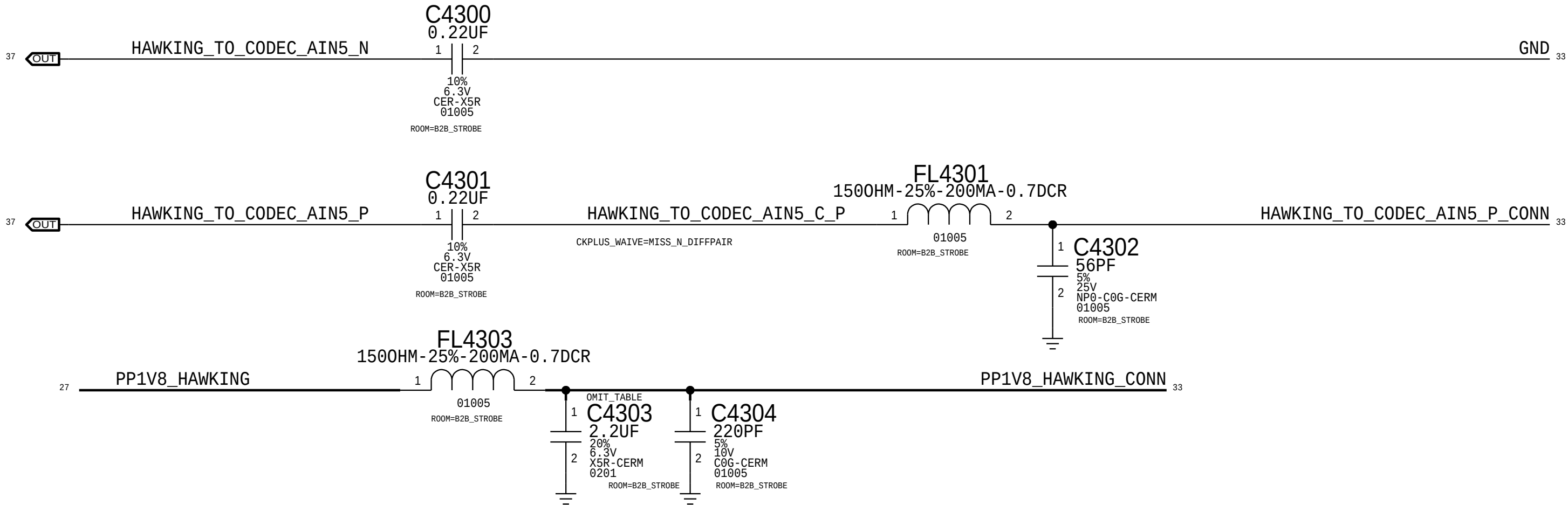
FCAM I/O



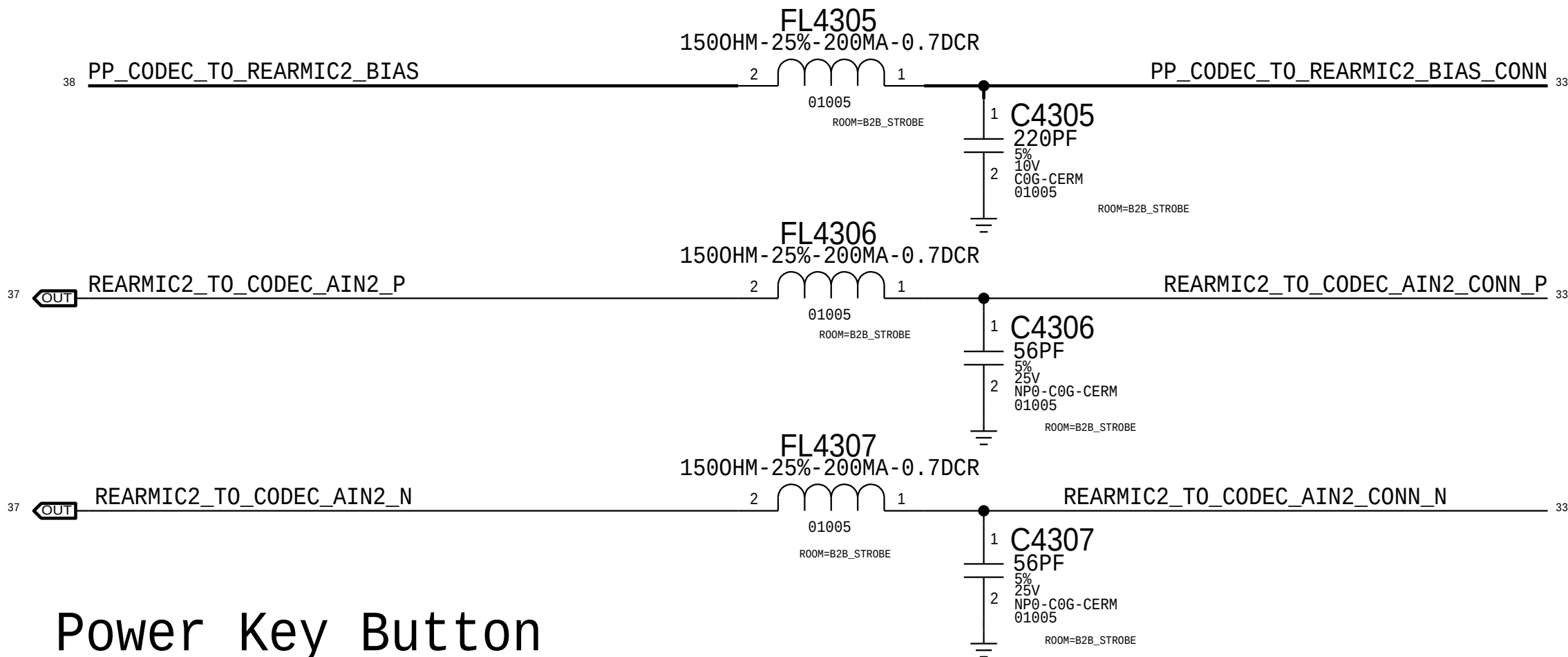
ISP I2C2



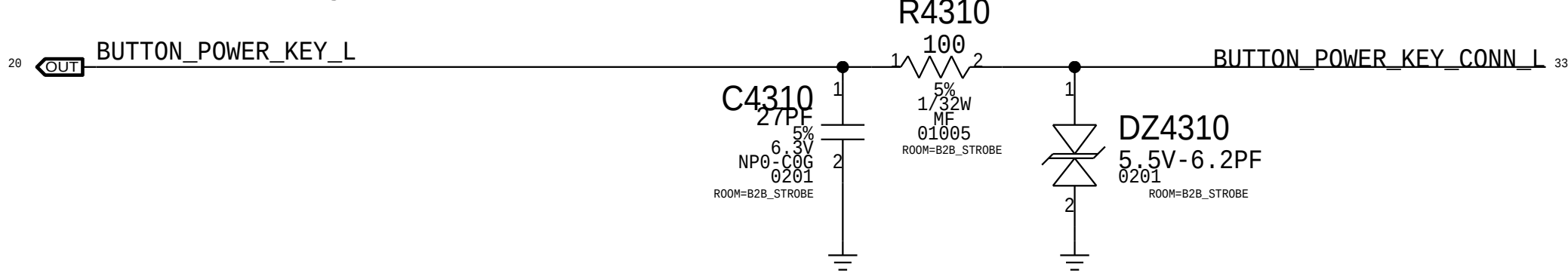
HAWKING



MIC2 (ANC REF)

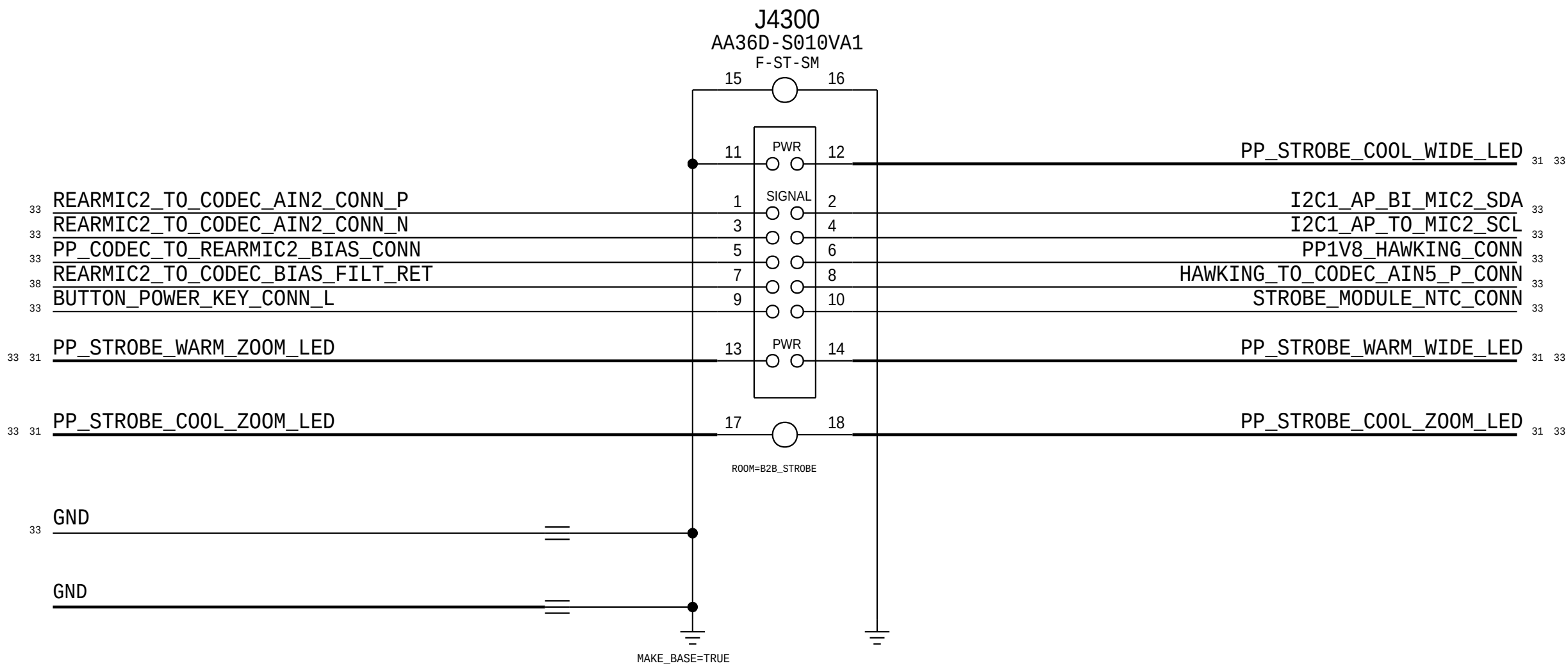


Power Key Button

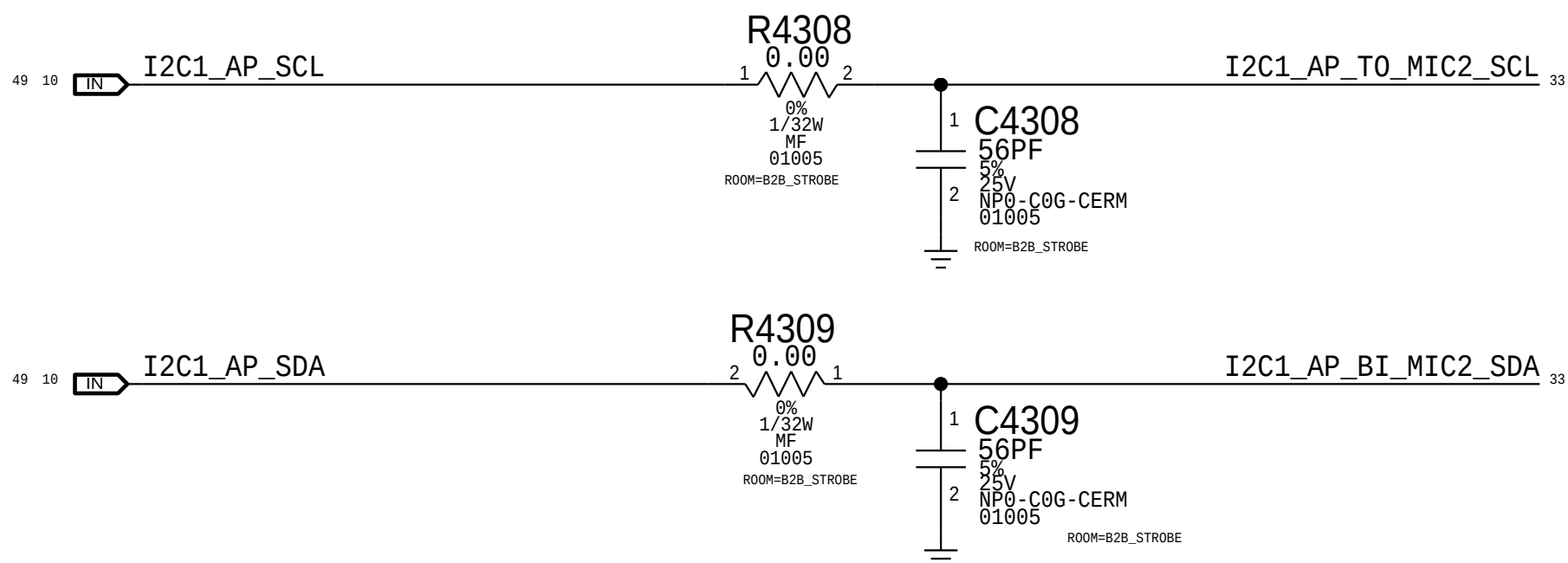
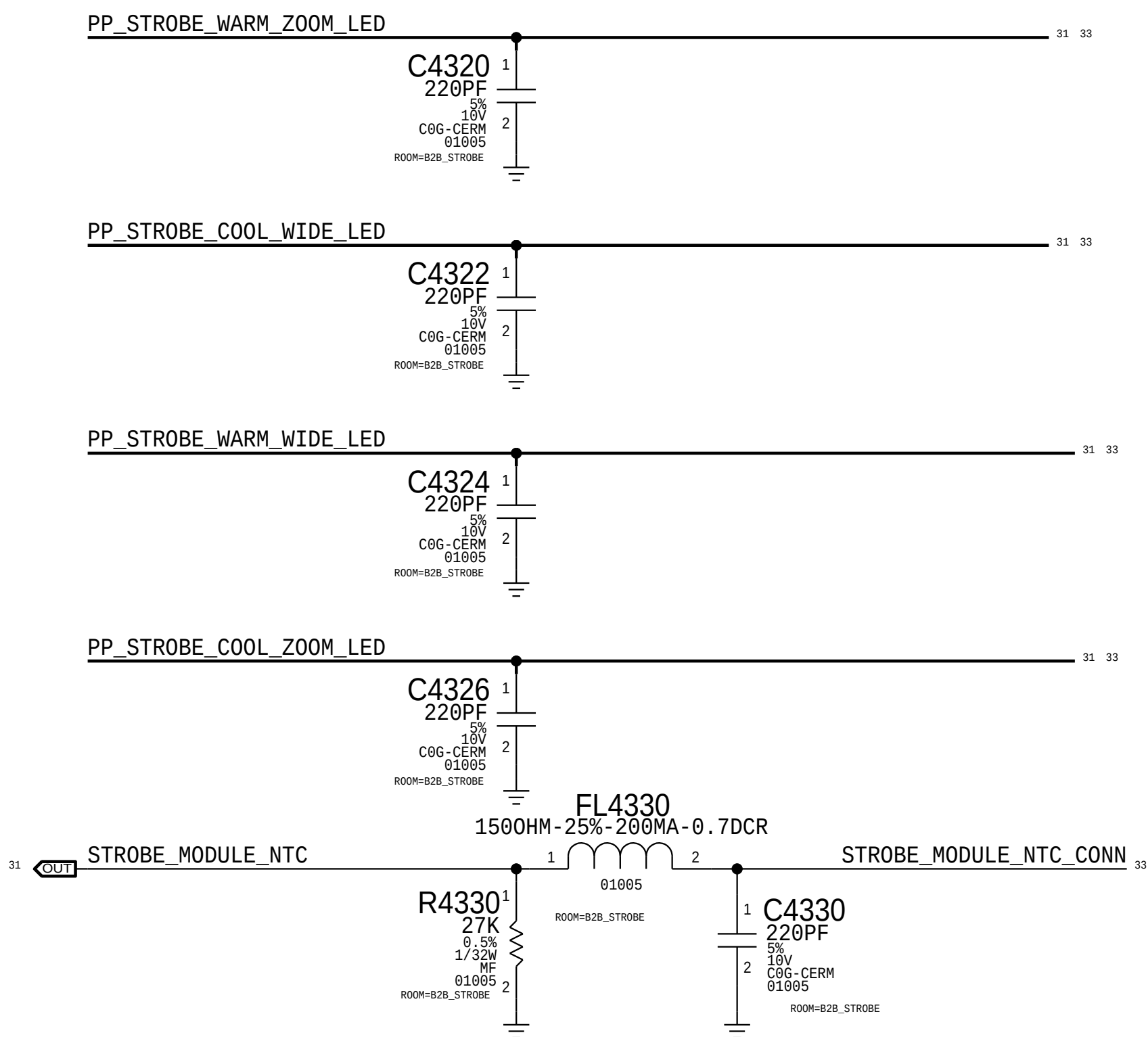


Strobe Connector

Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268



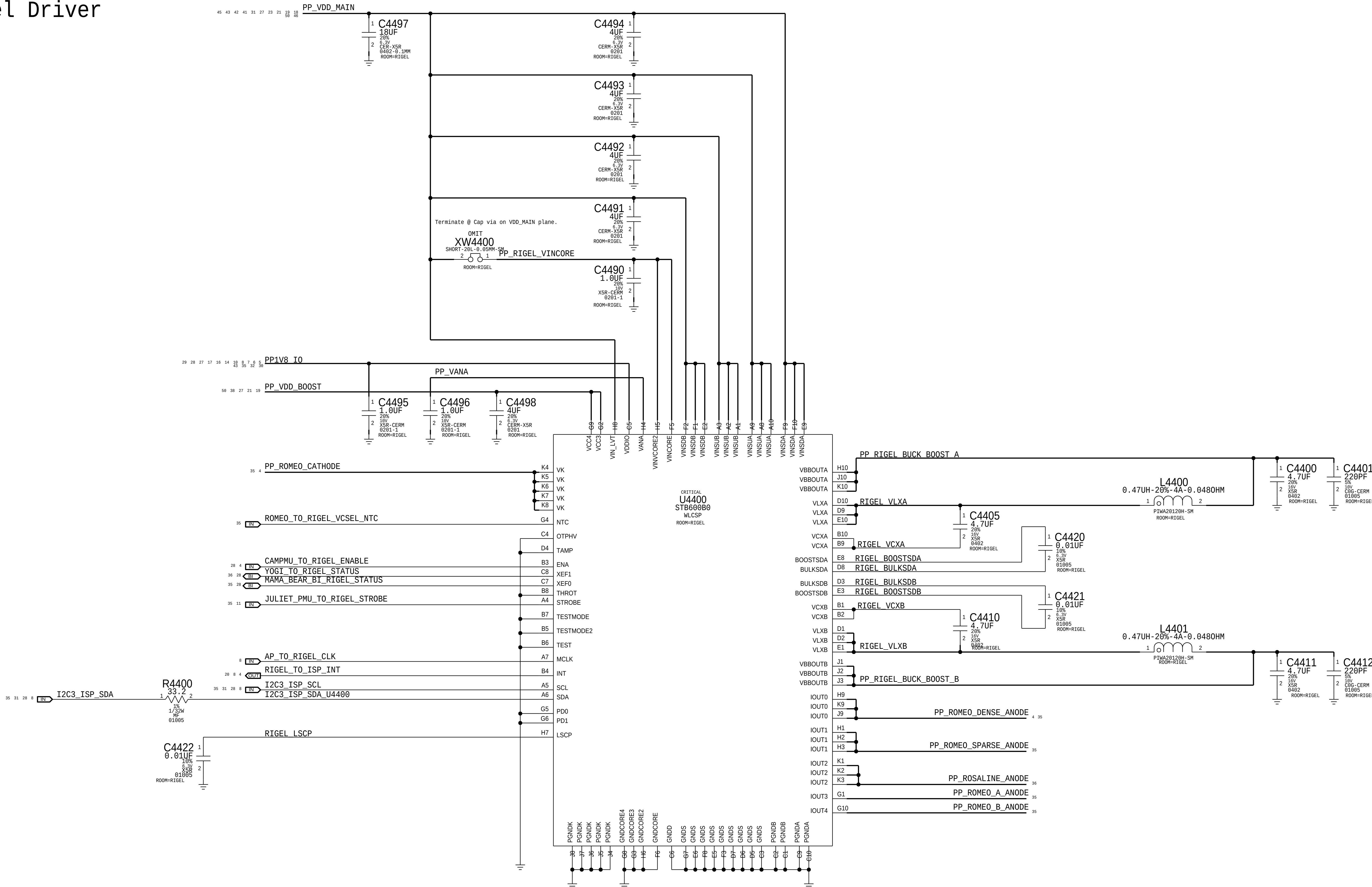
Strobe Filtering



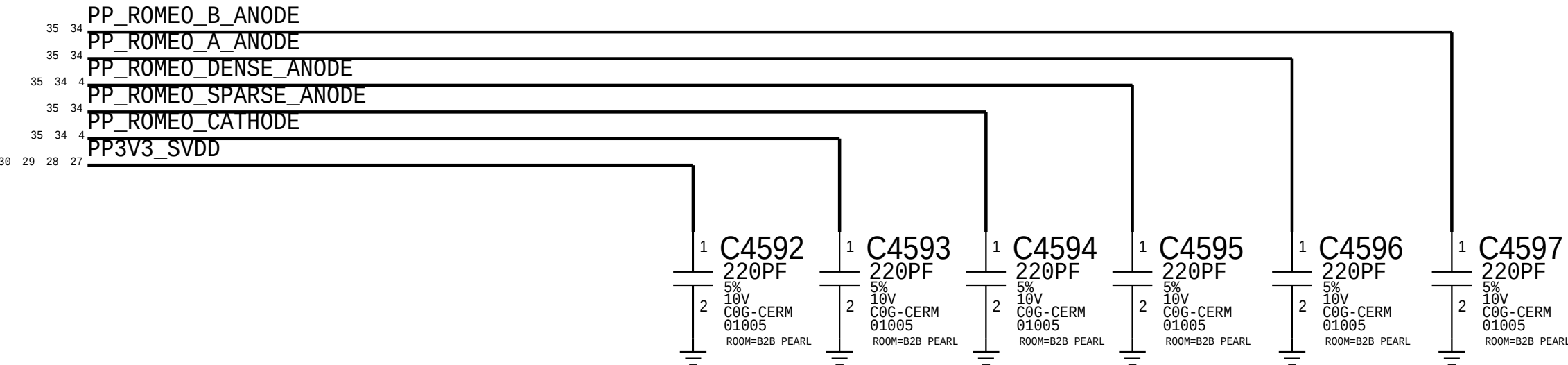
Rigel Driver

8 7 6 5 4 3 2 1

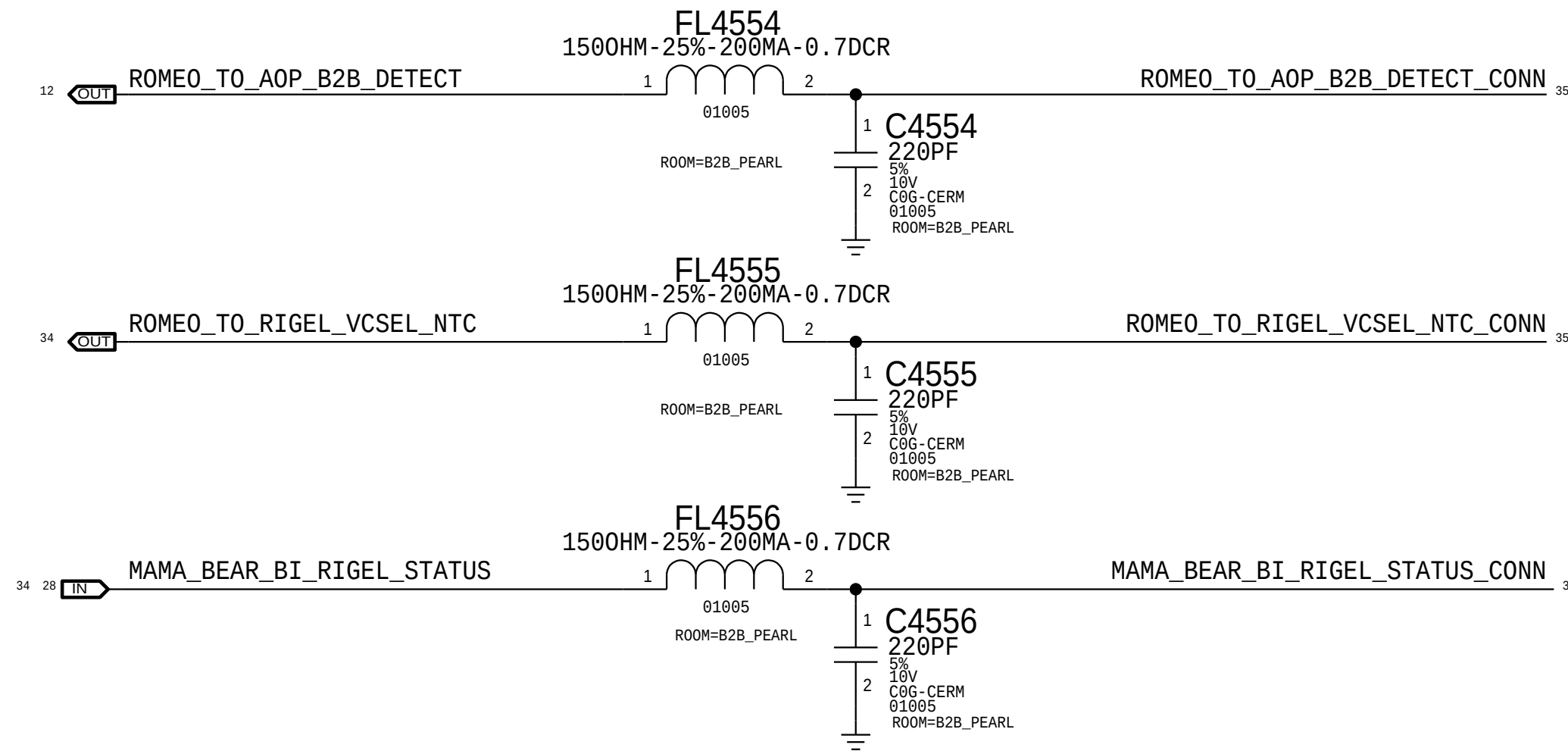
8 7 6 5 4 3 2 1



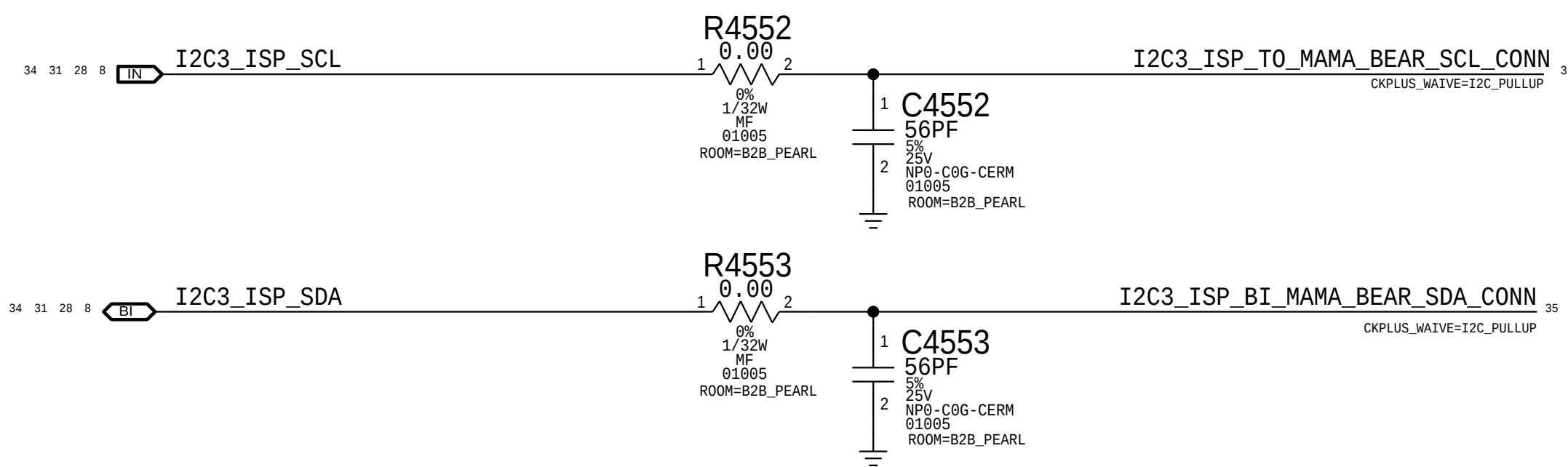
Romeo Power Filtering



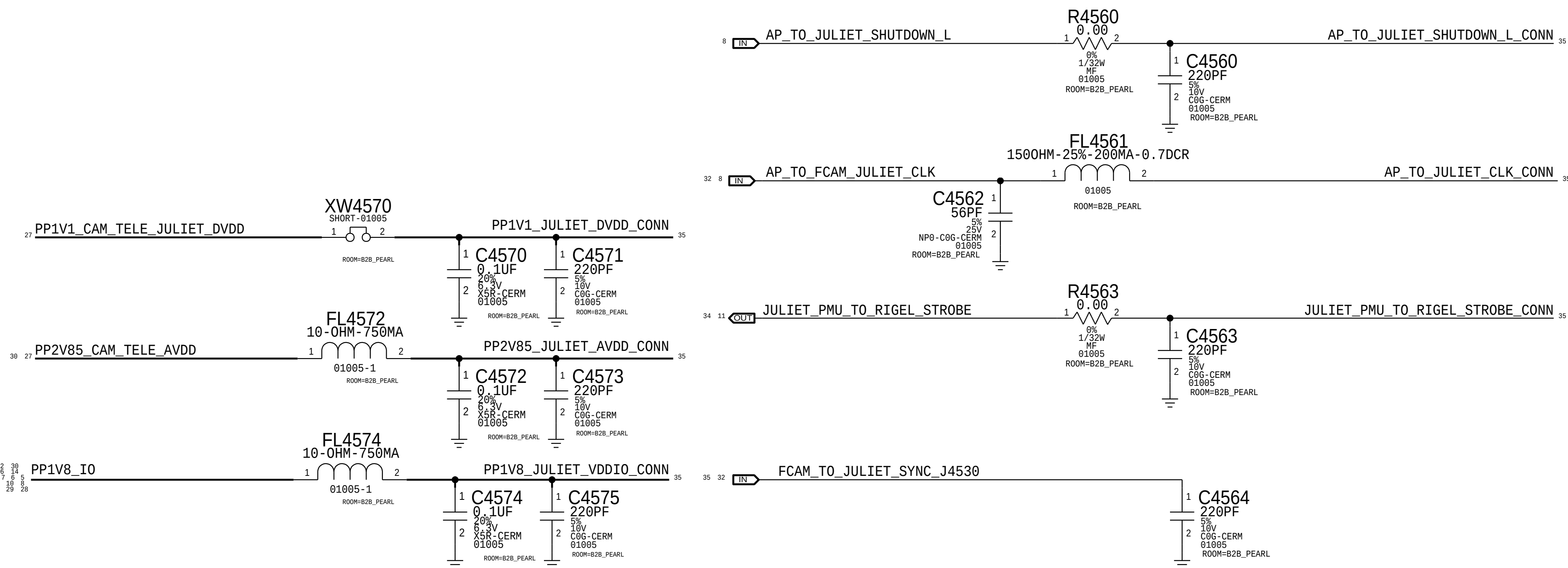
Romeo I/O



ISP I2C3

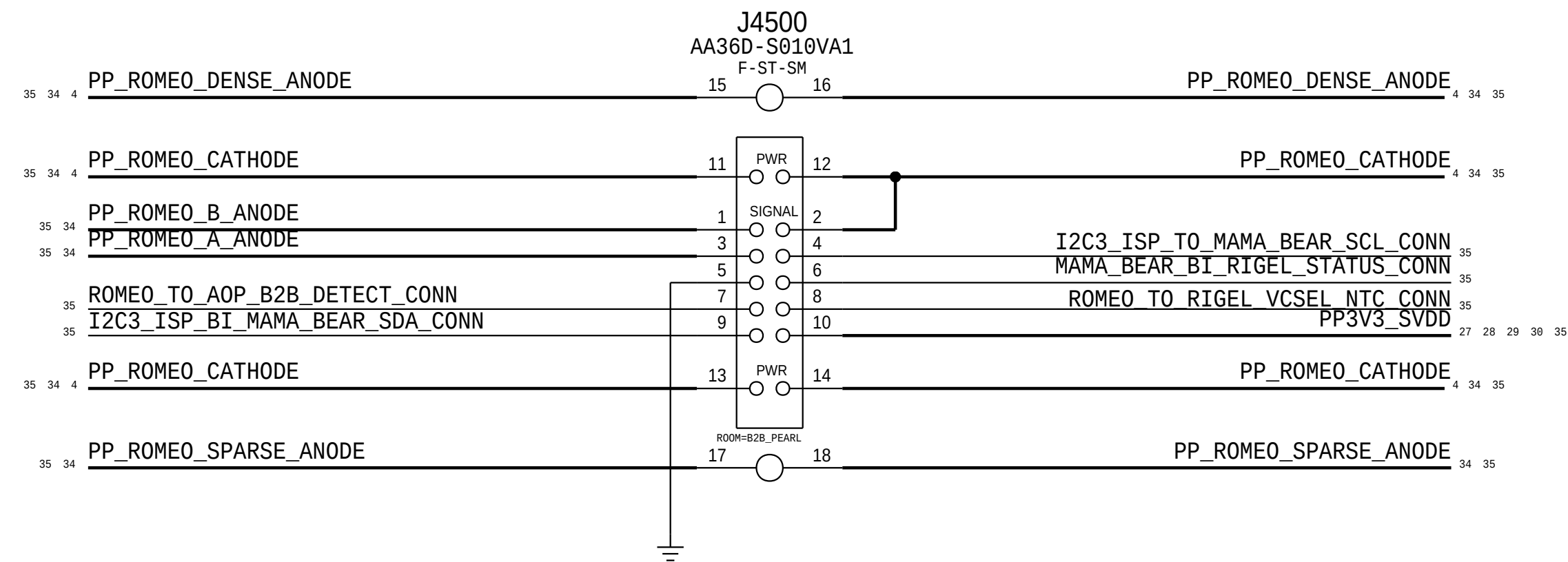


Juliet Power and I/O



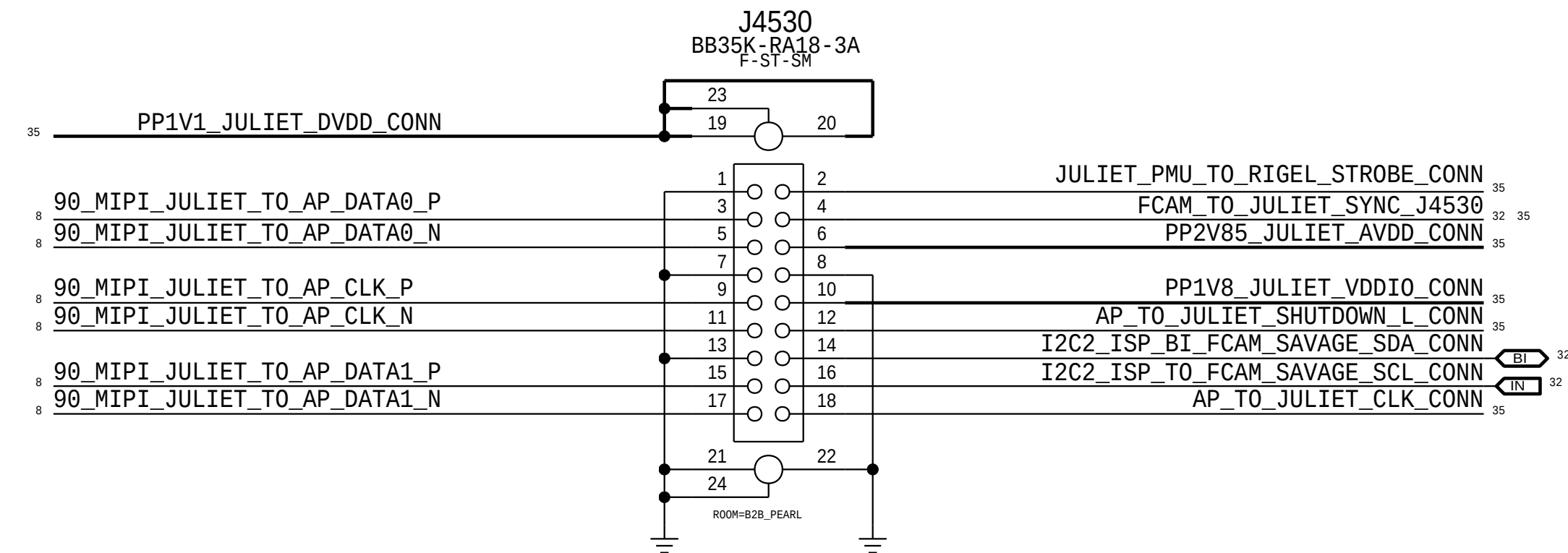
Romeo Connector

Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268

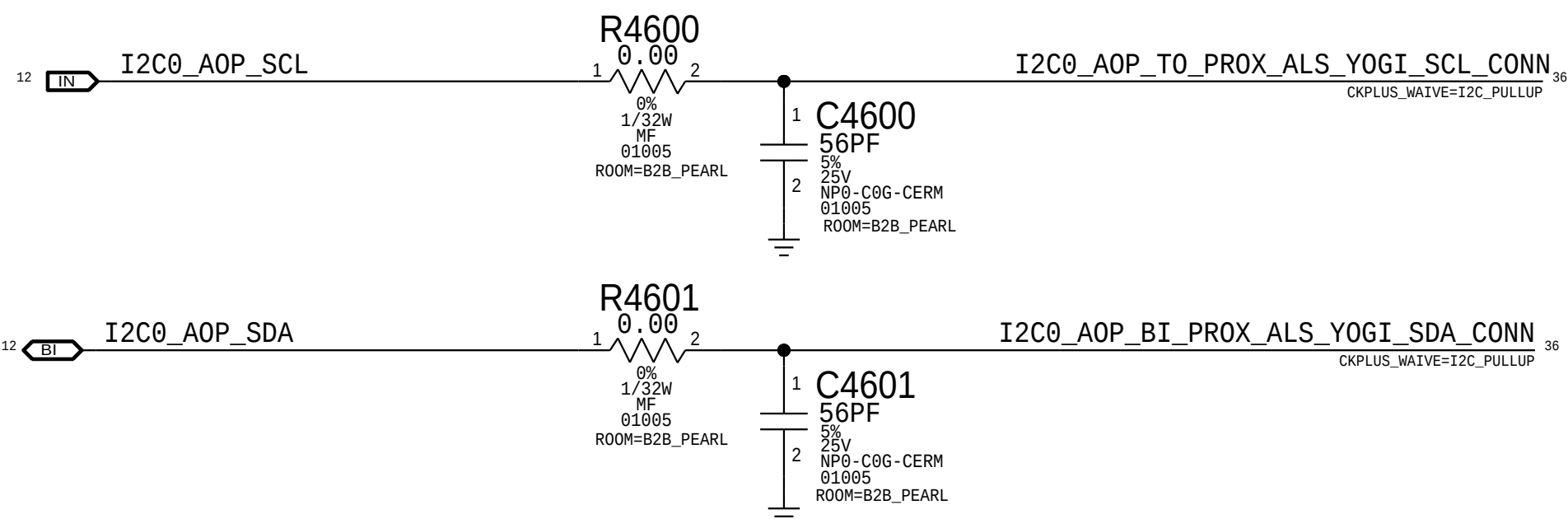


Juliet Connector

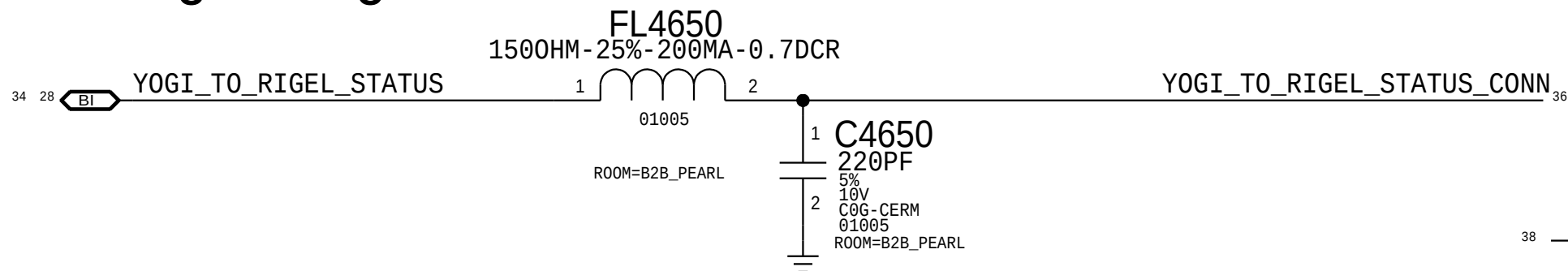
Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245



AOP I2C

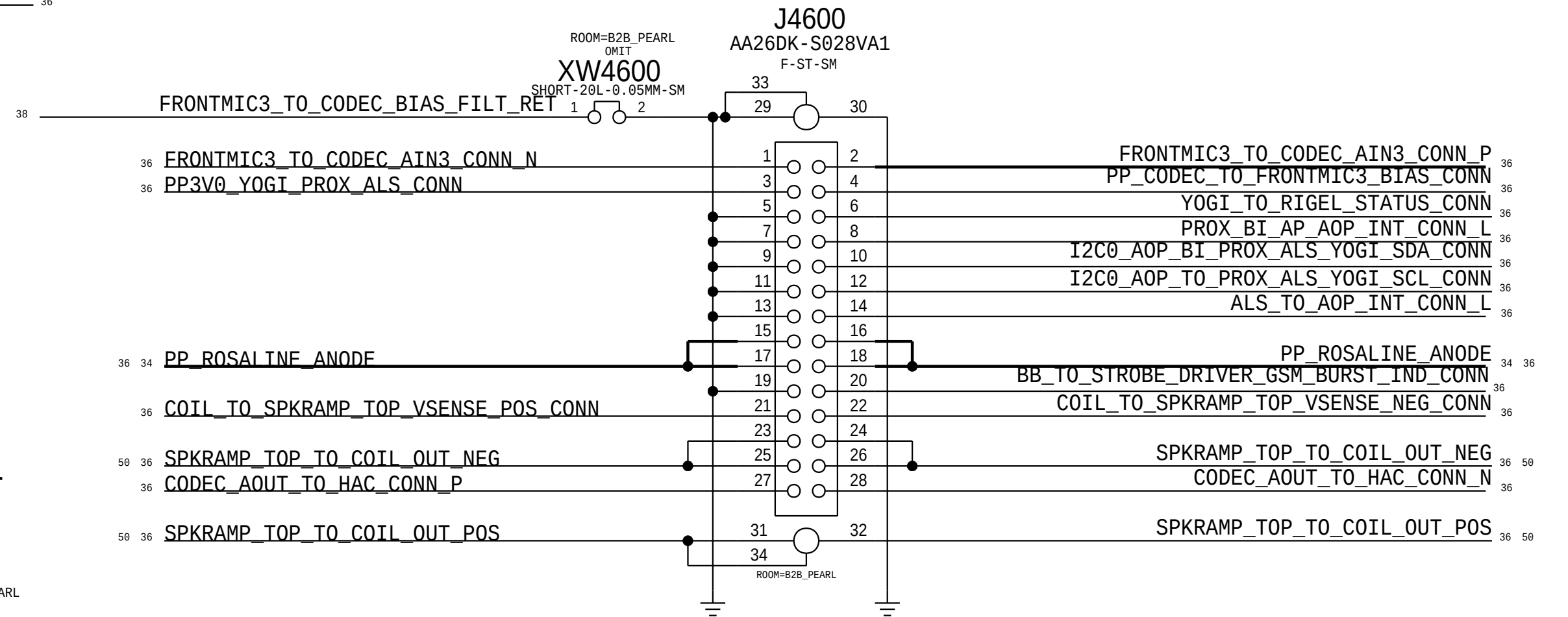


Yogi Signals

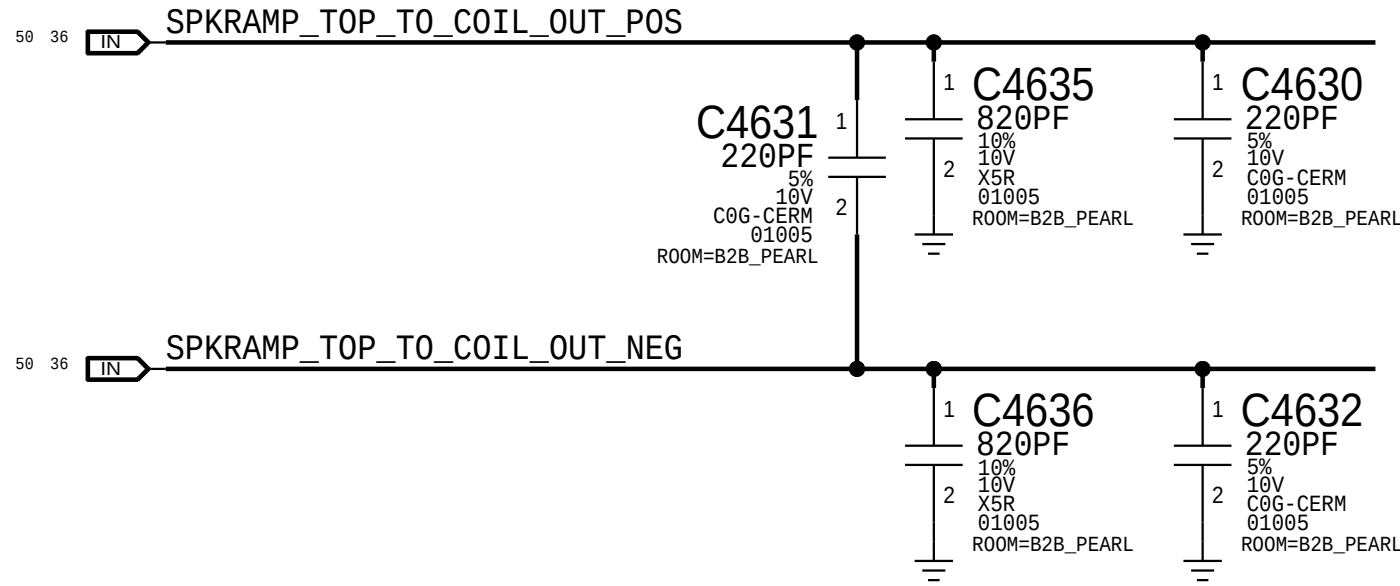


Rosaline + Misc Connector

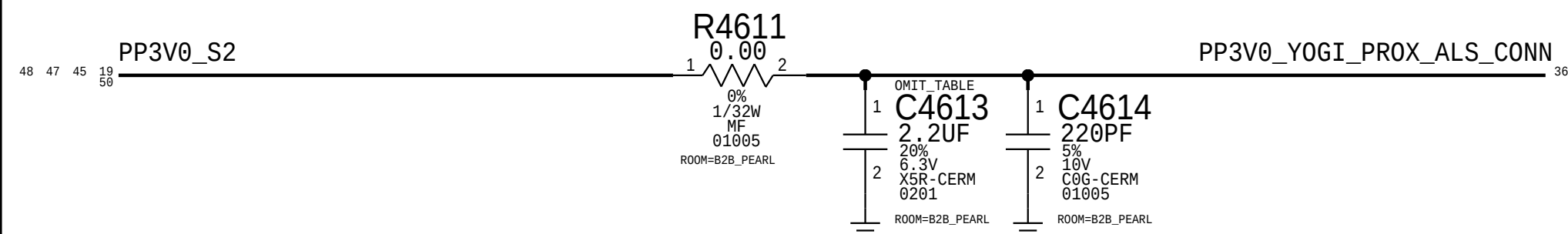
Rcpt: 516S00325 <-- This one on MLB
Plug: 516S00326



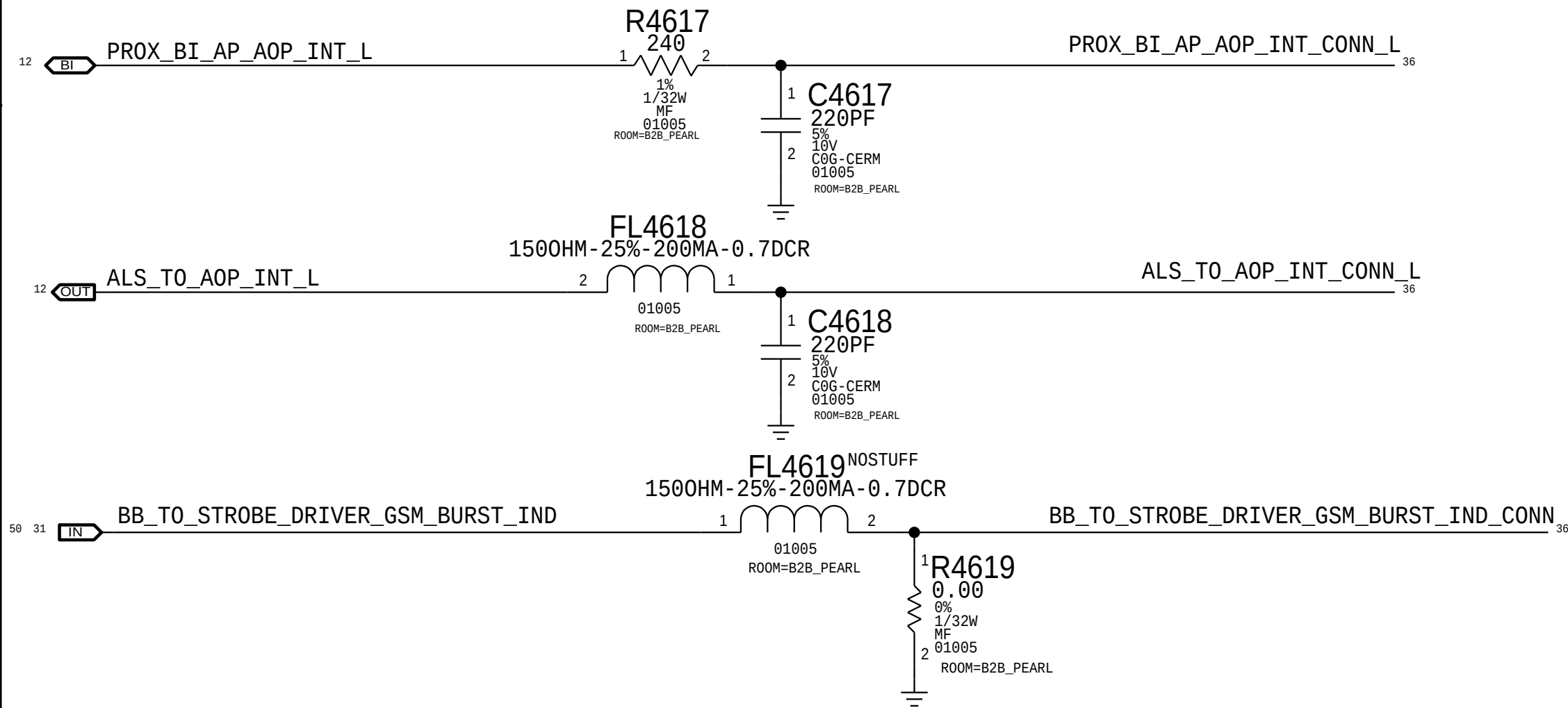
SPEAKER2



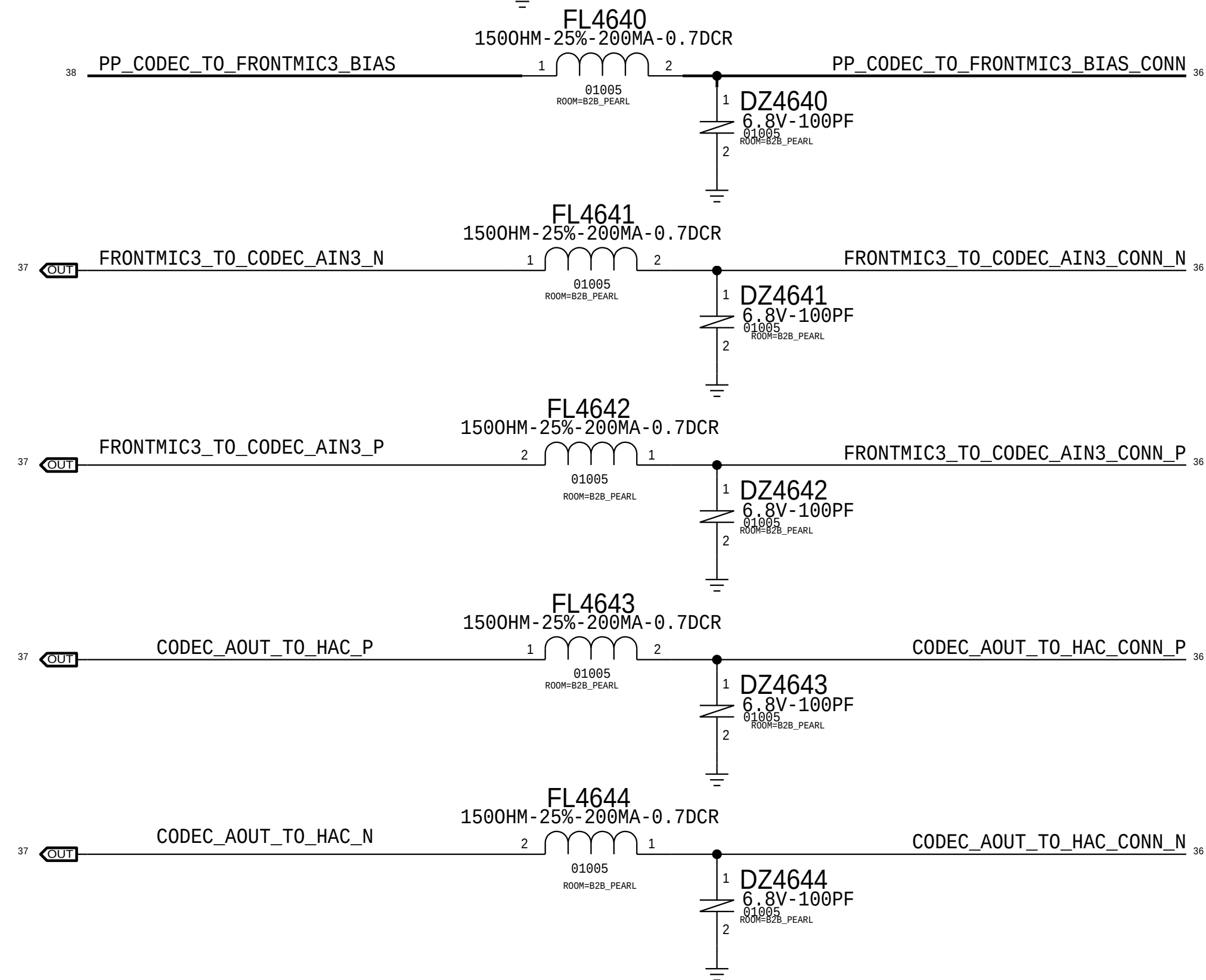
PROX & ALS POWER



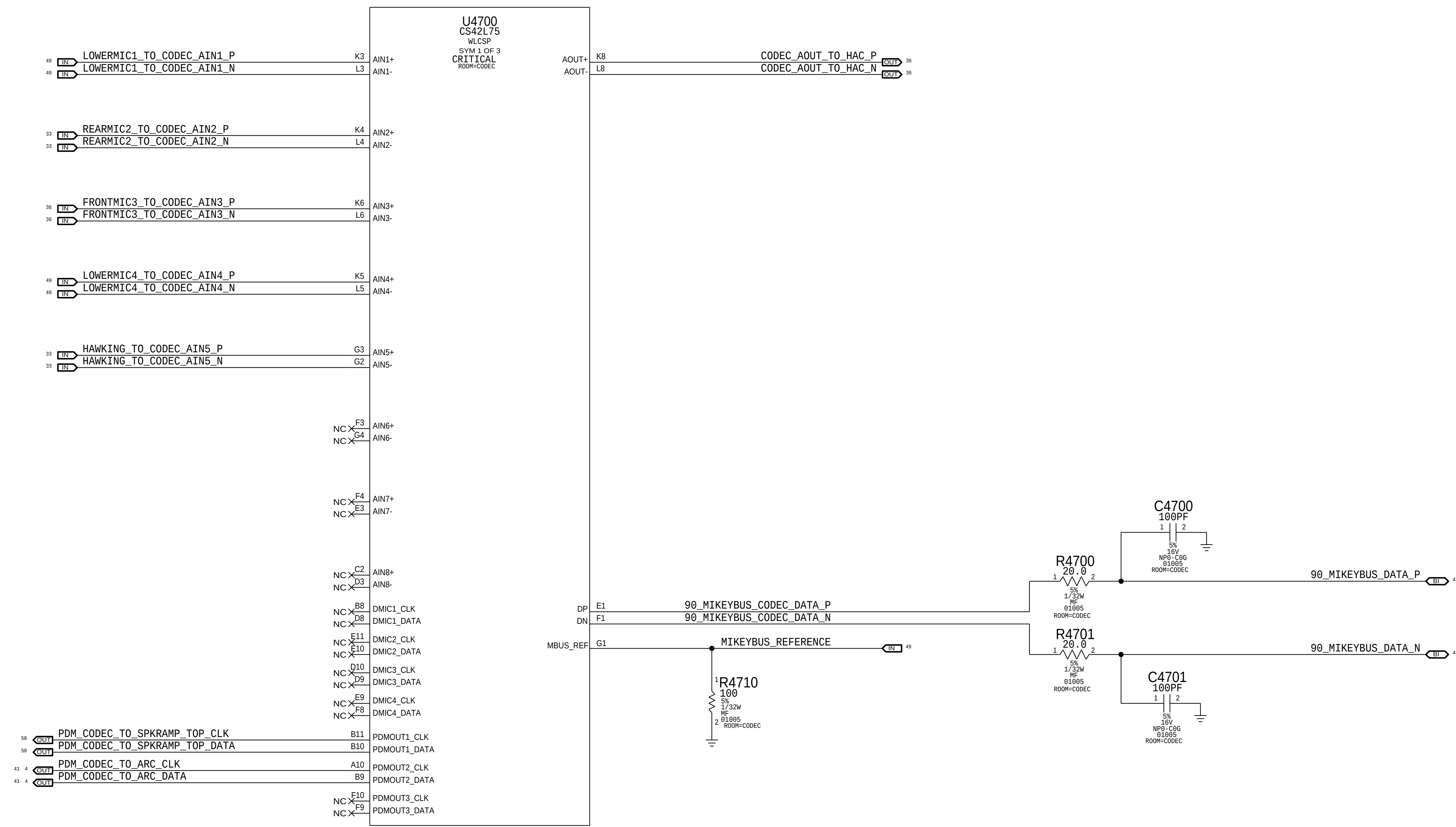
PROX/ALS I/O



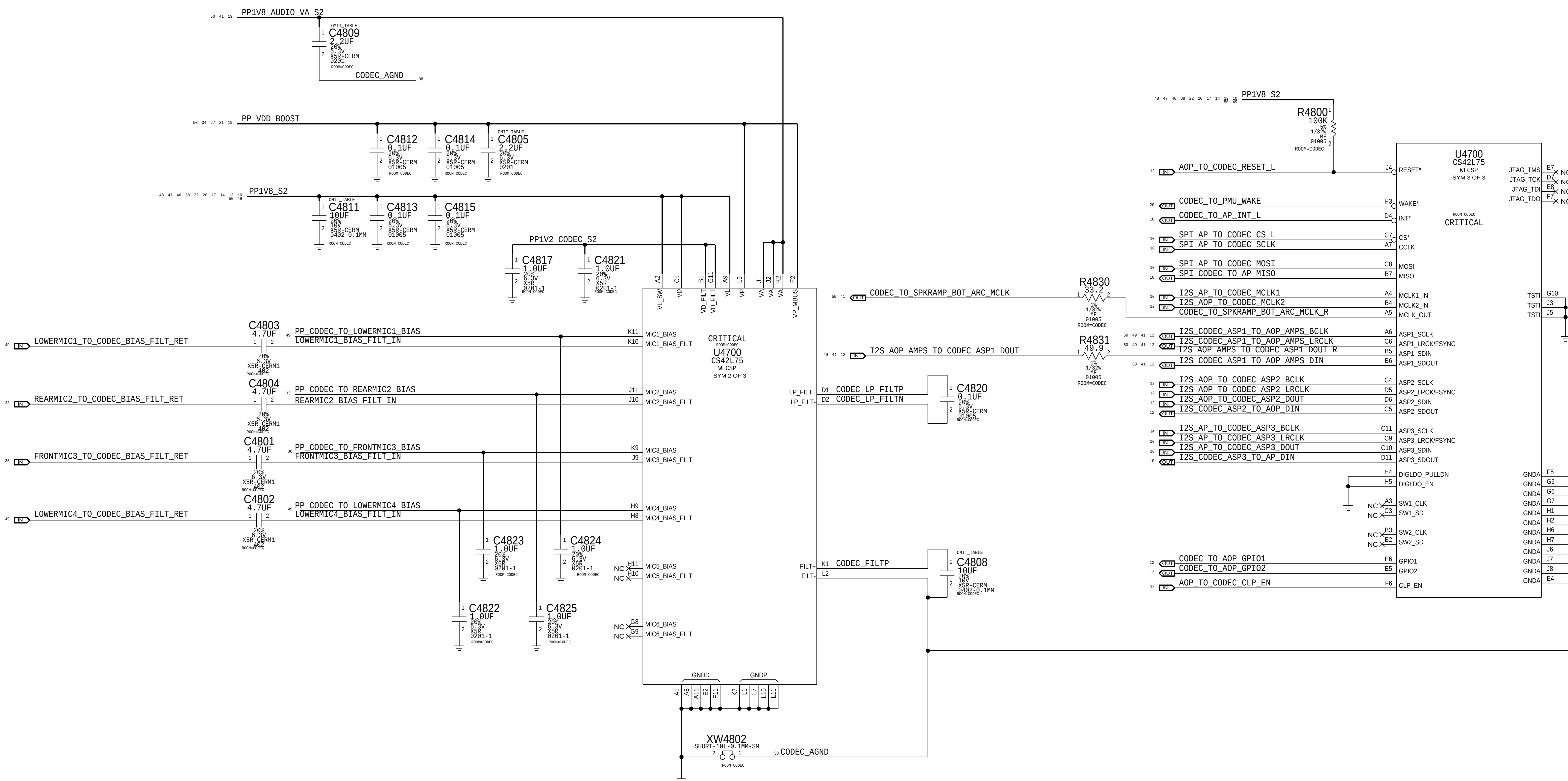
MIC3



CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



CALLAN AUDIO CODEC (POWER & I/O)





8

7

6

5

4

3

2

1

D

D

C

C

South Speaker Amplifier on MLB Bottom

B

B

A

A

8

7

6

5

4

3

2

1



8

7

6

5

4

3

2

1

D

D

C

C

North Speaker Amplifier on MLB Bottom

B

B

A

A

8

7

6

5

4

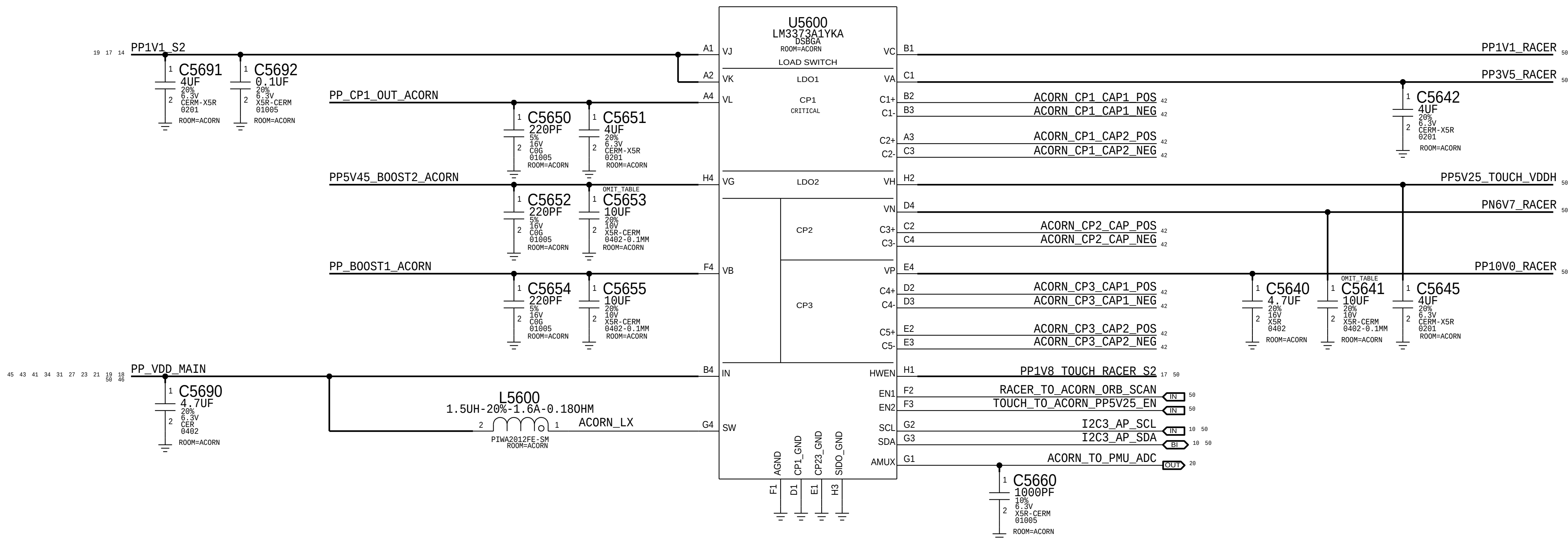
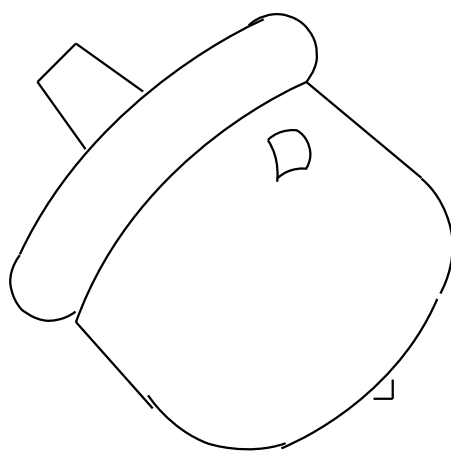
3

2

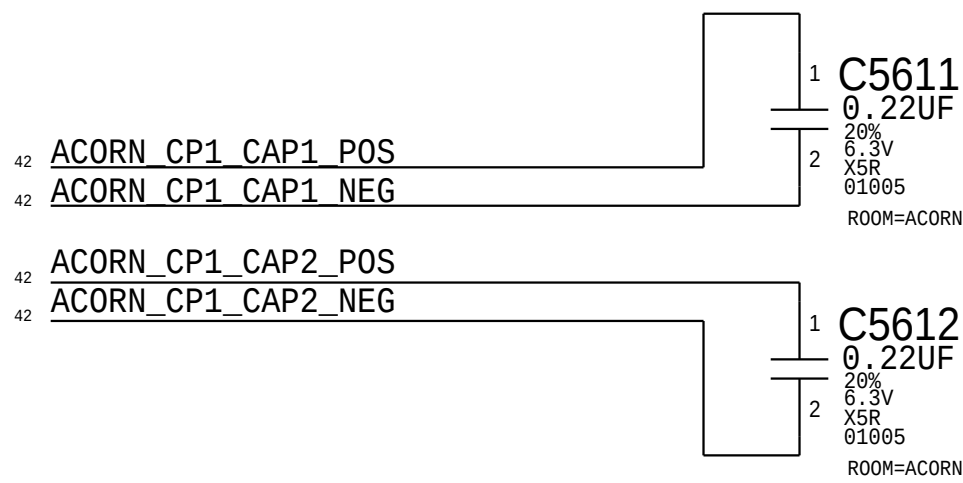
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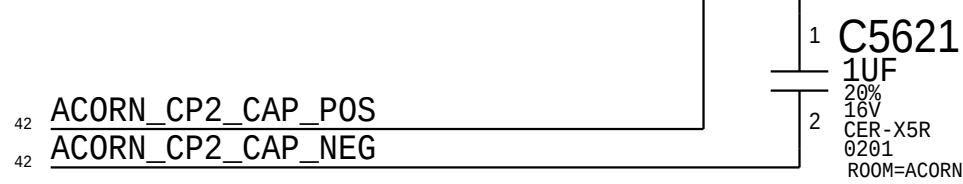
Acorn PMU



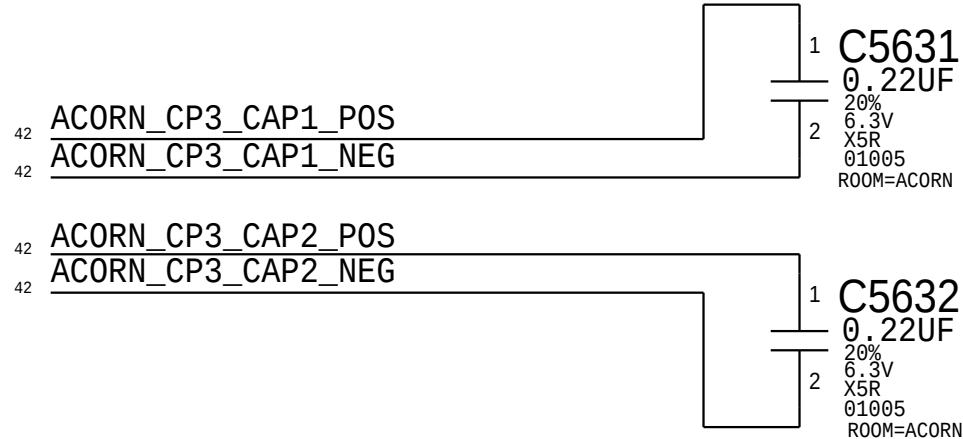
Charge Pump 1 Caps



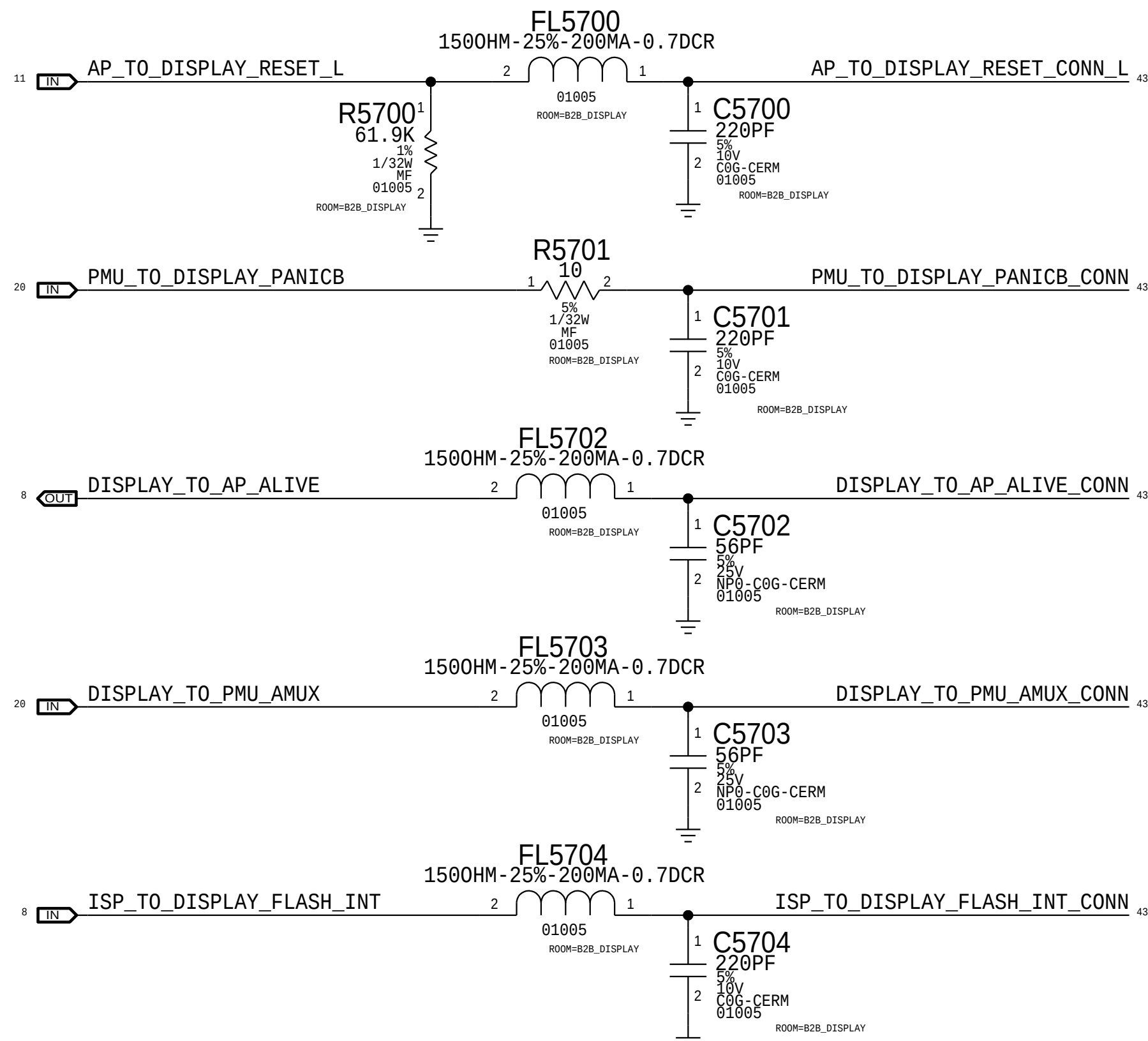
Charge Pump 2 Caps



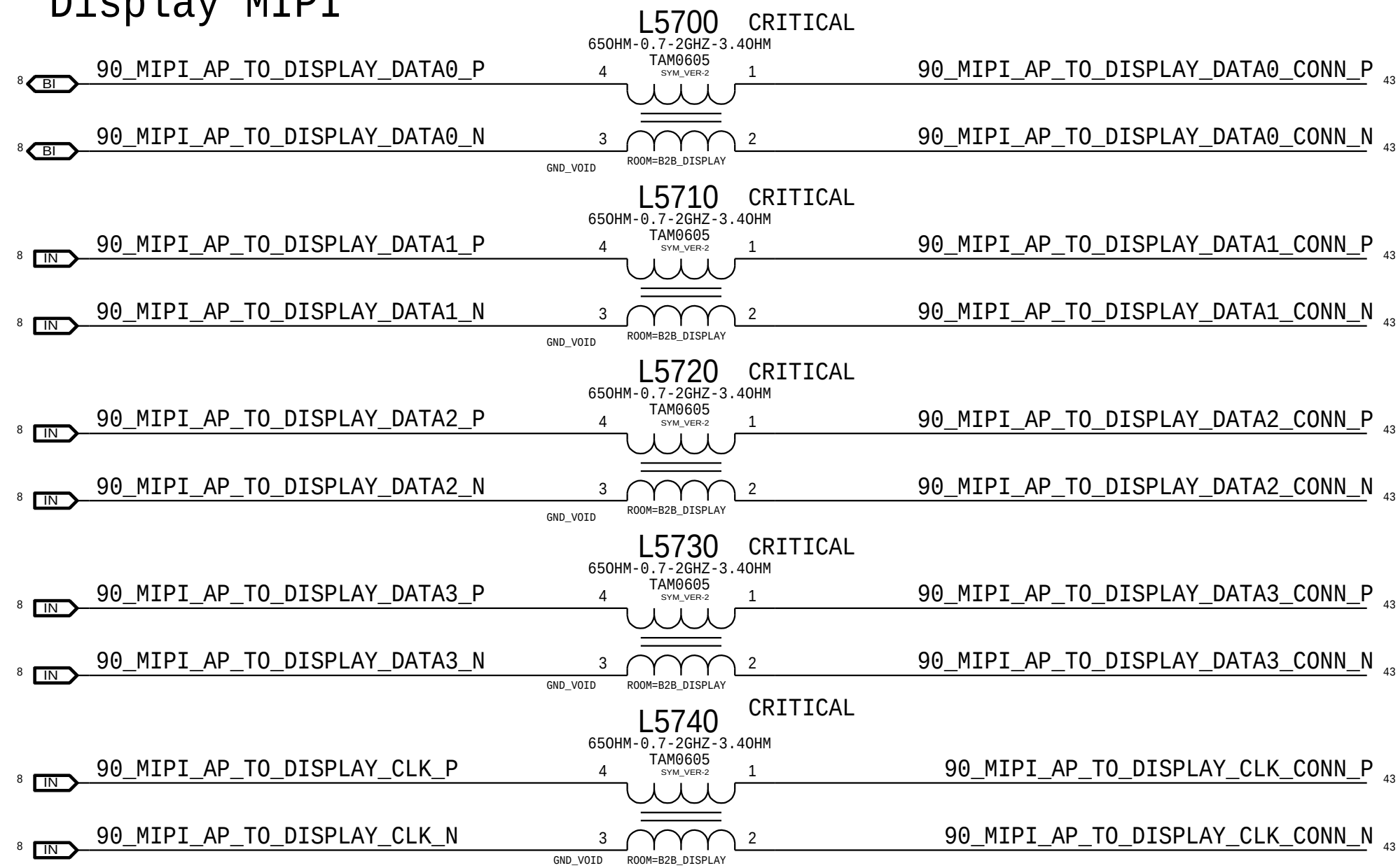
Charge Pump 3 Caps



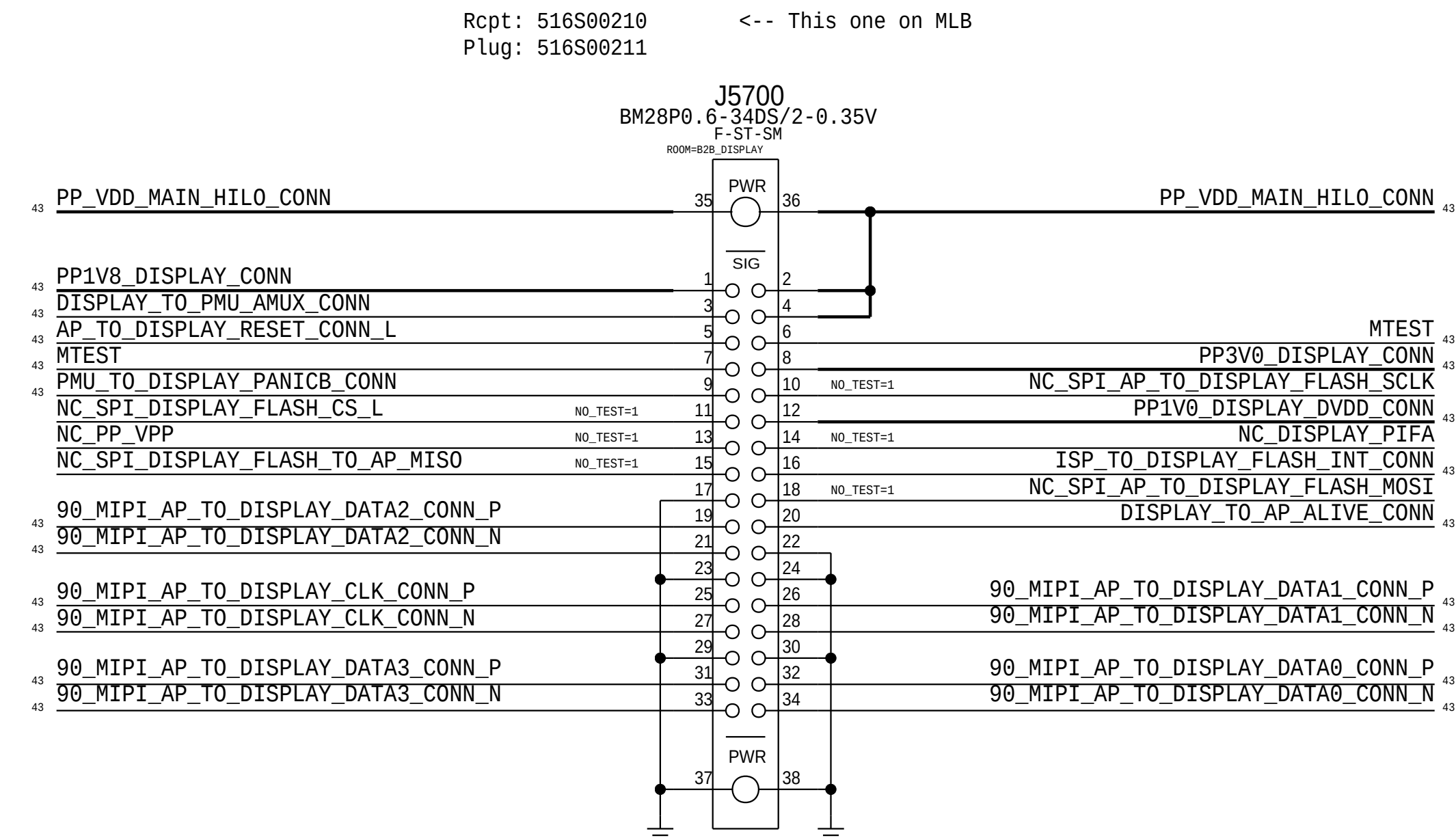
Display Control Signals



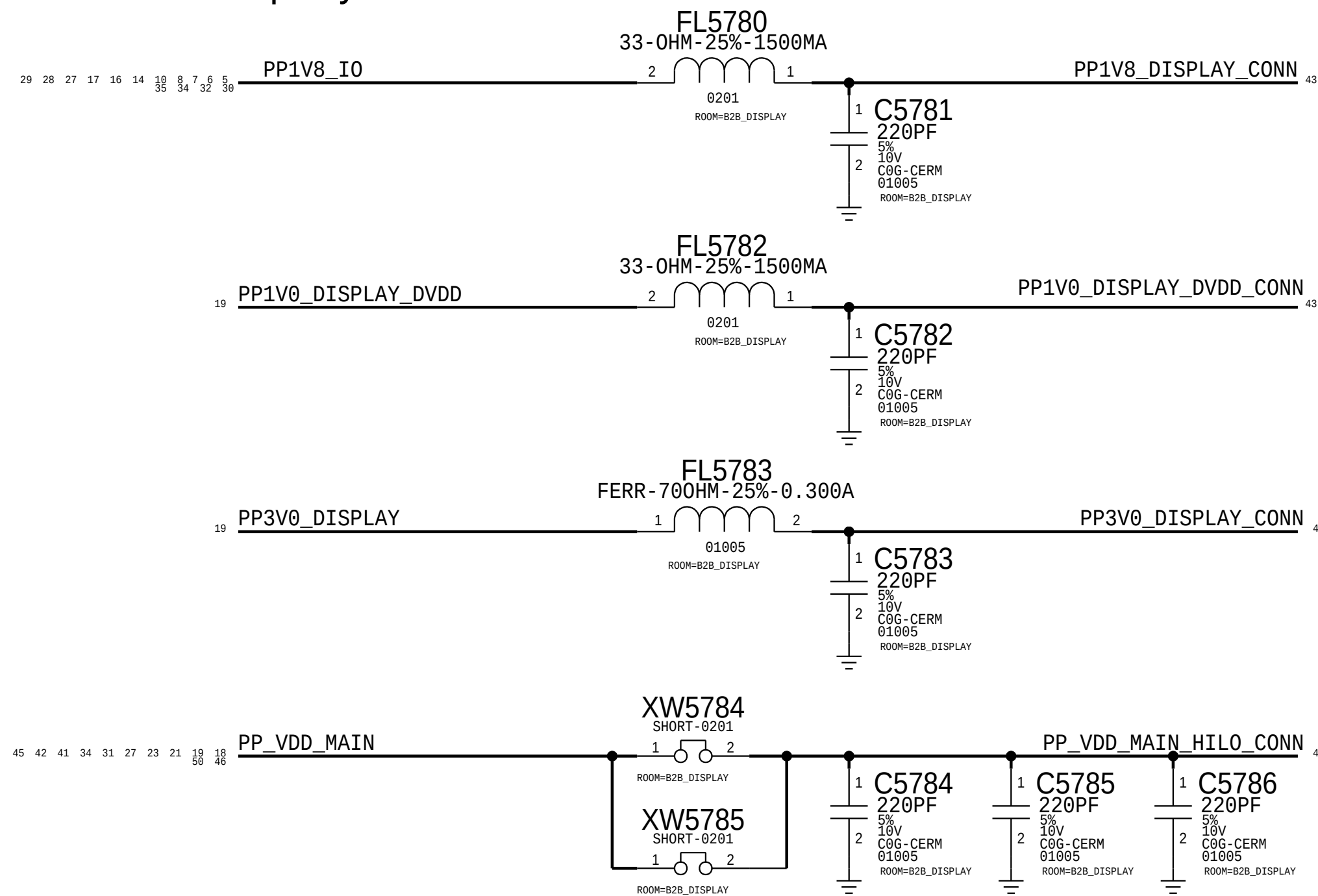
Display MIPI



Display Flex Connector



Display Power





8

7

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3

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1

D

D

C

C

Orb + Touch Connector on MLB Bottom

B

B

A

A

8

7

6

5

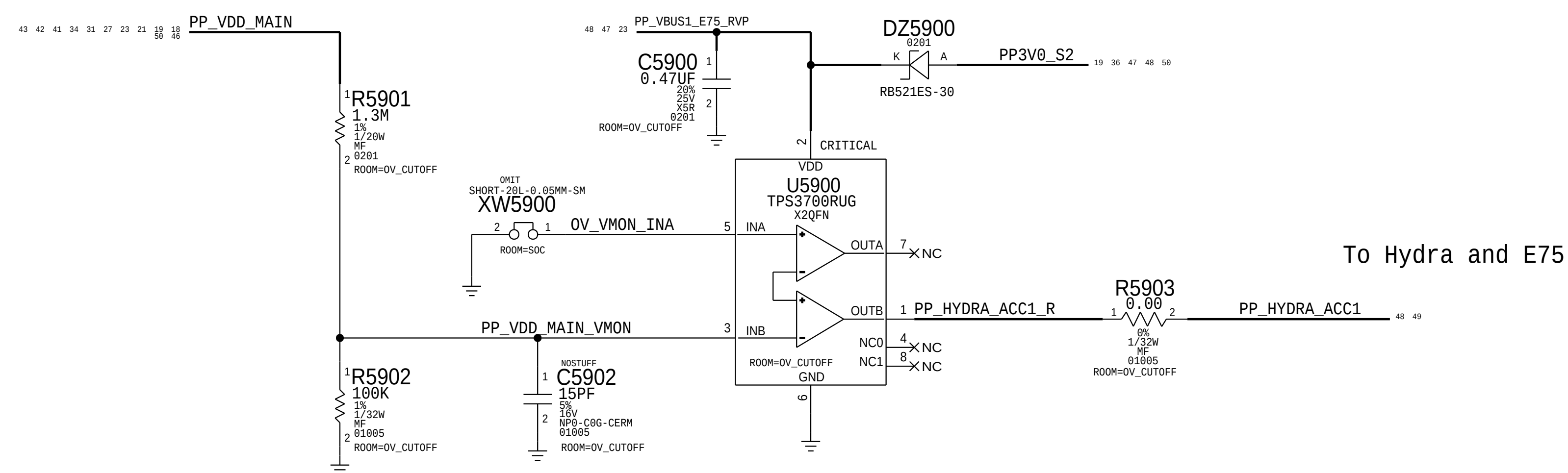
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3

2

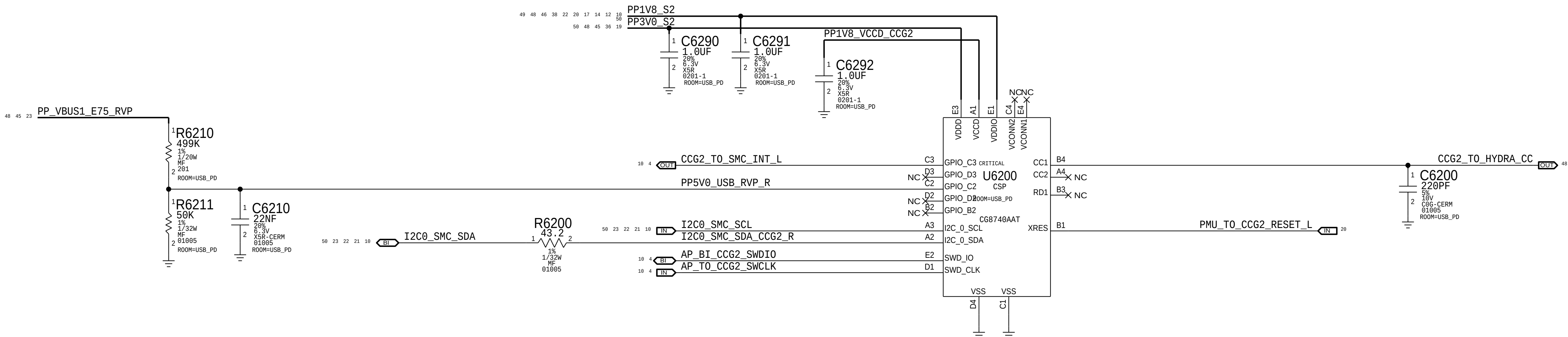
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VDD_MAIN 0V CUT-OFF CIRCUIT



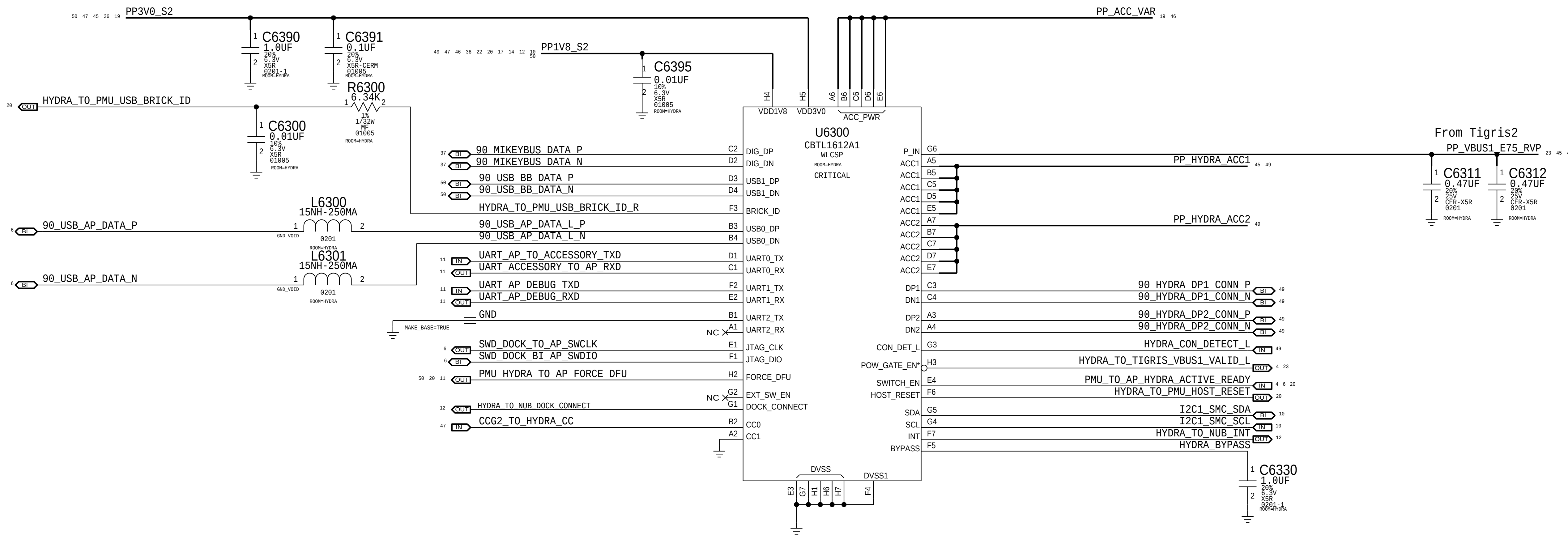


USB-PD



Hydra

I2C Address: 0011010X

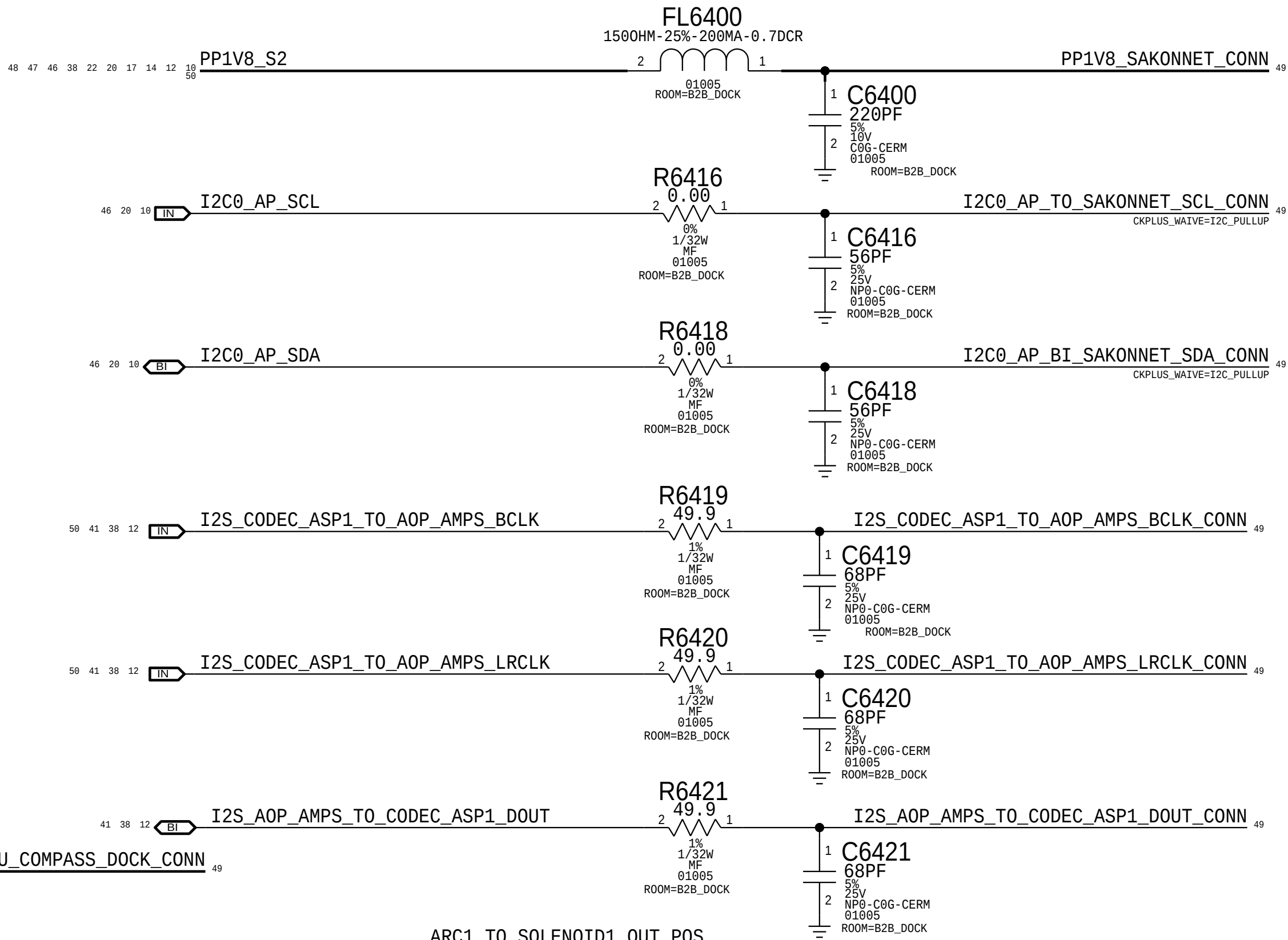




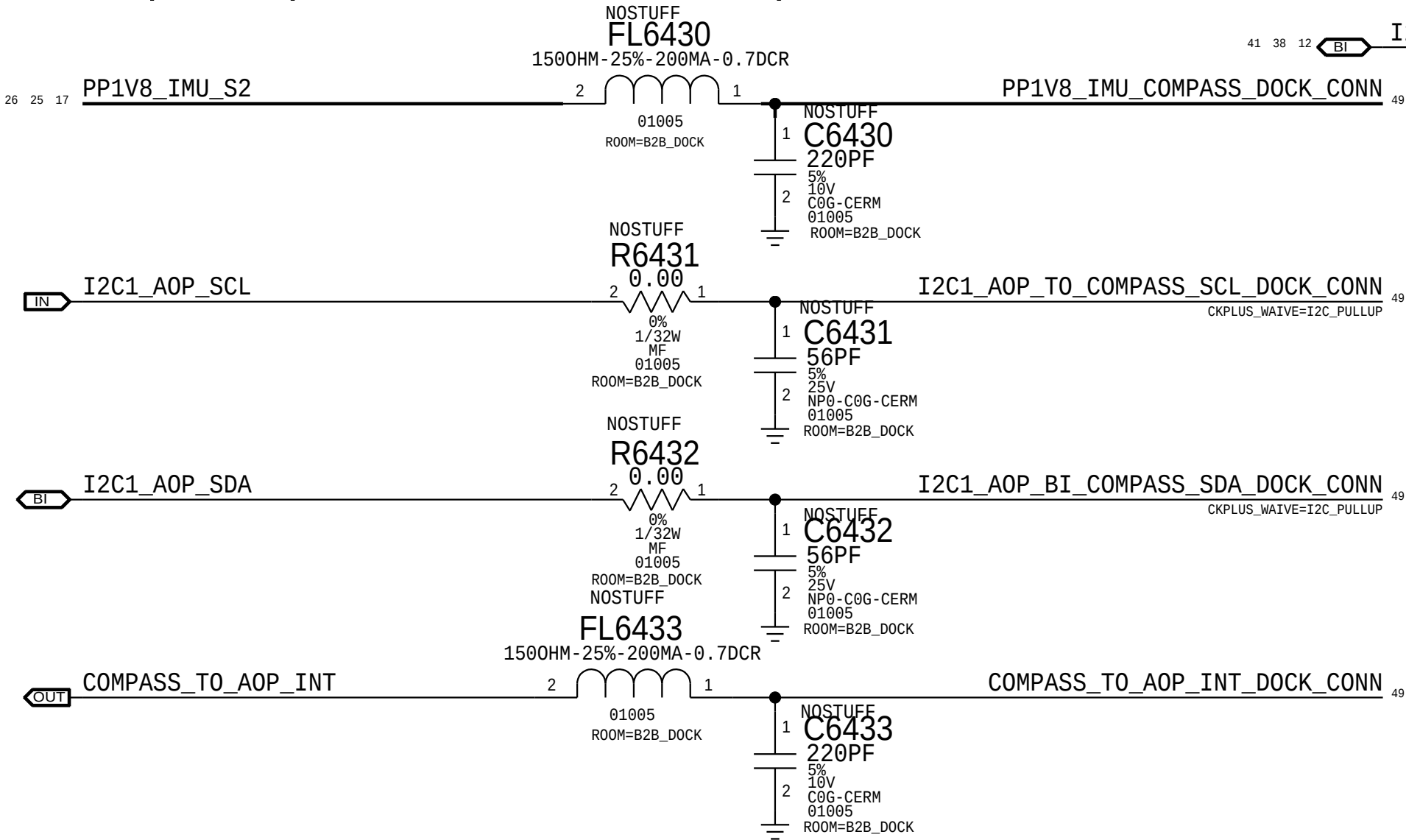
DOCK FLEX CONNECTOR

Rcpt: 516S00038 <-- This one on MLB
Plug: 516S00037

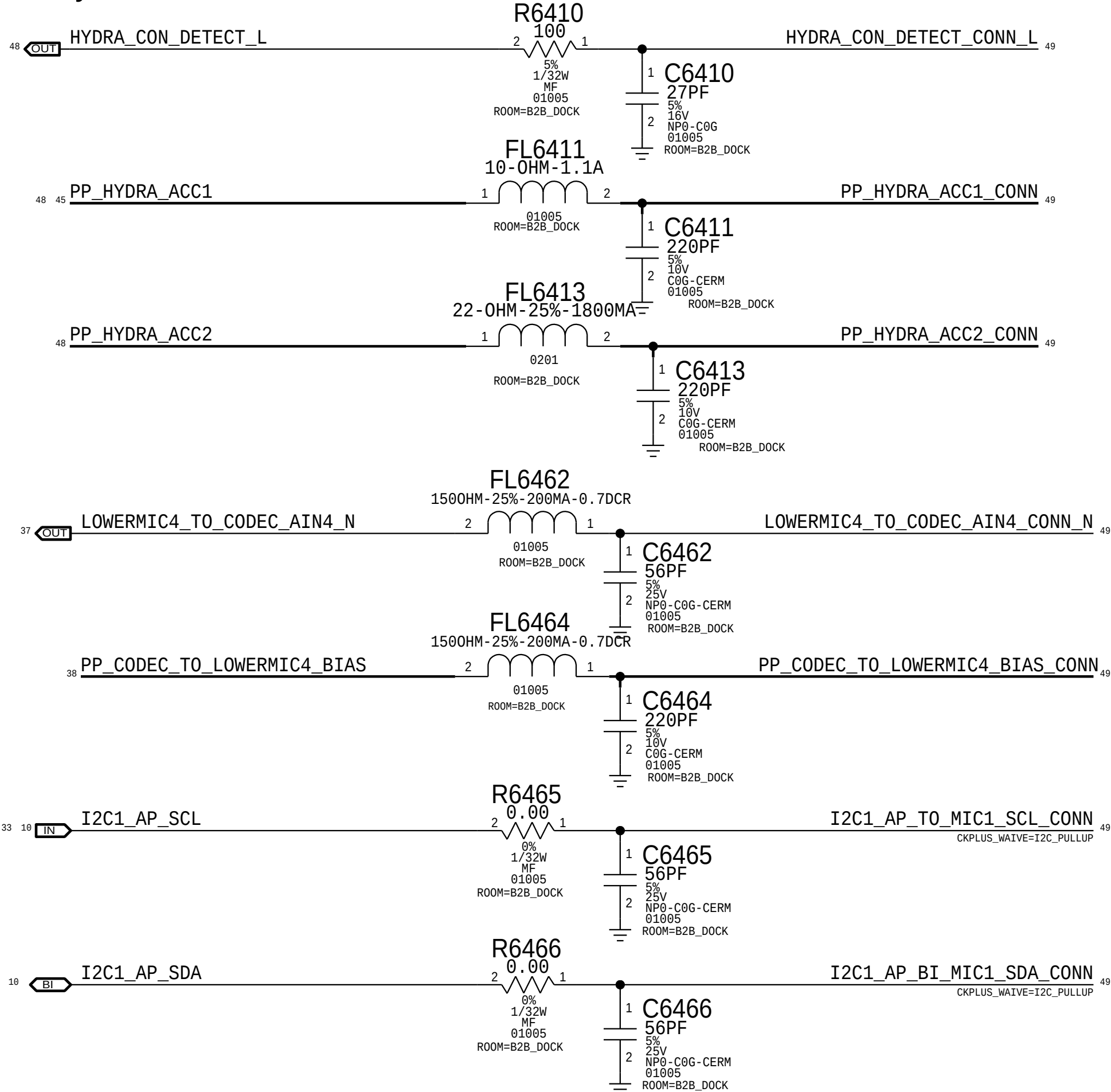
ARC



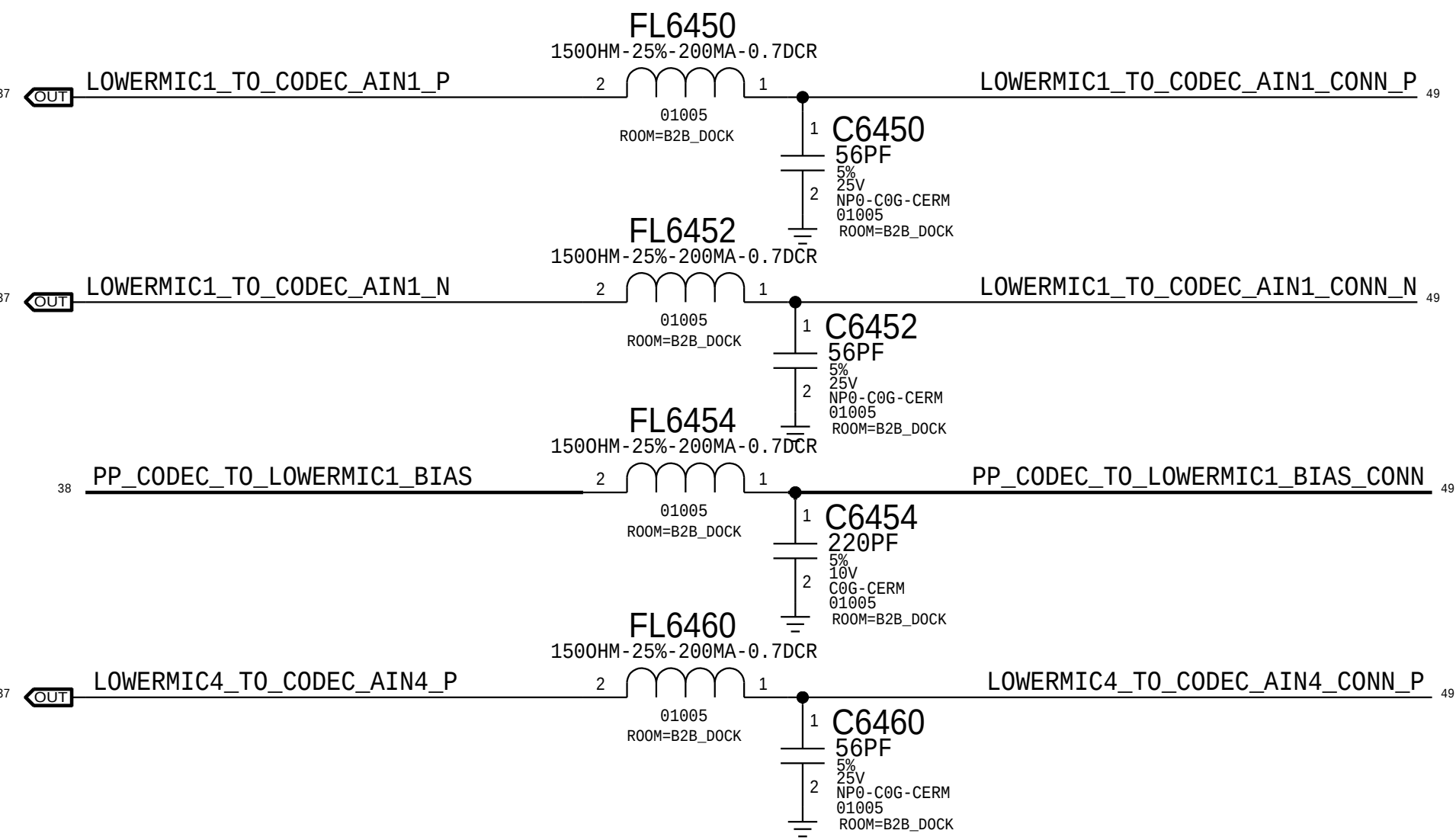
Compass (Dock Flex Location)



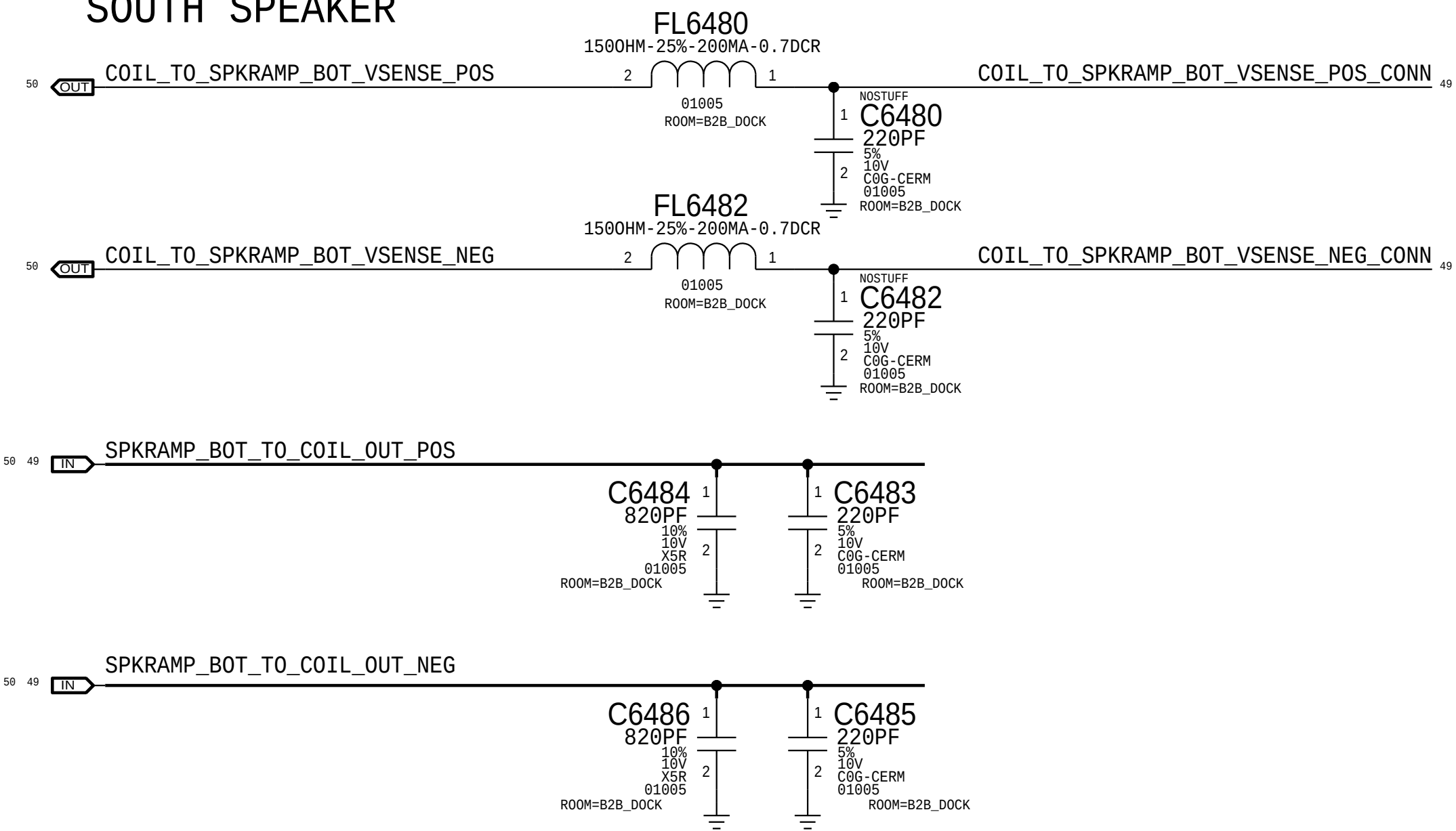
Hydra

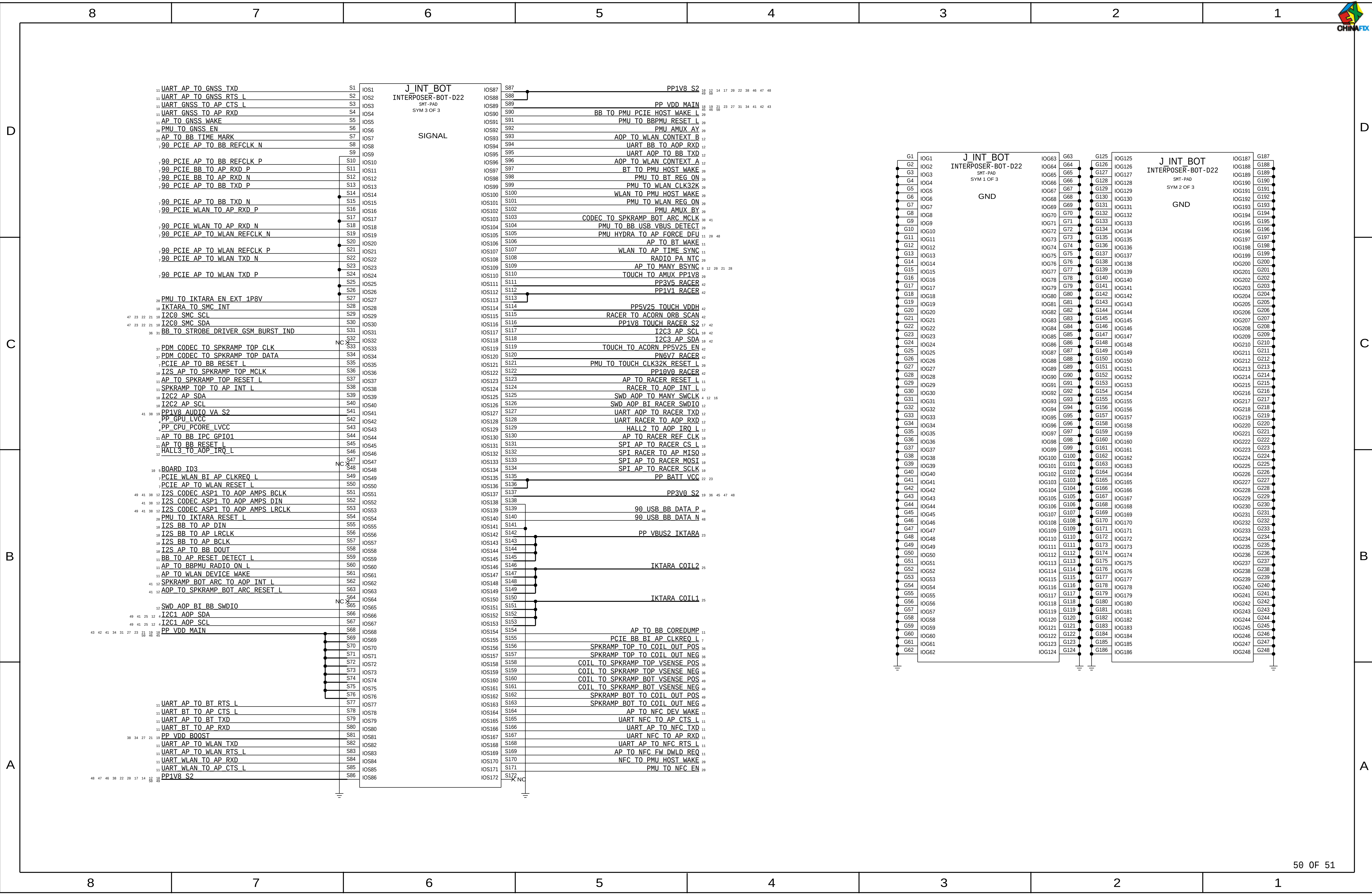


LOWER MIC1 + LOWER MIC4



SOUTH SPEAKER







8

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D

D

C

C

Radios on MLB Bottom

B

B

A

A

8

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6

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4

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2

1



1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	DATE
6	0008416016	ENGINEERING RELEASED	2017-04-06

X891 MLB Bottom: EVT

LAST_MODIFICATION=Wed Apr 5 18:45:28 2017

PAGE	CSA	CONTENTS	SYNC	DATE
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3	4	SYSTEM: Mechanical Components		11/03/2016
4	6	BOOTSTRAPPING		11/03/2016
5	7	SYSTEM: Testpoints (Bottom)	test_mlb	11/09/2016
6	34	SYSTEM POWER: Iktara	test_mlb	11/09/2016
7	49	AUDIO: Speaker Amp Bottom	test_mlb	11/09/2016
8	50	AUDIO: Speaker Amp Top	test_mlb	11/09/2016
9	55	HALL EFFECT	test_mlb	03/23/2017
10	58	CG: B2B Luna & Touch	test_mlb	11/09/2016
11	66	I/O: Interposer (Top)	test_mlb	11/10/2016
12	80	RADIOS	test_mlb	08/19/2016
13	1	page1		
14	2	BOM_OMIT_TABLE		
15	3	PMIC: BUCKS & LDOS		
16	4	PMIC: CLOCKS & CONTROL		
17	5	BB: POWER		
18	6	BB: CONTROL & HS PERIPHERALS		
19	7	BB: GPIOs & QLINK		
20	8	XCVR		
21	9	XCVR		
22	10	QPOET MODULE		
23	11	LB PAD		
24	12	HB PAD		
25	13	UHB PAD		
26	14	2G PA		
27	15	LB DSM		
28	16	HB DSM		
29	17	COUPLER2		
30	18	GNSS		
31	19	METROCIRC		
32	20	LOWER ANTENNA		
33	21	UPPER ANTENNA		
34	22	SIM, EUICC, DEBUG CONN	SIM TRAY	07/05/2016
35	23	TEST POINTS	PROBE POINTS	07/06/2016
36	24	RF FE TO GPO TRANSLATOR		
37	1	page1		
38	75	NFC		
39	1	SymbolPorts		
40	4	Guinness	WIFI	01/30/2014
41	5	WiFiANTFeeds		

[illegible]

BOM: 639-03222 (ROW)
BOM: 639-03410 (JPN)
MC0: 056-04079

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
651-02222	1	SCH, MLB_B0T, X891	SCH	NO	COMMON
828-00864	1	PCB, MLB_B0T, X891	PCB	NO	COMMON

Sub Designs

SOURCE PROJECT	SUB-DESIGN NAME	VERSION	HARD/ SOFT	SYNC_DATE/TIME
D22	RADIO_MLB	0.149.0	S	2017_04_05_18:45:19
D22	NFC_MLB	0.22.0	S	2017_03_22_22:10:24
D22	WIFI_MLB	0.20.0	S	2017_04_04_08:01:02

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB_BOT, 639-03418, JPN)	EEEE_HP27	NO	JPN
825-7691	1	EEEE FOR (MLB_BOT, 639-03222, ROW)	EEEE_HL39	NO	ROW

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, Kyocera
138S00150	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, SEMCO
138S00151	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Kyocera
138S00163	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Kyocera
138S00164	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00145	138S00146	BOM_TABLE_ALTS	ALL	0402,5.1uF@3V, Kyocera
138S00165	138S00146	BOM_TABLE_ALTS	ALL	0402,5.1uF@3V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Kyocera
138S00142	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, SEMCO
138S00166	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Taiyo

CRITICAL PART#	COMMENT
138S00149	0402-3T,10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402,16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201,3uF@1V

CRITICAL PART#	COMMENT
138S00146	0402,5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201,1.1uF@3V

Global Ferrites

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00558	152S00557	BOM_TABLE_ALTS	ALL	IND,MLD,0.47UH,20%,2.5A,80MD,1608
155S00194	155S00610	BOM_TABLE_ALTS	ALL	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005
155S00200	155S00610	BOM_TABLE_ALTS	ALL	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005

CRITICAL PART#	COMMENT
152S00557	IND,MLD,0.47UH,20%,2.5A,80MD,1608
155S00610	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0648	138S0652	BOM_TABLE_ALTS	ALL	CAP,XSR,4.7UF,6.3V,0.05MM,0402,TAIYO
138S00024	138S0986	BOM_TABLE_ALTS	ALL	CAP,C0G,3.300M,1% 50V,20%,4V,0305,MURATA
138S0706	138S0739	BOM_TABLE_ALTS	ALL	CAP,C0R,10F,20%,10V,XSR,0305,MURATA
138S0945	138S0739	BOM_TABLE_ALTS	ALL	CAP,C0R,10F,20%,10V,XSR,0305,KYOCERA
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP,C0R,XSR,0.22UF,20%,6.3V,20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP,C0R,XSR,0.22UF,20%,6.3V,01005
138S00049	138S0831	BOM_TABLE_ALTS	ALL	CAP,C0R,XSR,2.2UF,20%,6.3V,0201

CRITICAL PART#	COMMENT
138S0652	CAP,XSR,4.7UF,6.3V,0.05MM,0402,TAIYO
138S0986	CAP,CER,3-TERM,7.5UF,20%,4V,0402,TAIYO/TDK
138S0739	CAP,CER,1UF,20%,10V,XSR,0201,MURATA

138S0706	CAP,CER,XSR,0.22UF,20%,6.3V,20%
132S0400	CAP,CER,XSR,0.22UF,20%,6.3V,01005
138S0831	CAP,CER,XSR,2.2UF,20%,6.3V,0201

Global Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00653	152S00651	BOM_TABLE_ALTS	ALL	IND,1.2UH,3A,2016,0.6SZ
152S00654	152S00652	BOM_TABLE_ALTS	ALL	IND,1.2UH,3A,2016,0.8Z

CRITICAL PART#	COMMENT
152S00651	IND,1.2UH,3A,2016,0.6SZ
152S00652	IND,1.2UH,3A,2016,0.8Z

Multi-Vendor Criticals

CRITICAL PART#	COMMENT	CRITICAL PART#	COMMENT
138S0979	CAP,CER,XSR,10UF,20%,10V,0402,H=0.65MM	132S00008	CAP,CER,0.1UF,10%,50V,X7R,0402
138S0683	CAP,CER,XSR,1UF,10%,25V,0402	131S08084	CAP,CER,Z7PF,5%,C06,25V,0201
132S0663	CAP,CER,XSR,1UF,10%,25V,0402	131S0307	CAP,CER,NP0/C0G,100PF,5%,10V,01005
132S0288	CAP,CER,XSR,0.1UF,10%,10V,0201	131S00053	CAP,CER,C0G,220PF,5%,10V,01005
132S0275	CAP,CER,XSR,470PF,10%,10V,01005	117S00055	RES,ME,1/20W,2M OHM,5,0201,SMD
132S0245	CAP,CER,XSR,0.01UF,10%,6.3V,01005	107S0257	THERMISTOR,NTC,10K OHM,1%,B=3435,01005

Iktara

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
132S00021	8	CAP,CER,0.082UF,10%,50V,X7R,0402	C3451,C3402,C363,C365,C361,C362,C363,C364	NO	COMMON
132S0423	2	CAP,CER,0.022UF,50V,X7R,10%,0402	C3455,C3465	CRITICAL	COMMON
132S0423	2	CAP,CER,0.022UF,50V,X7R,10%,0402	C3456,C3466	CRITICAL	NOSTUFF

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00183	685-00182	BOM_TABLE_ALTS	SUBBOM_DS	SUBBOM,MLB,BOT,100051,020051,X001

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00182	1	SUBBOM,MLB,BOT,D100ES,D100ES,X001	SUBBOM_DS	CRITICAL	COMMON
371S00133	4	D100ES,SHOTTKY DIODE,30V,2A,0603	D3400,D3401,D3402,D3403	CRITICAL	D100ES_DS
371S00132	4	ONSEMI,SHOTTKY DIODE,30V,2A,0603	D3400,D3401,D3402,D3403	CRITICAL	ONSEMI_DS

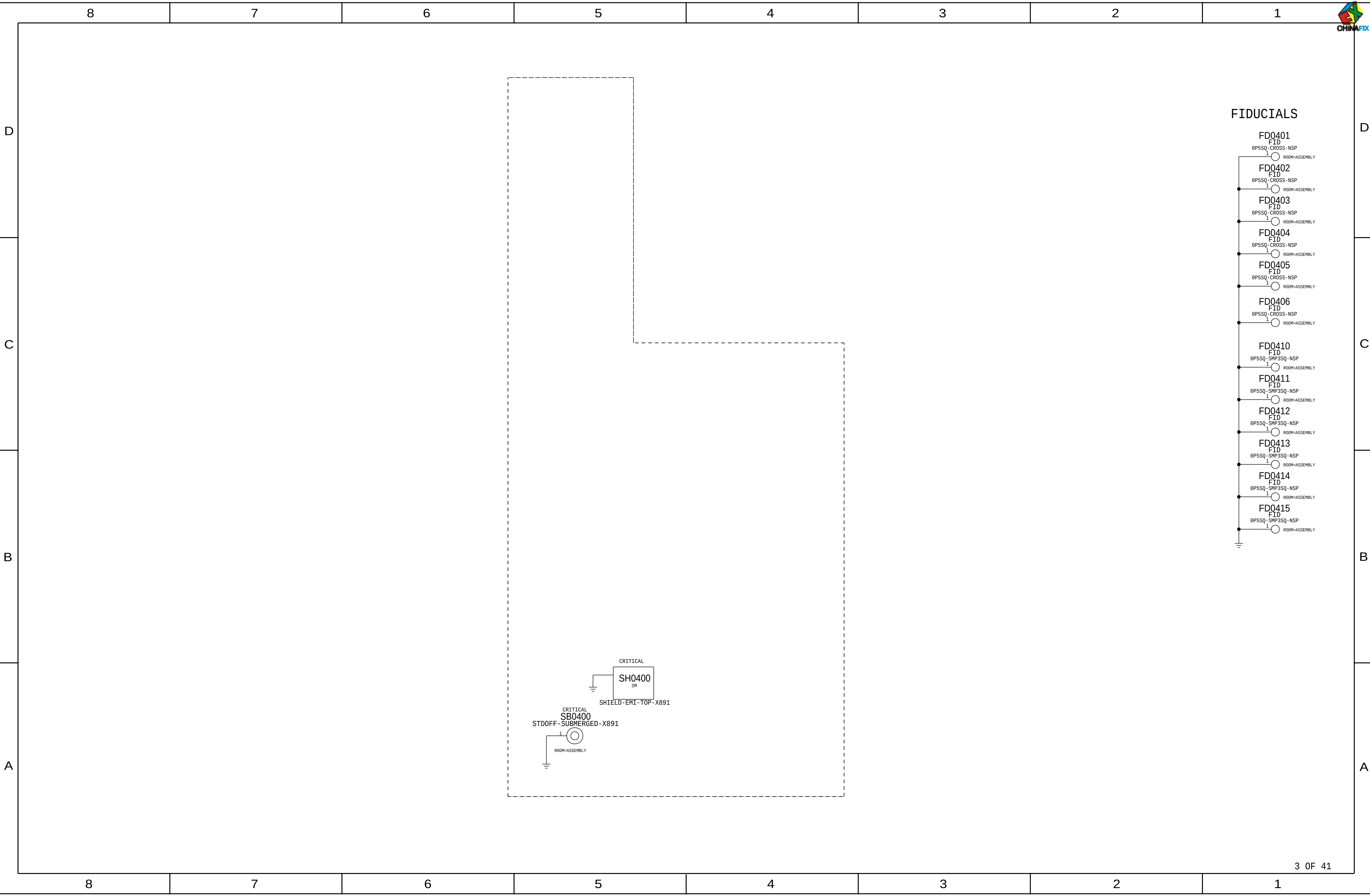
Soft-Term Cap Sub BOMs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00157	1	SUBBOM,MLB,BOT,CAP,TYPICAL,X001	SUBBOM_CAP	CRITICAL	COMMON

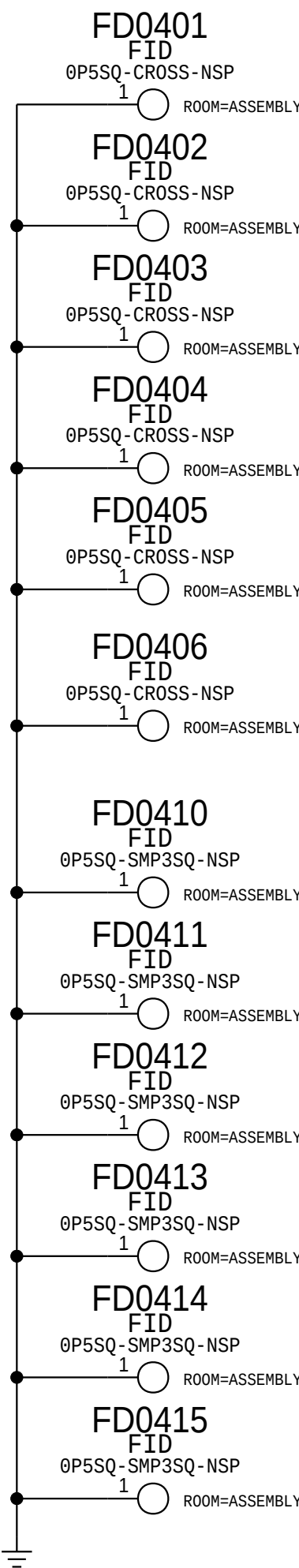
Touch/Luna B2B

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	1	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C5890	CRITICAL	SOFT_CAP
138S0831	1	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C5890	CRITICAL	TYPICAL_CAP

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00158	685-00157	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM,MLB,BOT,CAP,SOFT,X001



FIDUCIALS



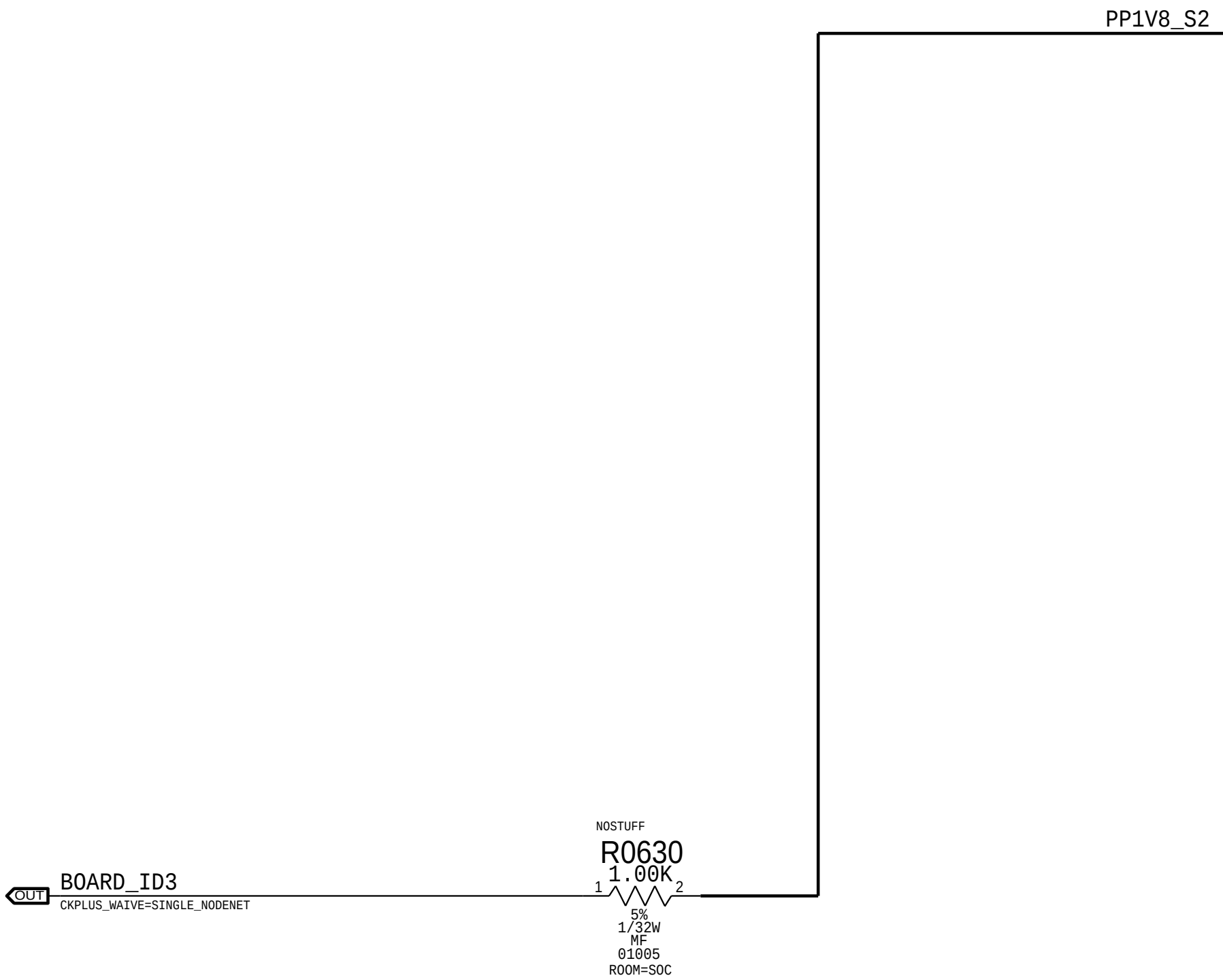
D

C

B

A

BOOTSTRAPPING:BOARD ID[3]



Board Rev[3:0]				
Float=Low I PU=High				
	3	2	1	0
	Build Major		Build Minor	
Proto 1	1	1	1	1
Proto 2	1	1	1	0
Proto 3	1	1	0	1
(Spare)	1	1	0	0
EVT	1	0	1	1
(Spare)	1	0	1	0
Carrier	0	1	1	1
(Spare)	0	1	1	0
DVT	0	0	1	1
(Spare)	0	0	1	0
(Spare)	0	0	0	1
PVT	0	0	0	0

Board ID[4:0]					
Float=Low I PU=High					
	4	3	2	1	0
	X	Baseband	Form Factor		MLB/Dev
D20 MLB	0	0	0	1	0
D20 Dev	0	0	0	1	1
D21 MLB	0	0	1	0	0
D21 Dev	0	0	1	0	1
D22 MLB	0	0	1	1	0
D22 Dev	0	0	1	1	1
D201 MLB	0	1	0	1	0
D201 Dev	0	1	0	1	1
D211 MLB	0	1	1	0	0
D211 Dev	0	1	1	0	1
D221 MLB	0	1	1	1	0
D221 Dev	0	1	1	1	1

Boot Config[2:0]			
Float=Low I PU=High			
	2	1	0
SPI0	0	0	0
SPI0 Test Mode	0	0	1
SPI0 NAND	0	1	0
SPI0 NAND Test	0	1	1
Slow SPI0 Test	1	1	0

Test Points

Probe Points

D

C

B

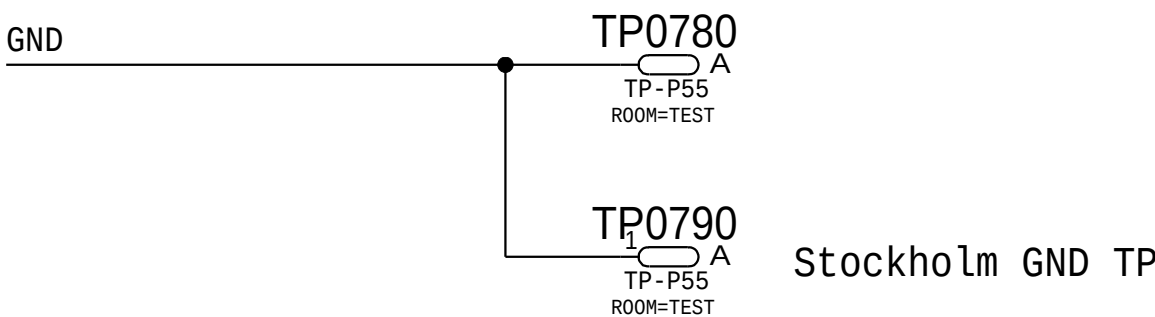
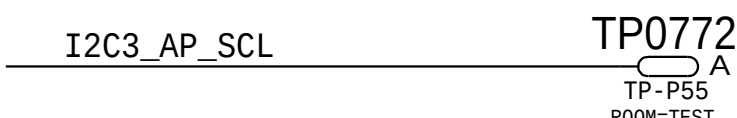
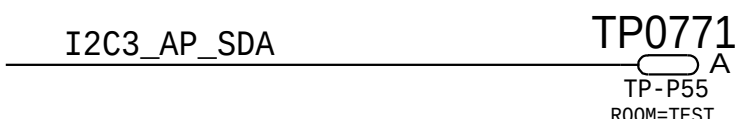
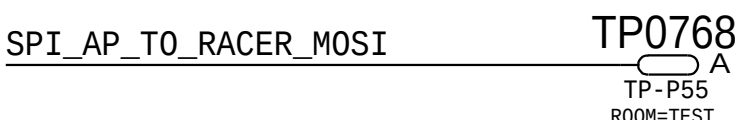
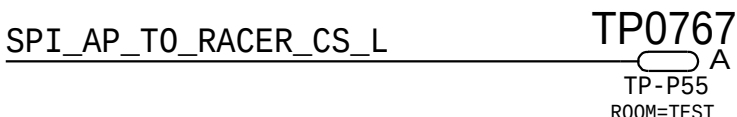
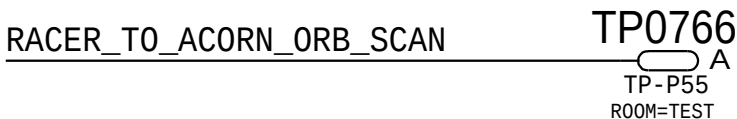
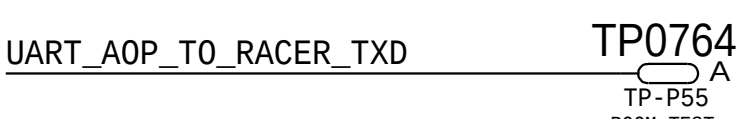
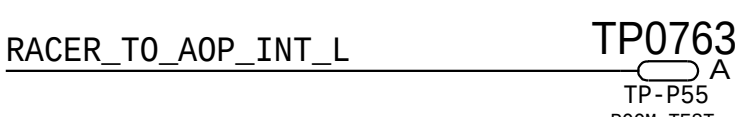
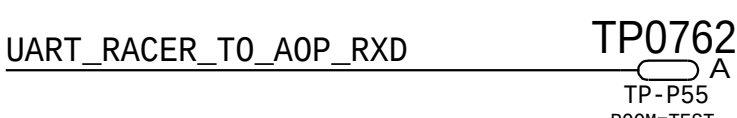
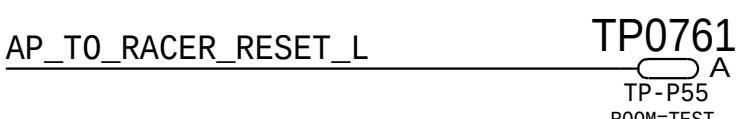
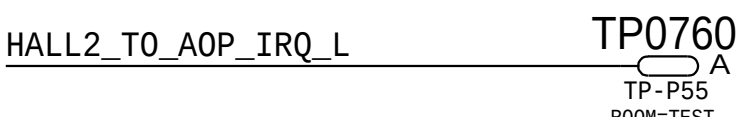
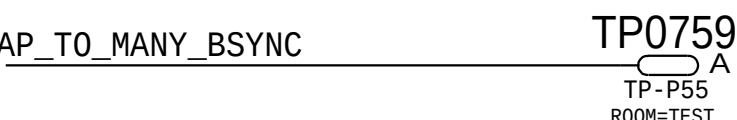
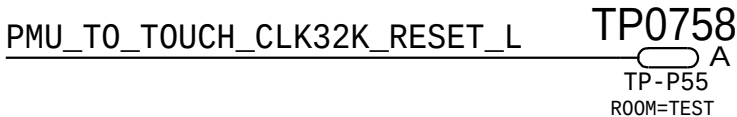
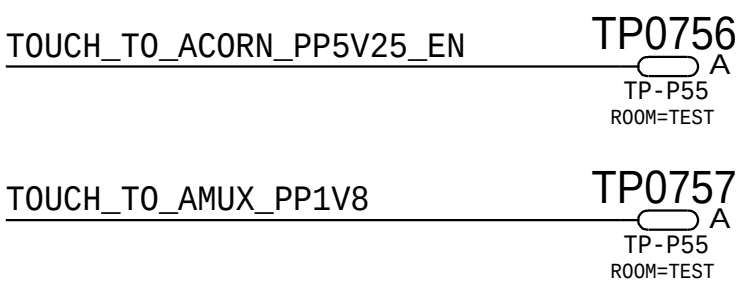
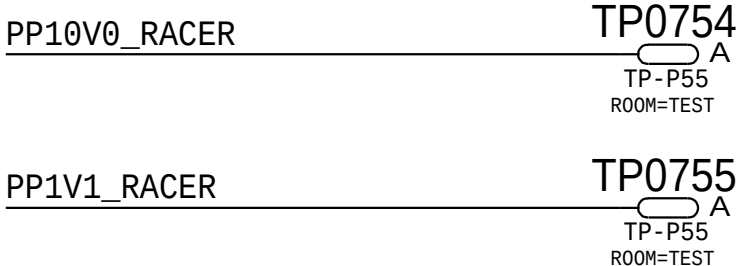
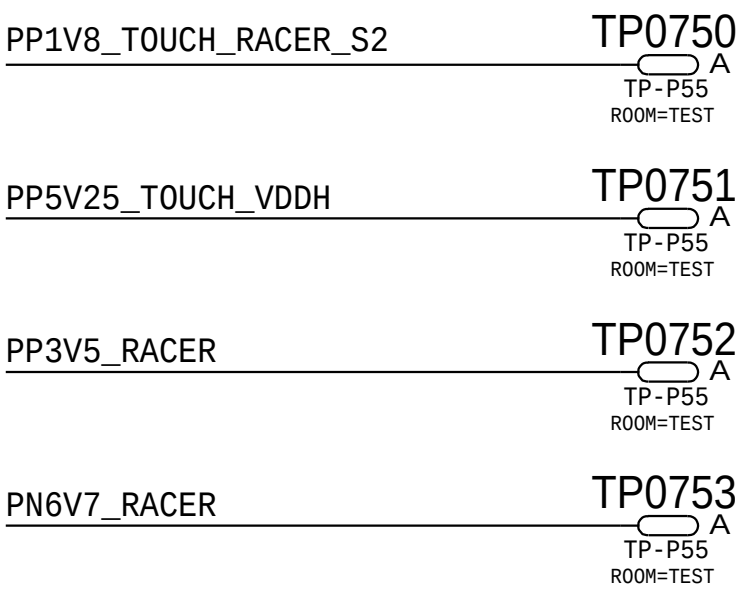
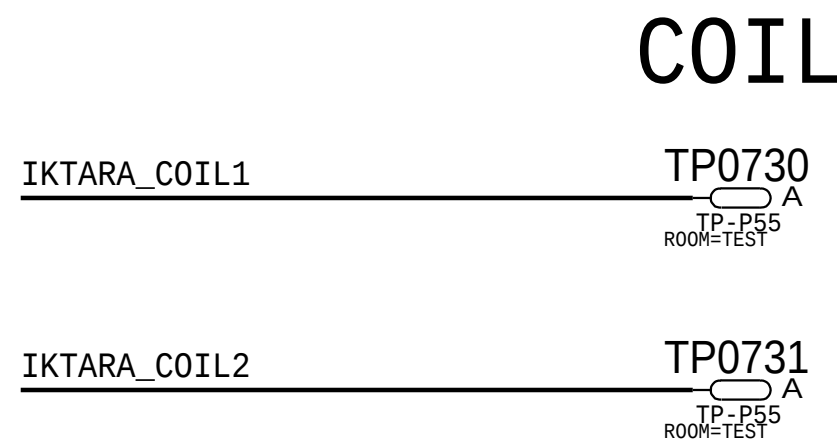
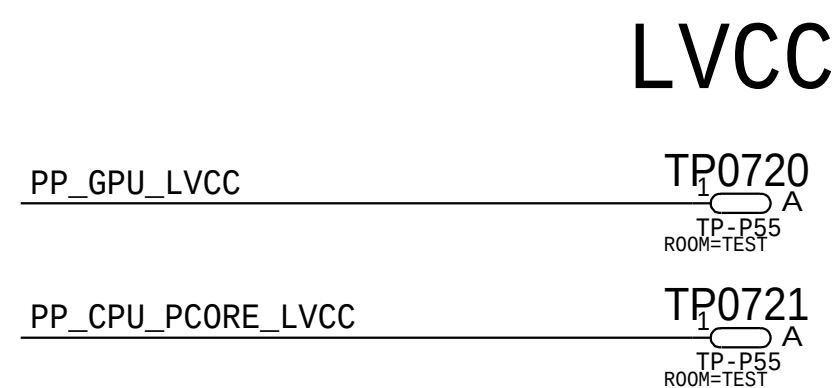
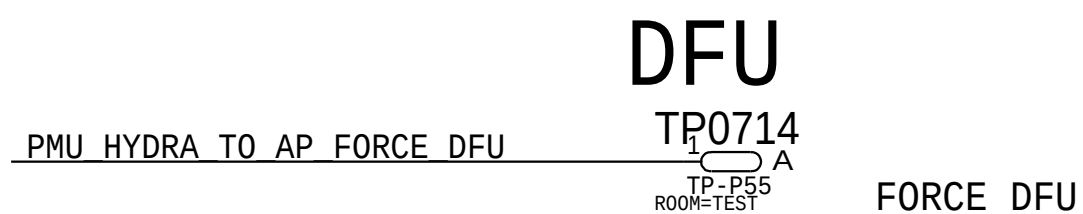
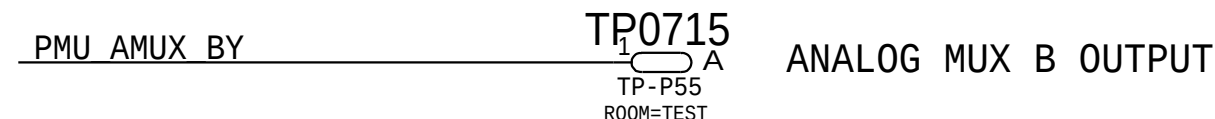
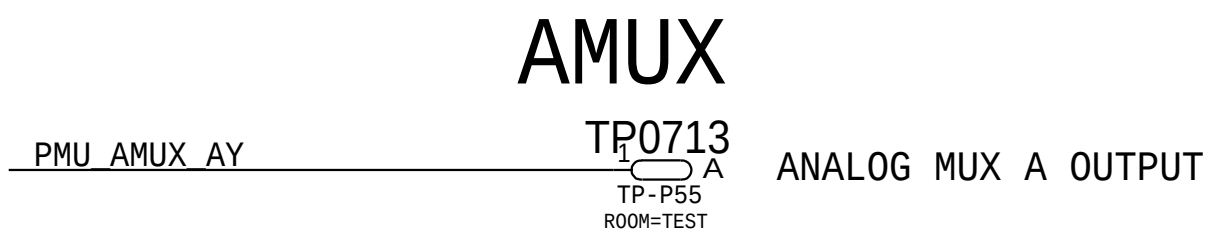
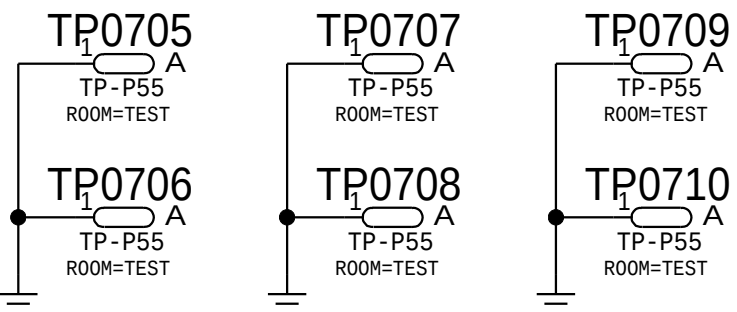
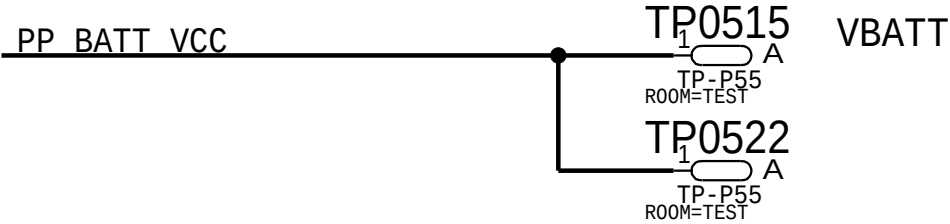
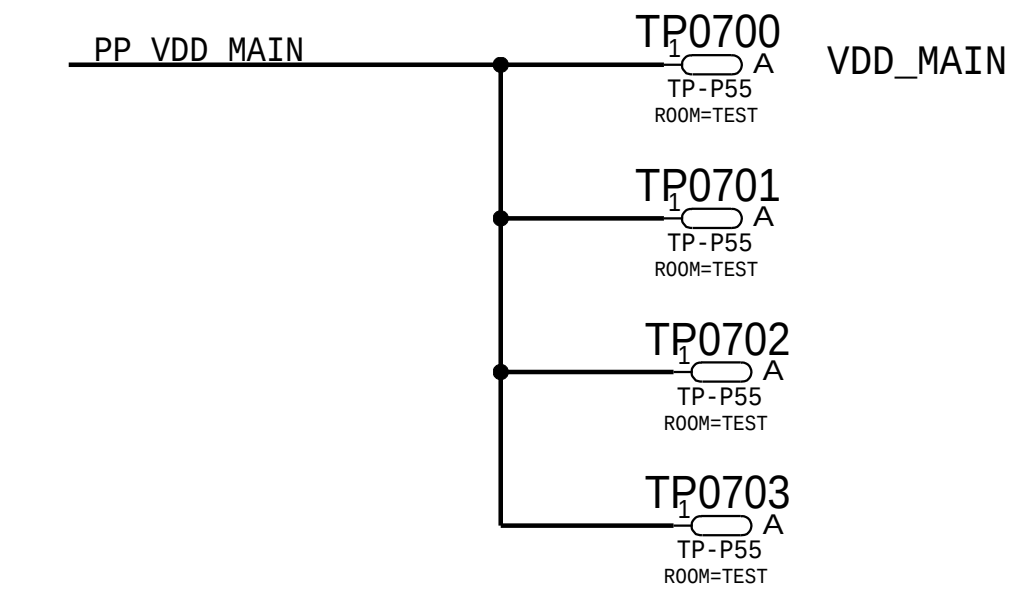
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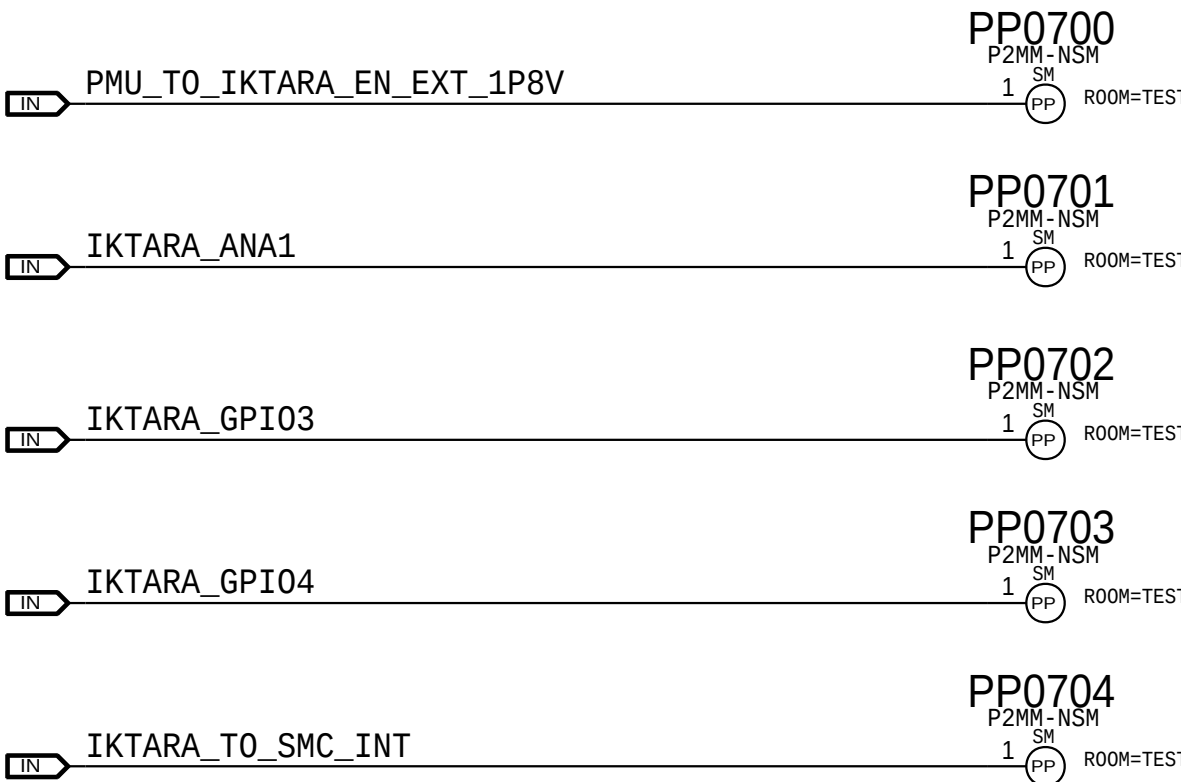
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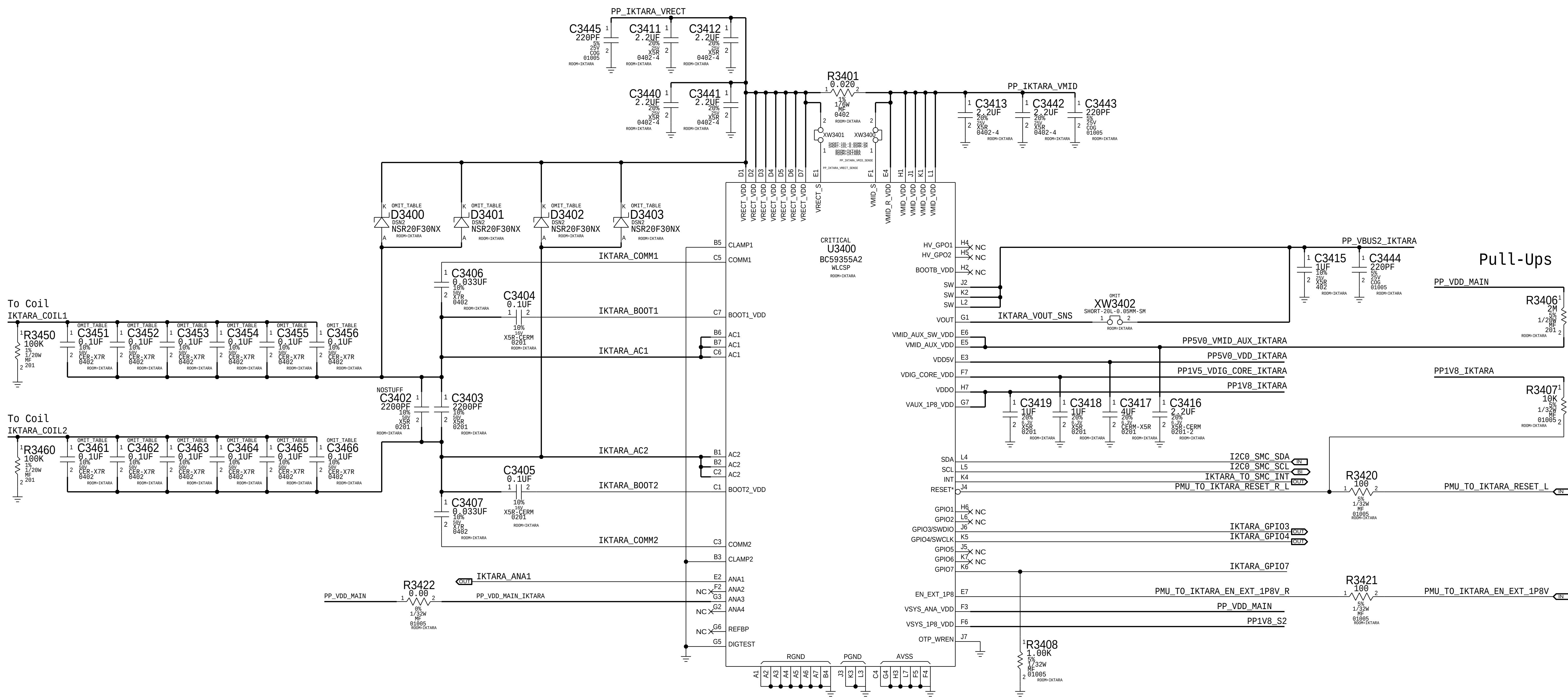
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Iktara Debug



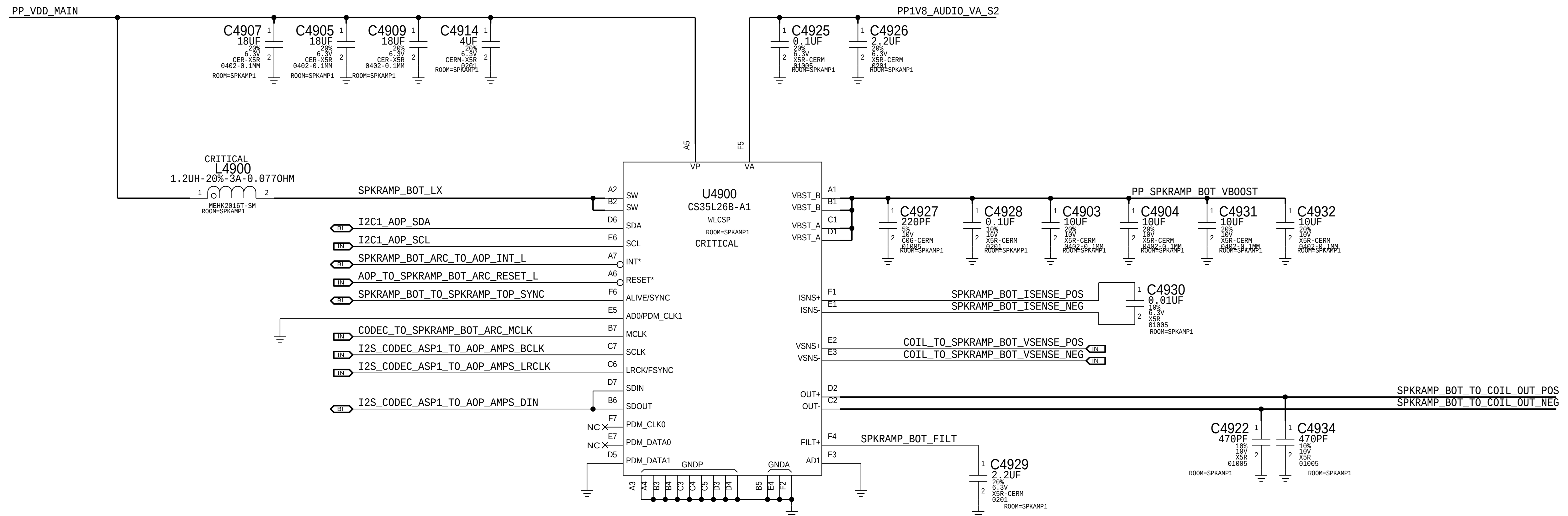
Iktara



South Speaker Amplifier

APN: 338S00295

I2C ADDRESS: 1000 000x
0x80



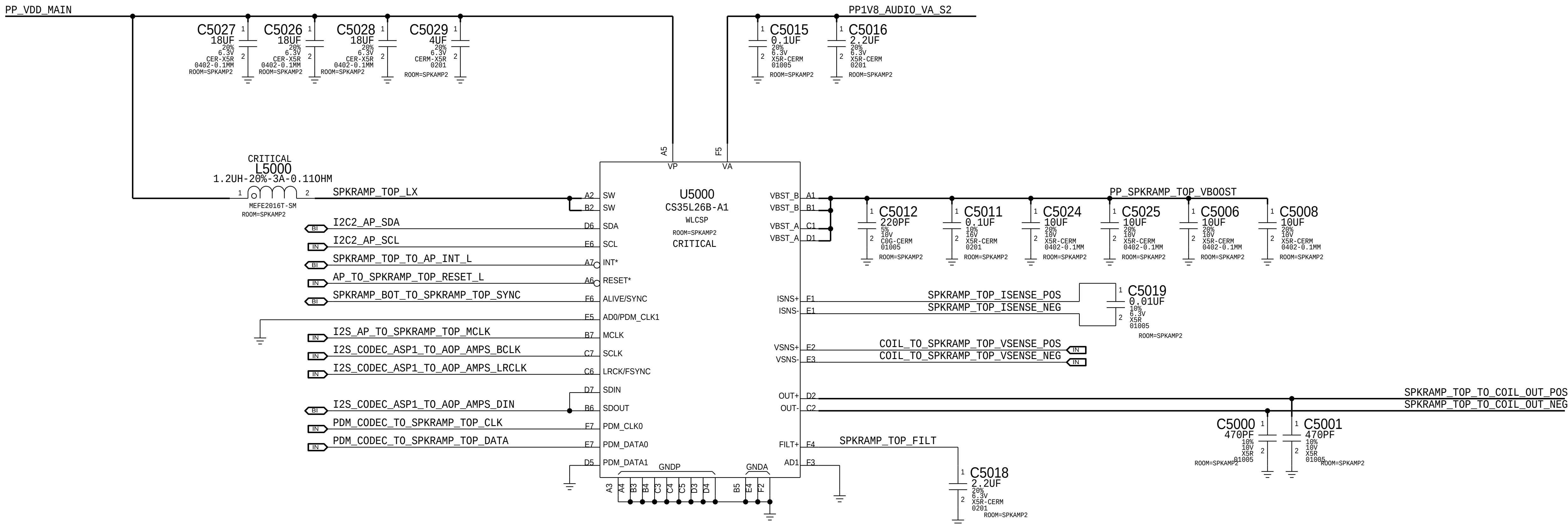
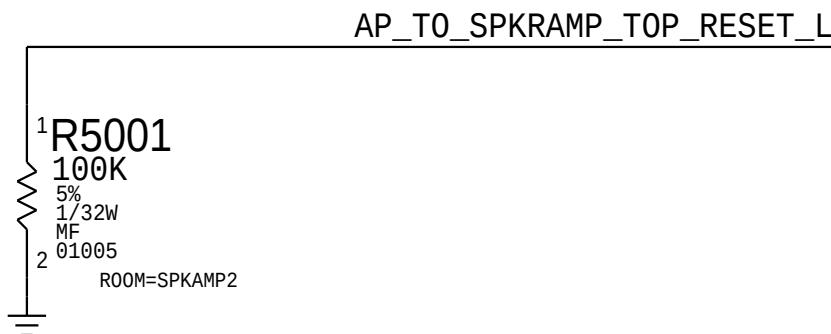
North Speaker Amplifier

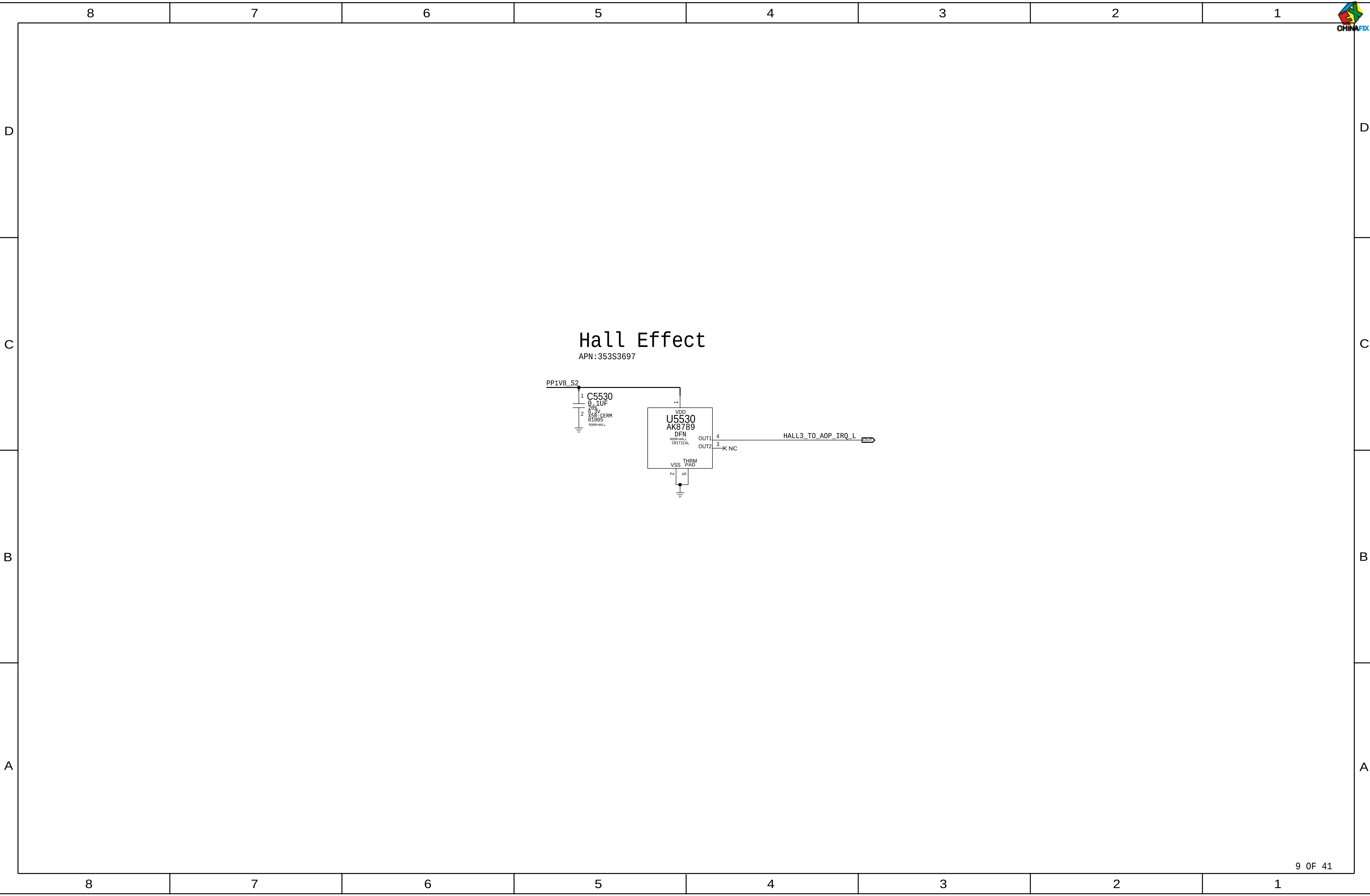
APN: 338S00295

I2C ADDRESS: 1000 000x

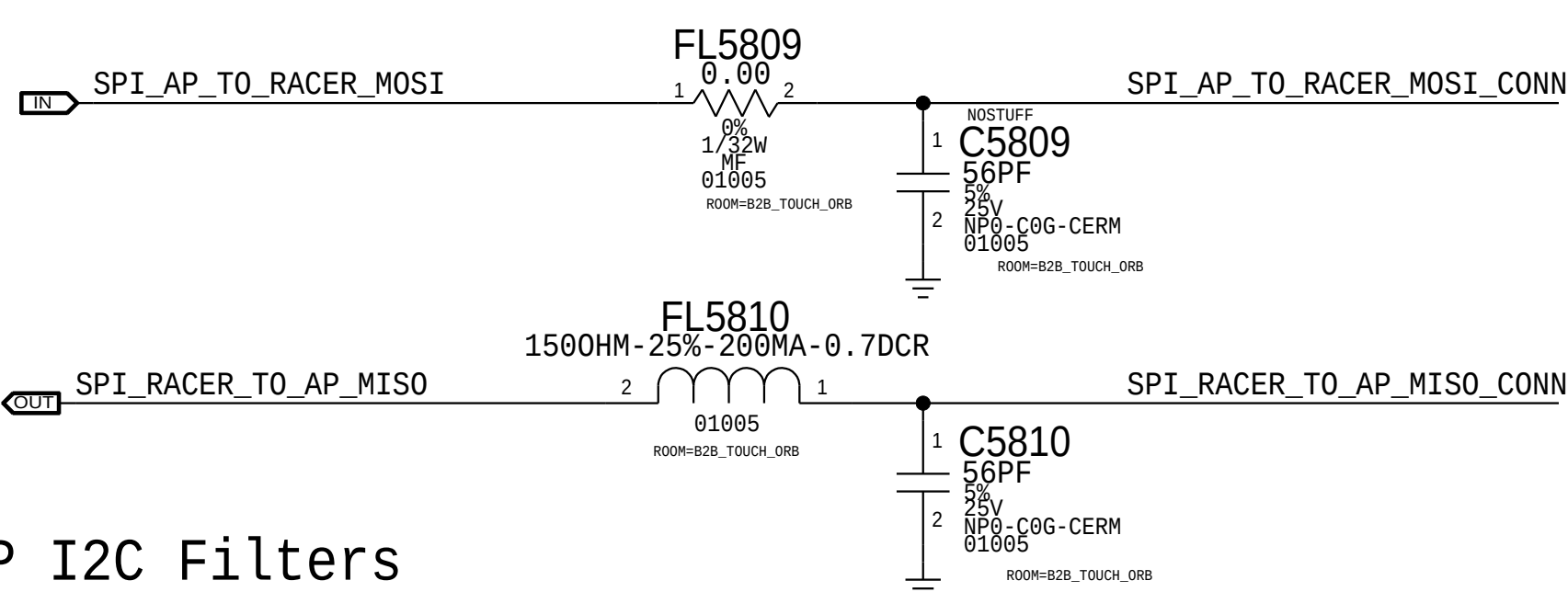
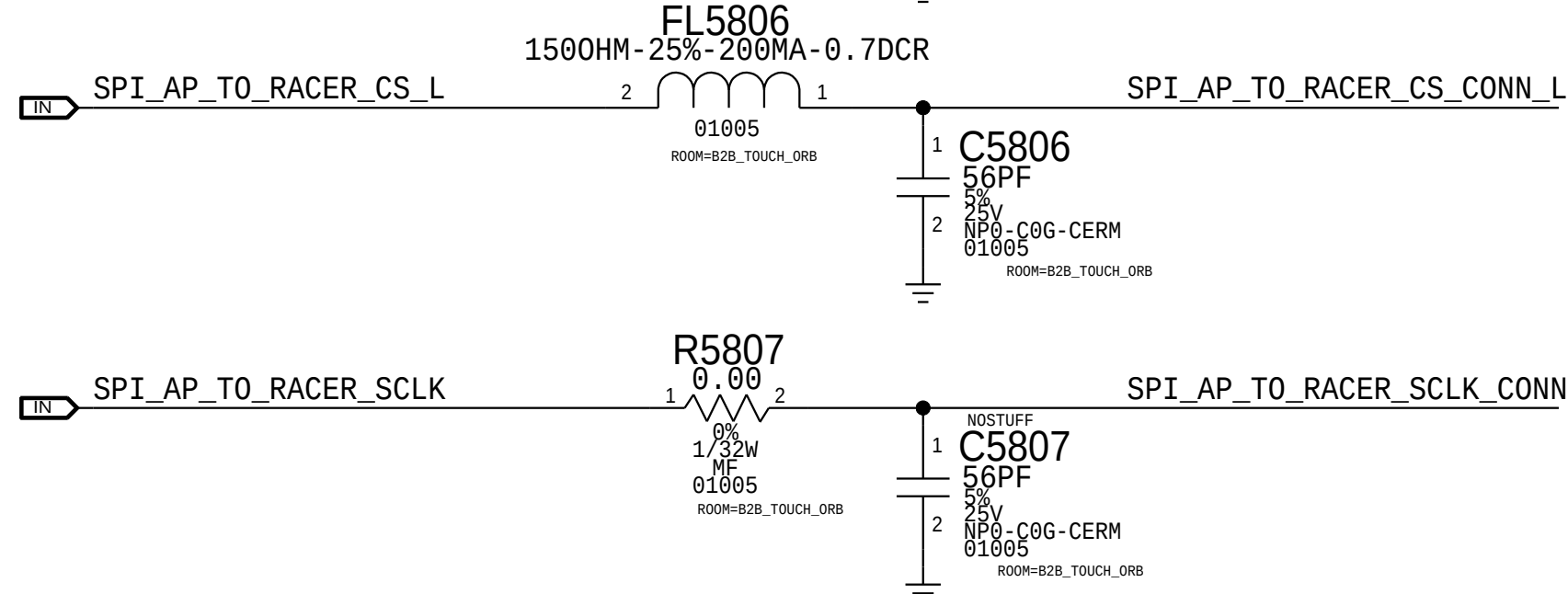
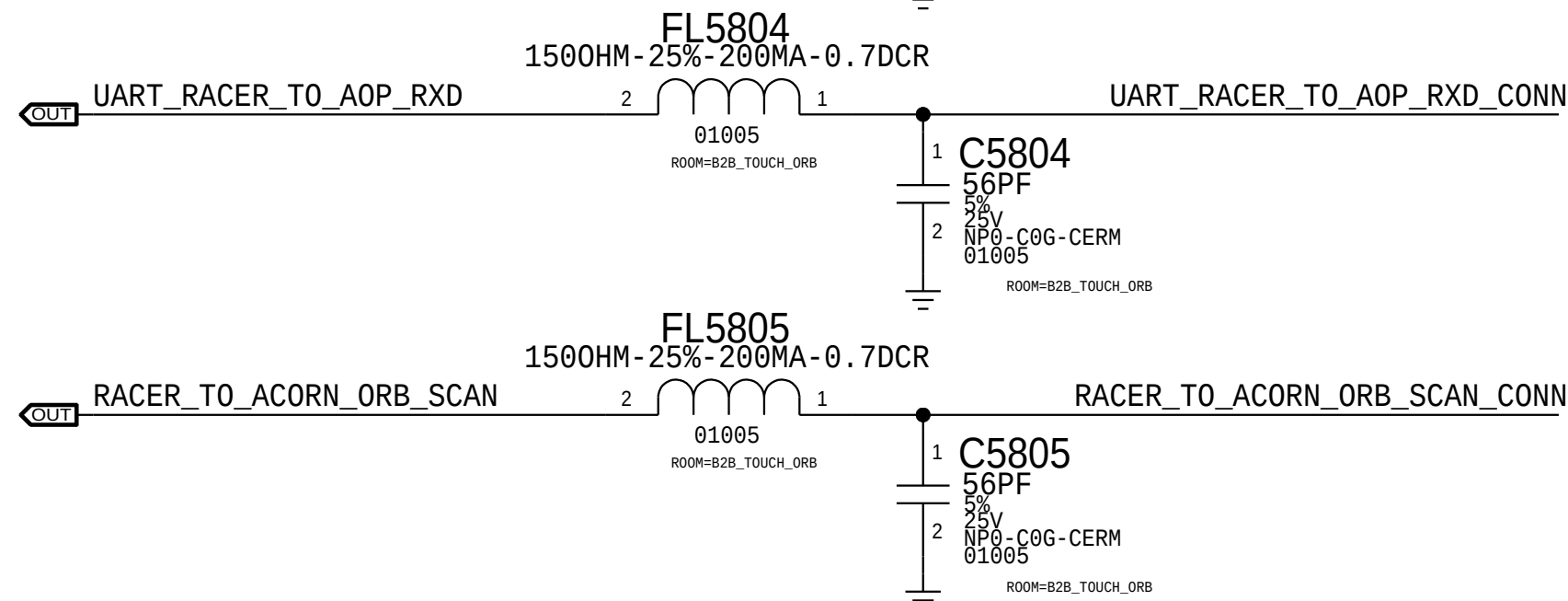
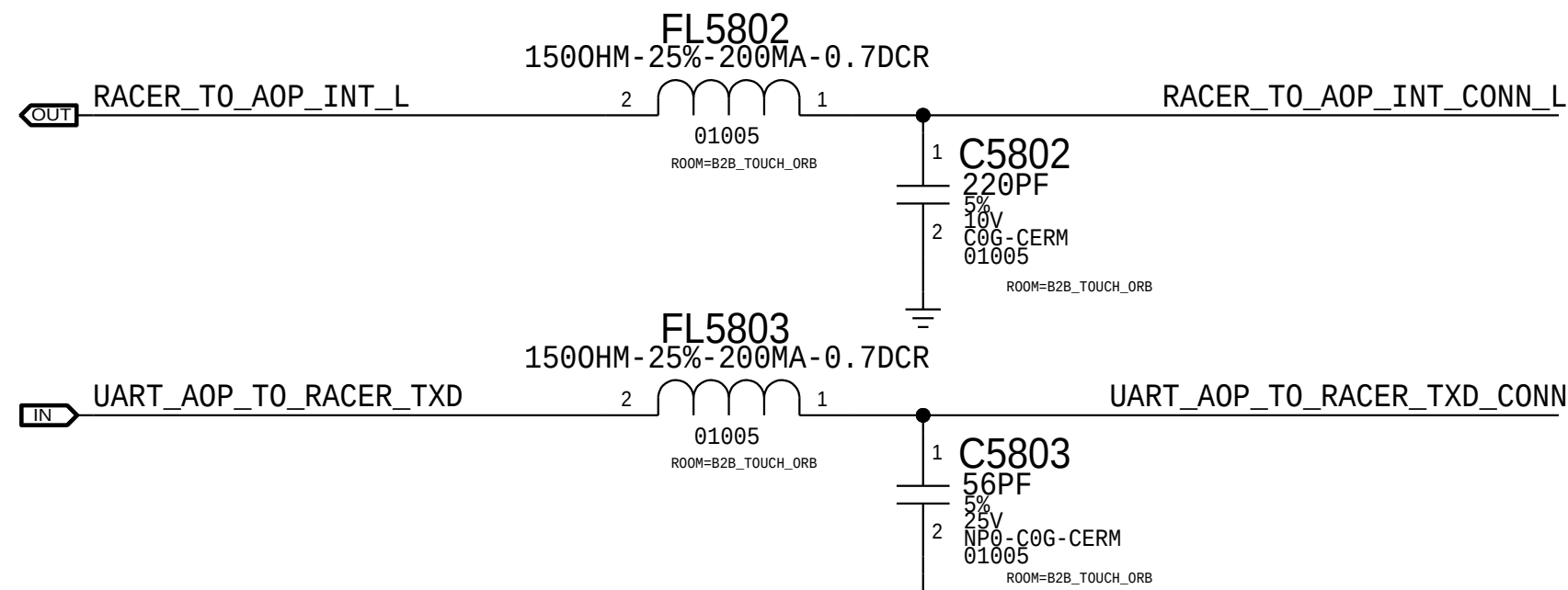
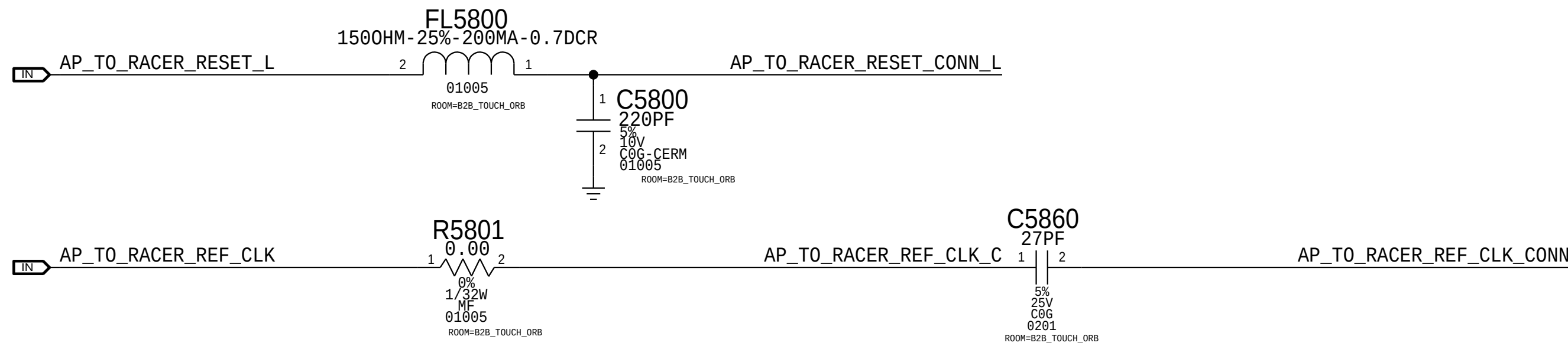
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Pull Downs

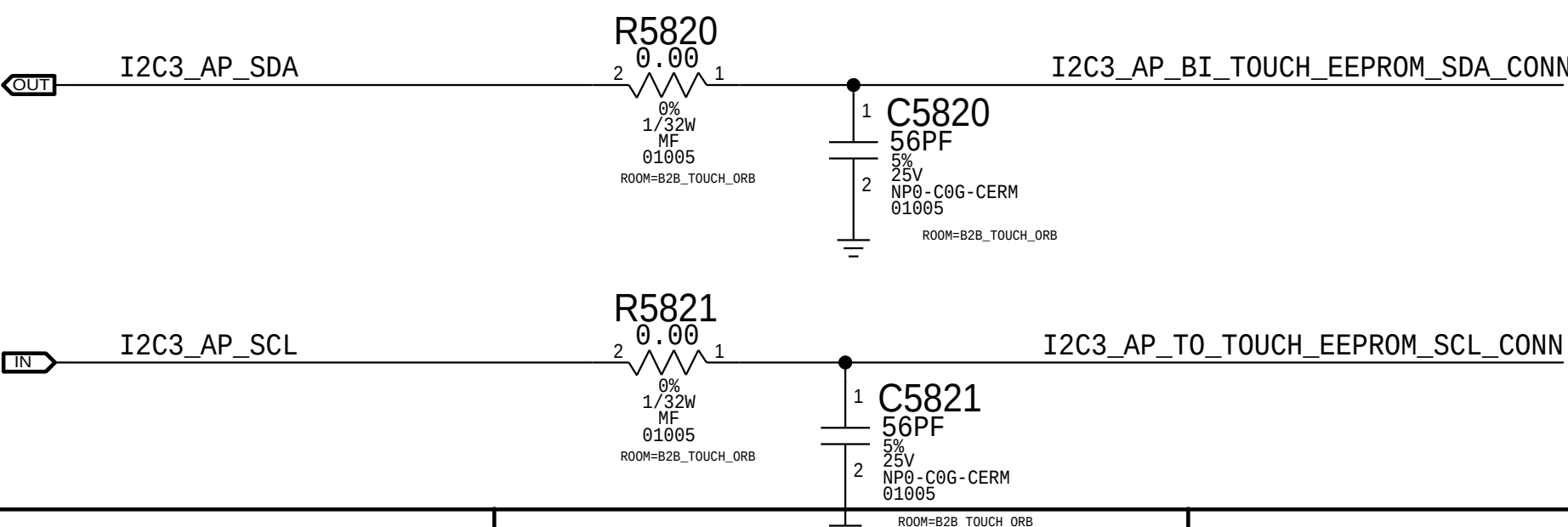




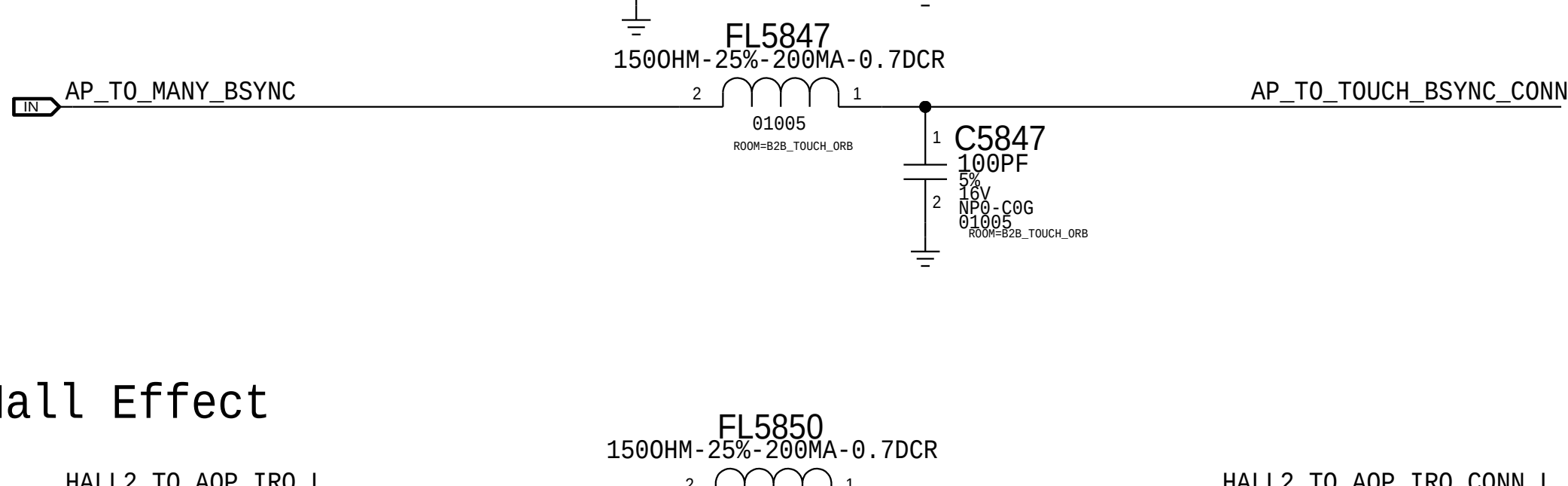
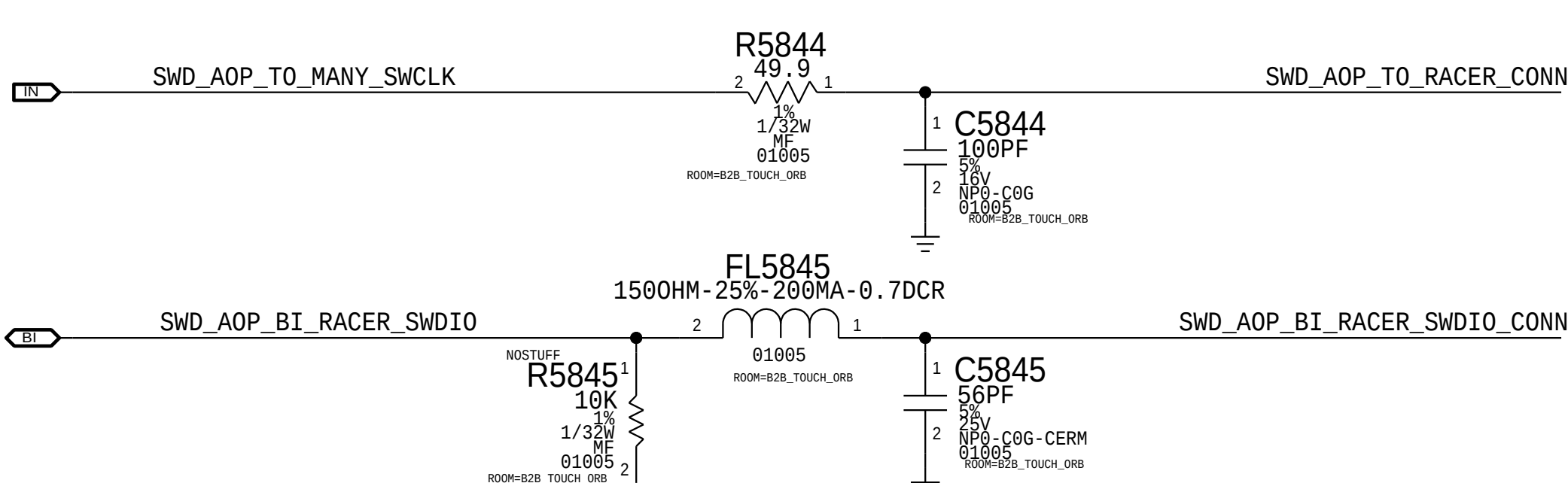
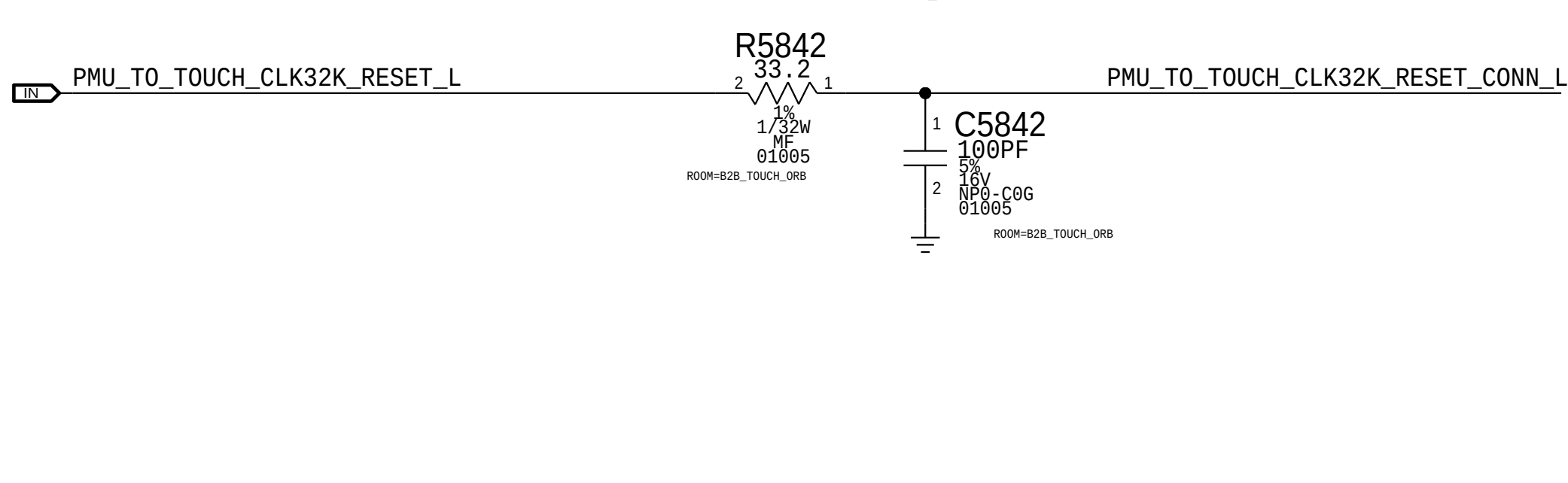
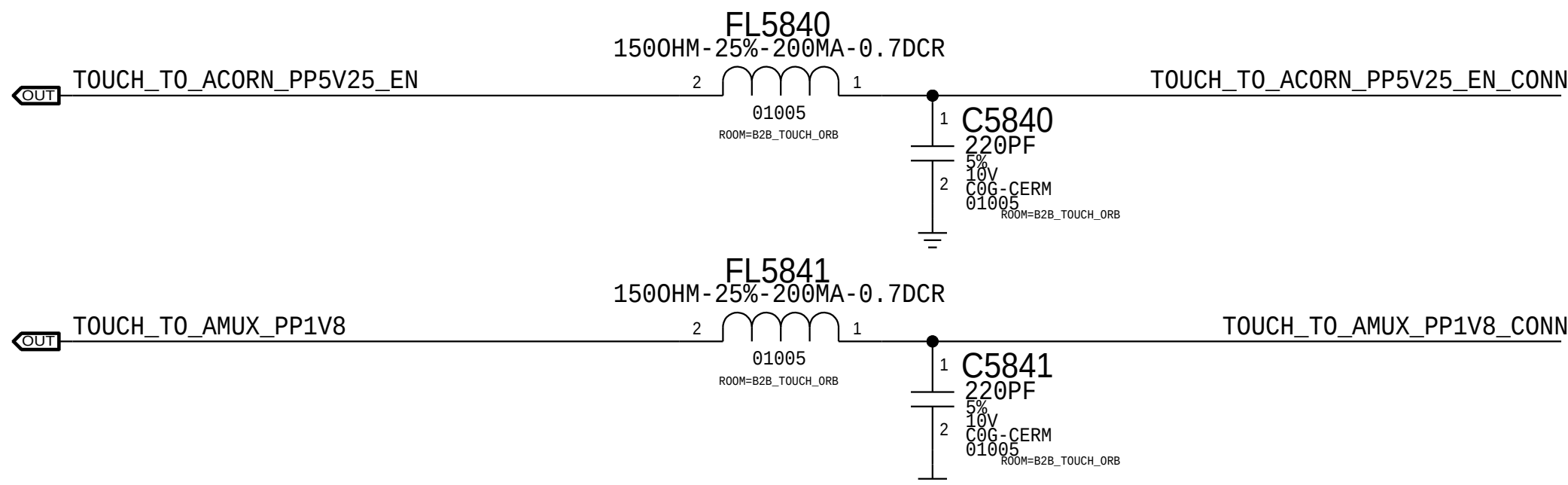
Racer I/O



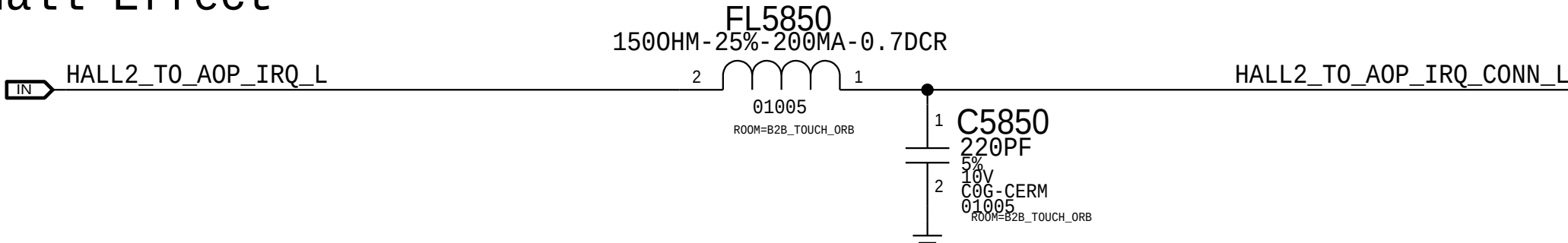
AP I2C Filters



Touch and Misc I/O

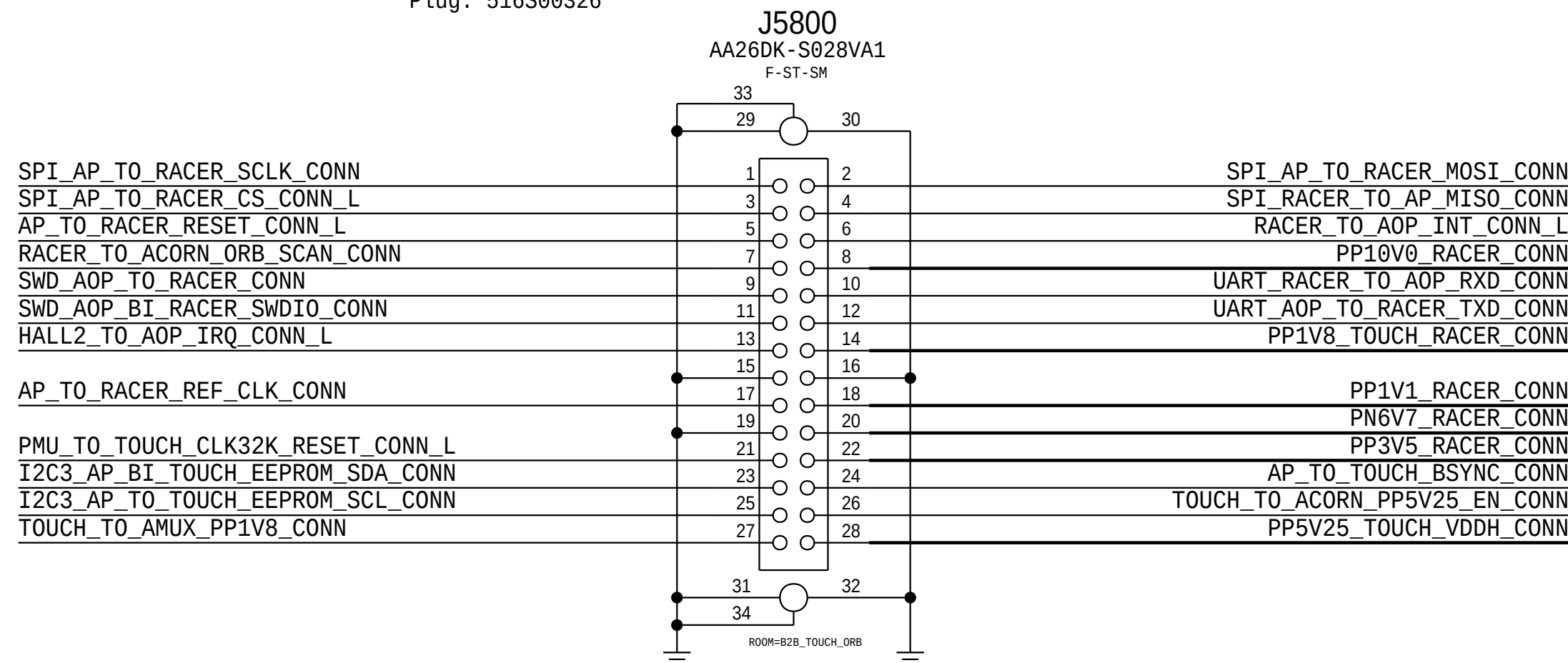


Hall Effect

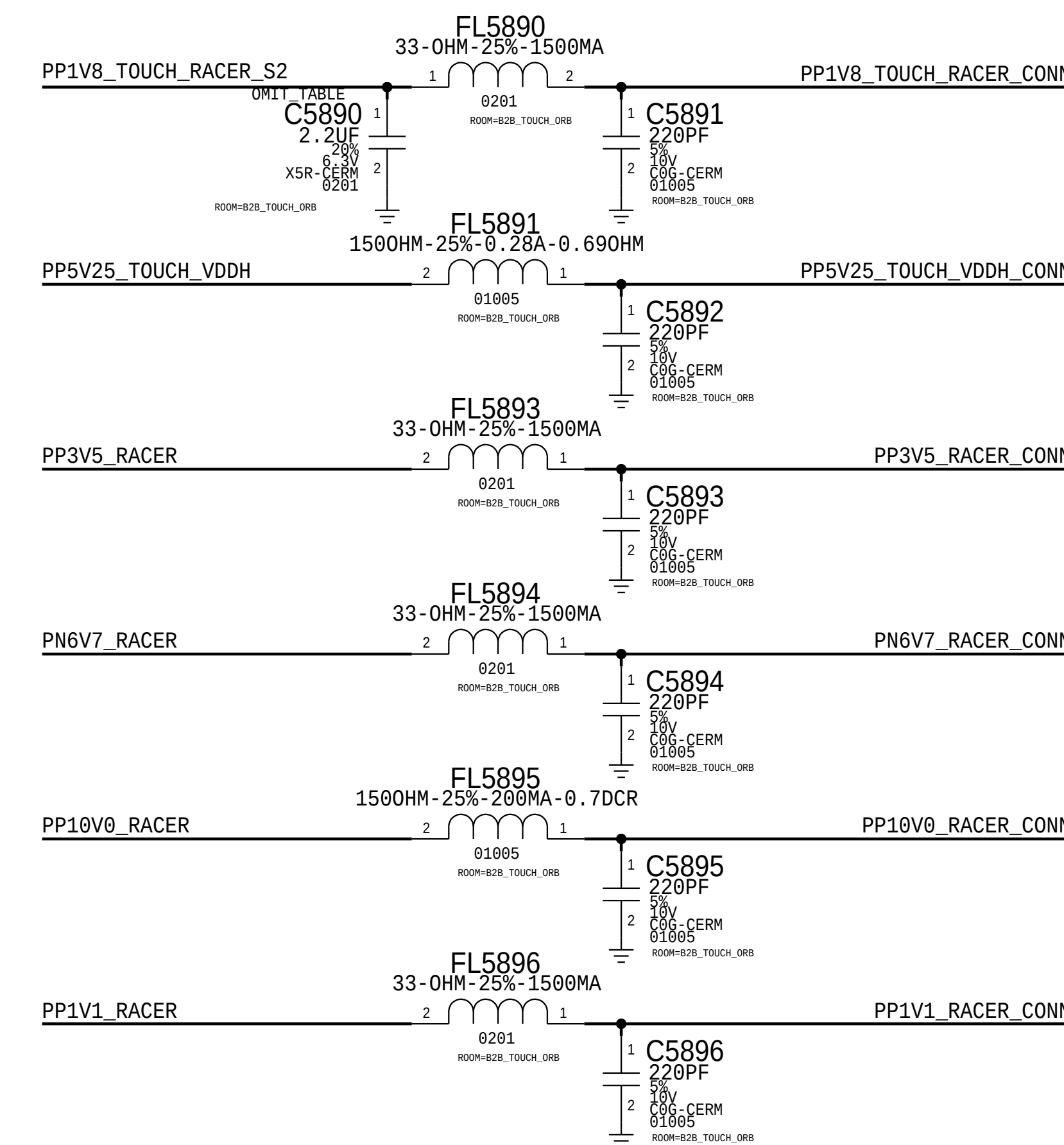


Luna + Touch Connector

Rcpt: 516S00325 <-- This one on MLB
Plug: 516S00326



Touch And Racer Power



D

C

B

A

D

C

B

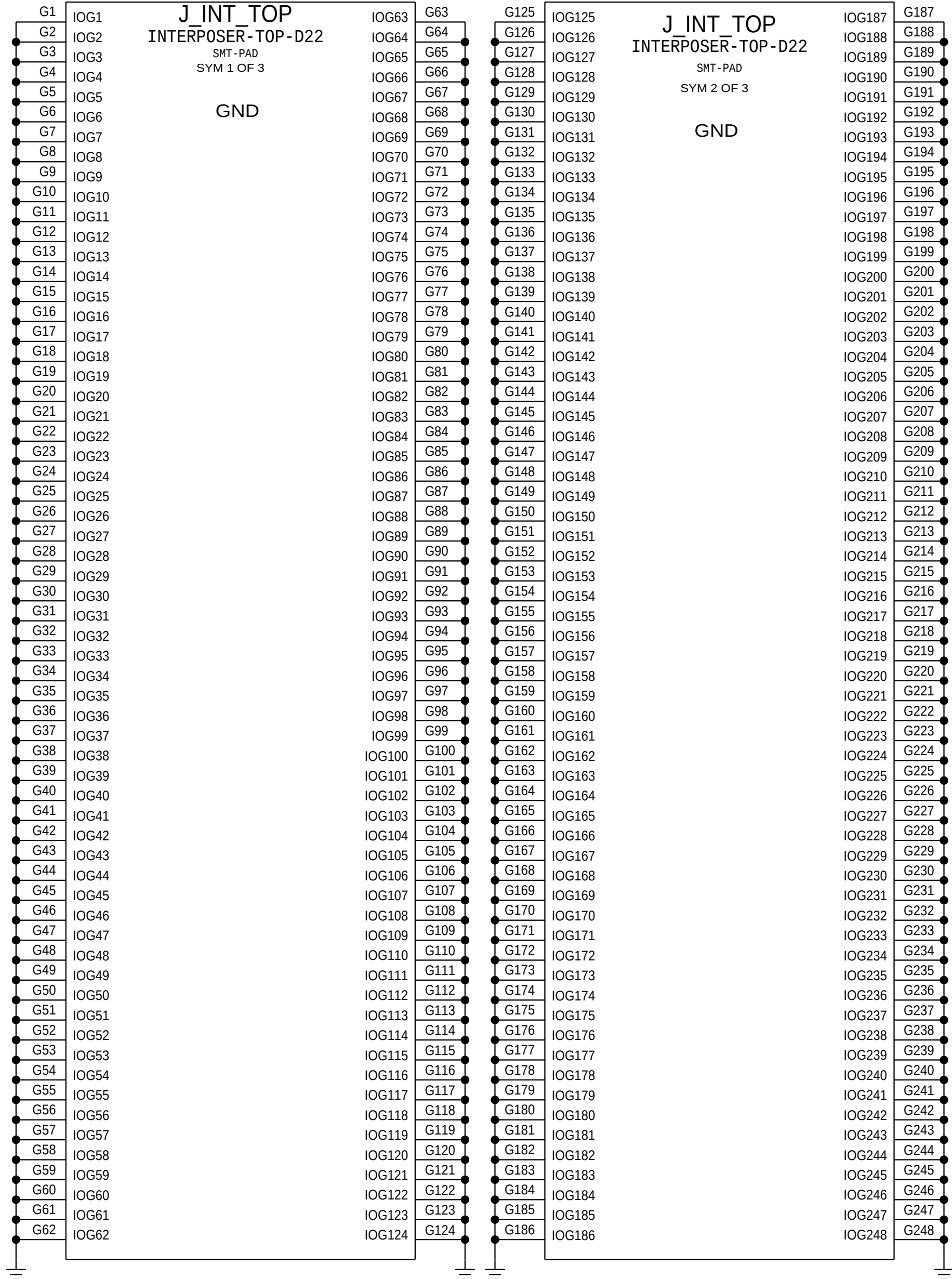
A

NC_UART_AP_TO_GNSS_TXD	S1	IOS1
NC_UART_AP_TO_GNSS_RTS_L	S2	IOS2
NC_UART_GNSS_TO_AP_CTS_L	S3	IOS3
NC_UART_GNSS_TO_AP_RXD	S4	IOS4
NC_AP_TO_GNSS_WAKE	S5	IOS5
NC_PMU_TO_GNSS_EN	S6	IOS6
AP_TO_BB_TIME_MARK	S7	IOS7
90_PCIE_AP_TO_BB_REFCLK_N	S8	IOS8
	S9	IOS9
90_PCIE_AP_TO_BB_REFCLK_P	S10	IOS10
90_PCIE_BB_TO_AP_RXD_P	S11	IOS11
90_PCIE_BB_TO_AP_RXD_N	S12	IOS12
90_PCIE_AP_TO_BB_TXD_P	S13	IOS13
	S14	IOS14
90_PCIE_AP_TO_BB_TXD_N	S15	IOS15
90_PCIE_WLAN_TO_AP_RXD_P	S16	IOS16
	S17	IOS17
90_PCIE_WLAN_TO_AP_RXD_N	S18	IOS18
90_PCIE_AP_TO_WLAN_REFCLK_N	S19	IOS19
	S20	IOS20
90_PCIE_AP_TO_WLAN_REFCLK_P	S21	IOS21
90_PCIE_AP_TO_WLAN_TXD_N	S22	IOS22
	S23	IOS23
90_PCIE_AP_TO_WLAN_TXD_P	S24	IOS24
	S25	IOS25
	S26	IOS26
PMU_TO_IKTARA_EN_EXT_1P8V	S27	IOS27
IKTARA_TO_SMC_INT	S28	IOS28
I2C0_SMC_SCL	S29	IOS29
I2C0_SMC_SDA	S30	IOS30
BB_TO_STROBE_DRIVER_GSM_BURST_IND	S31	IOS31
	S32	IOS32
	S33	IOS33
PDM_CODEC_TO_SPKRAMP_TOP_CLK	S34	IOS34
PDM_CODEC_TO_SPKRAMP_TOP_DATA	S35	IOS35
PCIE_AP_TO_BB_RESET_L	S36	IOS36
I2S_AP_TO_SPKRAMP_TOP_MCLK	S37	IOS37
AP_TO_SPKRAMP_TOP_RESET_L	S38	IOS38
SPKRAMP_TOP_TO_AP_INT_L	S39	IOS39
I2C2_AP_SDA	S40	IOS40
I2C2_AP_SCL	S41	IOS41
PP1V8_AUDIO_VA_S2	S42	IOS42
PP_GPU_LVCC	S43	IOS43
PP_CPU_PCORE_LVCC	S44	IOS44
AP_TO_BB_IPC_GPIO1	S45	IOS45
AP_TO_BB_RESET_L	S46	IOS46
HALL3_TO_AOP_IRQ_L	S47	IOS47
	S48	IOS48
BOARD_ID3	S49	IOS49
PCIE_WLAN_BI_AP_CLKREQ_L	S50	IOS50
PCIE_AP_TO_WLAN_RESET_L	S51	IOS51
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK	S52	IOS52
I2S_CODEC_ASP1_TO_AOP_AMPS_DIN	S53	IOS53
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK	S54	IOS54
PMU_TO_IKTARA_RESET_L	S55	IOS55
I2S_BB_TO_AP_DIN	S56	IOS56
I2S_BB_TO_AP_LRCLK	S57	IOS57
I2S_BB_TO_AP_BCLK	S58	IOS58
I2S_AP_TO_BB_DOUT	S59	IOS59
BB_TO_AP_RESET_DETECT_L	S60	IOS60
NC_AP_TO_BBPMU_RADIO_ON_L	S61	IOS61
AP_TO_WLAN_DEVICE_WAKE	S62	IOS62
SPKRAMP_BOT_ARC_TO_AOP_INT_L	S63	IOS63
AOP_TO_SPKRAMP_BOT_ARC_RESET_L	S64	IOS64
	S65	IOS65
SMD_AOP_BI_BB_SWDIO	S66	IOS66
I2C1_AOP_SDA	S67	IOS67
I2C1_AOP_SCL	S68	IOS68
PP_VDD_MAIN	S69	IOS69
	S70	IOS70
	S71	IOS71
	S72	IOS72
	S73	IOS73
	S74	IOS74
	S75	IOS75
	S76	IOS76
UART_AP_TO_BT_RTS_L	S77	IOS77
UART_BT_TO_AP_CTS_L	S78	IOS78
UART_AP_TO_BT_TXD	S79	IOS79
UART_BT_TO_AP_RXD	S80	IOS80
PP_VDD_BOOST	S81	IOS81
UART_AP_TO_WLAN_TXD	S82	IOS82
UART_AP_TO_WLAN_RTS_L	S83	IOS83
UART_WLAN_TO_AP_RXD	S84	IOS84
UART_WLAN_TO_AP_CTS_L	S85	IOS85
PP1V8_S2	S86	IOS86

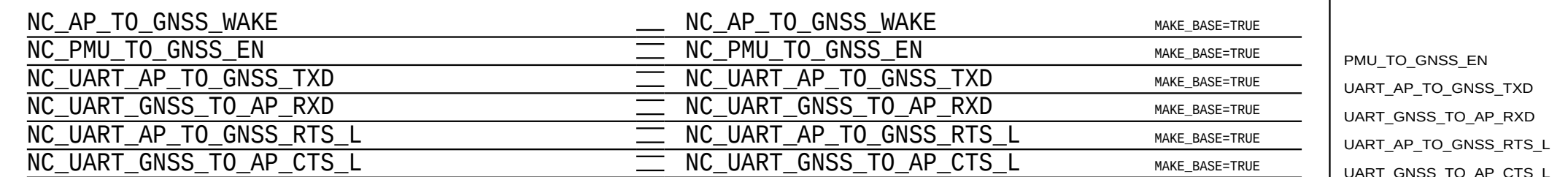
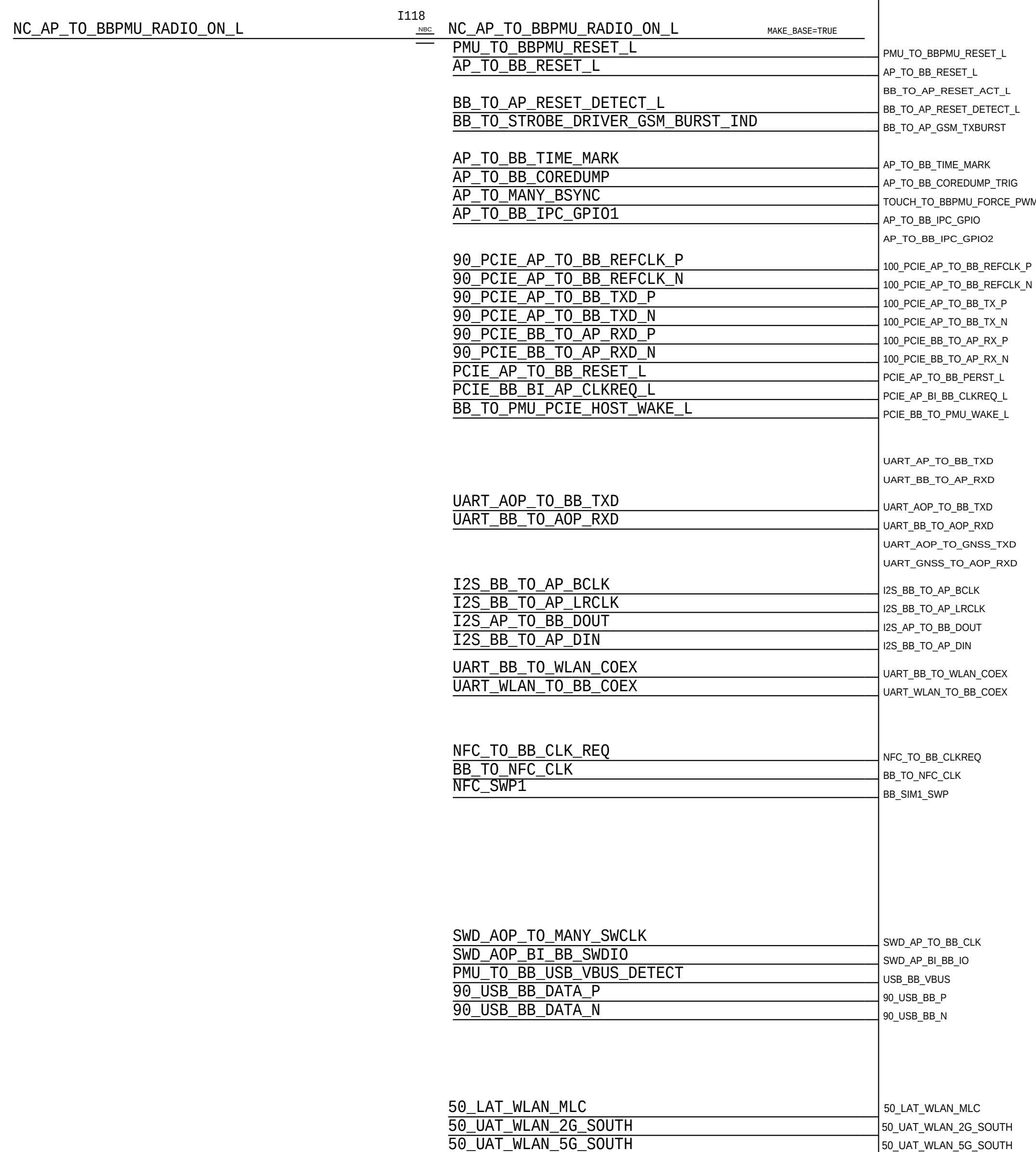
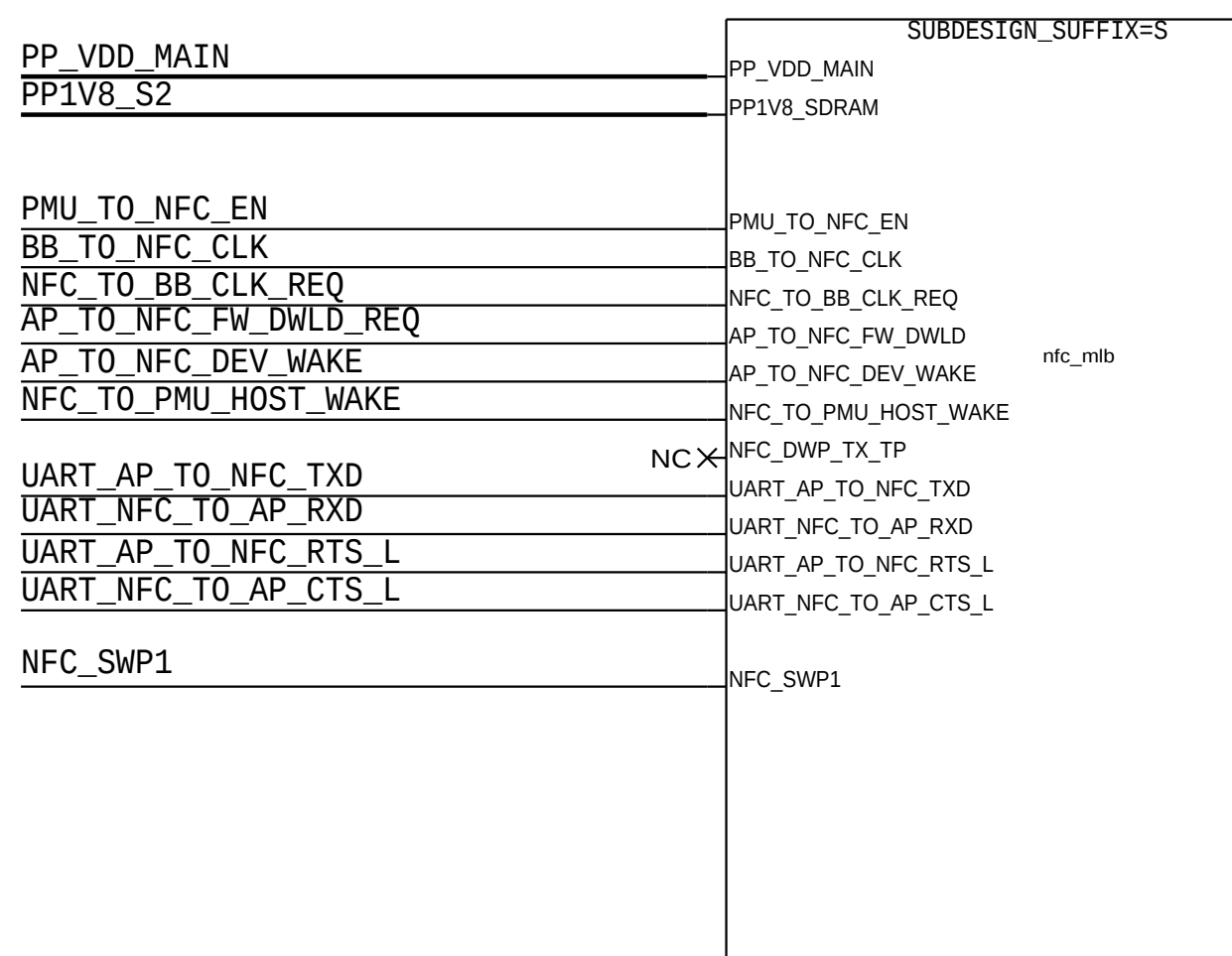
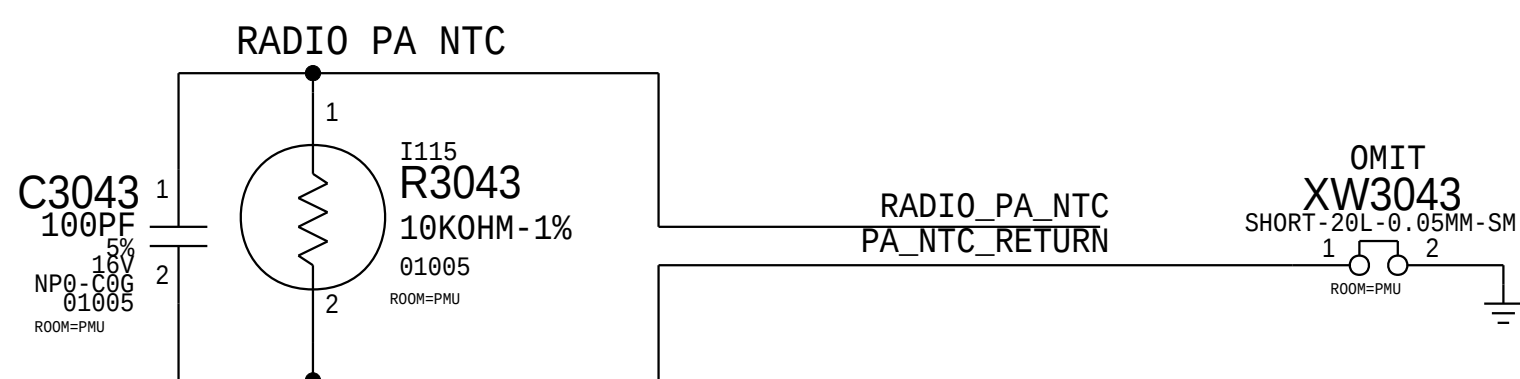
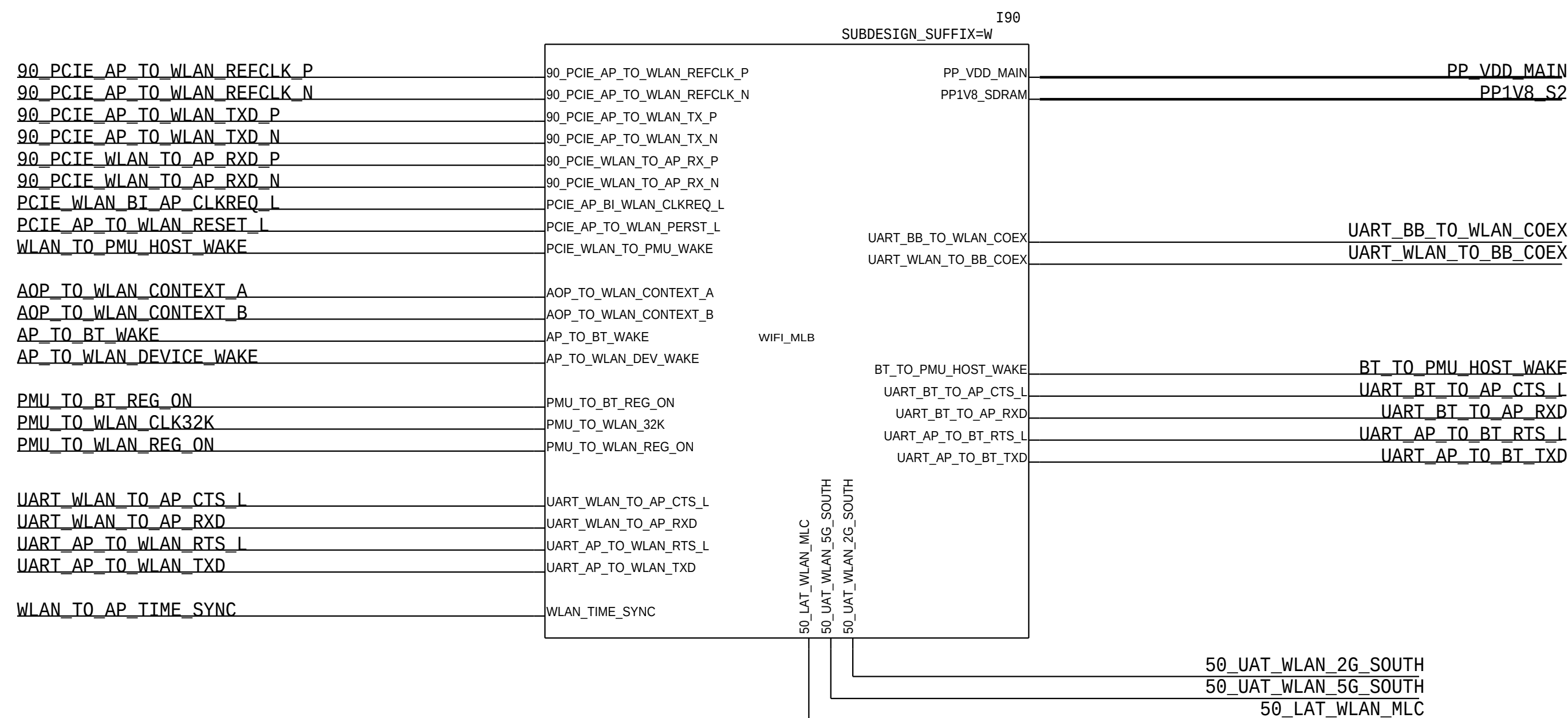
J INT TOP
INTERPOSER-TOP-D22
SMT-PAD
SYM 3 OF 3

SIGNAL

S87	PP1V8_S2
S88	
S89	PP_VDD_MAIN
S90	BB_TO_PMU_PCIE_HOST_WAKE_L
S91	PMU_TO_BBPMU_RESET_L
S92	PMU_AMUX_AY
S93	AOP_TO_WLAN_CONTEXT_B
S94	UART_BB_TO_AOP_RXD
S95	UART_AOP_TO_BB_TXD
S96	AOP_TO_WLAN_CONTEXT_A
S97	BT_TO_PMU_HOST_WAKE
S98	PMU_TO_BT_REG_ON
S99	PMU_TO_WLAN_CLK32K
S100	WLAN_TO_PMU_HOST_WAKE
S101	PMU_TO_WLAN_REG_ON
S102	PMU_AMUX_BY
S103	CODEC_TO_SPKRAMP_BOT_ARC_MCLK
S104	PMU_TO_BB_USB_VBUS_DETECT
S105	PMU_HYDRA_TO_AP_FORCE_DFU
S106	AP_TO_BT_WAKE
S107	WLAN_TO_AP_TIME_SYNC
S108	RADIO_PA_NTC
S109	AP_TO_MANY_BSYNC
S110	TOUCH_TO_AMUX_PP1V8
S111	PP3V5_RACER
S112	PP1V1_RACER
S113	
S114	PP5V25_TOUCH_VDDH
S115	RACER_TO_ACORN_ORB_SCAN
S116	PP1V8_TOUCH_RACER_S2
S117	I2C3_AP_SCL
S118	I2C3_AP_SDA
S119	TOUCH_TO_ACORN_PP5V25_EN
S120	PN6V7_RACER
S121	PMU_TO_TOUCH_CLK32K_RESET_L
S122	PP10V0_RACER
S123	AP_TO_RACER_RESET_L
S124	RACER_TO_AOP_INT_L
S125	SWD_AOP_TO_MANY_SMCCLK
S126	SWD_AOP_BI_RACER_SWDIO
S127	UART_AOP_TO_RACER_TXD
S128	UART_RACER_TO_AOP_RXD
S129	HALL2_TO_AOP_IRQ_L
S130	AP_TO_RACER_REF_CLK
S131	SPI_AP_TO_RACER_CS_L
S132	SPI_RACER_TO_AP_MISO
S133	SPI_AP_TO_RACER_MOSI
S134	SPI_AP_TO_RACER_SCLK
S135	PP_BATT_VCC
S136	
S137	PP3V0_S2
S138	
S139	90_USB_BB_DATA_P
S140	90_USB_BB_DATA_N
S141	
S142	PP_VBUS2_IKTARA
S143	
S144	
S145	
S146	IKTARA_COIL2
S147	
S148	
S149	
S150	IKTARA_COIL1
S151	
S152	
S153	
S154	AP_TO_BB_COREDUMP
S155	PCIE_BB_BI_AP_CLKREQ_L
S156	SPKRAMP_TOP_TO_COIL_OUT_POS
S157	SPKRAMP_TOP_TO_COIL_OUT_NEG
S158	COIL_TO_SPKRAMP_TOP_VSENSE_POS
S159	COIL_TO_SPKRAMP_TOP_VSENSE_NEG
S160	COIL_TO_SPKRAMP_BOT_VSENSE_POS
S161	COIL_TO_SPKRAMP_BOT_VSENSE_NEG
S162	SPKRAMP_BOT_TO_COIL_OUT_POS
S163	SPKRAMP_BOT_TO_COIL_OUT_NEG
S164	AP_TO_NFC_DEV_WAKE
S165	UART_NFC_TO_AP_CTS_L
S166	UART_AP_TO_NFC_TXD
S167	UART_NFC_TO_AP_RXD
S168	UART_AP_TO_NFC_RTS_L
S169	AP_TO_NFC_FW_DWLD_REQ
S170	NFC_TO_PMU_HOST_WAKE
S171	PMU_TO_NFC_EN
S172	NC



Cellular



8	7	6	5	4	3	2	1	
BOM OPTIONS:								
D								D
C								C
B								B
A								A
8	7	6	5	4	3	2	1	

RF PMIC: SWITCHERS & LDOS

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

D

D

C

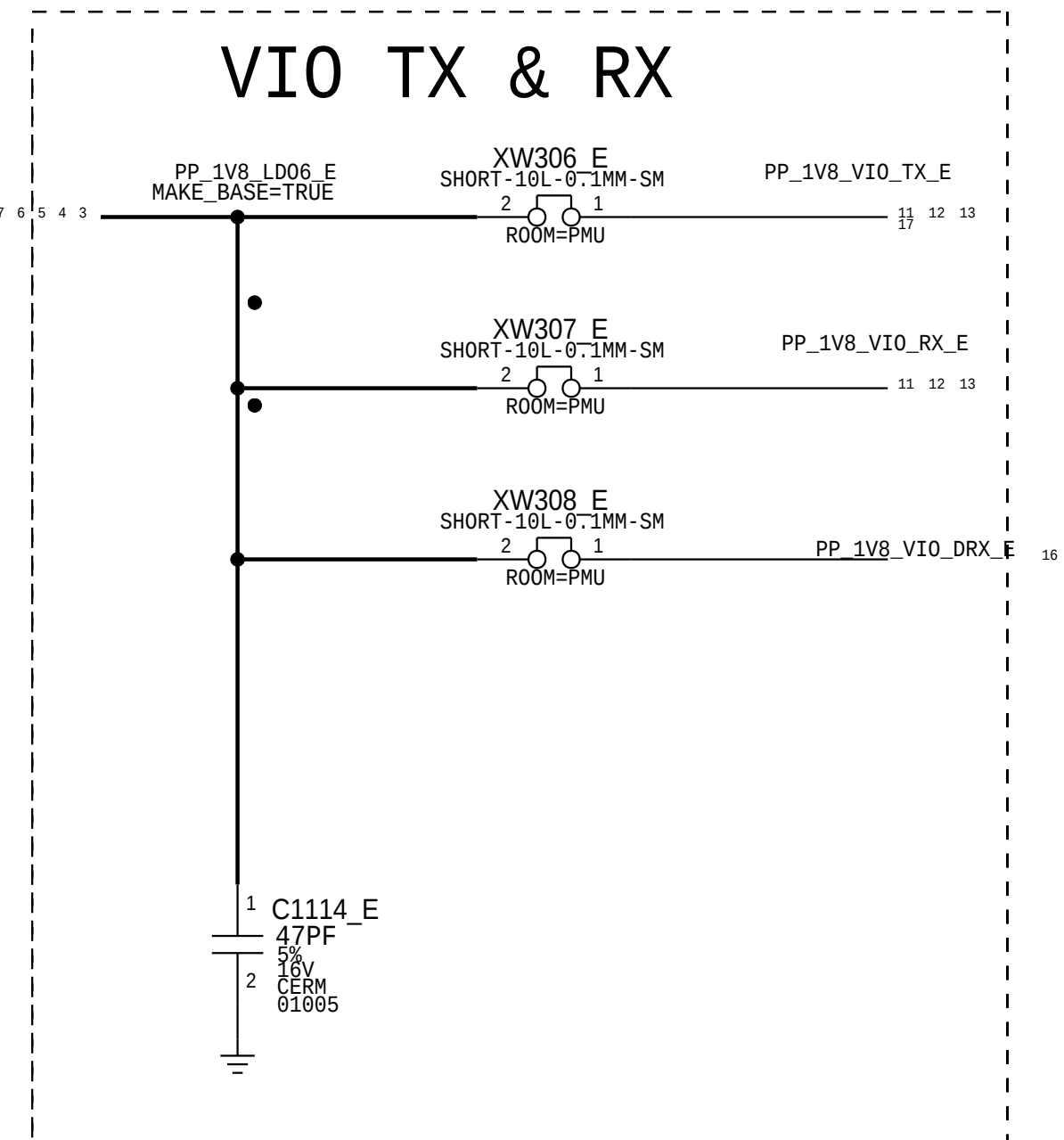
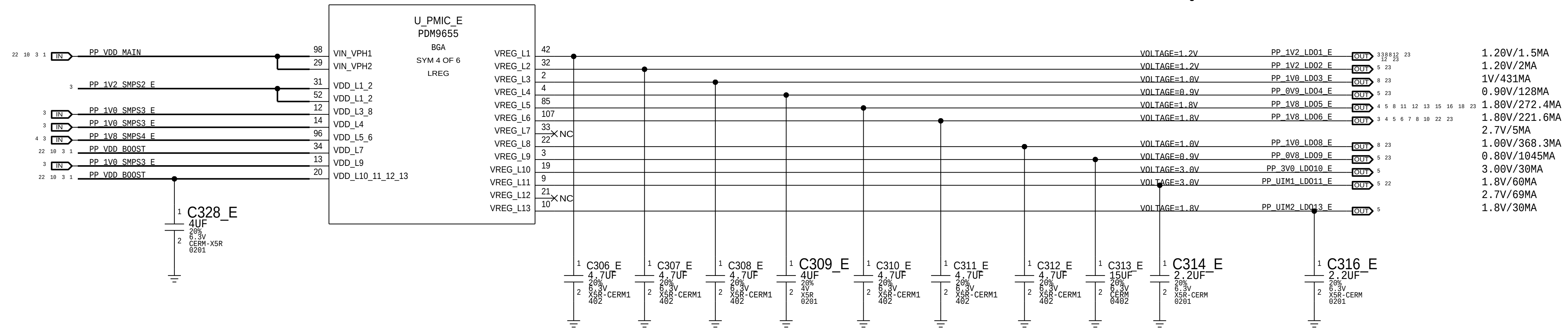
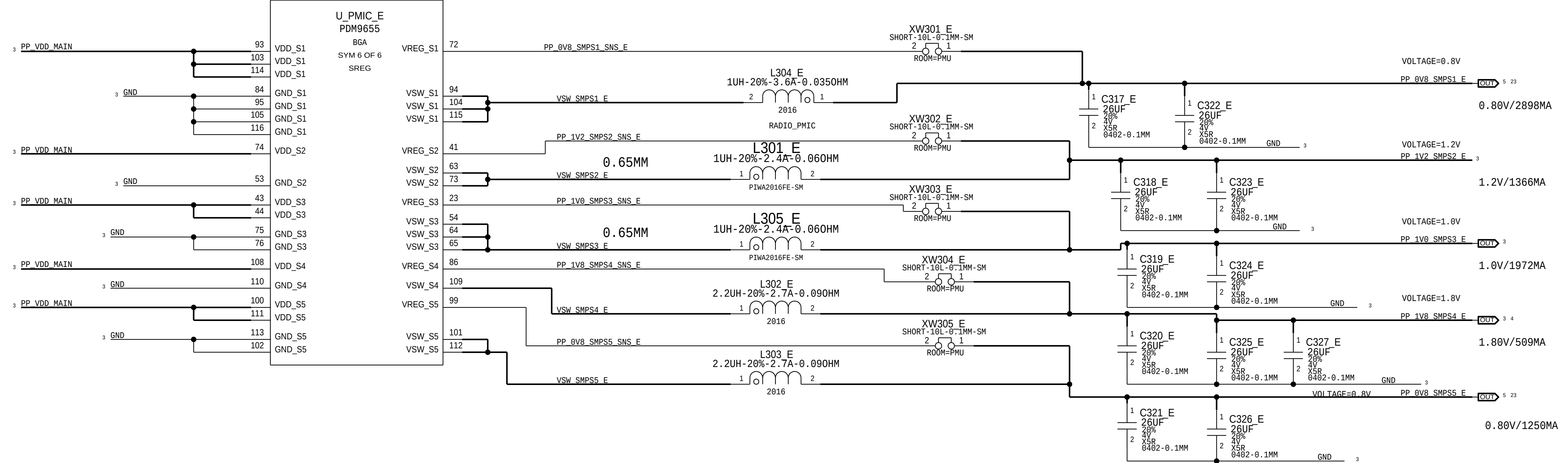
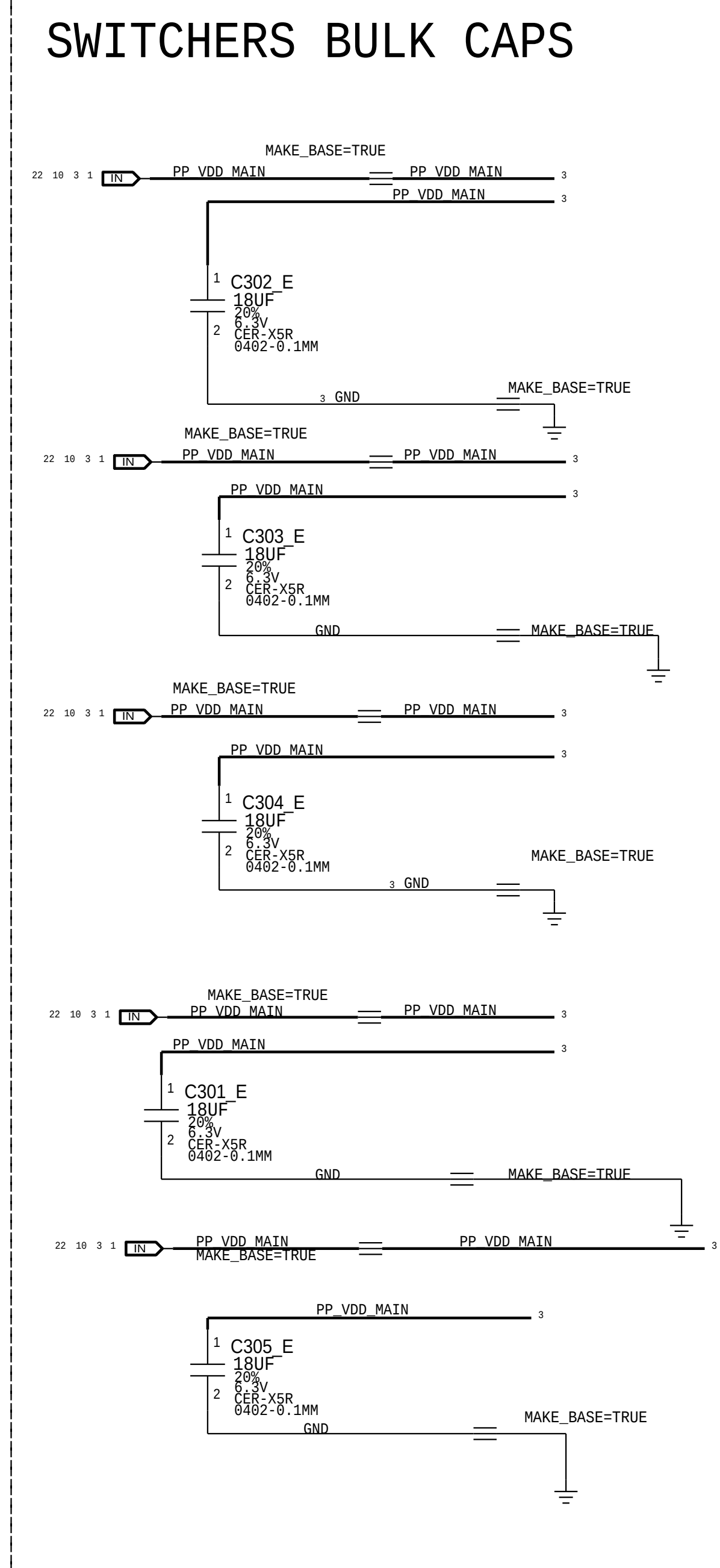
C

B

B

A

A





RF PMIC: XTAL, CLOCKS AND MISCELLANEOUS

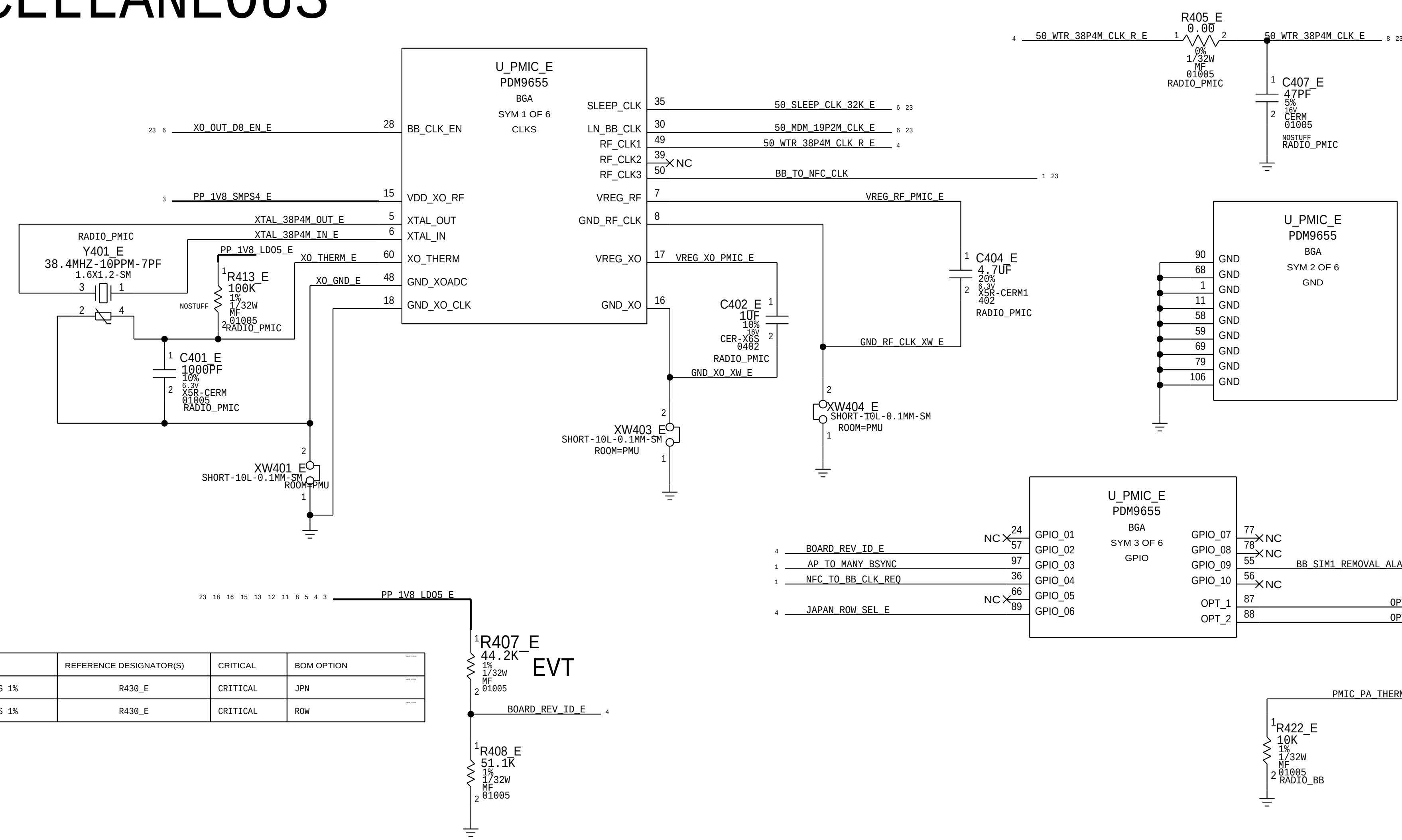
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<0.10V	NOSTUFF	51.1K	T500/DEV1
0.15V-0.25V	422K	51.1K	PROT01.5
0.35V-0.45V	180K	51.1K	PROT02.5/DEV2
0.55V-0.65V	105K	51.1K	PROT01/DEV3
0.75V-0.85V	66.5K	51.1K	PROT02/DEV4
0.95V-1.05V	44.2K	51.1K	EVT
1.15V-1.25V	28K	51.1K	CARRIER/DEV5
1.35V-1.45V	16.5K	51.1K	DVT/DEV6
1.55V-1.65V	7.87K	51.1K	PVT/DEV7
1.75V-1.85V	10K	NOSTUFF	PROT03

PA_THERM2	R409	R410	PRODUCT TYPE	Y
0.00V-0.10V	NOSTUFF	51.1K	RADIO DEV	1
0.15V-0.25V	422K	51.1K	3RD TYPE	2
0.35V-0.45V	180K	51.1K	POR MLB (FF)	3
0.55V-0.65V	105K	51.1K	SPARE	4
0.75V-0.85V	66.5K	51.1K	SPARE	5
0.95V-1.05V	44.2K	51.1K	SPARE	6
1.15V-1.25V	28K	51.1K	SPARE	7
1.35V-1.45V	16.5K	51.1K	SPARE	8
1.55V-1.65V	7.87K	51.1K	SPARE	9
1.75V-1.85V	10K	NOSTUFF	SPARE	A

JPN_ROW_SEL	R430	R431	MAV PLATFORM	Y	SKU
<0.10V	NOSTUFF	51.1K	MAV17.0	0	ROW
0.15V-0.25V	422K	51.1K	MAV17.1	1	JPN
0.35V-0.45V	180K	51.1K	MAV17.2	2	ROW
0.55V-0.65V	105K	51.1K	MAV17.3	3	JPN
0.75V-0.85V	66.5K	51.1K	MAV17.4	4	ROW
0.95V-1.05V	44.2K	51.1K	MAV17.5	5	JPN
1.15V-1.25V	28K	51.1K	MAV17.6	6	ROW
1.35V-1.45V	16.5K	51.1K	MAV17.7	7	JPN
1.55V-1.65V	7.87K	51.1K	MAV17.8	8	ROW
1.75V-1.85V	10K	NOSTUFF	MAV17.9	9	JPN

NOTE XTAL SYMBOL MIRRORED COMPARED TO PREVIOUS SCH

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
118S8743	1	RESISTOR 01005, 44.2K OHMS 1%	R430_E	CRITICAL	JPN
118S8612	1	RESISTOR 01005, 66.5K OHMS 1%	R430_E	CRITICAL	ROW



BB : POWER

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

D

C

B

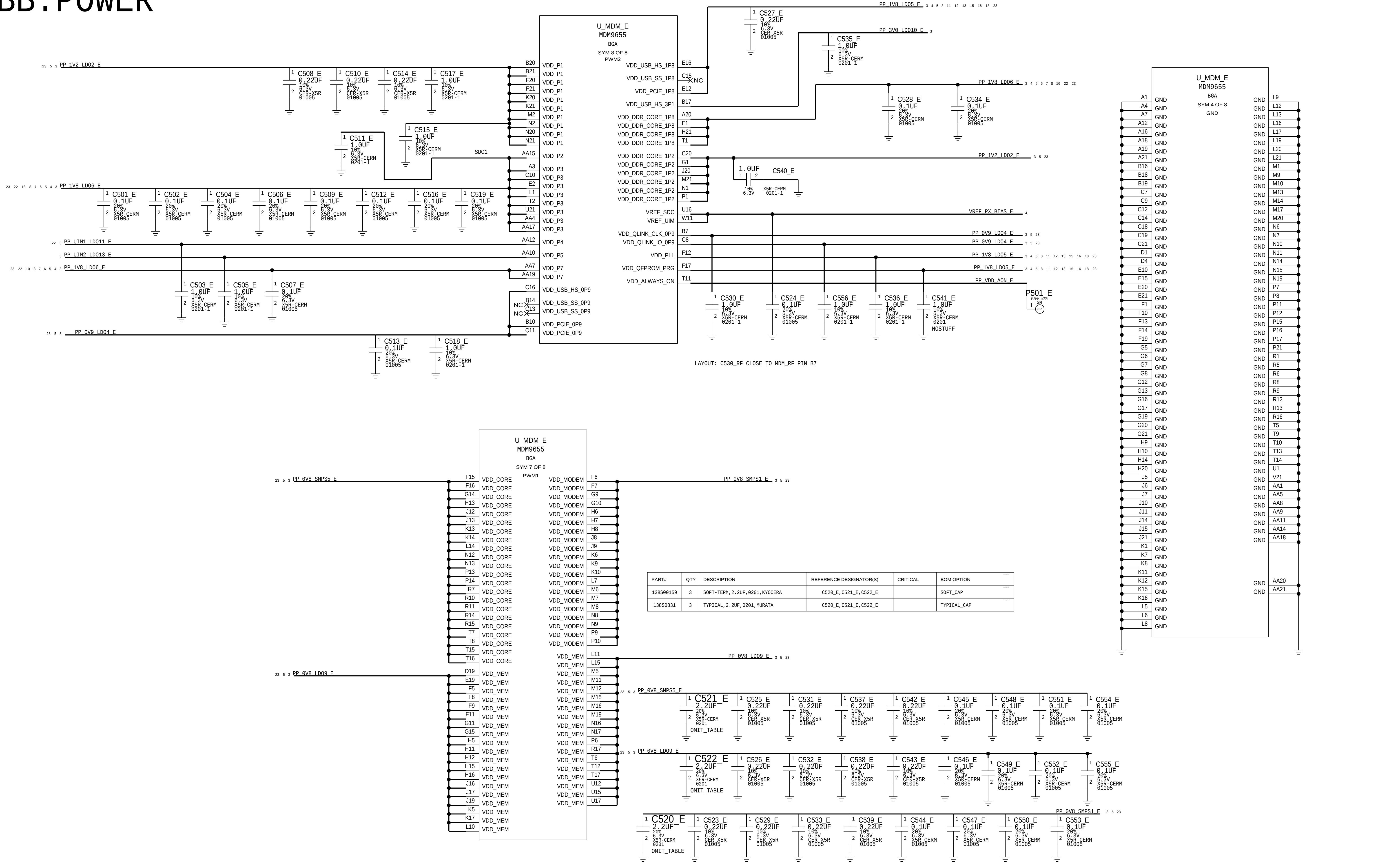
A

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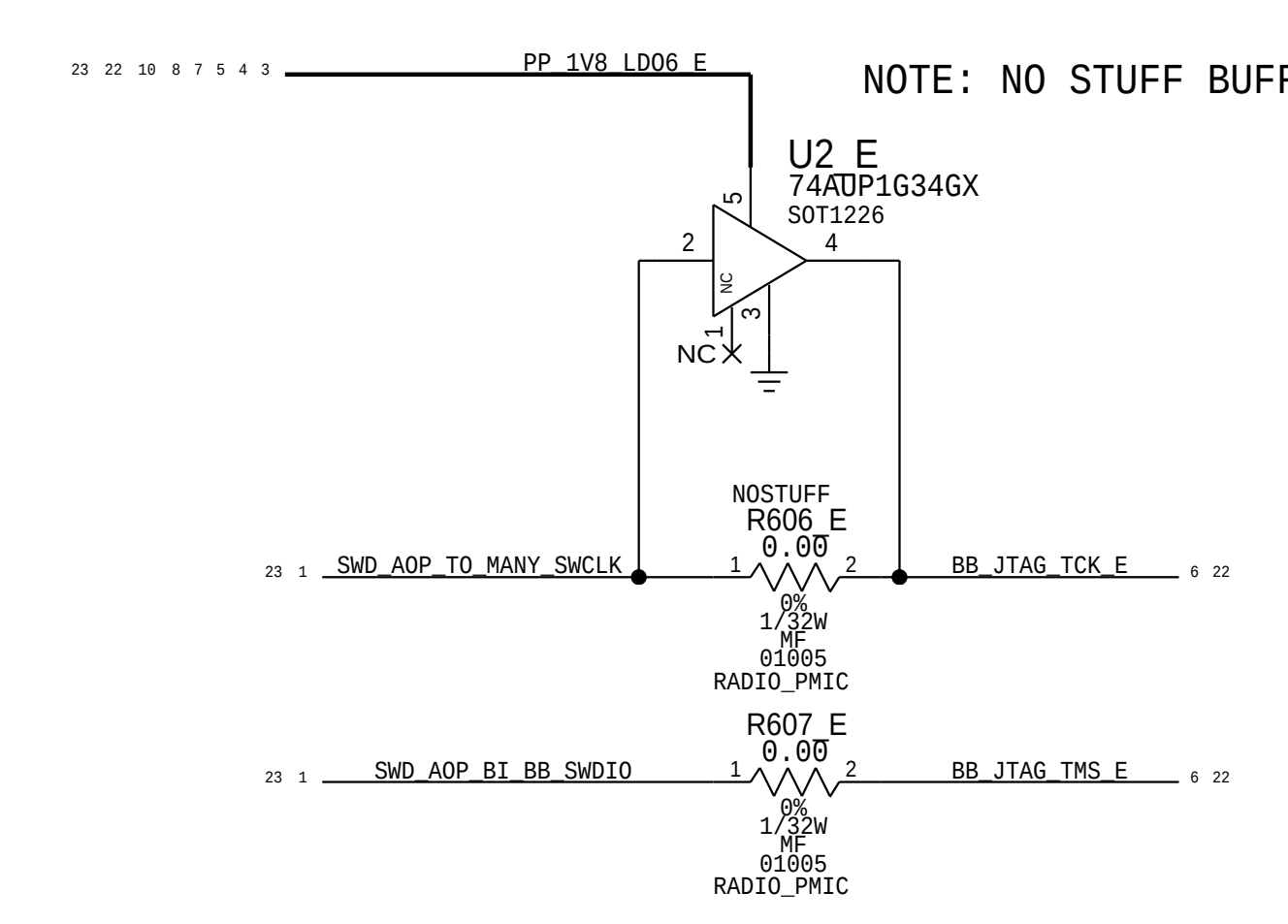
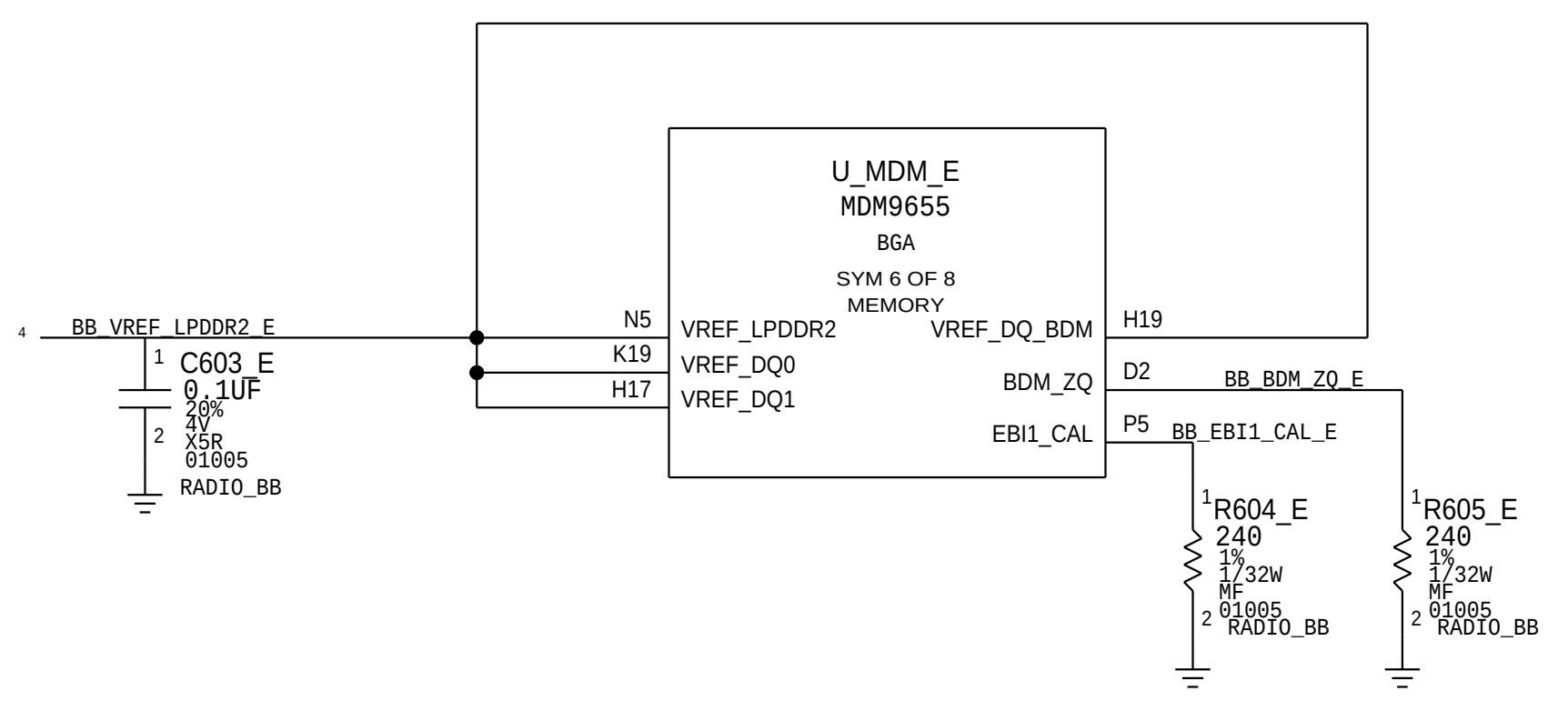
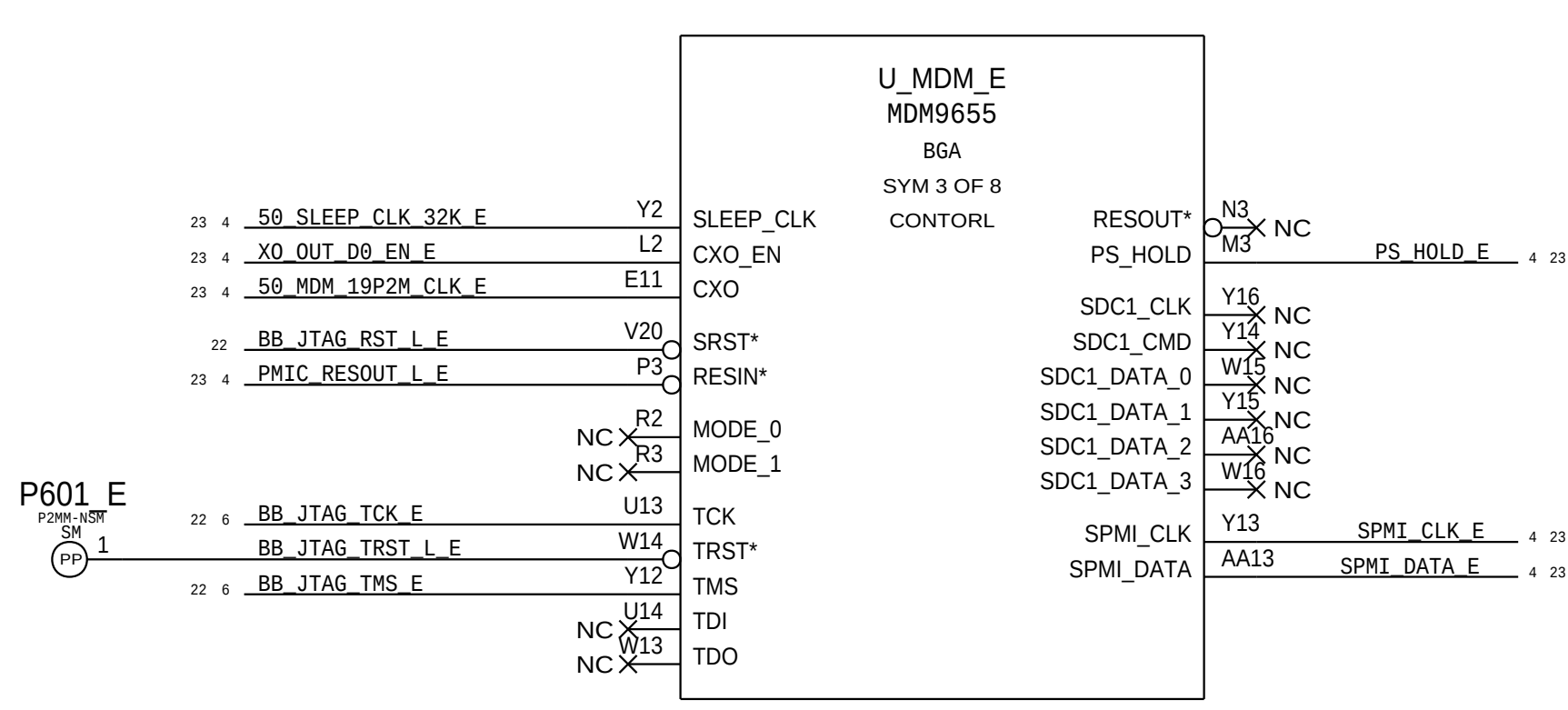
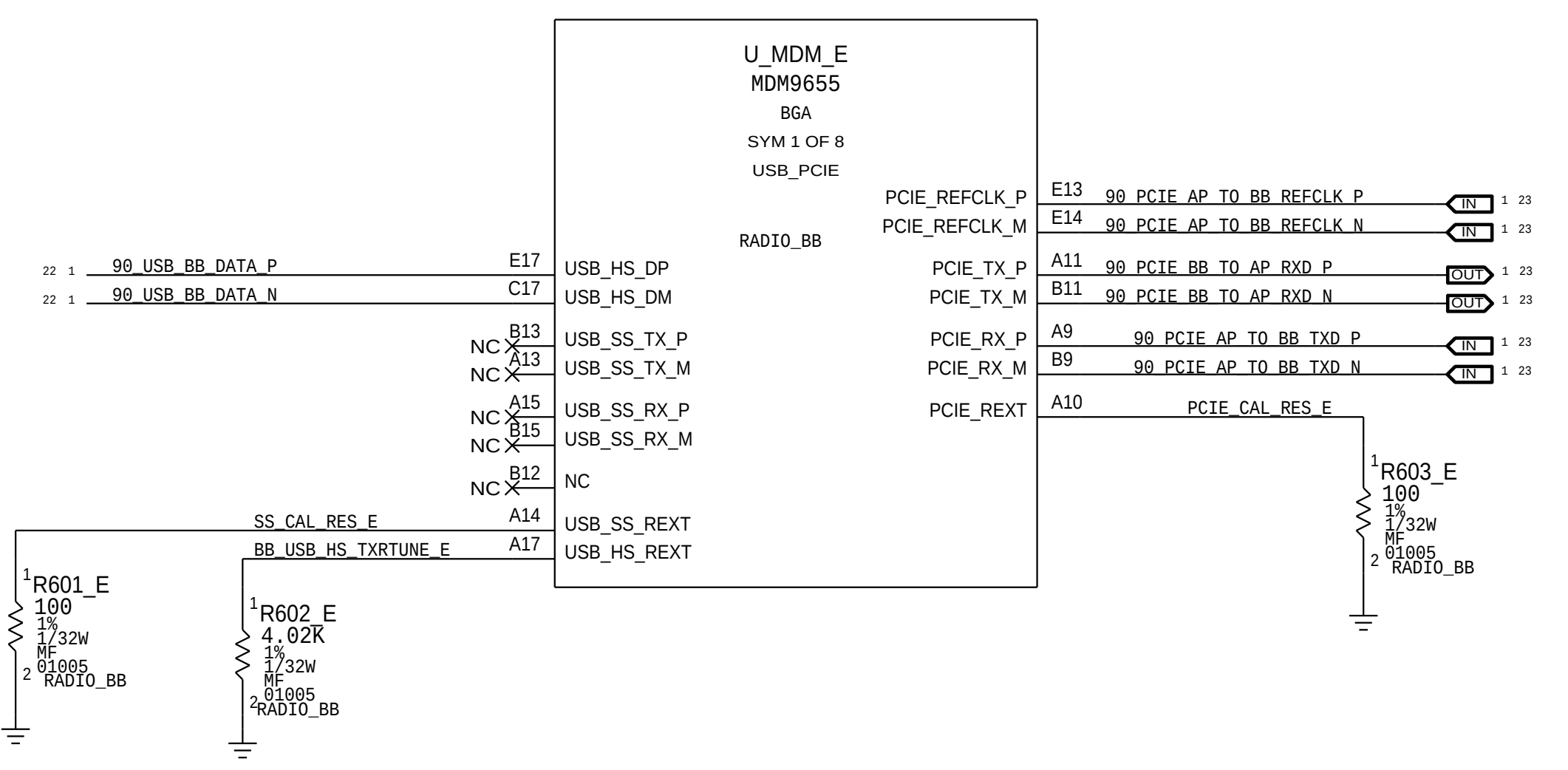
B

A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	3	SOFT-TERM, 2.2UF, 0201, KYOCERA	C520_E, C521_E, C522_E		SOFT_CAP
138S0831	3	TYPICAL, 2.2UF, 0201, MURATA	C520_E, C521_E, C522_E		TYPICAL_CAP

BB:CONTROL & HS PERIPHERALS



BB:GPIOS & QLINK

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

D

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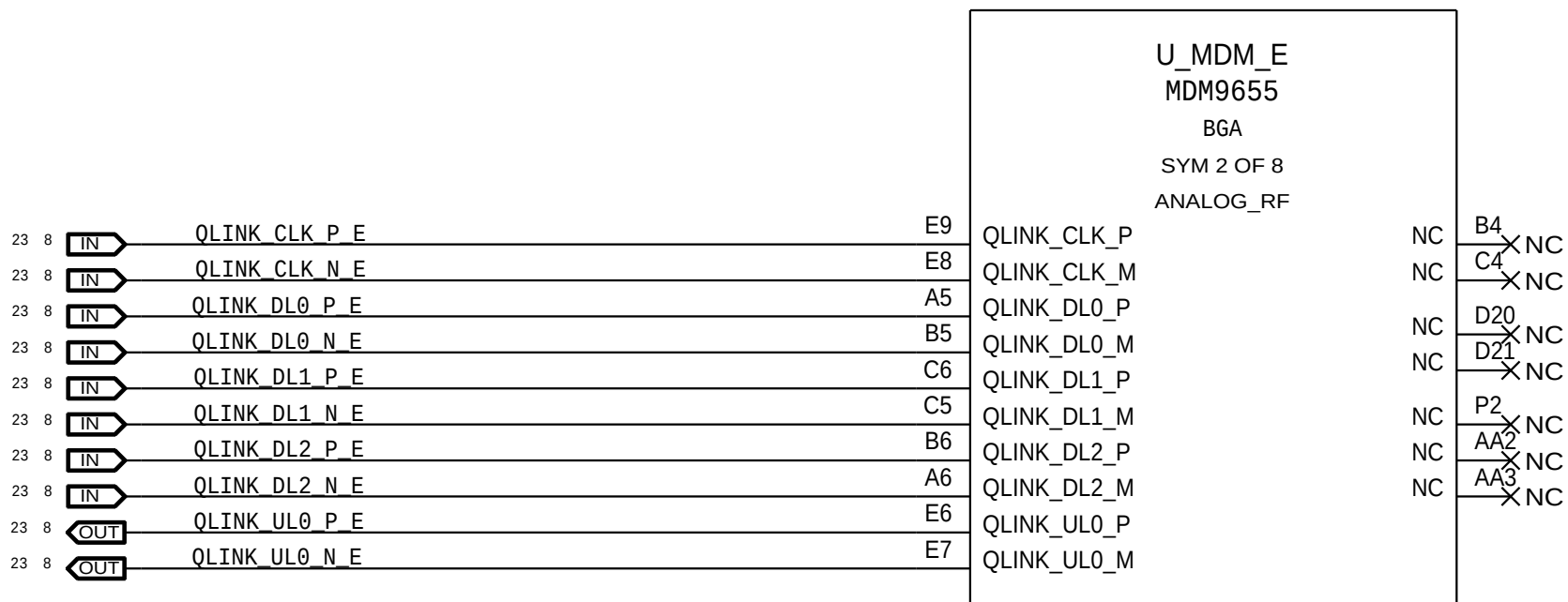
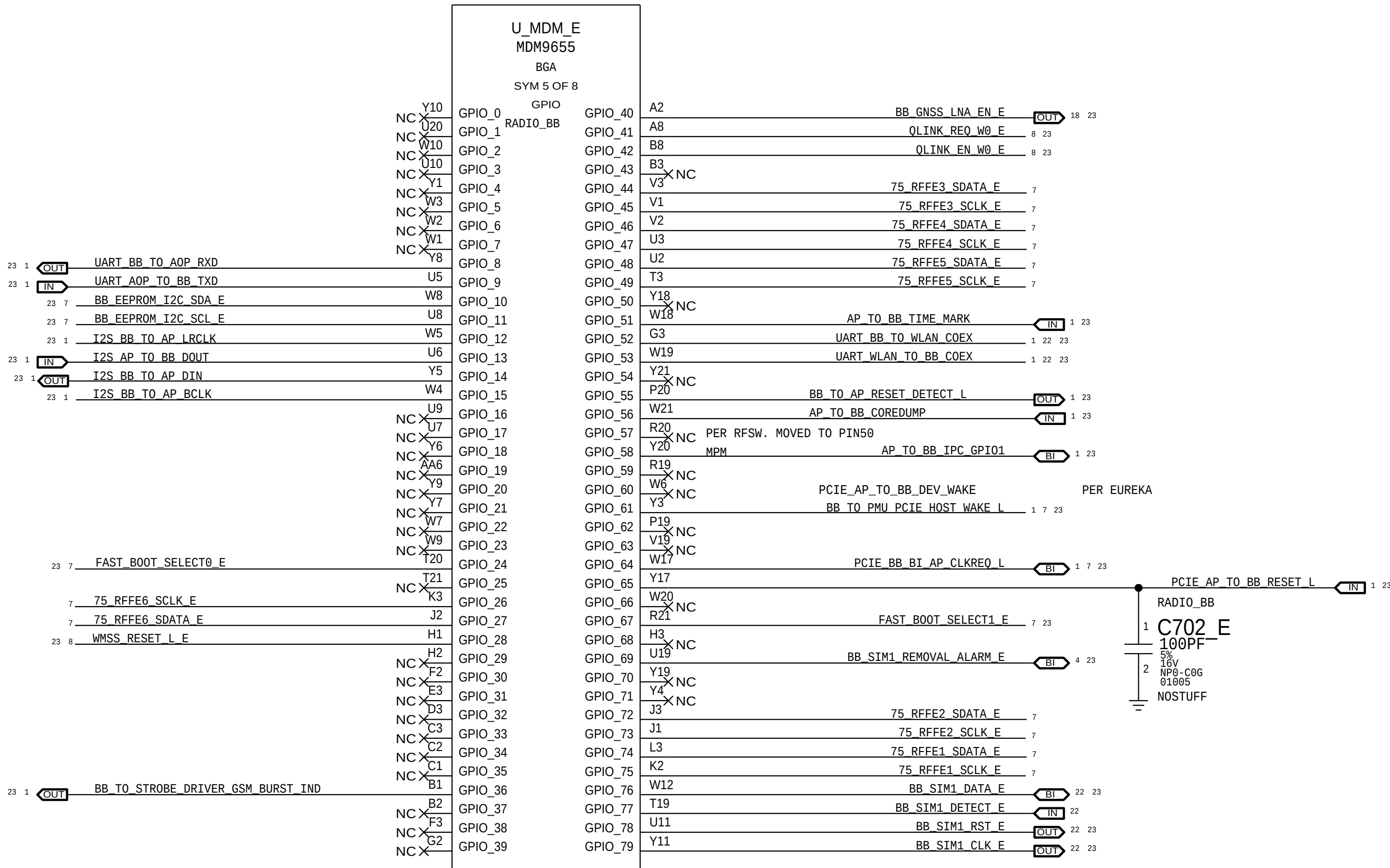
C

B

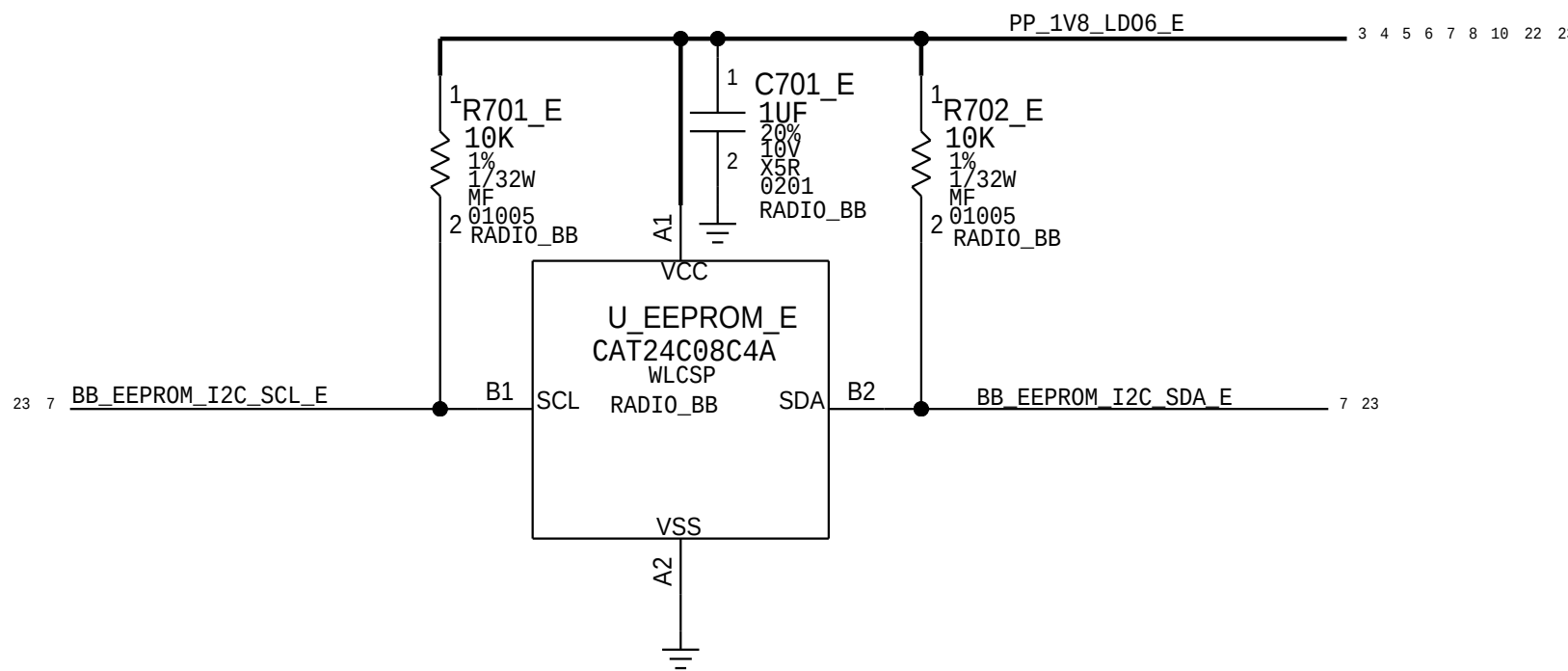
B

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A



BB EEPROM



RFFE MAP

RFFE1 - WTR, LATGP0, LATUNE

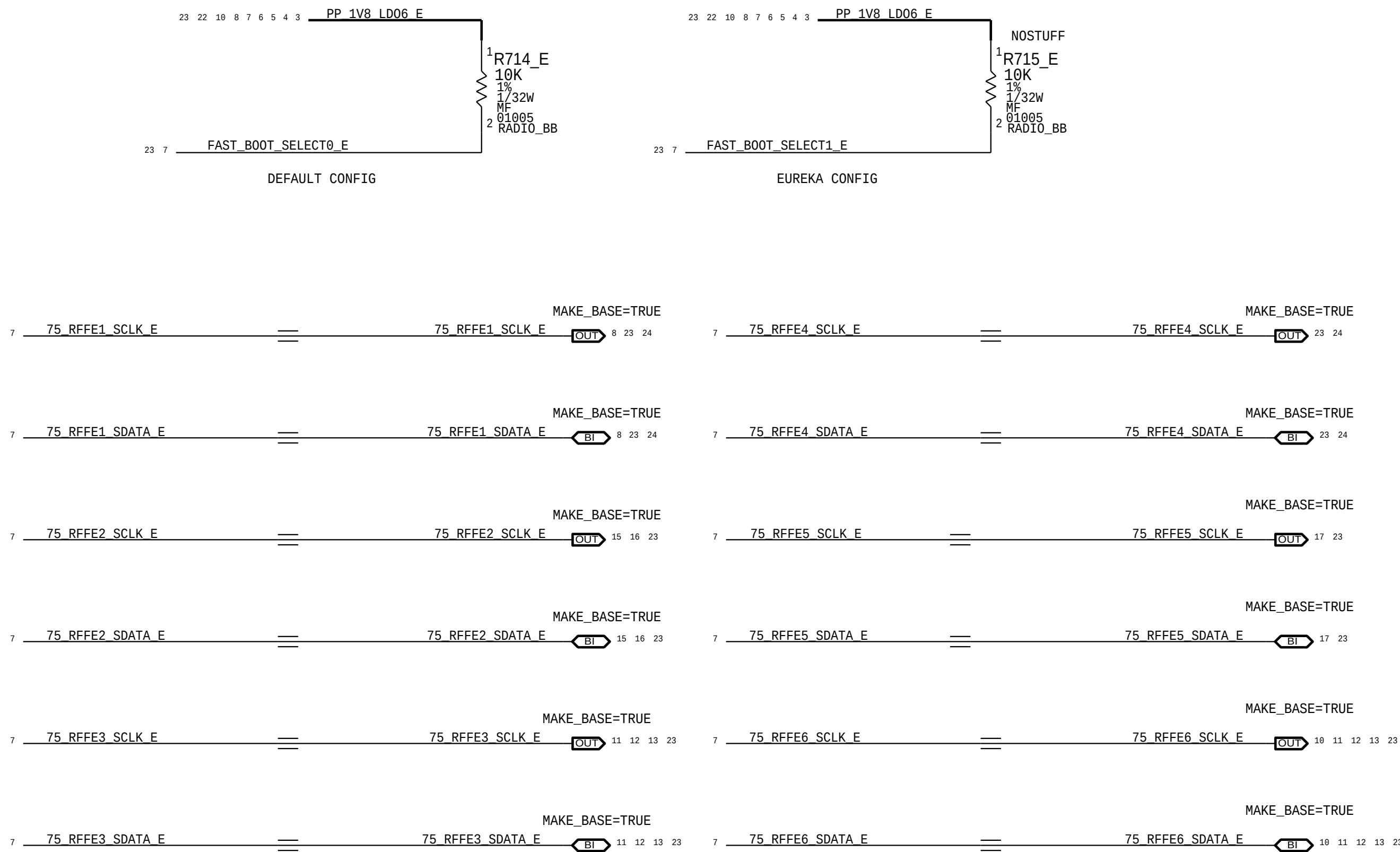
RFFE2 - HB DSM, LB DSM

RFFE3 - UHB, HB, LB PAD LNA

RFFE4 - UATGP0, UATUNE

RFFE5 - CPL

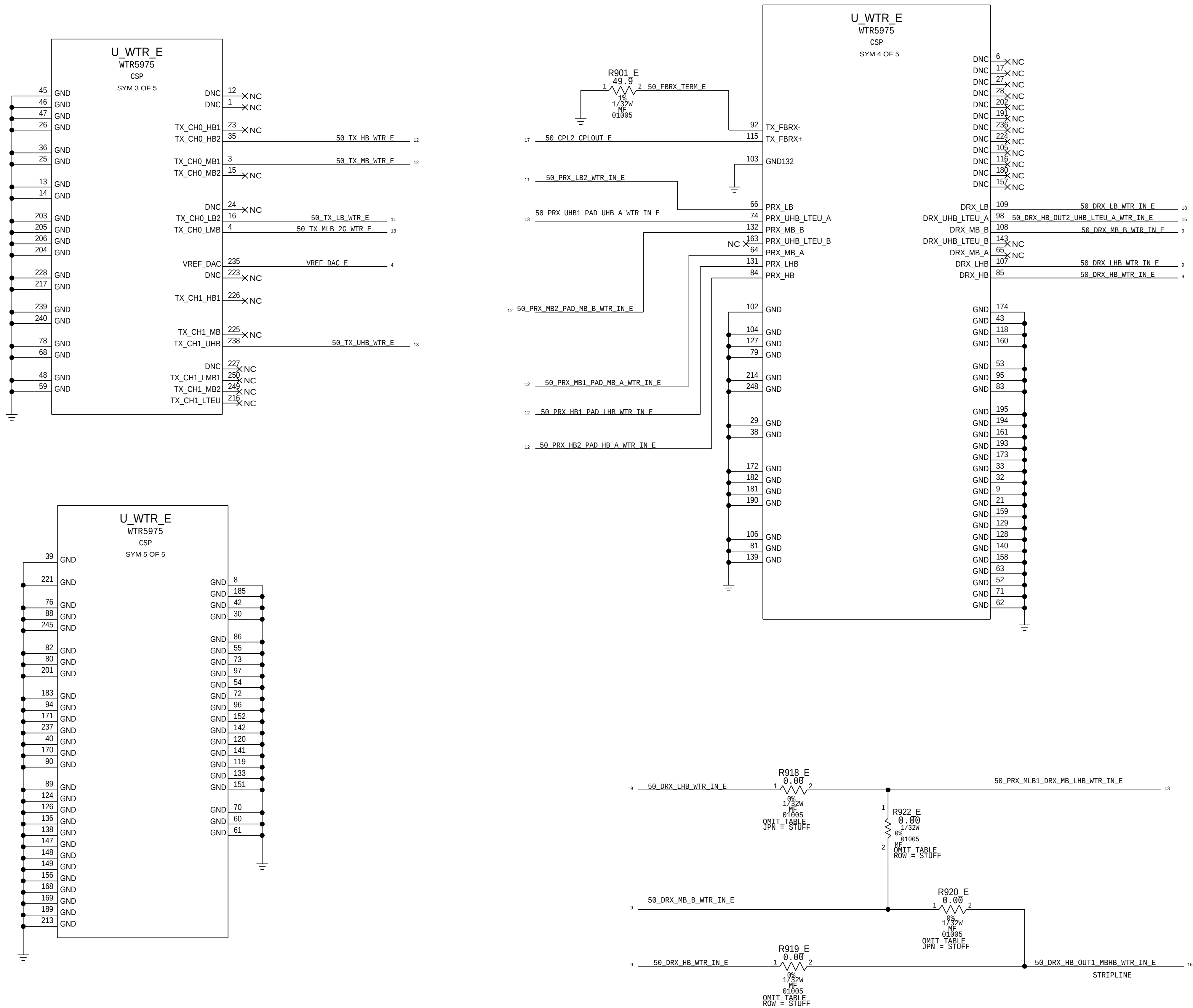
RFFE6 - QET, 2G PA/UHB, HB, LA PAD



TRANSCEIVER: TX & RX

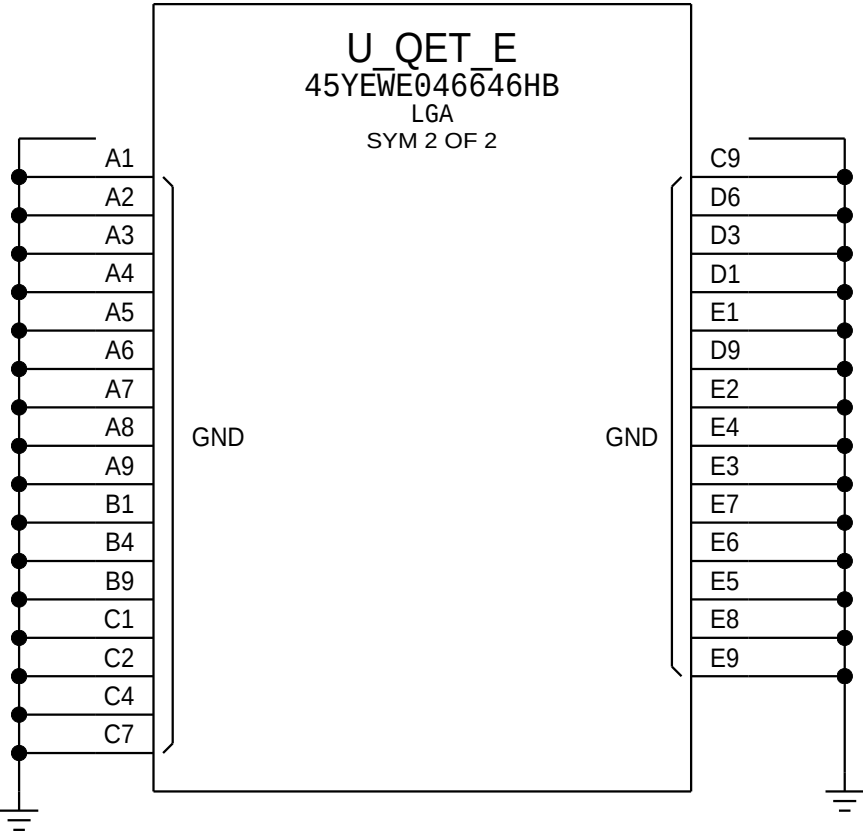
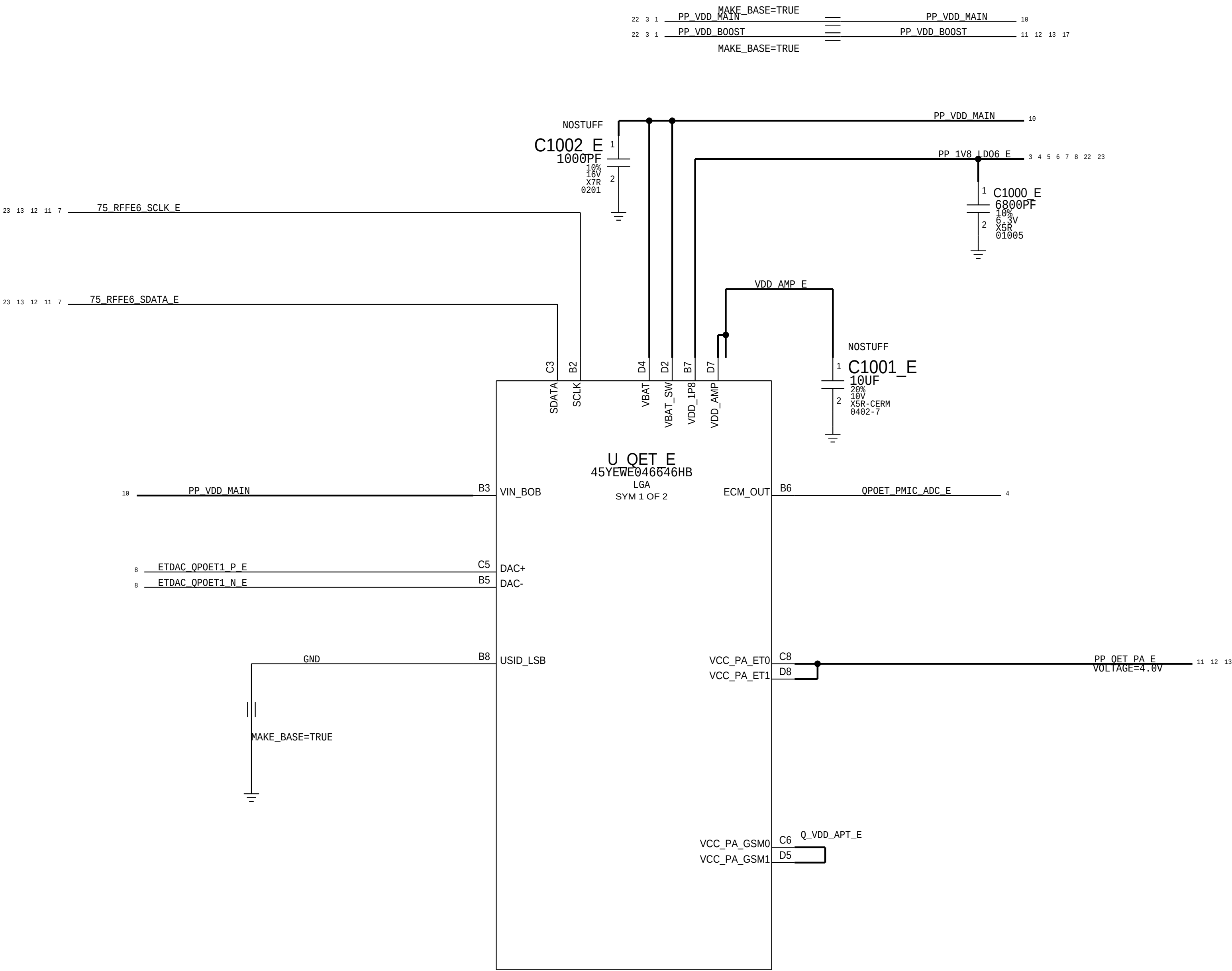
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R919_E	CRITICAL	ROW
117S0161	1	01005 0R	R922_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R918_E	CRITICAL	JPN
117S0161	1	01005 0R	R920_E	CRITICAL	JPN



QET:MODULE

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LB PAD

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
155S90316	1	2G CPLB JPN	CPL2GB_E	CRITICAL	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
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D

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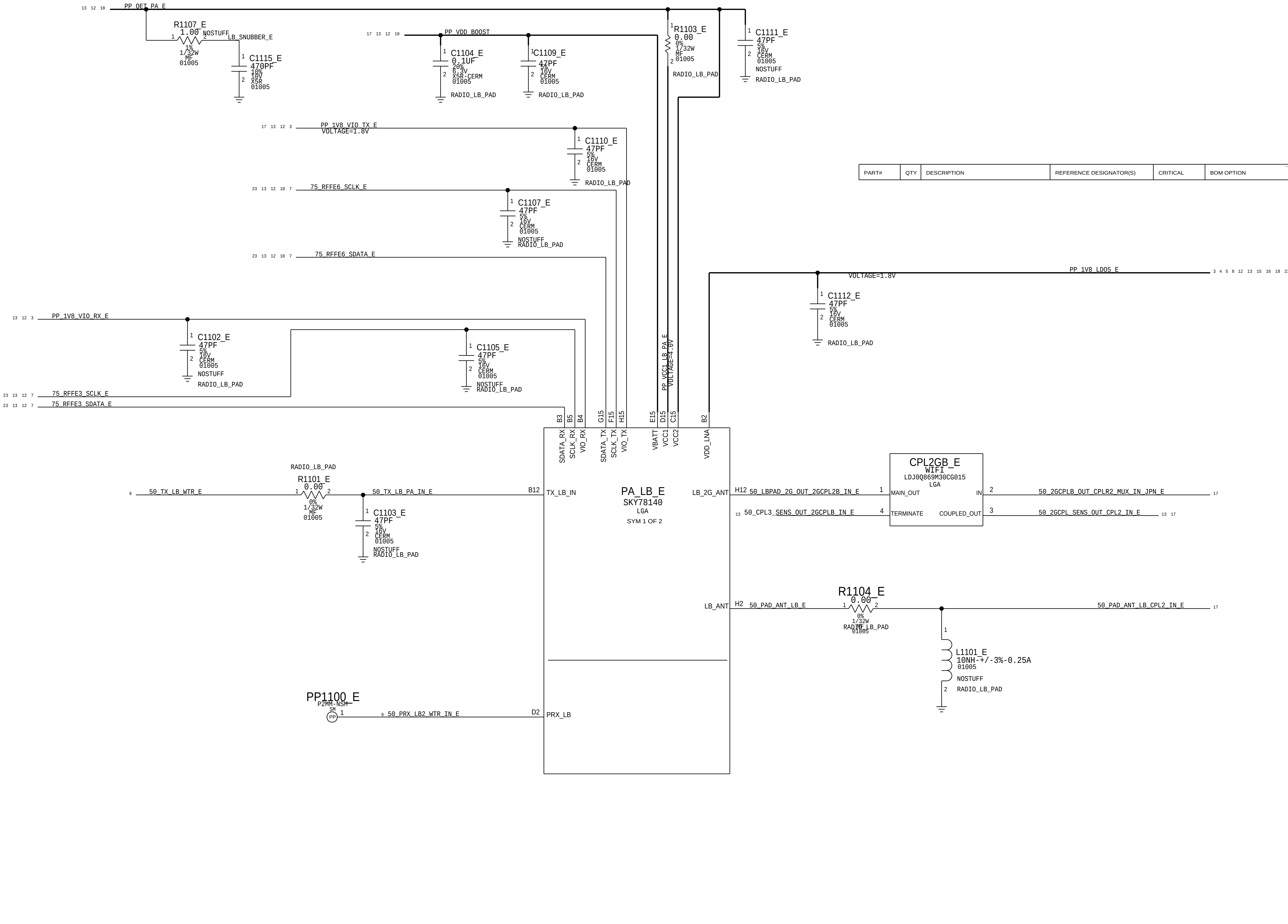
A

D

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1

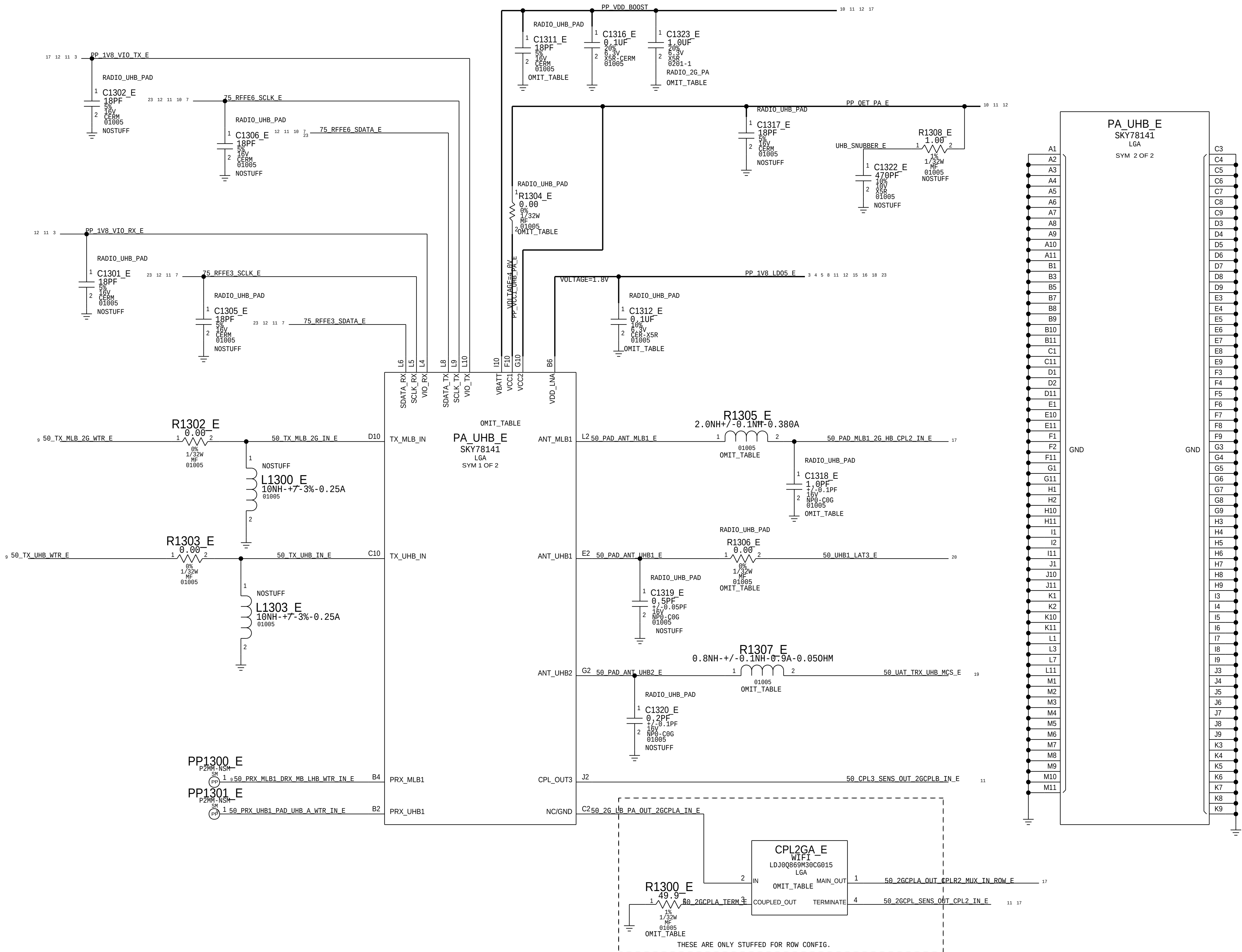


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
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UHB/2G PAD - JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01243	1	UHB PAD MP	PA_UHB_E	CRITICAL	JPN
11750161	1	MLB/2GTX IN MCH2, 0 OHMS	R1302_E	CRITICAL	JPN
11750161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	JPN
11750161	1	MLB/2G MB OUT MCH1 01005 0R	R1305_E	CRITICAL	JPN
11750161	1	UHB1 OUTPUT MCH1, 0 OHMS	R1306_E	CRITICAL	JPN
152500494	1	UHB2 OUT/DRX BP MCH1, 0.8NH 0.9A	R1307_E	CRITICAL	JPN
13250683	1	UHB LNA DECOUPLING, 0.1UF 6.3V 10%	C1312_E	CRITICAL	JPN
13150214	1	VBAT DECOUPLING, 18PF	C1311_E	CRITICAL	JPN
13850692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	JPN
11750161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	JPN
13150375	1	MLB OUTPUT MCH UAT, 01005 1.0PF	C1318_E	CRITICAL	JPN

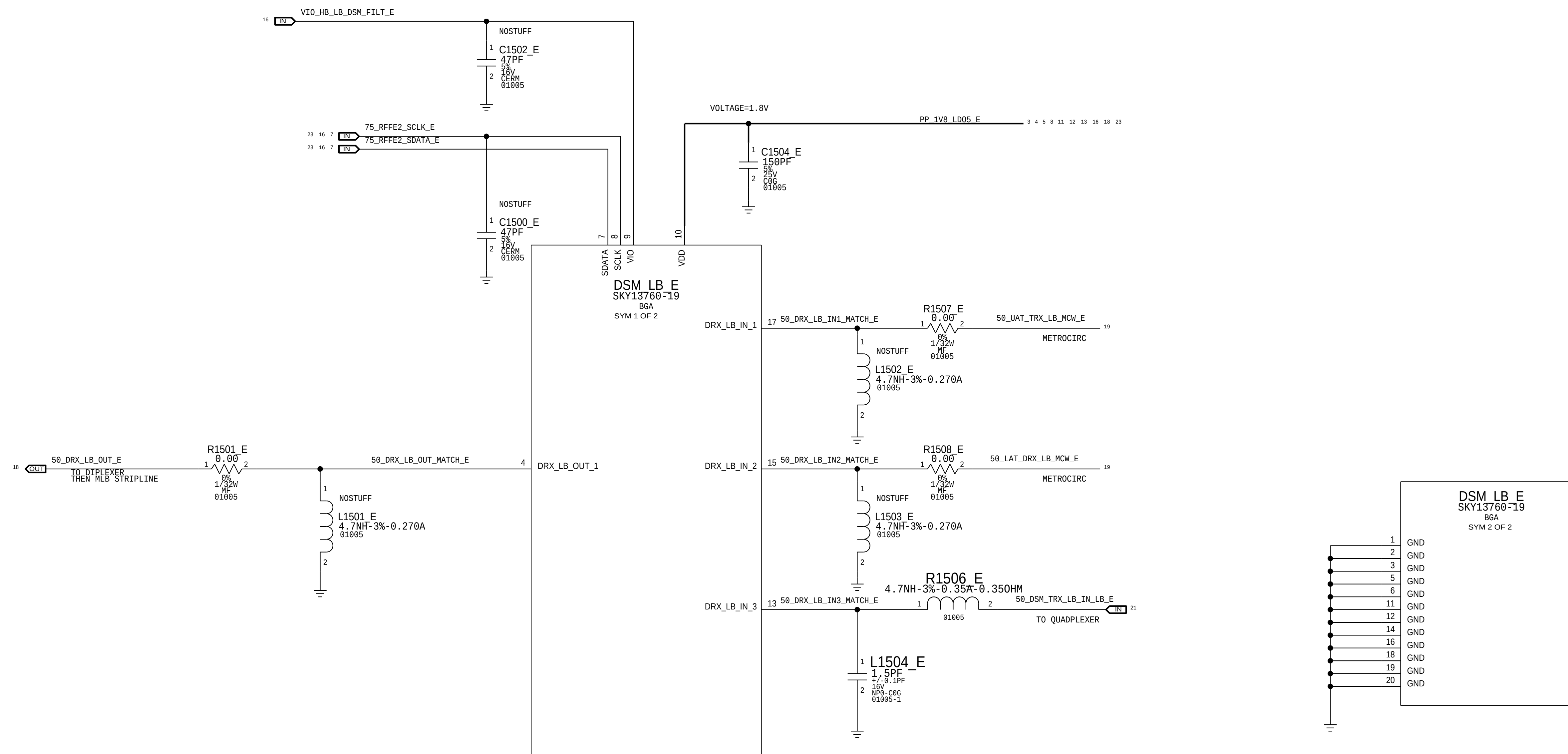


8		7		6		5		4		3		2		1		CHINAFOX	
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST																	
2G PA - ROW																	
D																D	
C																C	
B																B	
A																A	
8		7		6		5		4		3		2		1		26 OF 41	

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01247	1	2G PA MP	PA_UHB_E	CRITICAL	ROW
117S0161	1	MLB/2GTX IN MCH2, 0 OHMS	R1302_E	CRITICAL	ROW
117S0161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	ROW
152S1917	1	01005 2.0NH 0.6A	R1305_E	CRITICAL	ROW
117S0161	1	UHB2 OUT/DRX BP MCH1, 0 OHMS	R1307_E	CRITICAL	ROW
131S0214	1	VBAT DECOUPLING,18PF	C1311_E	CRITICAL	ROW
130S0092	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	ROW
117S0161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	ROW
155S00316	1	ROW CLPA	CPL2GA_E	CRITICAL	ROW
118S0052	1	01005 40.9R	R1300_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01247	1	2G PA MP	PA_UHB_E	CRITICAL	ROW
117S0161	1	MLB/26TX IN MCH2, 0 OHMS	R1302_E	CRITICAL	ROW
117S0161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	ROW
152S1917	1	01005 2.0NH 0.6A	R1305_E	CRITICAL	ROW
117S0161	1	UHB2 OUT/DRX BP MCH1, 0 OHMS	R1307_E	CRITICAL	ROW
131S0214	1	VBAT DECOUPLING,18PF	C1311_E	CRITICAL	ROW
138S0692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	ROW
117S0161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	ROW
155S00316	1	ROW CLPA	CPL26A_E	CRITICAL	ROW
118S0652	1	01005 49.9R	R1300_E	CRITICAL	ROW

LB DSM



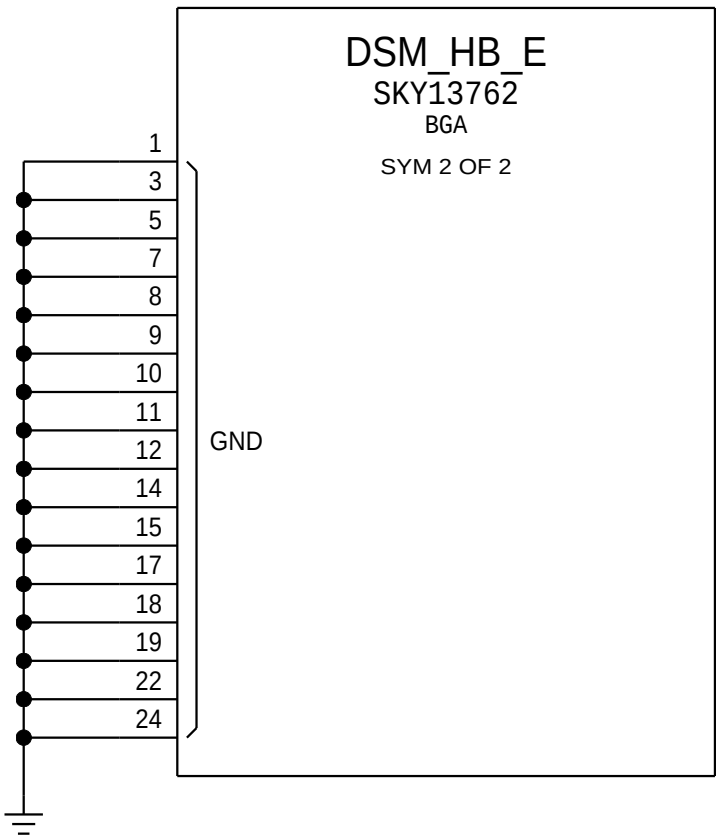
HB DSM

DECOUPLING SHARED WITH LB DSM

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R1600_E	CRITICAL	ROW
117S0161	1	01005 0R	R1602_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R1603_E	CRITICAL	JPN
117S0161	1	01005 0R	R1612_E	CRITICAL	JPN

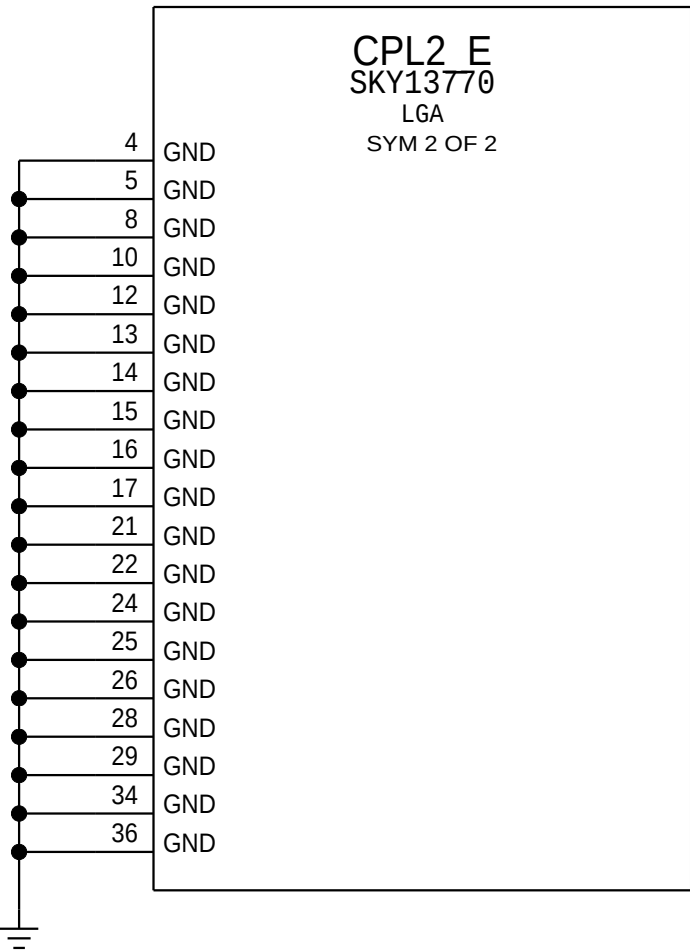
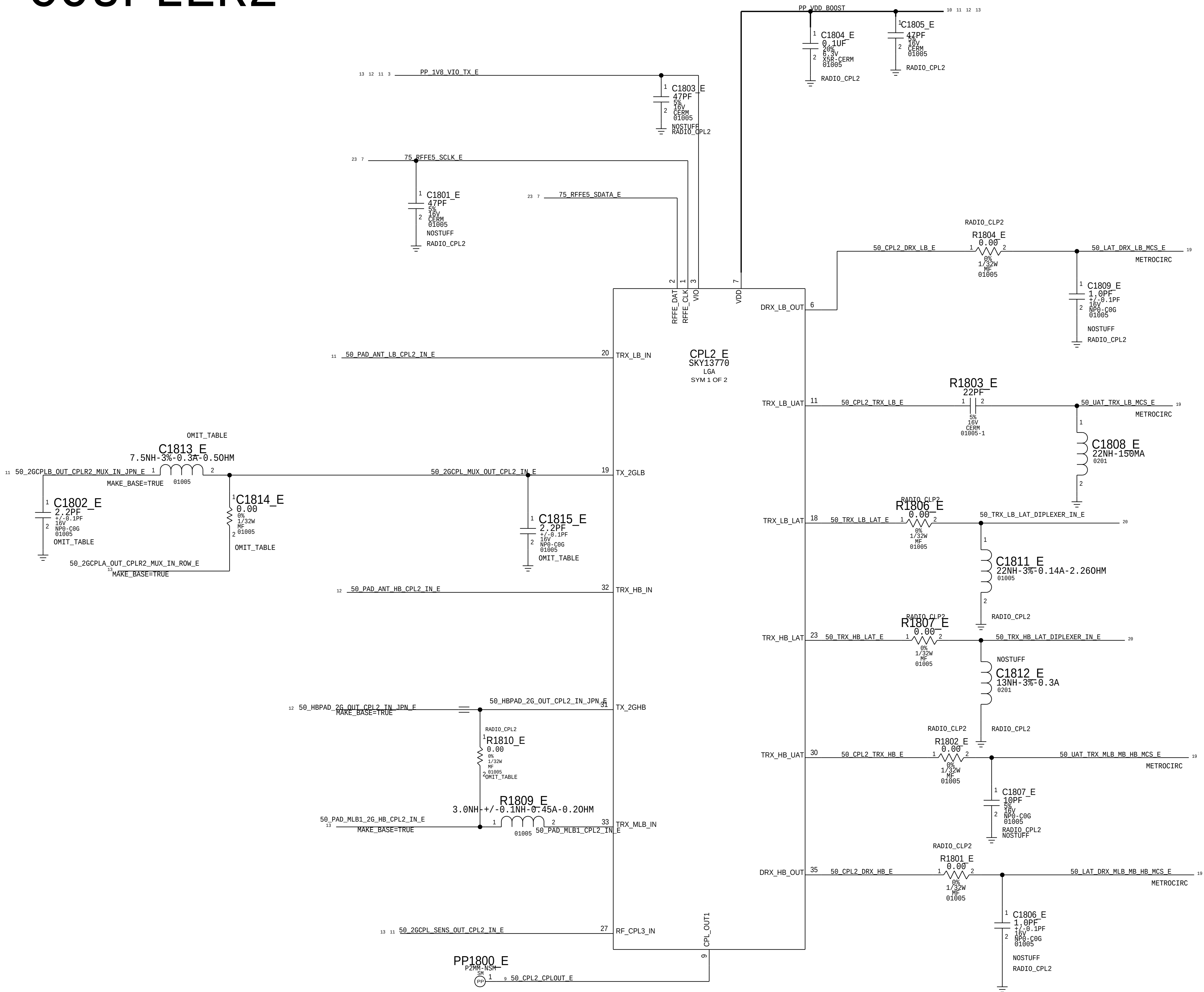


COUPLER2

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST

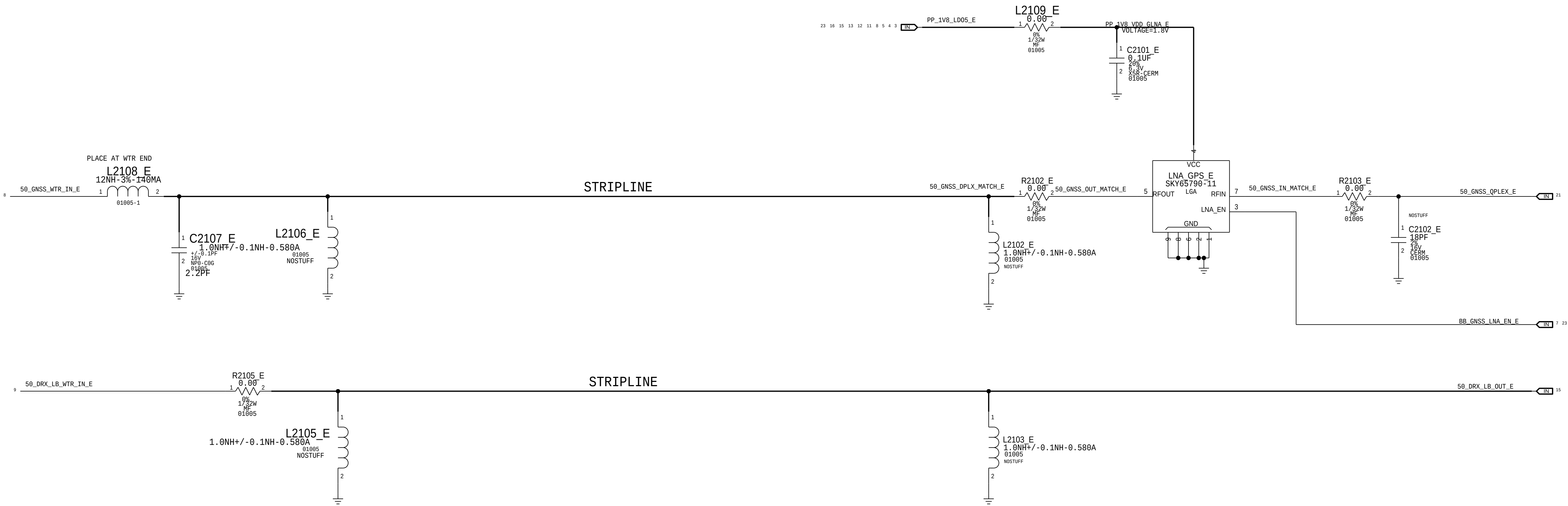
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00504	1	01005 3.0NH 0.45A	R1809_E	CRITICAL	JPN
152S00524	1	01005 7NSH 300MA	C1813_E	CRITICAL	JPN
131S0249	1	01005 2.2PF 0.1PF	C1815_E	CRITICAL	JPN
131S0249	1	01005 2.2PF 0.1PF	C1802_E	CRITICAL	JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R1810_E	CRITICAL	ROW
117S0161	1	01005 0R	C1814_E	CRITICAL	ROW



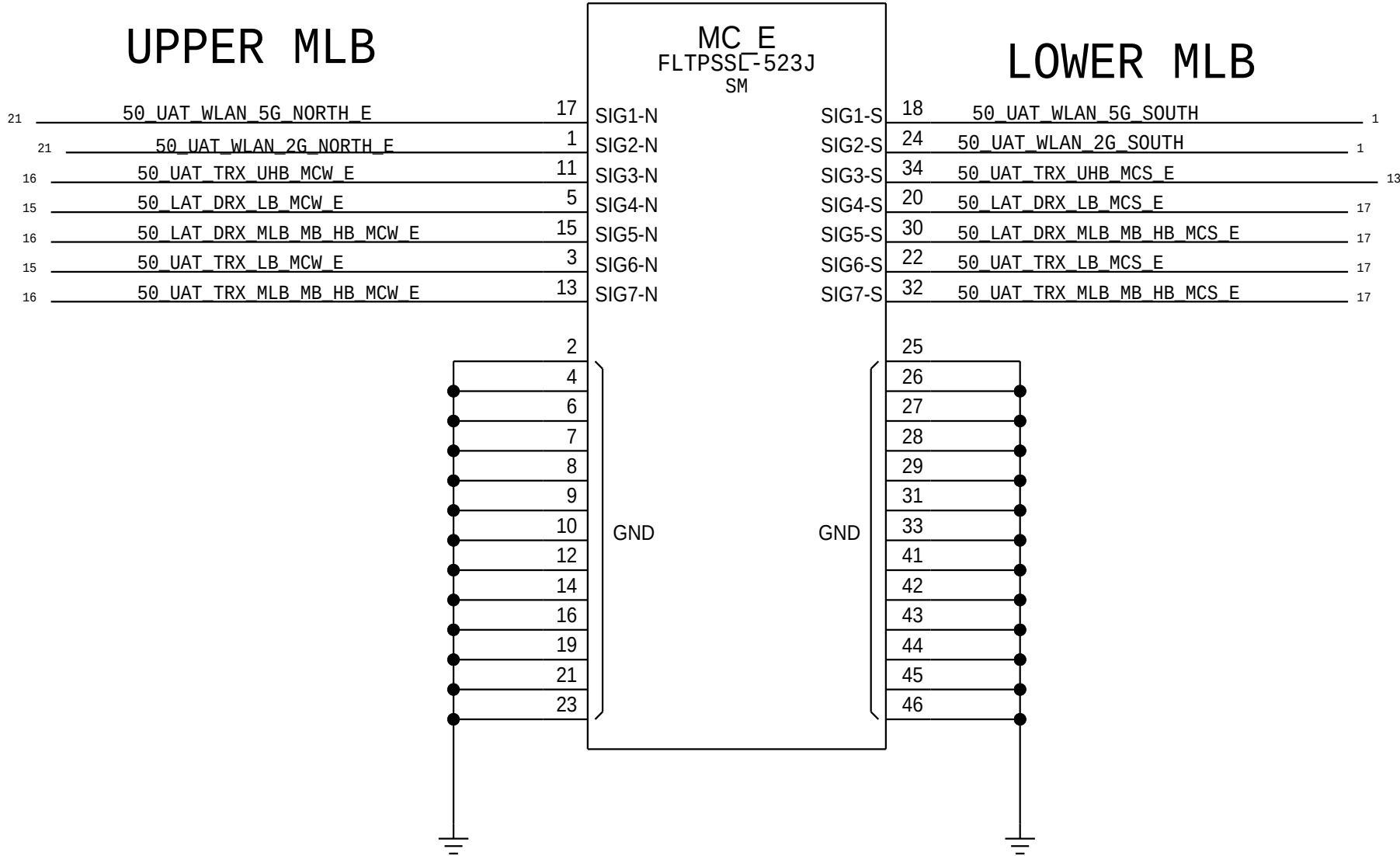


GPS & GPS/LB DIPLEXER



METROCIRC

METROCIRC



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GND

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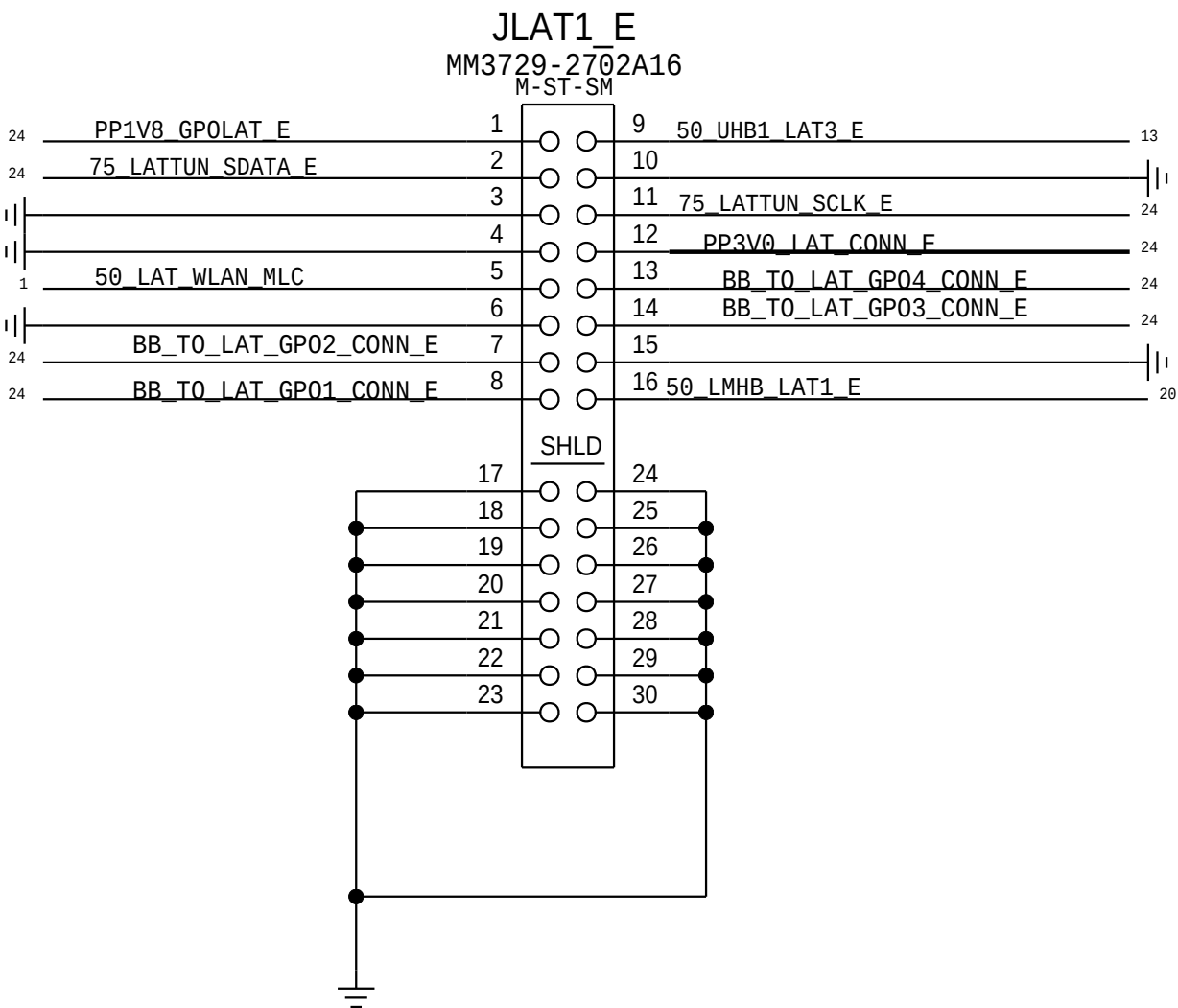
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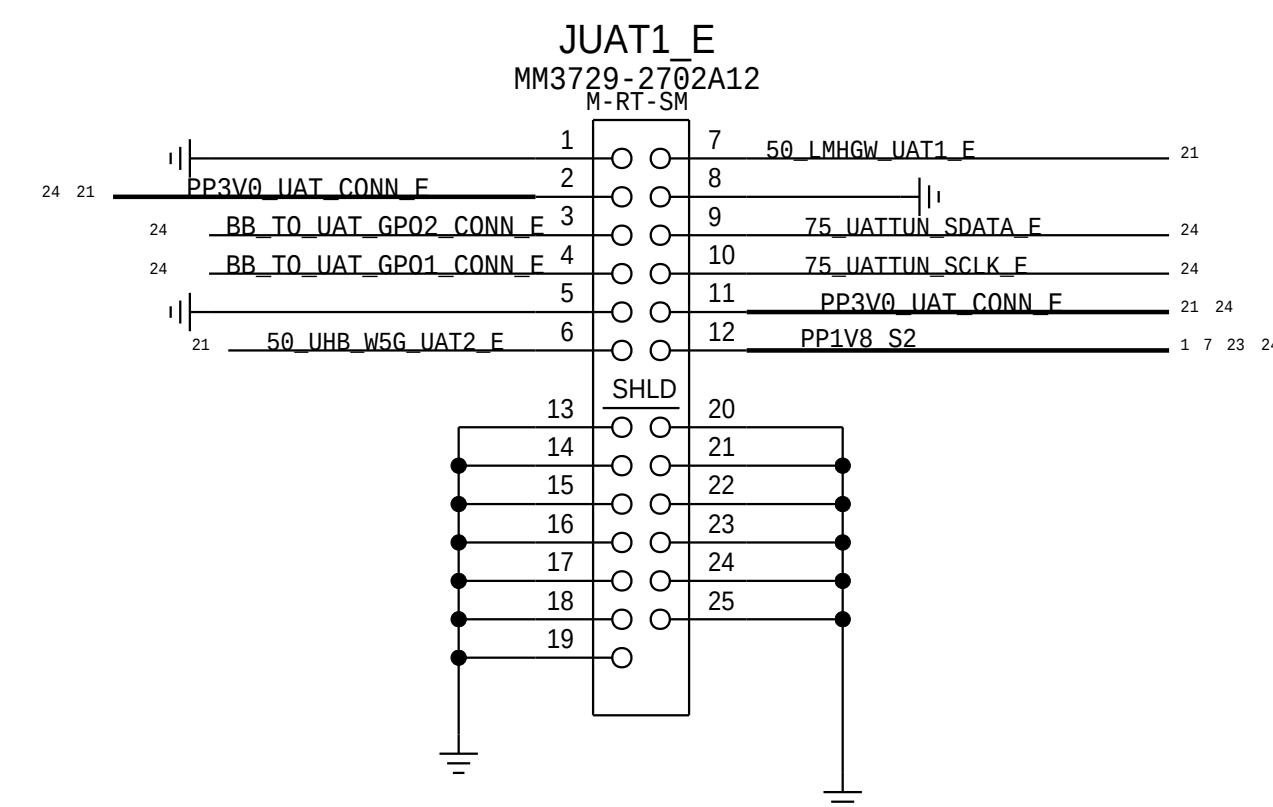
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46

GND

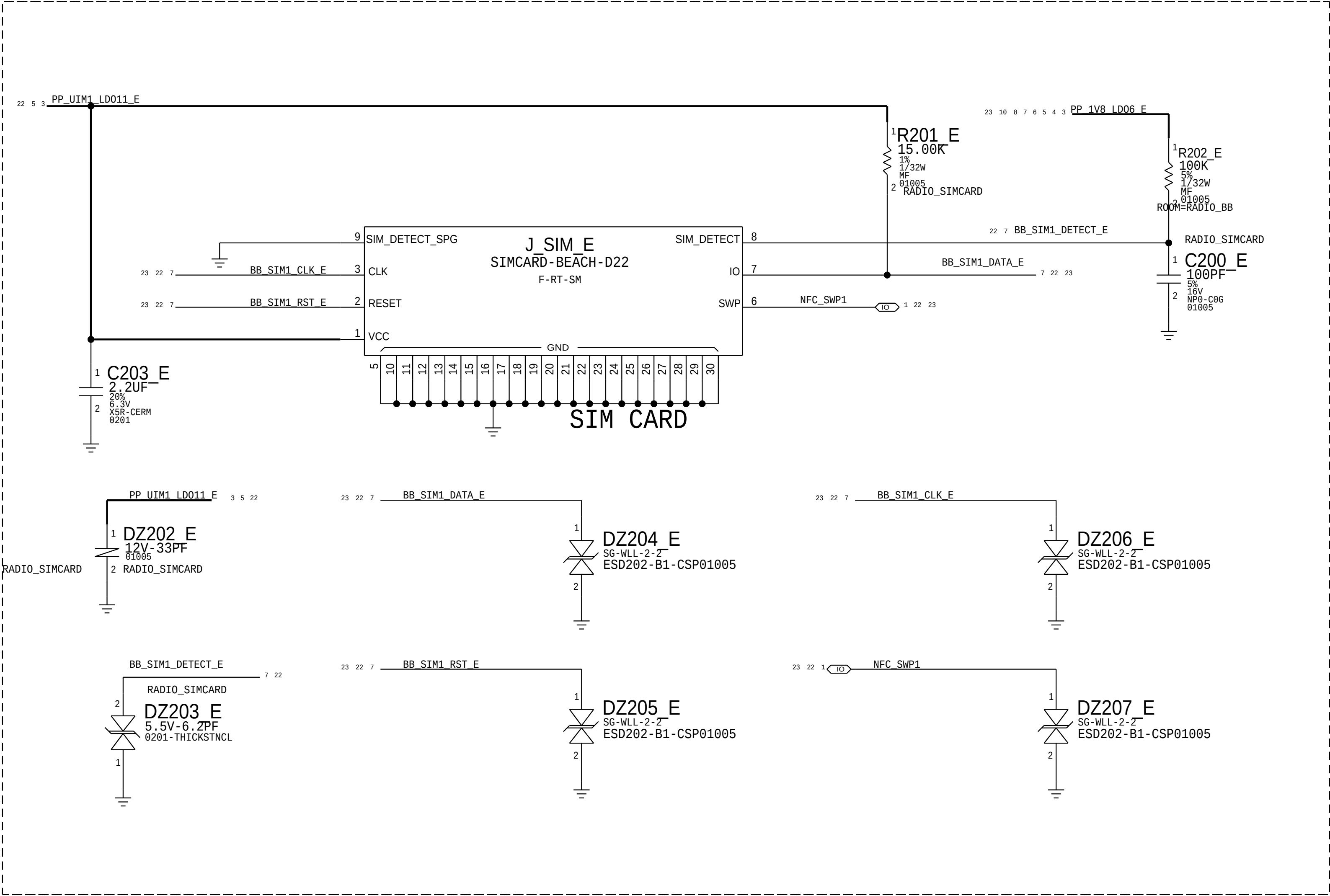
31 OF 41



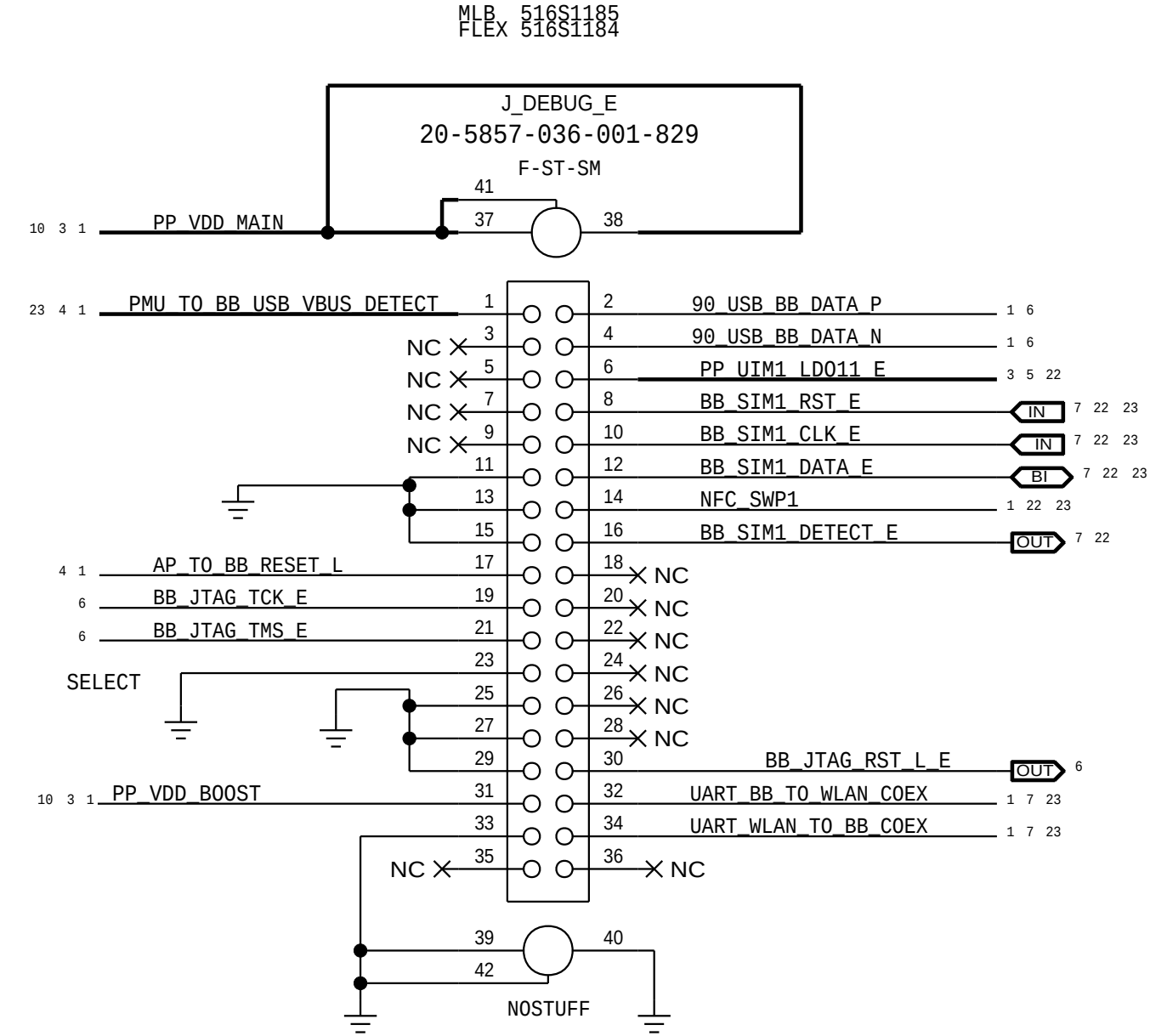


SIM CONN EUICC & DEBUG CONN

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE ONLY - NOT A CHANGE REQUEST



DEBUG CONNECTOR

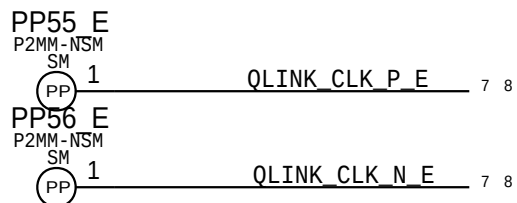
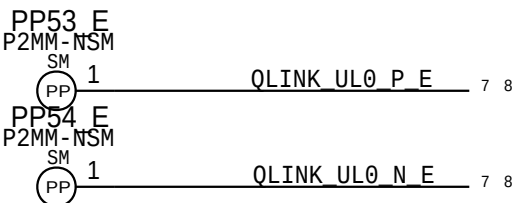




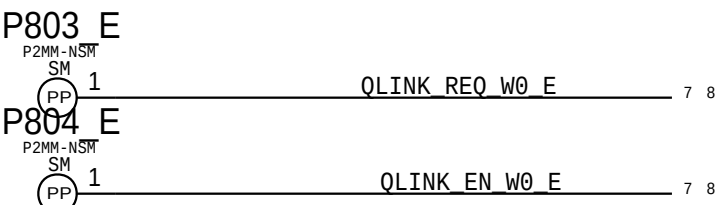
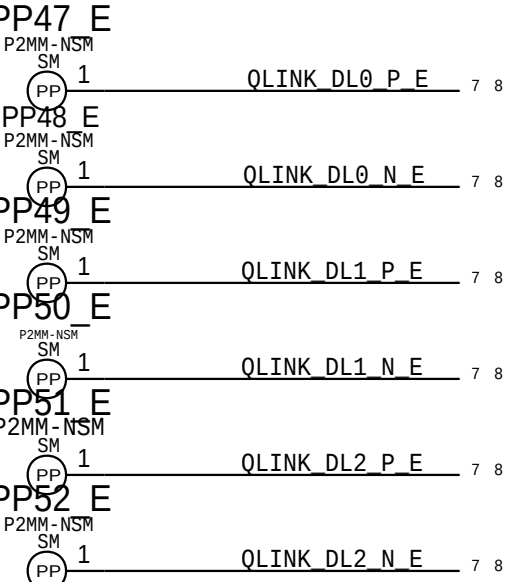
BB PROBE POINTS

QLINK

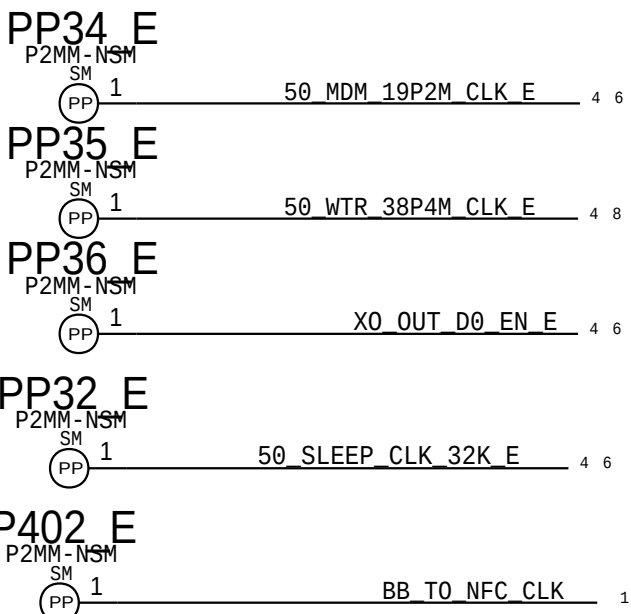
LAYOUT: VIA STUB AT WTR PINS



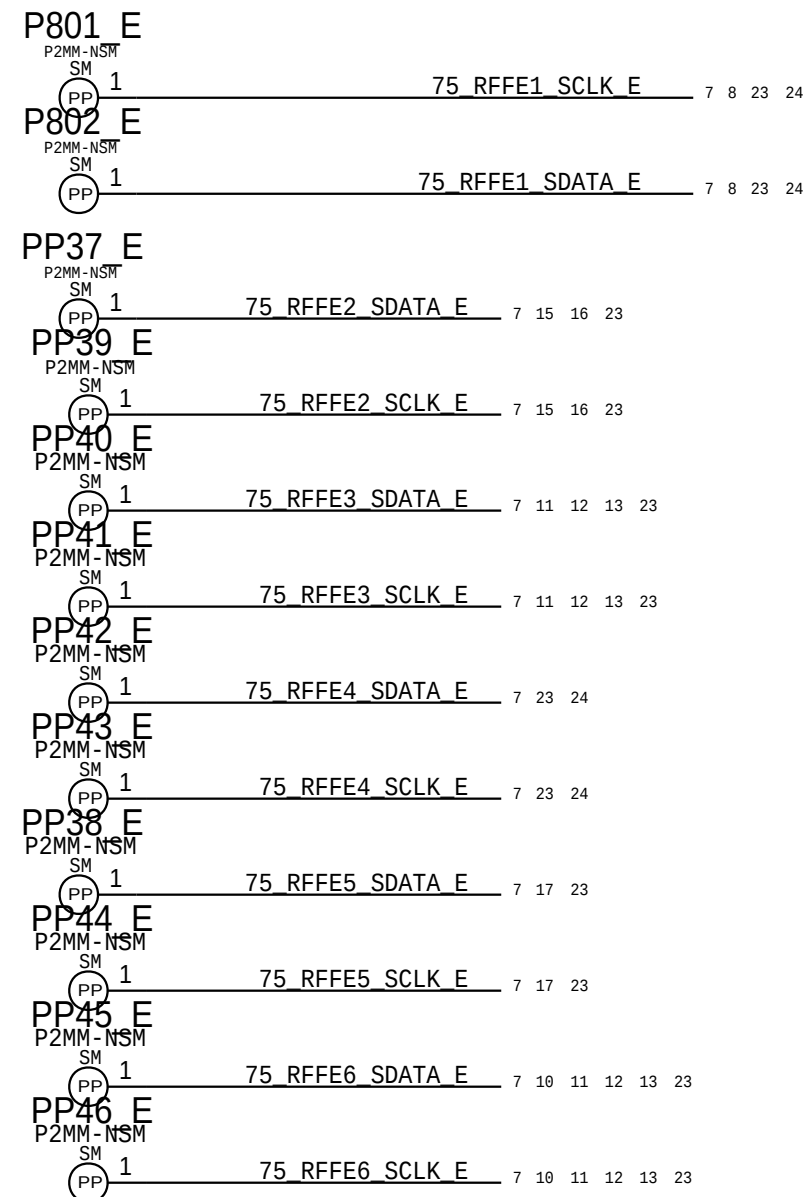
LAYOUT: VIA STUB AT BB PINS



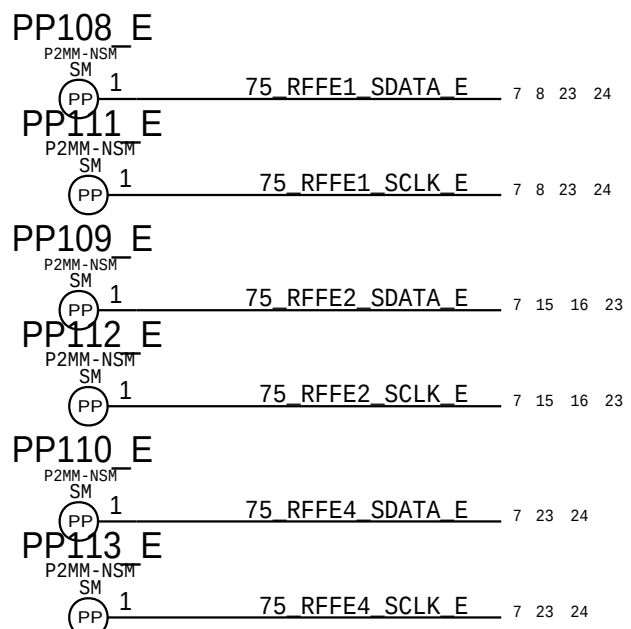
BASEBAND CLK



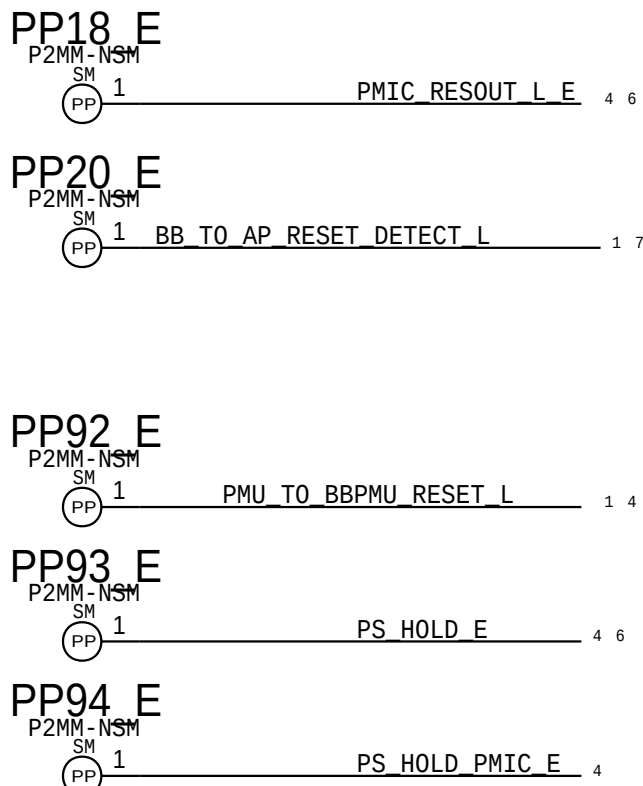
RFFE



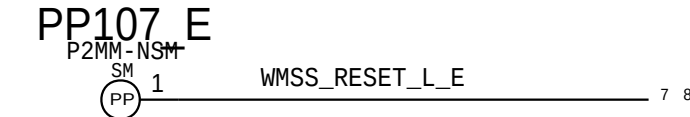
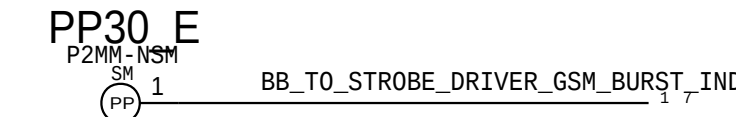
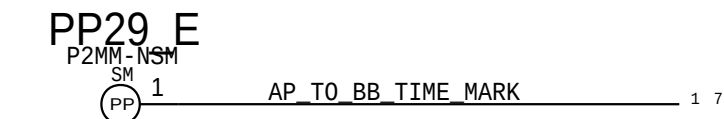
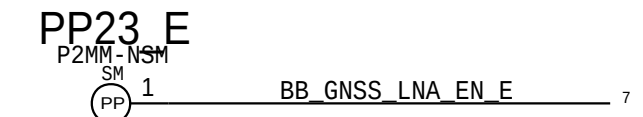
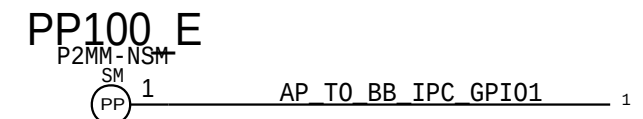
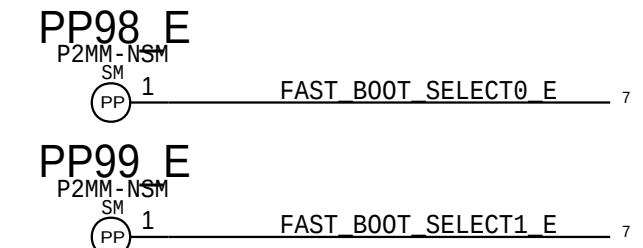
RFFE BB



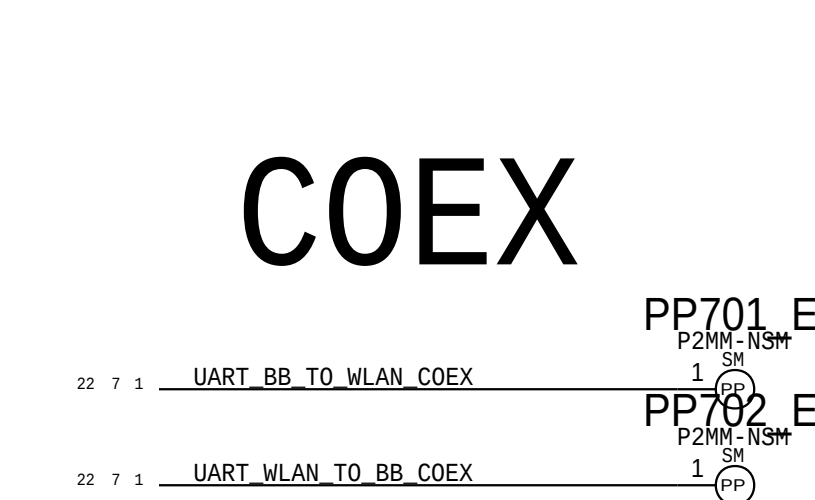
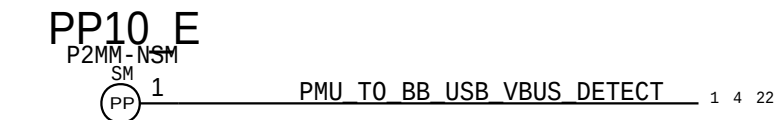
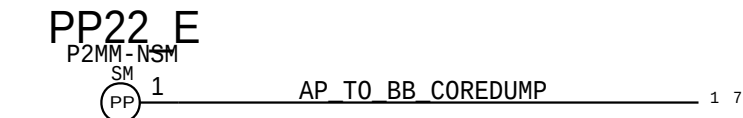
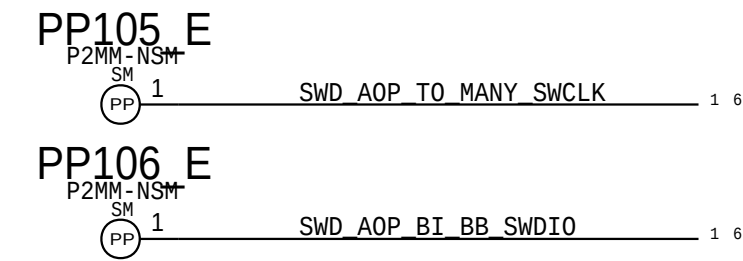
PWR SEQ



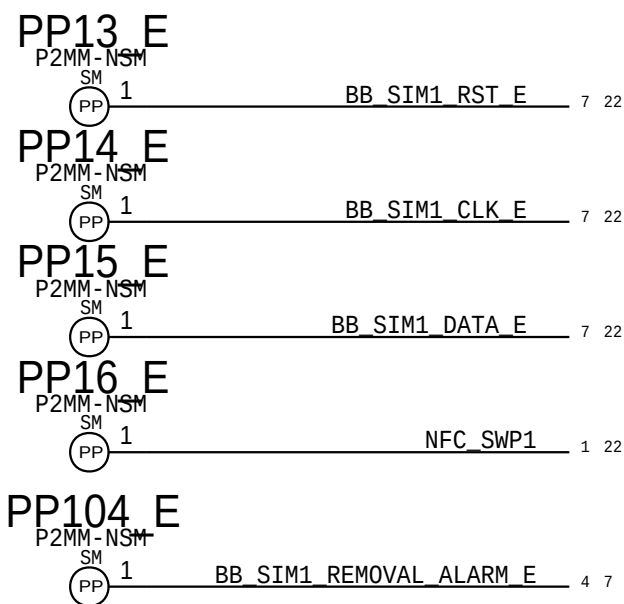
CTRL



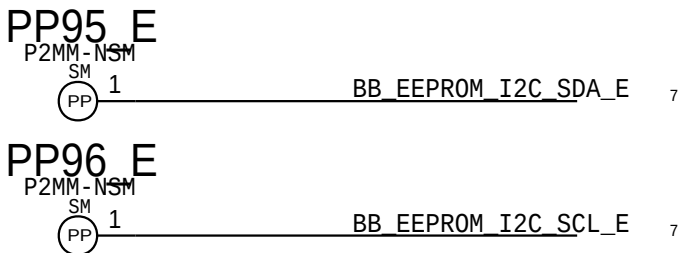
DEBUG



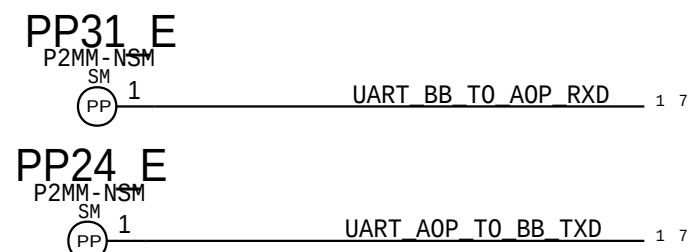
SIM



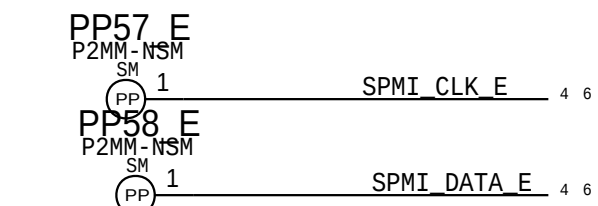
I2C



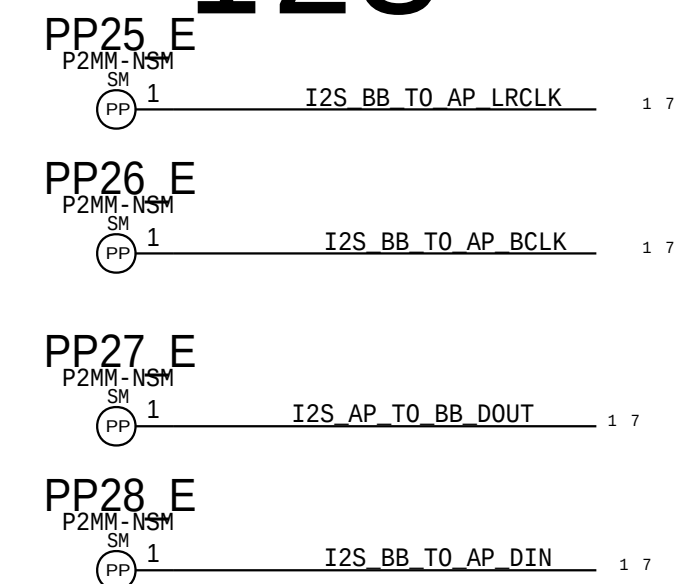
UART



SPMI

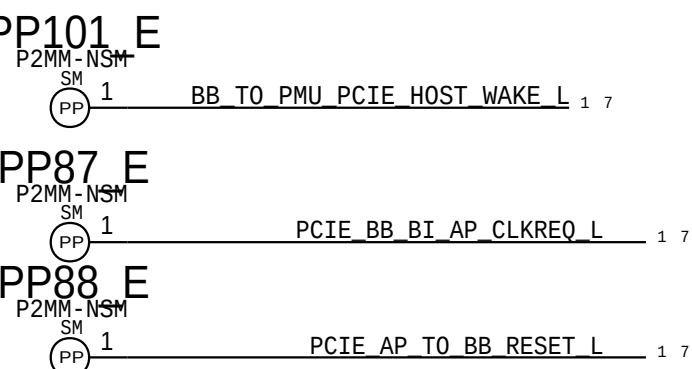


I2S

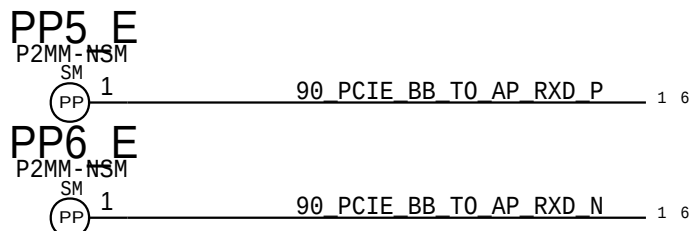


PCIE

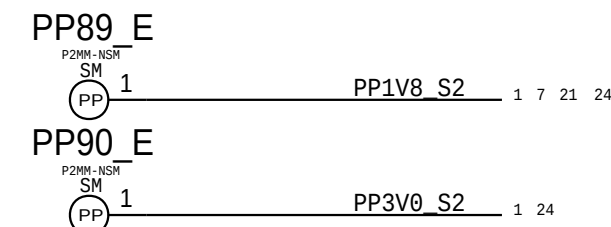
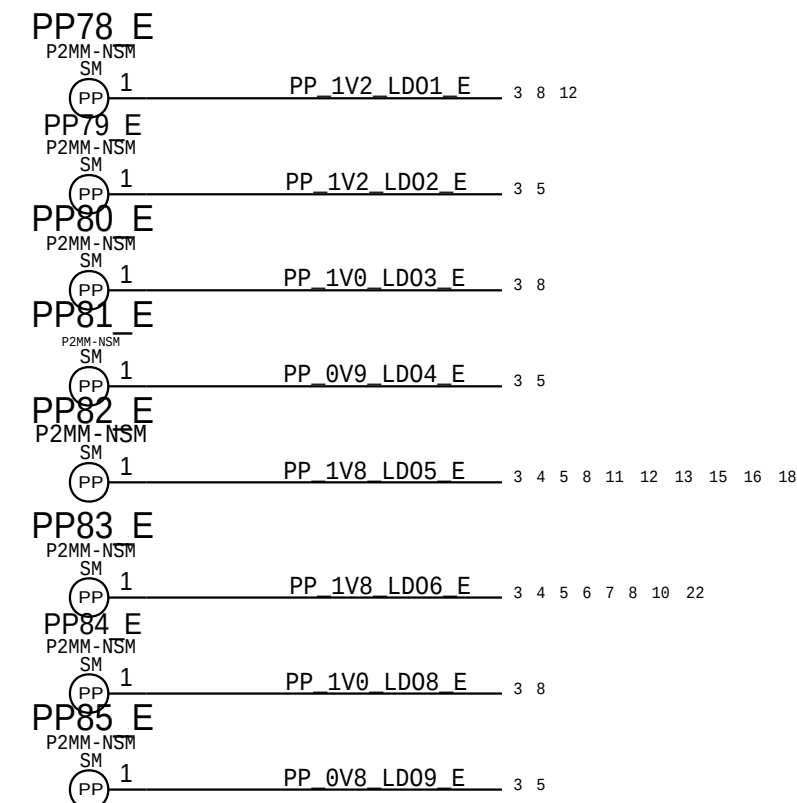
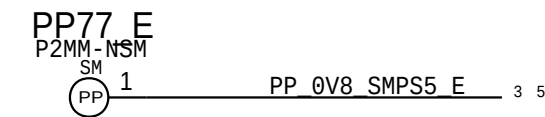
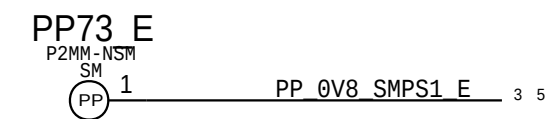
LAYOUT: VIA STUB AT BB PINS

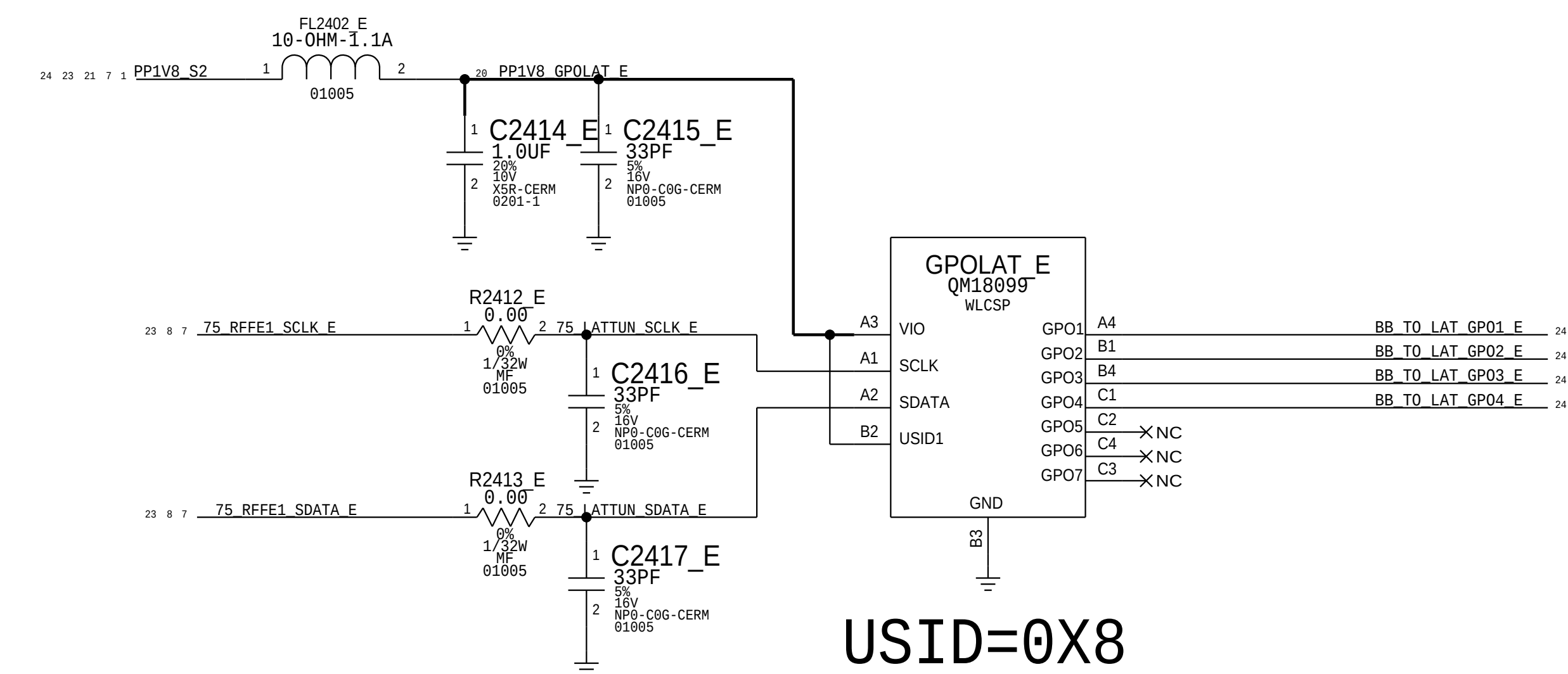


LAYOUT: VIA STUB AT PADS



PWR





PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	TYPICAL_CAP

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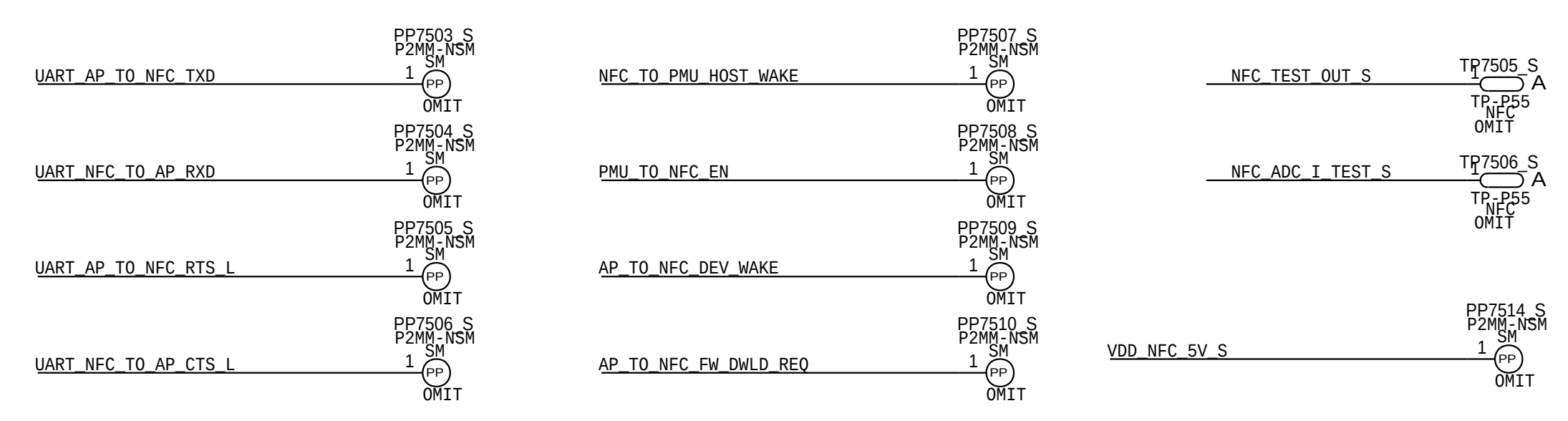
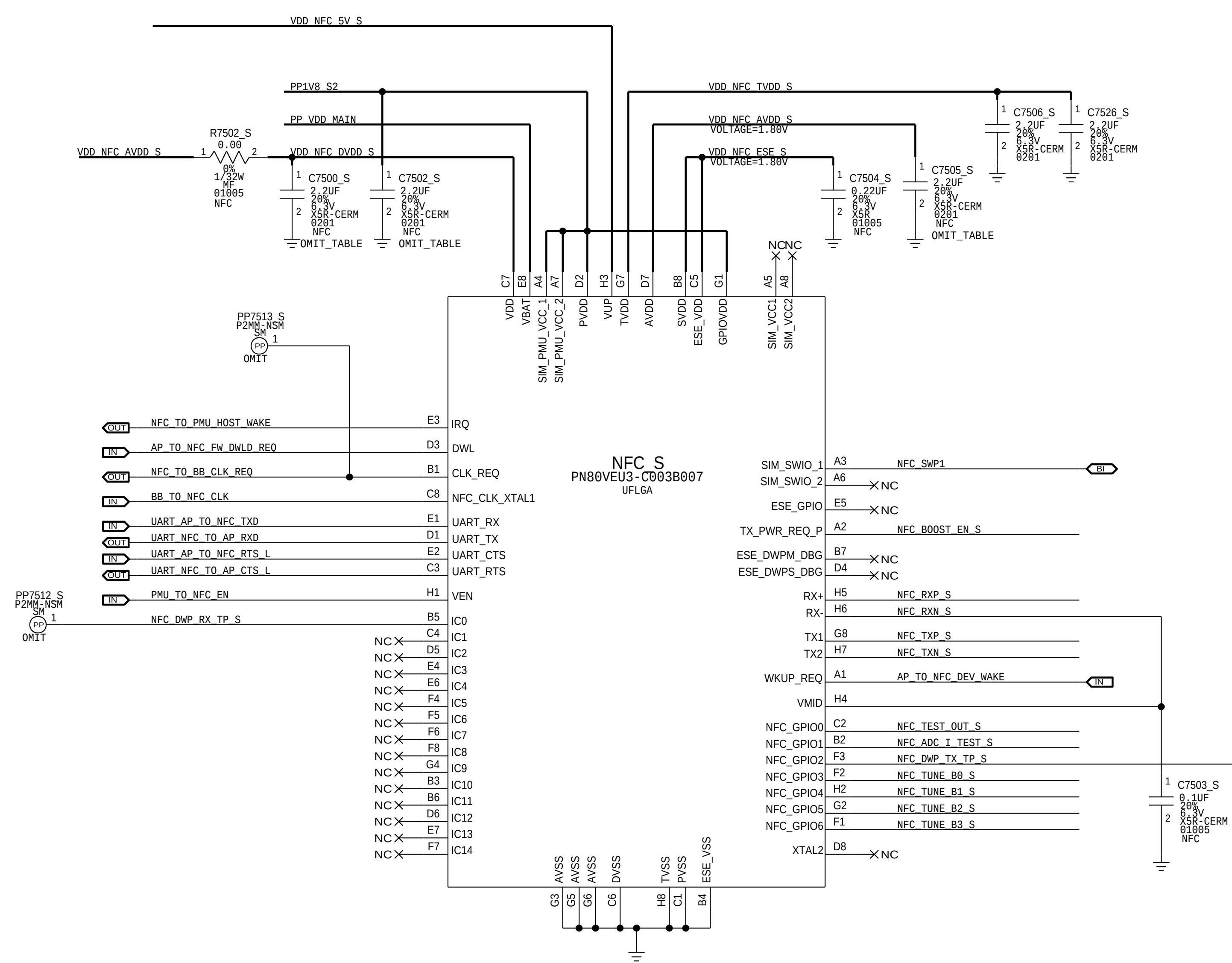
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2

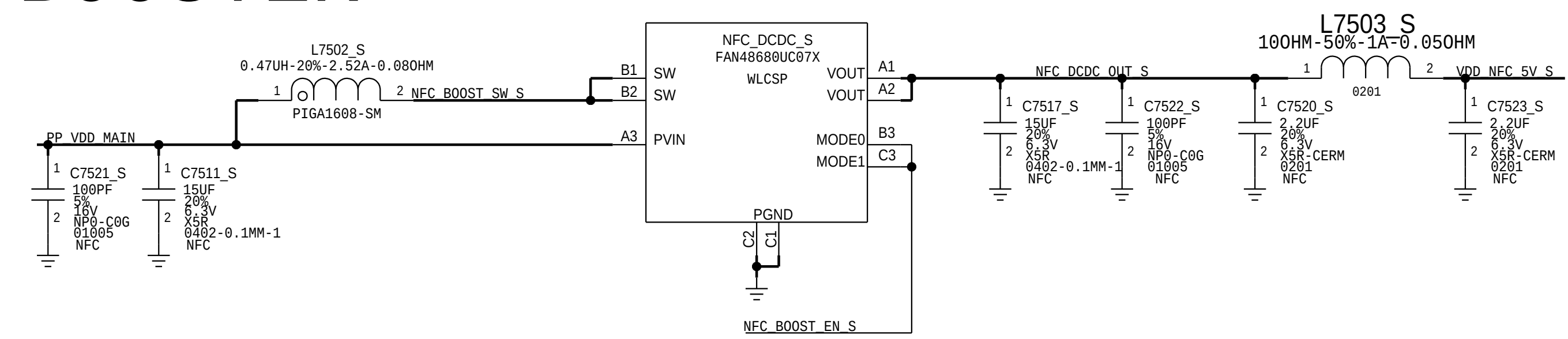
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STOCKHOLM

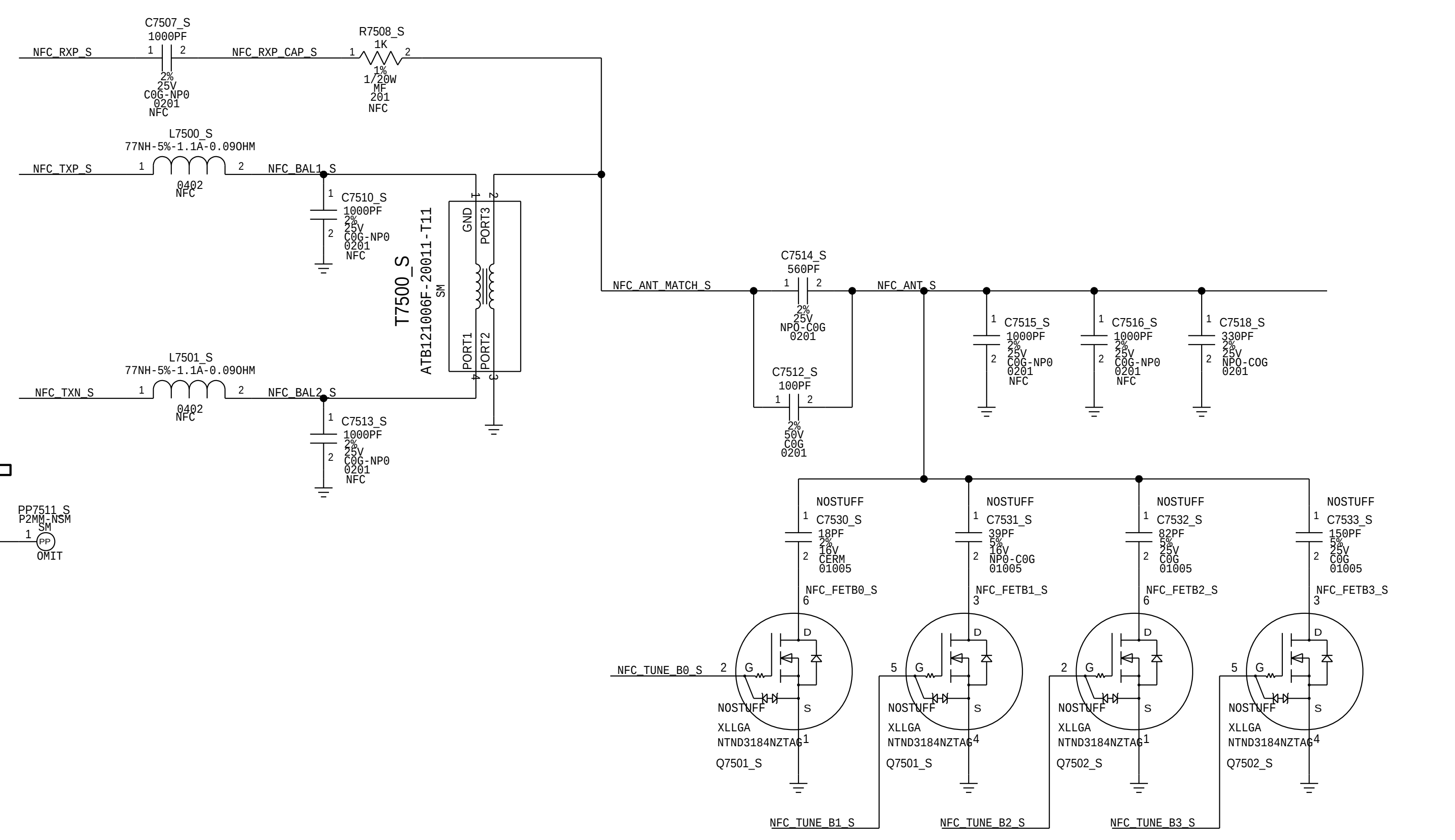
NFC CONTROLLER



5V BOOSTER



NFC FRONT END



CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSE ONLY - NOT A CHANGE REQUEST

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8

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3

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1

WIFI/BT

D

C

B

A

D

C

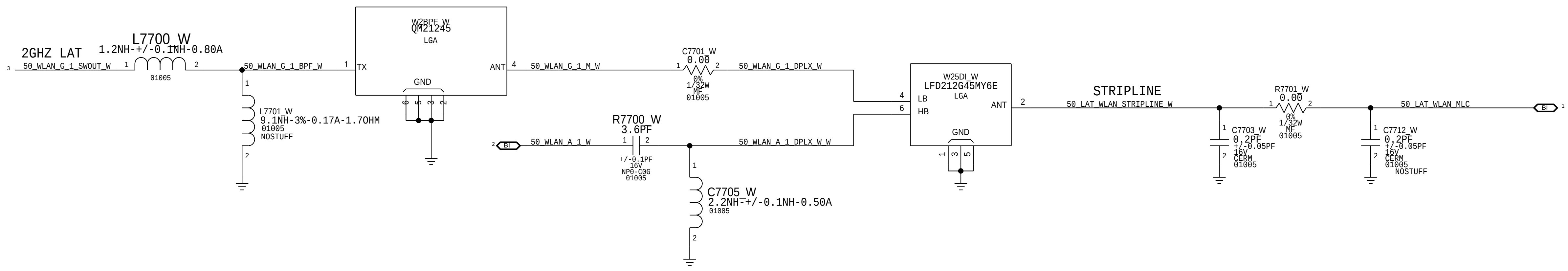
B

A

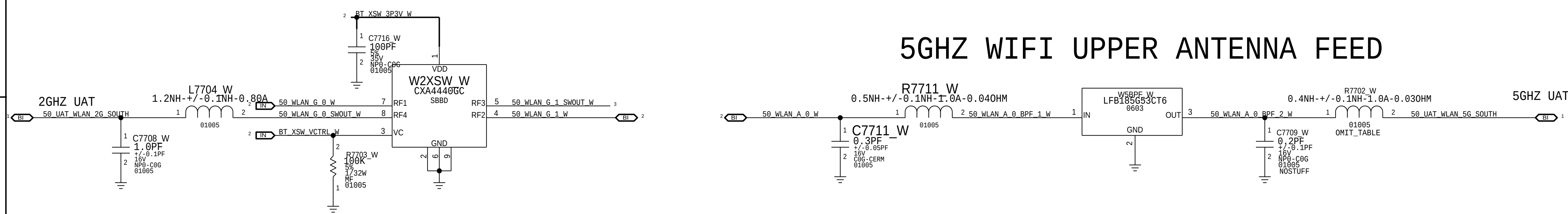


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
998-10376	998-10147	BOM_TABLE_ALTS	UWLAN_W	USI ES6.1 WIFI MODULE

WIFI LOWER ANTENNA FEED



5GHZ WIFI UPPER ANTENNA FEED



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
11750161	1	5GHZ C0 BPF OUTPUT MATCH	R7702_W	CRITICAL	D221
152S00498	1	5GHZ C0 BPF OUTPUT MATCH	R7702_W	CRITICAL	ROW
152S00498	1	5GHZ C0 BPF OUTPUT MATCH	R7702_W	CRITICAL	JPN